TENDRARA PROJECT

GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS

PLANT OPERATING MANUAL

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Contents

1 Introduction **1**
- 1.1 Purpose of the Manual . . . 1
- 1.2 Scope and Applicability . . . 2
- 1.3 Abbreviations and Definitions . . . 3
- 1.4 Units of Measurement . . . 5

2 Overview of the Tendrara Micro LNG Plant **6**
- 2.1 Process Plant . . . 6
- 2.2 PCS – Control System . . . 9
- 2.3 ESD - Emergency Shutdown System . . . 12
- 2.4 Communication Network . . . 15
- 2.5 Power Generation System . . . 16

3 Unit 200 - LNG Plant Inlet Unit **19**
- 3.1 Operating Summary . . . 19
- 3.2 General Arrangement and Risk Location . . . 20
- 3.3 Relevant Documents . . . 21
- 3.4 Process Description . . . 21
- 3.5 Technological Objects . . . 22
  - 3.5.1 Main Equipment List . . . 22
  - 3.5.2 Valves . . . 23
  - 3.5.3 Instrumentation . . . 23
  - 3.5.4 Junction Boxes . . . 24
  - 3.5.5 Controllers . . . 25
- 3.6 Operating Interface . . . 26
  - 3.6.1 PCS Operating Screens – Unit 200 . . . 26
  - 3.6.2 Critical Alarms and Trips . . . 28
  - 3.6.3 Unit Cause and Effect Matrix . . . 28

4 Unit 220 - LNG Storage, metering system and loading station **29**
- 4.1 Operating Summary . . . 29
- 4.2 General Arrangement and Risk Location . . . 30
- 4.3 Relevant Documents . . . 31
- 4.4 Process Description . . . 31
- 4.5 Technological Objects . . . 35
  - 4.5.1 Main Equipment List . . . 35
  - 4.5.2 Electrical Loads . . . 35
  - 4.5.3 Valves . . . 35
  - 4.5.4 Instrumentation . . . 38
  - 4.5.5 Junction Boxes . . . 40

4.5.6 Controllers . . . 40
4.6 Operating Interface . . . 42
4.6.1 PCS Operating Screens . . . 42
4.6.2 Critical Alarms and Trips . . . 45
4.6.3 Unit Cause and Effect Matrix . . . 46
4.7 Operation and Control . . . 47
4.7.1 Special Controllers XC-2212 and XC-2222 . . . 47
4.7.2 Loading Panel Operation . . . 49
4.7.3 LNG Transfer System . . . 50
4.7.4 Liquid Level . . . 52
4.7.5 Suction Line Cooling Down . . . 54
4.7.6 Discharge Cooling Down . . . 55
4.7.7 Truck Cooling Down . . . 56
4.7.8 Truck LNG Loading . . . 57
4.8 Troubleshooting . . . 60

5 Unit 310 - Gas Dehydration Unit (molecular sieve) 62
5.1 Operating Summary . . . 62
5.2 General Arrangement and Risk Location . . . 63
5.3 Relevant Documents . . . 64
5.4 Process Description . . . 64
5.5 Technological Objects . . . 67
5.5.1 Main Equipment List . . . 67
5.5.2 Electrical Loads . . . 67
5.5.3 Motors as Remote MCC . . . 68
5.5.4 Valves . . . 68
5.5.5 Instrumentation . . . 70
5.5.6 Junction Boxes . . . 71
5.5.7 Controllers . . . 71
5.6 Operating Interface . . . 73
5.6.1 PCS Operating Screens – Unit 310 . . . 73
5.6.2 Critical Alarms and Trips . . . 75
5.6.3 Unit Cause and Effect Matrix . . . 75

6 Unit 330 - Amine Sweetening Unit (CO2 removal) 76
6.1 Operating Summary . . . 76
6.2 General Arrangement and Risk Location . . . 77
6.3 Relevant Documents . . . 78
6.4 Process Description . . . 78
6.5 Technological Objects . . . 81
6.5.1 Main Equipment List . . . 81
6.5.2 Electrical Loads . . . 81
6.5.3 Motors as Remote MCC . . . 82
6.5.4 Valves . . . 83
6.5.5 Instrumentation . . . 84
6.5.6 Junction Boxes . . . 86
6.5.7 Controllers . . . 86

6.6 Operating Interface . . . 88
6.6.1 PCS Operating Screens – Unit 330 . . . 88
6.6.2 Critical Alarms and Trips . . . 90
6.6.3 Unit Cause and Effect Matrix . . . 90

7 Unit 340 - Refrigeration Unit (ammonia R1 loop) 92
7.1 Operating Summary . . . 92
7.2 General Arrangement and Risk Location . . . 93
7.3 Relevant Documents . . . 94
7.4 Process Description . . . 94
7.5 Technological Objects . . . 96
7.5.1 Main Equipment List . . . 96
7.5.2 Electrical Loads . . . 97
7.5.3 Motors as Remote MCC . . . 98
7.5.4 Valves . . . 101
7.5.5 Instrumentation . . . 102
7.5.6 Junction Boxes . . . 103
7.5.7 Controllers . . . 104
7.6 Operating Interface . . . 105
7.6.1 PCS Operating Screens – Unit 340 . . . 105
7.6.2 Critical Alarms and Trips . . . 110
7.6.3 Unit Cause and Effect Matrix . . . 111
7.7 Operation and Control . . . 112
7.7.1 Compressor Capacity Control . . . 112
7.7.2 Ammonia Receiver Pressure Control . . . 113
7.7.3 Compressor Panel Operation . . . 114
7.7.4 Ammonia Refrigeration Loop R1 . . . 116
7.7.5 Natural Gas Precooler Level Control LIC-3111 . . . 118
7.7.6 Ammonia MR Cooler Level Control LIC-3540 . . . 118
7.7.7 Lube Oil System . . . 119
7.7.8 Magnetic Coupling . . . 122
7.7.9 Compressor Start-Up Permissives . . . 123
7.7.10 Compressor Start-Up Sequence . . . 123
7.7.11 Compressor Shutdown Sequence . . . 124
7.8 Troubleshooting . . . 125

8 Unit 350 - Liquefaction Unit (MR compressors, cold box, NRU). 128
8.1 Operating Summary . . . 128
8.2 General Arrangement and Risk Location . . . 129
8.3 Relevant Documents . . . 130
8.4 Process Description . . . 130
8.5 Technological Objects . . . 134
8.5.1 Main Equipment List . . . 134
8.5.2 Electrical Loads . . . 136
8.5.3 Motors as Remote MCC . . . 137
8.5.4 Valves . . . 139
8.5.5 Instrumentation . . . 144

8.5.6 Junction Boxes 148
8.5.7 Controllers 149
8.6 Operating Interface 150
8.6.1 PCS Operating Screens – Unit 350 150
8.6.2 Critical Alarms and Trips 159
8.6.3 Unit Cause and Effect Matrix 161
8.7 Operation and Control 165
8.7.1 Compressor Capacity Control 165
8.7.2 MR Compression System Capacity Control 166
8.7.3 Compressor Panel Operation 167
8.7.4 Mixed Refrigerant Refrigeration Loop 170
8.7.5 Lube Oil System 172
8.7.6 Magnetic Coupling 174
8.7.7 Compressor Start-Up Permissives 175
8.7.8 Compressor Start-Up Sequence 176
8.7.9 Compressor Shutdown Sequence 177
8.8 Troubleshooting 178

9 Unit 360 - Flash Gas / BOG Recovery Unit (PK-361 and PK-362). 182
9.1 Operating Summary 182
9.2 General Arrangement and Risk Location 183
9.3 Relevant Documents 184
9.4 Process Description 184
9.5 Technological Objects 186
9.5.1 Main Equipment List 186
9.5.2 Electrical Loads 186
9.5.3 Motors as Remote MCC 187
9.5.4 Valves 187
9.5.5 Instrumentation 188
9.5.6 Junction Boxes 190
9.5.7 Controllers 190
9.6 Operating Interface 191
9.6.1 PCS Operating Screens – Unit 360 191
9.6.2 Critical Alarms and Trips 193
9.6.3 Unit Cause and Effect Matrix 194
9.7 Operation and Control 195
9.7.1 Compressor Capacity Control 195
9.7.2 BOG Compression System Capacity Control 196
9.7.3 BOG Recycle Compressor Capacity Control 197
9.7.4 Compressor Panel Operation 199
9.7.5 Lub Oil Systems (PK-361 and PK-362) 200
9.7.6 Magnetic Coupling 203
9.7.7 Compressor Start-Up Permissives 204
9.7.8 Compressor Start-Up Sequence 204
9.7.9 Compressor Shutdown Sequence 205
9.8 Troubleshooting 206

10 Unit 370 - Vaporizer Unit (E-371, E-372) 209
10.1 Operating Summary 209
10.2 General Arrangement and Risk Location 210
10.3 Relevant Documents 211
10.4 Process Description 211
10.5 Technological Objects 212
10.5.1 Main Equipment List 212
10.5.2 Electrical Loads 213
10.5.3 Valves 214
10.5.4 Instrumentation 214
10.5.5 Junction Boxes 215
10.5.6 Controllers 215
10.6 Operating Interface 217
10.6.1 PCS Operating Screens – Unit 370 217
10.6.2 Critical Alarms and Trips 218
10.6.3 Unit Cause and Effect Matrix 218

11 Plant Operating Procedures 219
11.1 Black Start 219
11.2 Normal Start-up Sequence 222
11.3 Normal Operation Tasks 222
11.4 Normal Shutdown 222

1 Introduction

Safety Notices and Hazard Symbols

Descriptions with this symbol contain warnings about hazards that can result in personal injury, or warnings about factors which could result in damage to the equipment if ignored.

Failure to comply may result in:

- Hazards to the safety of operators
- Breach of warranty
- Disclaimer of manufacturer's responsibility

The descriptions with this symbol concern explosion risk and have value for the equipment specifically produced to operate in the presence of potentially explosive atmospheres.

The non-observance of these instructions may lead to serious health risks to people and damage objects nearby the equipment described in this manual.

1.1 Purpose of the Manual

The purpose of this Plant Operating Manual is to provide operators, maintenance personnel, and supervisory staff with a comprehensive and authoritative reference describing the safe, reliable, and efficient operation of the Tendrara Micro LNG Plant. This manual consolidates the operational philosophy, process descriptions, control strategies, and procedures required throughout the full lifecycle of the facility, including:

- **Commissioning and Start-up:** Initial system energization, process conditioning, functional validation, and controlled start-up.
- **Normal Operation:** Daily operational tasks, monitoring requirements, process adjustments, and routine optimization to maintain production stability and product quality.
- **Shutdown and Emergency Response:** Controlled depressurization, system isolation,

and response actions to abnormal or hazardous conditions, aligned with the project's Cause & Effect matrix and safety design intent.

The manual ensures consistent operational practices aligned with the design basis and vendor documentation, and serves as the primary reference for training and competency development of plant personnel.

1.2 Scope and Applicability

This manual applies to the Tendrara Micro LNG Facility as a whole, including all process units, utility systems, and supporting infrastructure. The plant is composed of the following main areas:

Table 1.1 — List of Plant Units — Tendrara LNG Facility

Unit	Description
200	LNG Plant Inlet Unit.
220	LNG Storage, metering system and loading station.
230	Flare System (hot flare, cold flare, ammonia flare, client flare).
310	Gas Dehydration Unit (molecular sieve).
330	Amine Sweetening Unit (CO_2 removal).
340	Refrigeration Unit (ammonia R1 loop).
350	Liquefaction Unit (MR compressors, cold box, NRU).
360	Flash Gas / BOG Recovery Unit (PK-361 and PK-362).
370	Vaporizer Unit (E-371, E-372).
410	Hot Oil Package (fired heaters B-411A/B, pumps, expansion tank).
430	Inline Analyzers Unit.
480	Power Generator Unit (Jenbacher engines).
490	Secondary Power Generation Unit.
500	Instrument Air System — Screw compressors PK-501A/B/C.
600	Nitrogen Generator Package — PSA N_2 + V-602 receiver.
530	Process Water Treatment Package (coalescers, filters, degasser).
540	Raw Water Treatment Package.
550	Closed Drain System (V-551, pumps P-551A/B).
700	Fire & Gas Detection System.
730	Fire Fighting System.
900	Cabins (electrical, control, buildings).
1000	Living Camp / Accommodation.
120	Inlet Manifold for TE-6 / TE-7 wells (pipeline tie-in).

Operating Modes Covered

- Cold start-up / warm start-up
- Normal operation
- Temporary modes such as regeneration cycles, recycle operation, turndown conditions

- Shutdown (normal, LSD, PSD, and PSD with depressurization)
- Emergency operating conditions and utility failures

The manual is applicable to all personnel involved in Operations, Maintenance, HSE, and related activities.

1.3 Abbreviations and Definitions

Table 1.2 — Abbreviations and Definitions

Abbreviation	Definition
BDV	Blowdown valve; pneumatically actuated, typically with restriction orifice for depressurizing a system to flare in emergencies.
BOG	Boil-off gas.
CRA	Corrosion Resistant Alloy.
CK	Operator checkpoint
CSC	Car sealed closed.
CS	Carbon Steel materials (suitable to MDMT –28°C).
CSO	Car sealed open.
ESD	Emergency Shutdown System.
FAH	Flow alarm high.
FAL	Flow alarm low.
HC	Hydrocarbon.
HMB	Heat and Material Balance.
ITF	Italfluid Geoenergy S.R.L.
LAH	Level alarm high.
LAHH	Level alarm high high.
LAL	Level alarm low.
LALL	Level alarm low low.
LC	Locked closed.
LG	Level gauge.
LNG	Liquefied natural gas.
LO	Locked open.
LT	Level transmitter.
LTCS	Low Temperature Carbon Steel (suitable to MDMT –45°C).
MDMT	Minimum Design Metal Temperature.
MR	Mixed refrigerant.
MSD	Material Selection Diagram.
NLG	Natural gas liquids.
NRU	Nitrogen Rejection Unit.
OC	Operator command (field or PCS)
PAH	Pressure alarm high.
PAHH	Pressure alarm high high.
PAL	Pressure alarm low.
PALL	Pressure alarm low low.
PCS	Plant Control System; PLC-based control and operation system of the facility.
PDG	Pressure differential gauge.

Abbreviation	Definition
P&ID	Piping and Instrumentation Diagram.
PFD	Process Flow Diagram.
PFHE	Plate and Fin Heat Exchanger.
PI	Pressure indicator.
PSD	Process Shutdown System.
PSV	Pressure safety valve.
SC	System command (PCS or ESD).
SDV	Shutdown valve.
TAD	Temperature alarm high.
TAHH	Temperature alarm high high.
TAL	Temperature alarm low.
TALL	Temperature alarm low low.
TAM	Technical America Inc.
TRV	Thermal relief valve.
UFD	Utility Flow Diagram.

1.4 Units of Measurement

Table 1.3 — Units of Measurement Used in This Manual

Quantity	Unit(s)
Area	m^2
Concentration	ppm_{mol}, ppmv, ppmw, ppbw, mg/L, %wt
Density	kg/m^3
Enthalpy	kJ/kg, kcal/kg
Gas and vapour flow	Sm^3/d, Sm^3/h, Nm^3/h, am^3/h
Gas production capacity	$MMSm^3/d$, MMSCFD
Heat	kW, kcal/h, BTU/h
Heat transfer coefficient	$W/(m^2 \cdot {}^\circ C)$, $kcal/(h \cdot {}^\circ C \cdot m^2)$, $Btu/(h \cdot {}^\circ F \cdot ft^2)$
Length	km, m, mm
Liquid production capacity	m^3/d
Mass, weight	kg, ton
Mass / molar flow	kmol/h, m^3/h
Molecular weight	kg/kmol, g/mol
Power	MW, kW
Pressure (absolute)	bara
Pressure (gauge)	barg
Pressure drop	bar
Temperature	$^\circ C$
Viscosity (dynamic)	cP
Volume	m^3, bbl

For the definition of reference conditions used in the project:

- **Normal conditions:** $p_N = 1.013$ bar_a and $T_N = 0\ ^\circ C$.
- **Standard conditions:** $p_S = 14.696$ $psi_a \approx 1.013$ bar_a and $T_S = 60\ ^\circ F \approx 15.5\ ^\circ C$.

2 Overview of the Tendrara Micro LNG Plant

2.1 Process Plant

The Tendrara Micro LNG Plant is a modular, skid-mounted gas treatment and liquefaction facility designed to process raw natural gas from wells TE-6 and TE-7 and produce specification-grade liquefied natural gas (LNG) for road tanker export. The plant has been specifically engineered to operate reliably in a remote field environment, combining a compact footprint with a high level of modularization, factory-tested process skids, and simplified field integration.

From a process standpoint, the facility implements a fully integrated pretreatment, liquefaction, and LNG handling scheme. Raw natural gas is progressively conditioned through dedicated separation, sweetening, dehydration, and refrigeration stages prior to cryogenic processing. Each process step is designed to protect downstream equipment, maintain product quality, and ensure compliance with cryogenic operating limits. The plant layout and process sequencing are optimized to maintain continuous operation while allowing safe and controlled shutdowns during abnormal or emergency conditions.

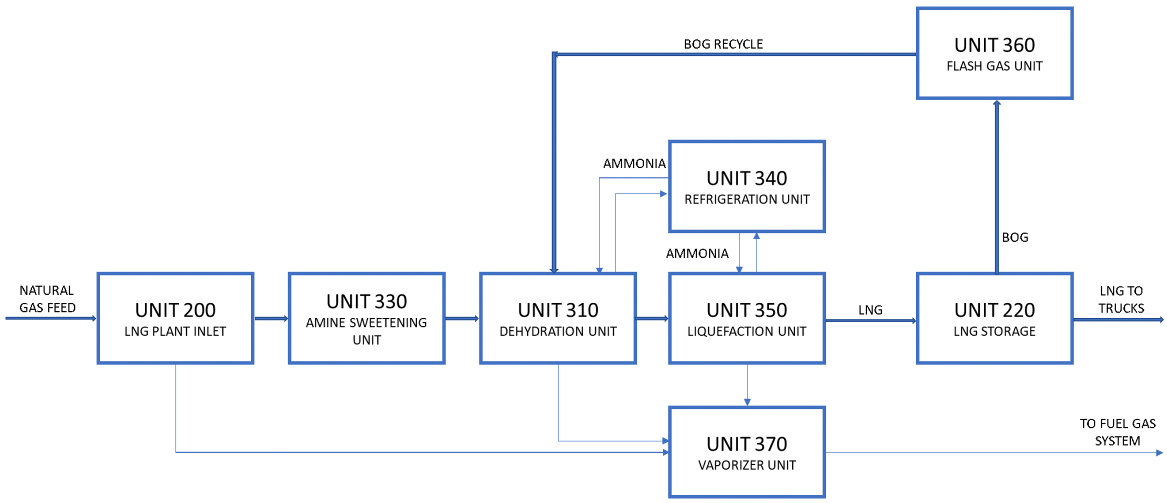


Figure 2.1: Process Overview Diagram — Tendrara LNG Facility

The plant is composed by the following main areas:

- **Unit 200 — Inlet Separation and Preheating:** Raw gas from the wells enters the facility through the inlet battery limit and is routed to a three-phase separator where free liquids and entrained solids are removed. Separated condensate and produced water are routed to their respective handling systems, while the gas phase is conditioned using hot oil heating. This temperature control step is essential to prevent hydrate formation and ensure stable operation of downstream acid gas removal and dehydration units.

- **Unit 330 — Amine Sweetening:** Acid gas removal is achieved through a regenerable amine absorption system designed primarily for CO removal. This unit ensures that the treated gas meets cryogenic specifications by reducing CO content to levels that prevent freezing in the low-temperature sections of the plant. Regeneration and solvent management are fully integrated into the process, with utilities such as hot oil supplied from centralized systems.
- **Unit 310 — Molecular-Sieve Dehydration:** Following sweetening, the gas is dehydrated to cryogenic quality using molecular sieve adsorption beds operating in a cyclic regeneration mode. The dehydration unit reduces the water content of the gas to typically below 1 ppmv, eliminating the risk of hydrate formation, ice deposition, or plugging in heat exchangers and cold boxes. Regeneration gas management is integrated with the overall fuel and flash gas handling philosophy of the plant.
- **Unit 340 — Refrigeration / Precooling:** A single-stage ammonia refrigeration system provides controlled precooling of the dry, sweet gas prior to liquefaction. This step improves the thermal efficiency of the liquefaction process and stabilizes inlet conditions to the mixed-refrigerant system. The refrigeration unit is designed as a closed-loop system, with ammonia flaring limited to emergency conditions only.
- **Unit 350 — Liquefaction and Nitrogen Rejection:** Liquefaction is achieved using a mixed-refrigerant (MR) cycle that cools the treated gas to cryogenic temperatures within the cold box. An integrated Nitrogen Rejection Unit (NRU) removes excess nitrogen dissolved in the LNG, ensuring that the final product meets sales specifications with controlled nitrogen content. The liquefaction and NRU systems are designed to operate in coordination with upstream and downstream units to maintain stable LNG production.
- **Units 360/370 — Flash Gas and BOG Compression:** Flash gas generated during pressure letdown and boil-off gas (BOG) produced in storage and loading operations are recovered through dedicated compression systems. These units automatically balance the distribution of recovered gas between the fuel gas network and process recycle streams. Excess flash gas can be routed upstream for reuse or employed as regeneration gas for the dehydration unit, significantly reducing the need for routine flaring during normal plant operation.
- **Unit 220 — LNG Storage and Truck Loading:** Liquefied natural gas is routed to a low-pressure atmospheric cryogenic storage tank designed in accordance with applicable LNG standards. LNG is subsequently transferred to road tankers via metered loading skids, ensuring accurate custody transfer and controlled loading operations. LNG flow rate and quality are continuously monitored to verify compliance with contractual and operational requirements.

All process units are fully integrated with plant-wide utility, control, and safety systems. Supporting systems include hot oil heating, instrument air and nitrogen generation, fuel gas conditioning, diesel and power generation systems, water treatment facilities, flare and blowdown systems, and comprehensive fire and gas detection. The interaction between process units and auxiliary systems is designed to ensure safe, automated, and coordinated operation under normal, upset, and emergency scenarios, with clear functional separation between process control and safety shutdown systems.

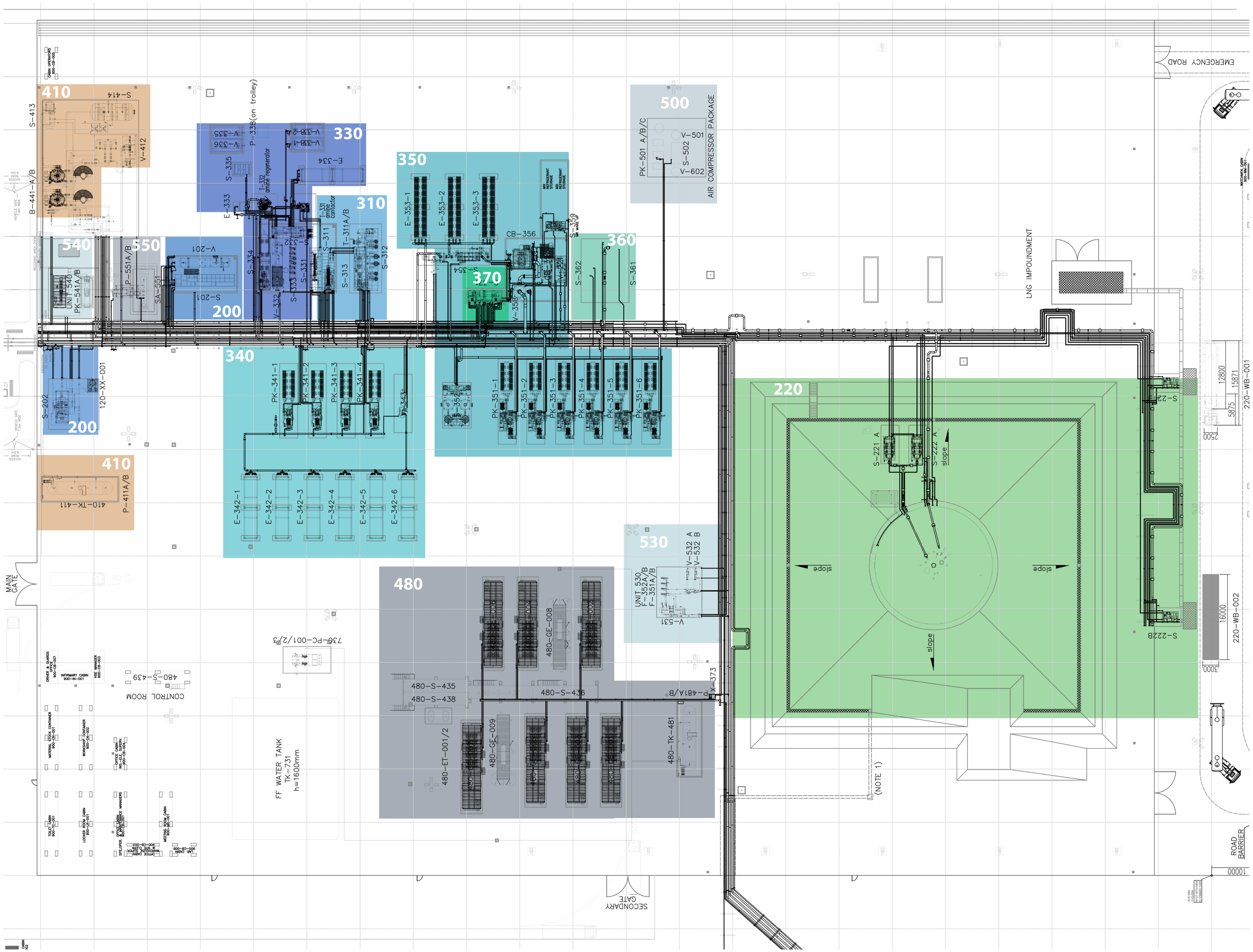


Figure 2.2: Plant General Arrangemenrt

2.2 PCS – Control System

The Plant Control System (PCS) is implemented on a Siemens S7-1500 PLC platform engineered using TIA Portal V20 and designed to provide reliable, deterministic, and high performance control of all normal operating functions of the LNG facility. The PCS constitutes the primary automation layer for continuous process control, equipment sequencing, interlocks, monitoring, and operator interaction, ensuring safe and efficient plant operation across all operating modes.

The overall automation architecture is structured into two fully integrated but functionally segregated layers: the Process Control System (PCS), which governs normal plant operation, and a dedicated Emergency Shutdown System (ESD), implemented using S7-1500F safety PLC technology in accordance with IEC functional safety requirements. High system availability is achieved through a fully redundant communication architecture and a clear separation between control and safety functions.

All field instrumentation, electrical systems, and packaged-unit controls are interfaced using a PROFINET IO–based network architecture. The plant communication network is implemented in dual fiber-optic PROFINET rings, providing redundant communication paths between the PCS, remote I/O stations, and skid-mounted systems. This configuration ensures communication continuity in the event of a single cable or network device failure and represents a significant enhancement in availability and robustness compared to traditional star or single-ring topologies.

ET 200SP remote I/O panels are installed on each modular skid and distributed throughout the plant to interface directly with local instruments, on–off valves, control valves, analyzers, and rotating equipment. This distributed I/O philosophy minimizes field wiring, reduces installation and commissioning effort, and improves maintainability and troubleshooting during plant operation.

The PCS interfaces with intelligent electrical and mechanical systems through PROFINET as the primary communication protocol. Key integrations include the Power Generation System, the Hot Oil Unit, and the Emergency Shutdown System (ESD), all connected via PROFINET. This unified communication strategy establishes PROFINET as the backbone of the plant automation architecture, positioning the facility at a state-of-the-art technological level in terms of performance, diagnostics, and system integration.

Control application development is based on a standardized Process Library approach, ensuring modular, reusable, and fully structured control logic comparable to a DCS-style implementation. The use of process objects, standardized function blocks, and unified engineering conventions guarantees high-quality programming, consistent behavior across all units, and simplified long-term maintenance. This approach enables advanced control strategies, clear separation of concerns, and predictable system behavior during both normal operation and abnormal conditions.

Supervisory control and operational monitoring are implemented through a SIMATIC WinCC SCADA system deployed in a client–server architecture. A dedicated WinCC Server hosts the process database, alarm and event management system, historian, and graphical runtime services. Operator and maintenance access is provided through fixed control room stations and ATEX-certified mobile field tablets connected via the plant's industrial wireless network, allowing secure mobile access to real-time process data.

The SCADA system is developed in accordance with High-Performance HMI principles using standardized faceplates and object libraries aligned with the PCS process library. Alarm management follows a structured philosophy with clear prioritization, classification, and time-stamped event logging, ensuring effective operator response and full traceability of operational events.

A dedicated historian system is implemented to collect, store, and manage long-term process data. Historical information is made available to the Client's information systems and is prepared for integration with cloud-based platforms, enabling advanced analytics, reporting, and performance monitoring beyond the control domain. This architecture supports both operational optimization and corporate-level data utilization.

The PCS architecture is designed with scalability, cybersecurity, and lifecycle support in mind. Network segmentation, controlled access levels, and centralized configuration ensure secure and maintainable operation, while the modular PLC, I/O, and software structure allows future plant expansions or modifications with minimal impact on existing operations.

Before operating any equipment described in this manual, operators shall ensure that they are familiar with document **21044-AU-HMI-00002 – HMI Philosophy**. This document describes the operating modes of the system and their graphical representation within the control system interface. Correct understanding of the HMI indications, operating modes, and status representations is essential for the safe operation of the plant. Misinterpretation of the displayed information may lead to incorrect actions, potentially affecting process safety, equipment integrity, and personnel safety. Operators shall verify that they clearly understand the meaning of all operating modes, symbols, and status indications used in the HMI prior to performing any operation described in this manual.

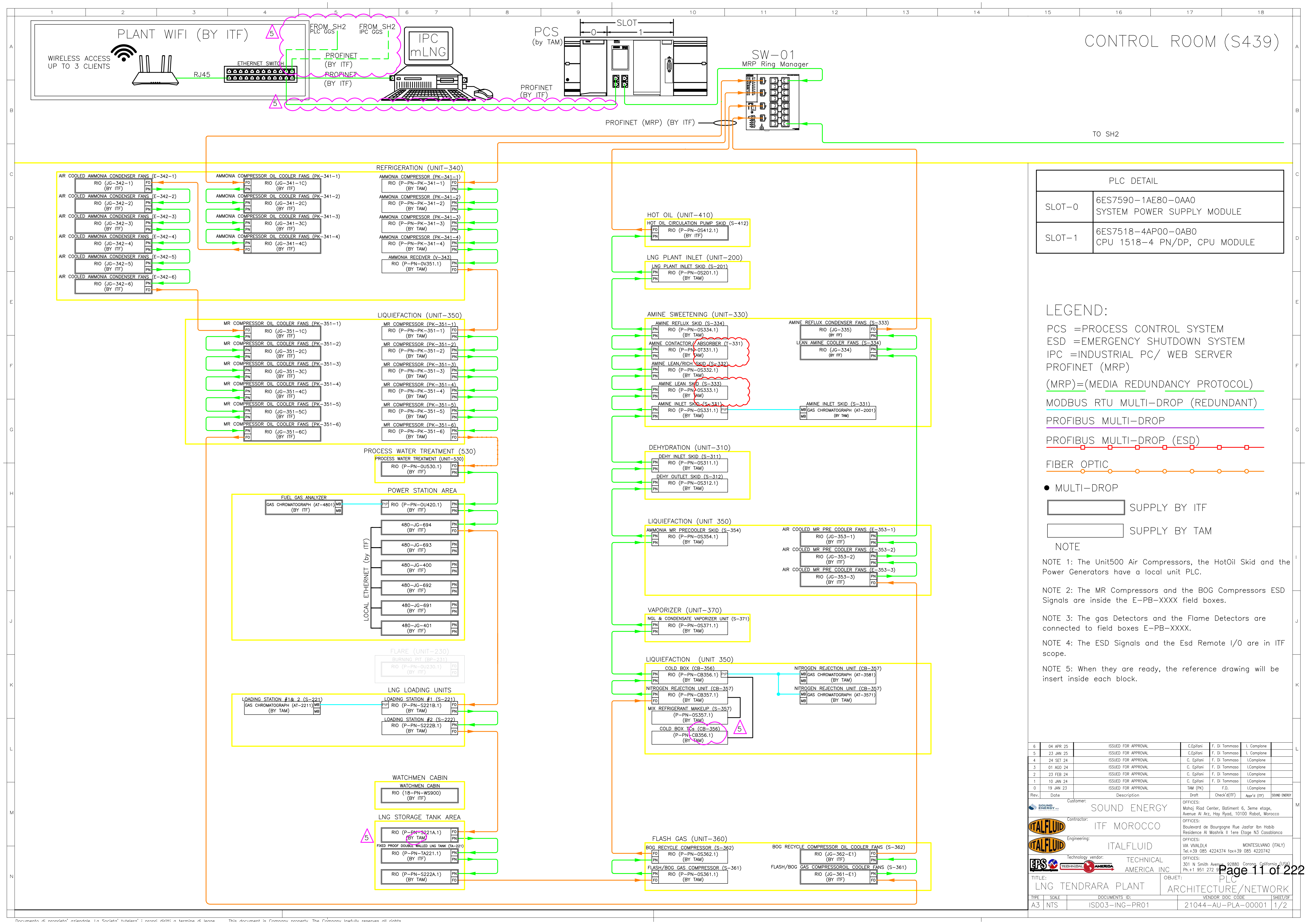


Figure 2.3: PCS Architecture

2.3 ESD - Emergency Shutdown System

The Emergency Shutdown System (ESD) of the Tendrara LNG Facility provides a dedicated, independent, and highly reliable safety layer designed to protect personnel, equipment, and the environment during abnormal or hazardous operating conditions. The ESD system is fully independent from the Plant Control System (PCS) and is implemented in accordance with the Project Process Safety and Isolation Philosophy and applicable international standards.

ESD Control Architecture.

The ESD is implemented using a Siemens SIMATIC S7-1500F fail-safe PLC installed in the Control Room. This safety PLC executes all emergency, process, and local shutdown logic and ensures deterministic and rapid response to critical events. The PLC hardware configuration includes safety-rated digital I/O modules and certified fail-safe communication over PROFINET, ensuring high availability and system integrity.

Distributed Remote I/O System.

The ESD architecture incorporates ET 200SP distributed remote I/O stations installed in the field, close to the protected process units. These remote stations are connected to the main ESD PLC through a high-speed PROFINET network, minimizing field wiring, improving diagnostics, and ensuring fast propagation of safety signals. Remote I/O panels are identified in the documentation with tags such as PB_OS201, PB_OS221, PB_OS312, and PB_OS371, among others.

Field Signals and Safety Functions.

The remote I/O stations collect signals from emergency shutdown pushbuttons, fire and gas detectors, critical process transmitters (pressure, level, and temperature), and valve status feedback. All safety-related signals are processed within the S7-1500F PLC using certified fail-safe logic, triggering predefined cause-and-effect actions such as isolation valve closure, depressurization, and equipment shutdown.

Shutdown Levels.

The ESD system manages multiple shutdown levels, including Emergency Shutdown (ESD), Process Shutdown (PSD), Local Shutdown (LSD), and Abandon Shutdown (ASD) for the LNG loading bay. Each shutdown level is activated by specific manual or automatic inputs and executes predefined outputs in accordance with the Cause and Effect Matrix, ensuring a controlled and safe response to each scenario.

Operator Interface and Visualization.

The system includes a local HMI for operation, monitoring, diagnostics, and maintenance of the ESD. In addition, a dedicated industrial PC with its own visualization system is provided, allowing enhanced monitoring, event logging, and detailed analysis of shutdown events and system status.

Integration with the Plant Control System.

The ESD system is interconnected with the PCS through a high-speed PROFINET link. This interface allows the PCS to receive near real-time information from the ESD system, including shutdown status, active trips, alarms, and valve positions. The communication is strictly informational, and no safety logic depends on the PCS, preserving the independence and integrity of the ESD system.

Summary.

The Emergency Shutdown System is based on a modern fail-safe PLC architecture with distributed remote I/O, high-speed deterministic communication, and clear separation from the PCS. This design ensures fast response, high reliability, and full compliance with the project safety philosophy, providing an effective and robust protection layer for the Tendrara LNG Facility.

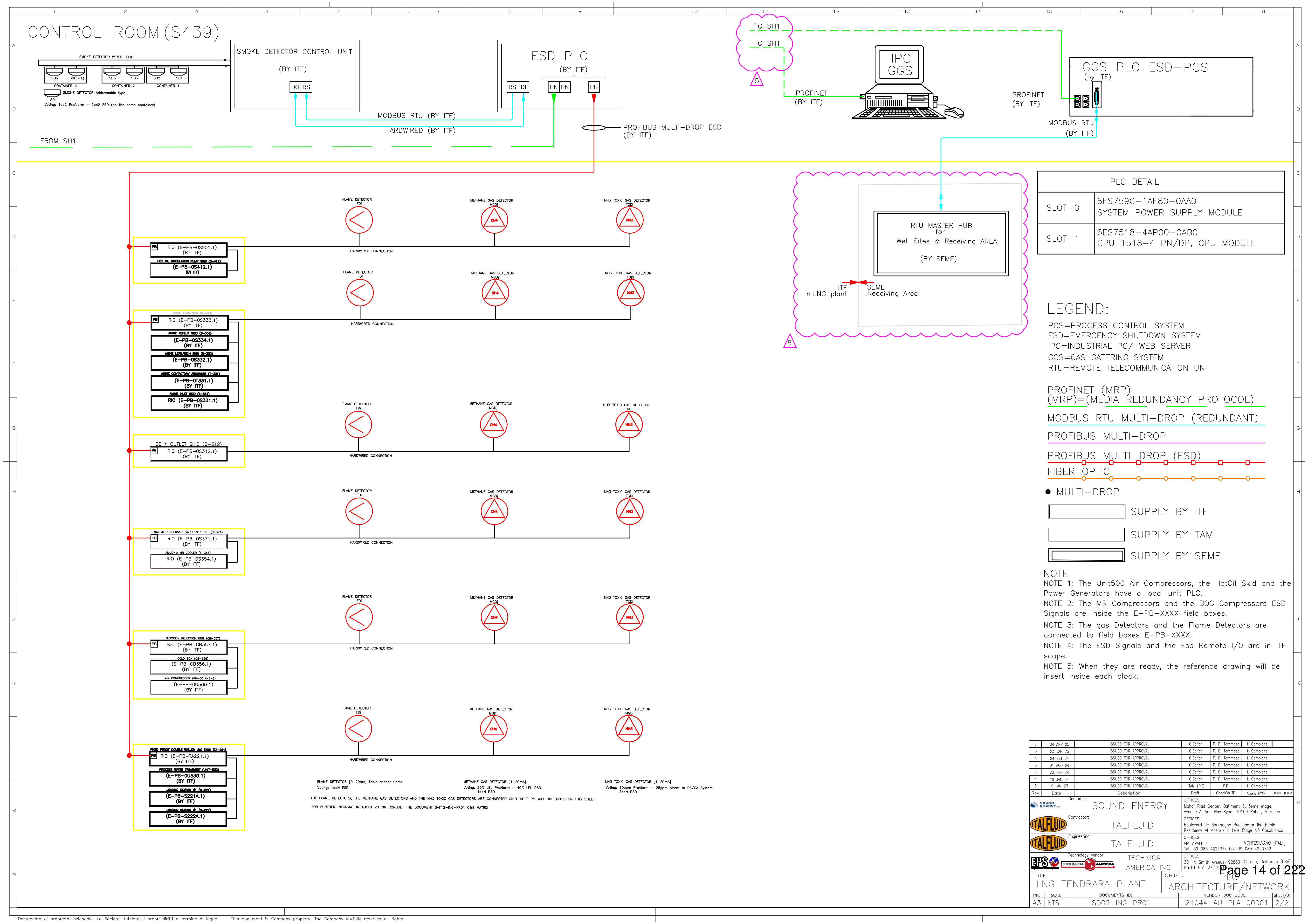


Figure 2.4: ESD Architecture

2.4 Communication Network

As shown in Figure 2.3, the plant communication network for the Tendrara LNG Facility is based on **PROFINET IO**, an internationally recognized industrial Ethernet standard defined under IEC 61158 and IEC 61784. PROFINET IO is a high-speed, deterministic and highly reliable communication protocol, widely adopted in modern process and manufacturing industries for real-time control, data acquisition, diagnostics and asset management.

PROFINET IO enables cyclic and acyclic data exchange with deterministic behavior, supporting typical update times in the order of milliseconds while maintaining high data integrity. The protocol allows seamless integration of controllers, remote I/O, intelligent field devices, drives and safety systems over standard Ethernet infrastructure, using copper, fiber optic and wireless media, depending on distance, electromagnetic environment and availability requirements.

For the Tendrara Project, a highly reliable network topology has been implemented, based on a **dual redundant ring architecture** using the **Media Redundancy Protocol (MRP)**, as defined in IEC 62439-2. This architecture consists of two independent PROFINET rings, managed by a central MRP-capable managed switch acting as Ring Manager for both communication loops.

The MRP-based ring topology provides fast fault recovery and high availability. In the event of a single cable break, device failure or network interruption, the communication path is automatically reconfigured, ensuring continuous data exchange through the remaining available path. Typical network recovery times are below 200 ms and, depending on the configuration and number of nodes, can be significantly lower, ensuring uninterrupted control and monitoring of the process.

This redundant communication architecture guarantees a high level of operational reliability for all plant units, including process systems, utilities, power generation, safety-related signals and auxiliary systems. In addition to redundancy, the PROFINET-based design allows advanced diagnostics, real-time network monitoring, remote troubleshooting and predictive maintenance, enabling rapid identification of communication faults and reducing downtime.

The selected architecture ensures that the plant remains fully operable even in the presence of physical network faults or individual equipment failures, always selecting the available communication path and maintaining control integrity. This approach aligns with modern industrial best practices and provides a robust, scalable and future-proof communication backbone for the Tendrara LNG Facility.

2.5 Power Generation System

The LNG facility is equipped with an autonomous on-site power generation system designed to ensure safe, reliable, and flexible electrical supply under all operating conditions. The power plant operates permanently in island mode and is fully independent from any external electrical grid.

System Scope and Functional Boundaries

The scope of the power generation system includes power production, low-voltage distribution, busbar sectionalization, generator synchronization, load sharing, and integration with plant control and safety systems. The system supplies all process units, utilities, auxiliary systems, and safety-critical consumers, including those requiring continuous availability during abnormal or emergency conditions.

Detailed functional logic, start-up and shutdown sequences, and load prioritization are defined in the functional analysis document and are not duplicated herein. This section provides a high-level technical description to support process understanding and operational context.

Installed Generation Capacity

The facility is equipped with seven Jenbacher gas-engine generator modules together with two additional diesel generator units.

The installed electrical capacity of the power generation system is based on seven gas engines rated at 1,807 kW each and two diesel generators rated at 1,280 kW each, resulting in a total installed capacity of approximately 15.21 MW (15,210 kW) at a power factor of $\cos\varphi = 0.8$, corresponding to approximately 19.0 MW at $\cos\varphi = 1.0$.

The gas-engine generators represent the primary source of electrical power under normal operating conditions, while the diesel generators are dedicated to emergency, black-start, and backup operation.

Electrical Architecture and Busbar Configuration

Electrical power is generated and distributed at a nominal voltage of 690 V, TN-C configuration. The power generation units are connected to a main low-voltage distribution panel subdivided into four independent busbar sections. These busbars are interconnected by three motorized and synchronizable tie breakers, allowing flexible system reconfiguration.

The busbar arrangement enables the creation of two electrically independent section groups, limiting fault propagation and ensuring compliance with short-circuit constraints. The switchboard is rated for a short-circuit withstand level of 85 kA at 690 V, which allows a maximum of four gas engine generators to be connected in parallel on the same energized busbar section.

A Single Line Diagram (SLD) is provided to illustrate the physical and functional interconnection of generators, busbars, tie breakers, and major loads, and shall be used as the primary electrical reference for system configuration.

Power Generation Units and Operating Modes

The power generation system comprises seven gas engine generator sets and two diesel generator units. Under normal operating conditions, five gas engine generators are expected to operate in parallel, with the remaining units available as standby. The system is designed to allow all gas engine generators to operate simultaneously if required by plant load conditions.

Diesel generators are provided in a $2 \times 100\%$ configuration and are dedicated to emergency operation and black-start capability. Diesel generators may also operate in parallel with gas engine generators to provide maximum operational flexibility during abnormal conditions or transitional operating phases.

Normal Operating Configuration and Load Sharing

During normal plant operation, the gas engine generators operate in parallel with automatic load sharing under the supervision of the Master Control Panel (MCP). The operator selects the optimal combination of running and standby generators based on actual plant demand, generator availability, and minimum load constraints recommended by the generator manufacturer.

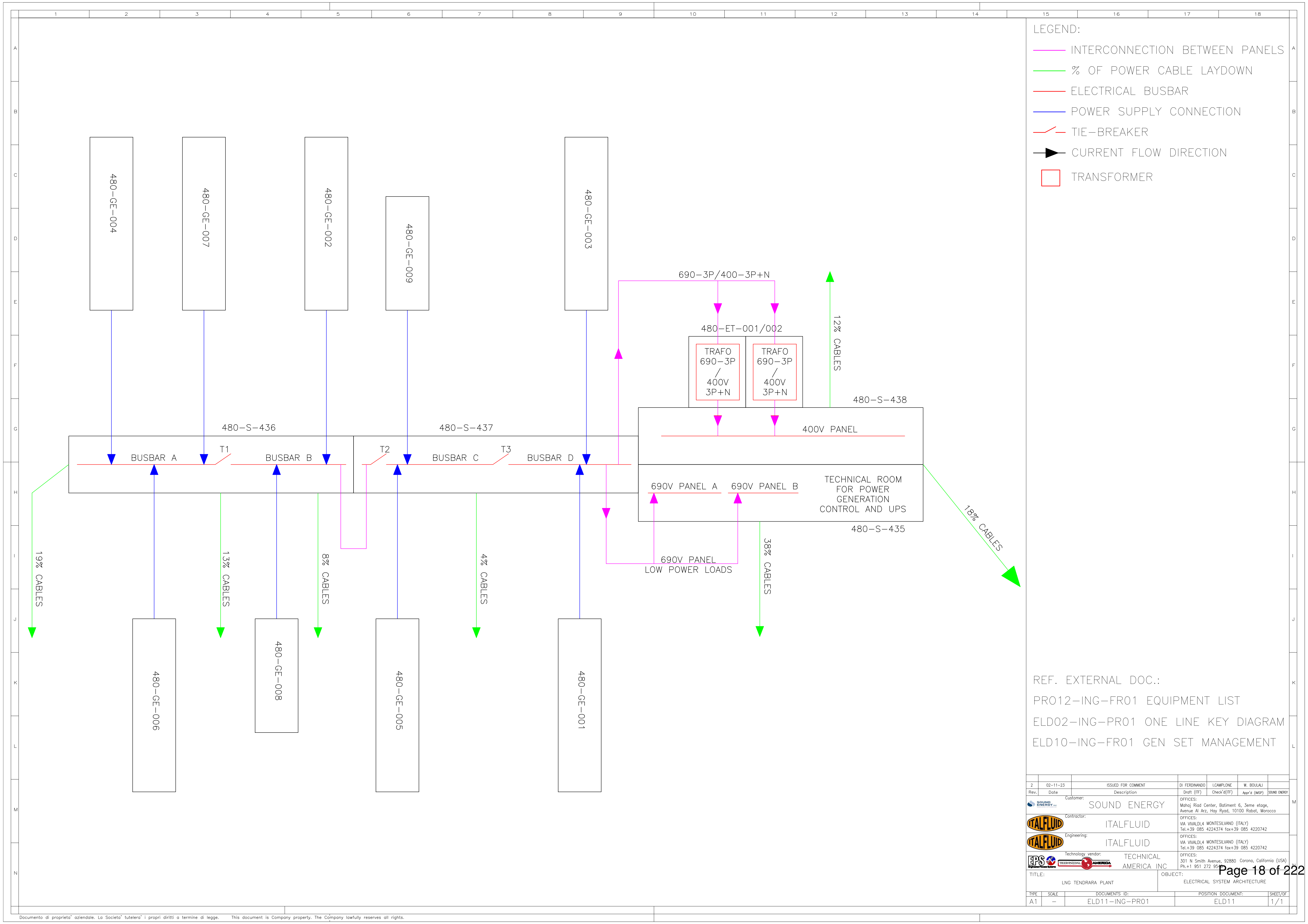


Figure 2.5: Electrical System Architecture

3 Unit 200 - LNG Plant Inlet Unit

3.1 Operating Summary

CATEGORY	DESCRIPTION
Inputs	Raw natural gas from TMF-6/TE-7 wells entering the plant through the Plant Inlet SDV (SDV-2024), with arrival pressure up to 55 barg.
	Corrosion inhibitor injection upstream of the inlet separation stage.
Outputs	Conditioned raw gas routed to UNIT 330, with flow regulated by FV-2001.
	Separated process water from V-202 and V-201 routed to UNIT 530.
	Condensates separated in V-201 routed to the Hot Oil System buffer tank for use as fuel.
Key Operating Specs	Hydrate prevention ensured by feed gas temperature control in E-201 via hot oil (TIC-2002).
	Inlet gas pressure regulated by PV-2001 to maintain stable downstream operation.
	Feed gas flow measured by FT-2001 and controlled by FIC-2001.
	Overpressure protection provided by PSVs; PAHH/TAHH/LALL/LAHH alarms initiate PSD/LSD actions and blowdown to flare via BDV-2004.
Major Equipment	V-202 inlet three-phase separator, E-201 feed gas heater (hot oil).
	F-201A/B particulate filters (5 µm), V-201 inlet raw gas separator.
	FV-2001 feed gas flow control valve, PV-2001 inlet pressure control valve.
	Associated SDVs, BDVs, and pressure relief valves.

3.2 General Arrangement and Risk Location

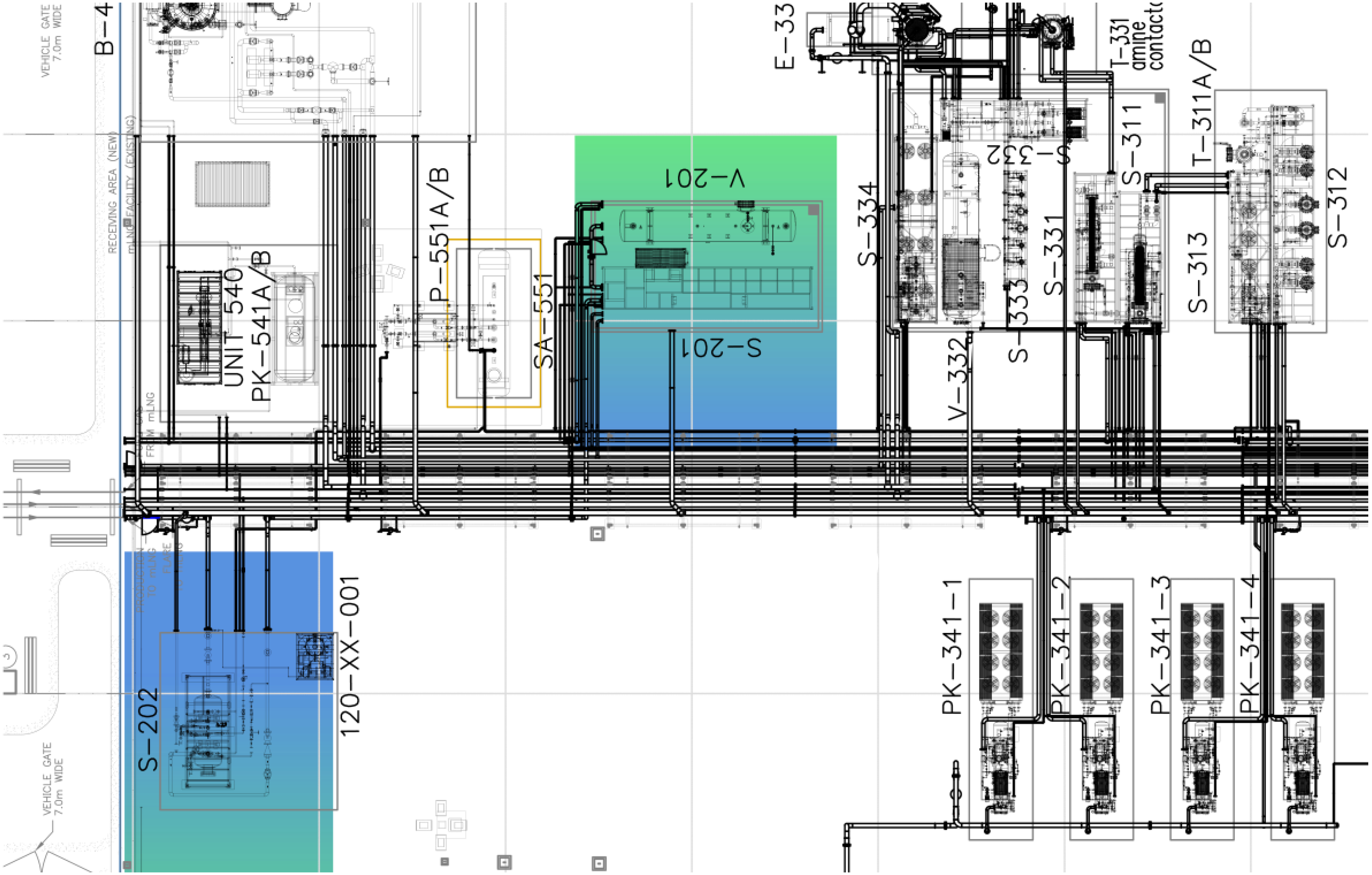


Figure 3.1: Unit 200 GA - Risk location

3.3 Relevant Documents

Table 3.2 — Unit 200 – Relevant Documents

ID	DOC NO	TITLE
1	PRO02-ING-FR02	PROCESS DESCRIPTION
2	PRO10-ING-PR14	PROCESS FLOW DIAGRAM (INLET SEPARATOR)
3	PRO13-ING-PR01	UNIT 200 – SKID S-201 / V-201
4	PRO13-ING-PR50	UNIT 200 – V-202
5	SAF01-ING-FR01	PROCESS SAFETY AND ISOLATION PHILOSOPHY
6	SAF12-ING-FR01	CAUSE AND EFFECT MATRIX
7	SAF05-ING-FR01	RELIEF AND BLOWDOWN PHILOSOPHY
8	21044-AU-HMI-00002	HMI PHILOSOPHY

3.4 Process Description

The Gas Reception Area (Unit 200) is the initial processing stage for raw gas entering the LNG plant. Its function is to perform primary separation and conditioning of the feed gas to ensure suitable conditions for downstream treatment and liquefaction. The unit consists of two main separation stages, a heating system, and filtration, all integrated to remove water and condensates, control pressure, and prevent hydrate formation. 3

The raw gas stream enters the plant through the PLANT INLET SDV (SDV-2024), which serves as the primary emergency shutdown valve for isolating the incoming pipeline. After passing through the Plant Inlet Shutdown Valve (SDV-2024), the raw gas stream is routed to the first-stage separation unit (V-202). Before entering the vessel, a corrosion inhibitor is injected into the gas line to prevent internal corrosion due to acidic compounds, moisture, and CO_2 presence. This chemical treatment ensures the integrity and longevity of the downstream equipment.

From there, the gas flows directly into the inlet three-phase separator V-202, where the first stage of separation occurs. In this vessel, water and condensates are separated from the gas stream. The water is routed to the Process Water Package, meanwhile the condensates are commingled again with the gas exiting the separator and together are routed to the downstream equipment.

The gas and condensates stream from V-202 is directed toward the heater E-201, passing through the INLET SKID SHUTDOWN VALVE (SDV-2001). This valve allows isolation of the Unit 200 skid in the event of local shutdown or maintenance. The gas heater E-201 is a shell-and-tube exchanger that uses hot oil to raise the gas temperature and prevent hydrate formation during winter operation. Gas outlet temperature is regulated via the hot oil flow using temperature control valve TV-2002.

After heating, the stream passes through the Gas Inlet Pressure Control Valve (PV-2001), which maintains the necessary pressure for downstream processing. The gas then flows through particulate filters F-201A/B, arranged in duty/standby configuration, to remove solid particles larger than 5 microns and protect subsequent equipment.

The filtered and conditioned gas and condensates stream is fed into the inlet raw gas separator V-201, where final three-phase separation occurs. Free water from this stage is again sent to the Process Water Package, while all the condensates are routed to the Condensates buffer tank (S-413) inside the Hot Oil System to be used as fuel for boilers.

The dry gas leaving V-201 is measured by FT-2001 and regulated via flow control valve FV-2001, with pressure and temperature monitored by PT-2003 and TT-2001, respectively. A bypass line allows part of the stream to be redirected to E-371 in the Vaporizer Unit (Unit 370) for fuel gas supply.

Finally, the gas—now stripped of liquids but still containing CO_2—is sent to the Gas Pre-treatment Section (Unit 330) for CO_2 removal via amine absorption, followed by dehydration in Unit 310, preparing the gas for liquefaction. Several safety and shutdown systems are integrated throughout the unit, including pressure alarms (PAHH), temperature alarms (TAHH), and level alarms (LAHH, LALL), which interface with both the Process Control System (PCS) and the Emergency Shutdown System (ESD) to ensure safe and reliable operation under all conditions.

3.5 Technological Objects

3.5.1 Main Equipment List

Table 3.3 — Unit 200 – Main Equipment List

ID	TAG	EQUIPMENT
1	V-202	Inlet three-phase separator
2	E-201	Feed gas heater (hot oil)
3	F-201A	Feed gas particulate filter A
4	F-201B	Feed gas particulate filter B
5	V-201	Inlet raw gas separator

3.5.2 Valves

Table 3.4 — Unit 200 – Valves

ID	TAG	SERVICE	SYSTEM	NOTES
1	BDEV-2004	Plant inlet blowdown	ESD	-
2	BDV-2004	Plant inlet blowdown	-	-
3	FEV-2001	Feed gas flow control	ESD	-
4	FV-2001	Feed gas flow control	-	-
5	LV-2011	V-201 heavy liquid level control	-	-
6	LV-2012	V-201 light liquid level control	-	-
7	PSV-2001A	V-201 pressure relief	-	Set 50 barg
8	PSV-2001B	V-201 pressure relief	-	Set 50 barg
9	PSV-2002A	Inlet section overpressure protection	-	Set
10	PSV-2002B	Inlet section overpressure protection	-	Set
11	PSV-2003A	F-201A pressure relief	-	Set 50 barg
12	PSV-2003B	F-201B pressure relief	-	Set 50 barg
13	PV-2001	Feed gas pressure control	-	-
14	SDEV-2001	Plant inlet emergency shutdown	ESD	-
15	SDEV-2002	V-201 HC outlet shutdown	ESD	-
16	SDEV-2003	V-201 water outlet shutdown	ESD	-
17	SDV-2001	Plant inlet emergency shutdown	-	-
18	SDV-2002	V-201 HC outlet shutdown	-	-
19	SDV-2003	V-201 water outlet shutdown	-	-
20	TV-2002	E-201 outlet temperature control	-	-

3.5.3 Instrumentation

Table 3.5 — Unit 200 – Instrument List

ID	TAG	SERVICE	RANGE	OP	UNITS
1	AE-2001	Feed gas analyzer	-	–	-
2	AT-2001	Feed gas analyzer	-	–	%mol
3	FE-2001	Feed gas flow	-	–	-
4	FE-2011	Water outlet from V-201 flow	-	–	-
5	FE-2012	Condensate outlet from V-201 flow	-	–	-
6	FT-2001	Feed gas flow	0..25	–	MMSCFD
7	FT-2011	Water outlet from V-201 flow	0..0.1	–	m^3/h

ID	TAG	SERVICE	RANGE	OP	UNITS
8	FT-2012	Condensate outlet from V-201 flow	0..1	–	m^3/h
9	FVY-2001	Feed gas flow control	-	–	-
10	LSLL-2004	V-201 low low liquid level switch	-	–	-
11	LT-2001	V-201 light liquid level	0..100	–	%
12	LT-2002	V-201 heavy liquid level	-	–	mm
13	LT-2003	V-201 total liquid level	-	–	mm
14	LVY-2011	V-201 heavy liquid level control	0..100	–	%
15	LVY-2012	V-201 light liquid level control	0..100	–	%
16	PDG-2001	F-201A/B pressure drop	0..5	–	bar
17	PG-2001A	F-201A vent pressure	0..50	–	barg
18	PG-2001B	F-201B vent pressure	0..50	–	barg
19	PG-2002	V-201 pressure	0..50	–	barg
20	PT-2001	Feed gas pressure	0..50	–	barg
21	PT-2002	Feed gas pressure	0..50	–	barg
22	PT-2003	Gas to dehydration pressure	0..50	–	barg
23	PVY-2001	Feed gas pressure control	0..100	–	%
24	RD-2005	Hot oil system overpressure protection	-	–	-
25	RO-2004	BDV-2004 blowdown restriction	-	–	-
26	TT-2001	Feed gas temperature	-15..150	–	°C
27	TT-2002	E-201 outlet temperature	-15..150	–	°C
28	TVY-2002	E-201 outlet temperature control	0..100	–	%
29	ZSH-2001	SDV-2001 open position indication	-	–	-
30	ZSH-2002	SDV-2002 open position indication	-	–	-
31	ZSH-2003	SDV-2003 open position indication	-	–	-
32	ZSH-2004	BDV-2004 open position indication	-	–	-
33	ZSH-2005	RD-2005 open position indication	-	–	-
34	ZSL-2001	SDV-2001 closed position indication	-	–	-
35	ZSL-2002	SDV-2002 closed position indication	-	–	-
36	ZSL-2003	SDV-2003 closed position indication	-	–	-
37	ZSL-2004	BDV-2004 closed position indication	-	–	-

3.5.4 Junction Boxes

Table 3.6 — Unit 200 – Junction Boxes

ID	TAG	SERVICE	CONNECTION	TYPE
1	P-PN-0S201.1	LNG PLANT INLET	COPPER-COPPER	STANDARD

3.5.5 Controllers

Table 3.7 — Unit 200 – Controllers

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
FIC-2001	FEED GAS FLOW CONTROLLER	FV-2001	0..25 MMSCFD	16.0
LIC-2001	V-201 LIGHT LIQUID LEVEL CONTROLLER	LV-2012	0..100 %	—
LIC-2002	V-201 HEAVY LIQUID LEVEL CONTROLLER	LV-2011	0..100 %	—
LIC-2021	V-202 LIGHT LIQUID LEVEL CONTROLLER	LV-2021	0..100 %	—
LIC-2022	V-202 HEAVY LIQUID LEVEL CONTROLLER	LV-2022	0..100 %	—
PIC-2001	FEED GAS PRESSURE CONTROLLER	PV-2001	0..50 barg	45.0
TIC-2002	E-201 OUTLET TEMPERATURE CONTROLLER	TV-2002	-15..150 °C	—

3.6 Operating Interface

3.6.1 PCS Operating Screens – Unit 200

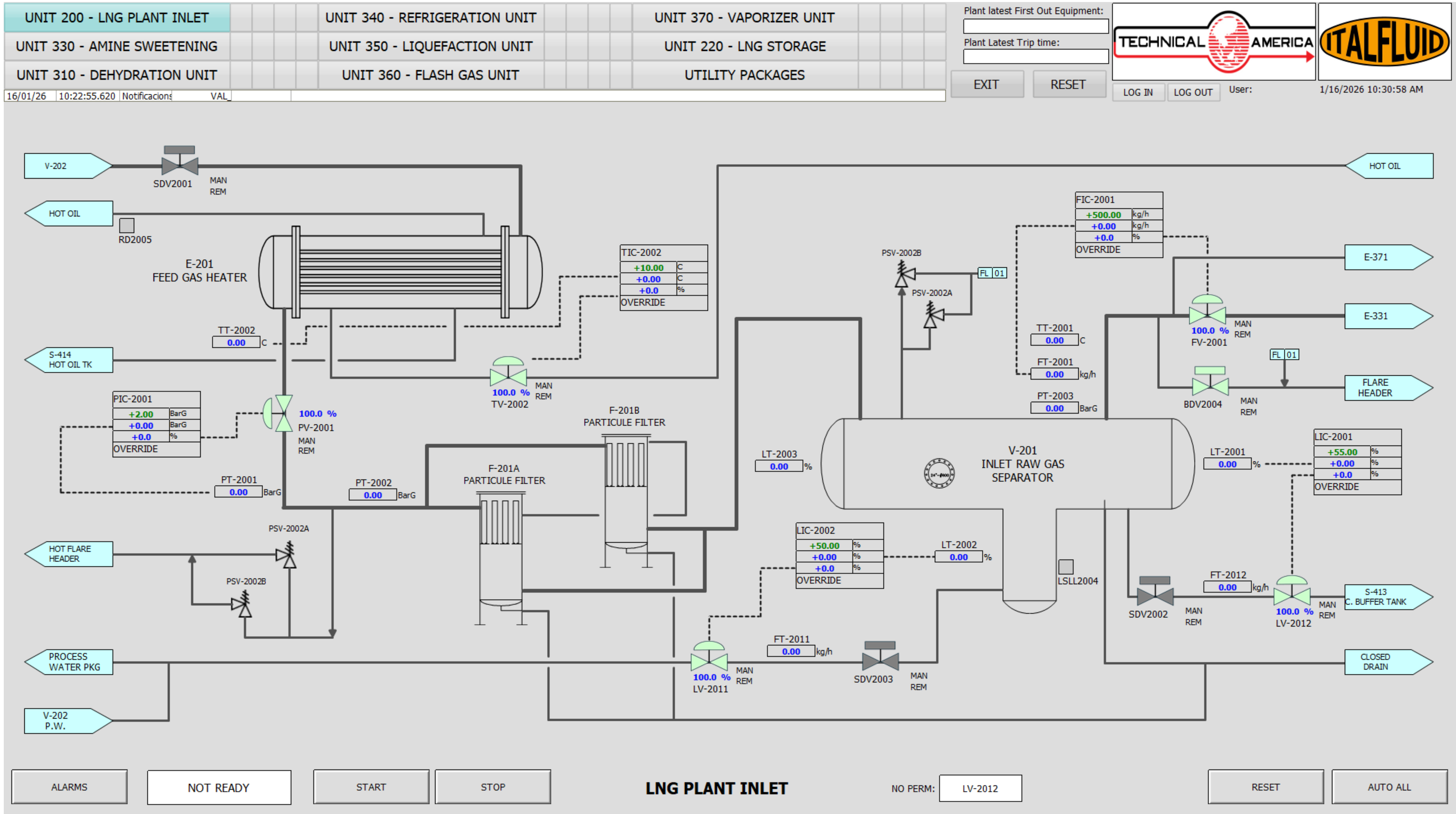


Figure 3.2: Area 200 - HMI Operating Screen

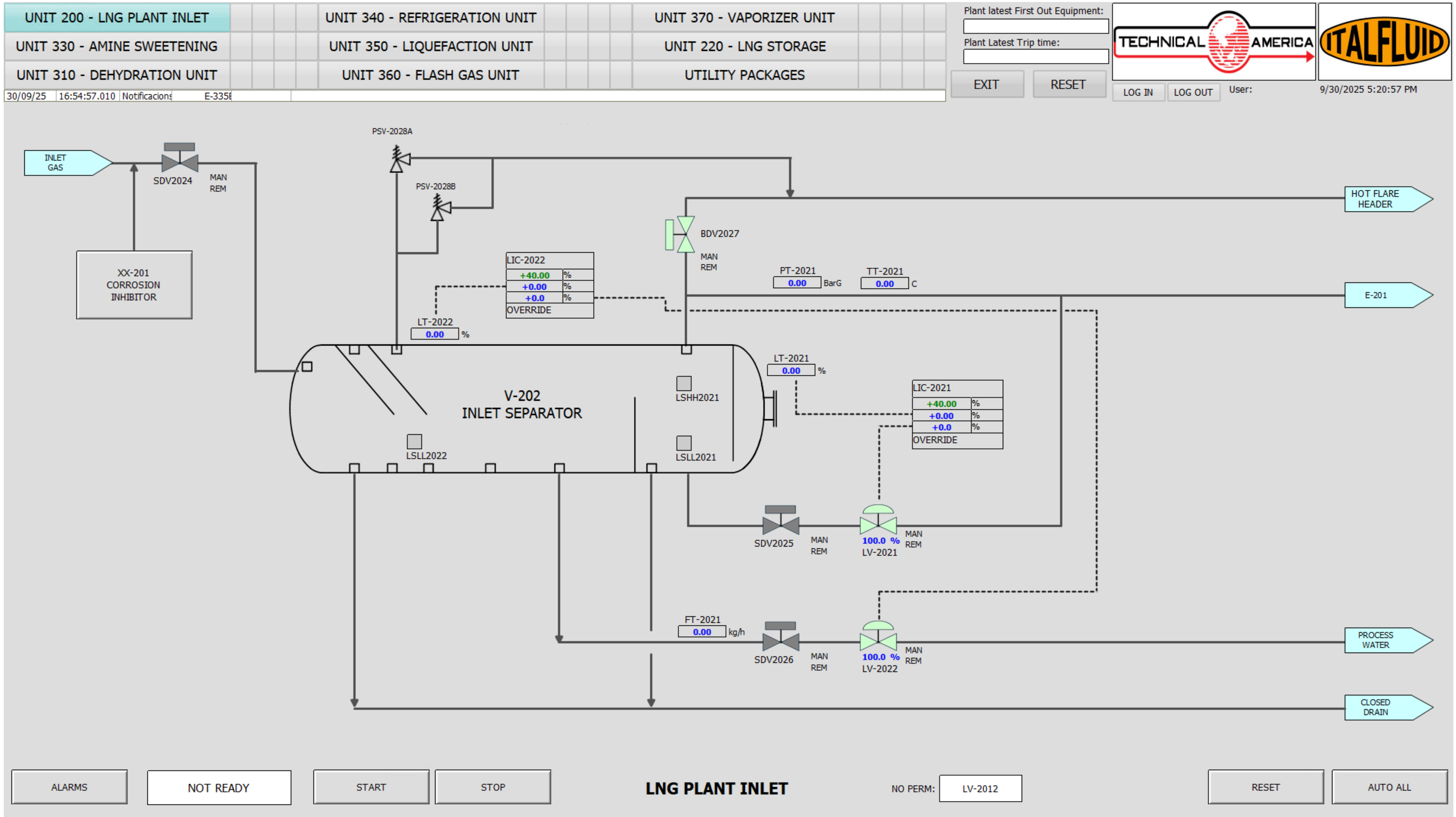


Figure 3.3: Area 200 - HMI Operating Screen

3.6.2 Critical Alarms and Trips

Table 3.8 — Unit 200 – Critical Alarms

ALARM	INSTRUMENT	TYPE	SP	DESCRIPTION
LALL-2004	LS-2004	LSD	—	LOW LOW HEAVY LIQUID LEVEL ON V-201
LAHH-2021	LSHH-2021	PSD	—	HIGH HIGH LIQUID LEVEL ON V-202
LALL-2021	LSLL-2021	LSD	—	LOW LOW LIGHT LIQUID LEVEL ON V-202
LALL-2022	LSLL-2022	LSD	—	LOW LOW HEAVY LIQUID LEVEL ON V-202
LAHH-2003	LT-2003	PSD	—	HIGH HIGH LIGHT LIQUID LEVEL ON V-201
LALL-2003	LT-2003	LSD	—	LOW LOW LIGHT LIQUID LEVEL ON V-201
PAHH-2021	PIT-2021	PSD	—	HIGH HIGH PRESSURE V-202
PAHH-2002	PT-2002	PSD	—	HIGH HIGH INLET PRESSURE
TAHH-2002	TT-2001	PSD	—	HIGH HIGH TEMPERATURE ON V-201
ZSH-2005	ZS-2005	PSD	—	LIMIT SWITCH ON RUPTURE DISK RD-2005

3.6.3 Unit Cause and Effect Matrix

Table 3.9 — Unit 200 – Cause & Effect Matrix

ALARM	INSTRUMENT	SDV-2002	SDV-2003	SDV-2026	SDV-2025
LALL-2003	LT-2003	C			
LALL-2004	LT-2004		C		
LALL-2021	LSLL-2021				C
LALL-2022	LSLL-2022			C	

4 Unit 220 - LNG Storage, metering system and loading station

4.1 Operating Summary

CATEGORY	DESCRIPTION
Inputs	Liquefied Natural Gas (LNG) from the Nitrogen Rejection Unit (UNIT 350), expanded across LV-2211 and routed to the LNG Separator V-221.
	Returned boil-off and flash vapors from downstream handling systems routed to the BOG recovery header.
Outputs	Stabilized LNG transferred from V-221 to the atmospheric LNG Storage Tank TA-221 via level control valve LV-2232.
	Separated vapor from V-221 routed to the BOG recovery system for compression and reuse.
	LNG delivered from TA-221 to road tankers through loading skids S-221 and S-222, with custody-transfer metering and controlled loading flow.
Key Operating Specs	Phase separation in V-221 ensured by pressure control PIC-2231 and level control LIC-2232 to provide stable LNG transfer to storage.
	LNG storage at near-atmospheric pressure in TA-221, with tank pressure controlled by PIC-2202 via controlled venting through TSV-2202.
	High-high and low-low level protections in TA-221 (LAHH-2201 / LALL-2201) initiate PSD or LSD actions to prevent overfilling and protect cryogenic pumps.
	LNG truck loading flow regulated by VFD-controlled cryogenic pumps (P-221A/B, P-222A/B) and pressure control valves PV-2211 / PV-2221 for smooth and safe loading.
	Product quality verified by online gas chromatographs AT-2211 and AT-3571 prior to storage and truck loading.
	Overpressure protection provided by dedicated PSVs and thermal relief valves; PAHH conditions initiate LSD actions isolating tanks, skids, and pumps in accordance with the shutdown philosophy.
Major Equipment	V-221 LNG separator, TA-221 atmospheric LNG storage tank.
	Cryogenic truck loading pumps P-221A/B and P-222A/B with associated loading skids S-221 and S-222.
	Custody-transfer metering systems FT-2212 and FT-2222, LNG quality analyzers AT-2211 and AT-3571.
	Associated pressure, level, and temperature instrumentation, SDVs, XVs, PSVs, and thermal safety valves.

4.2 General Arrangement and Risk Location

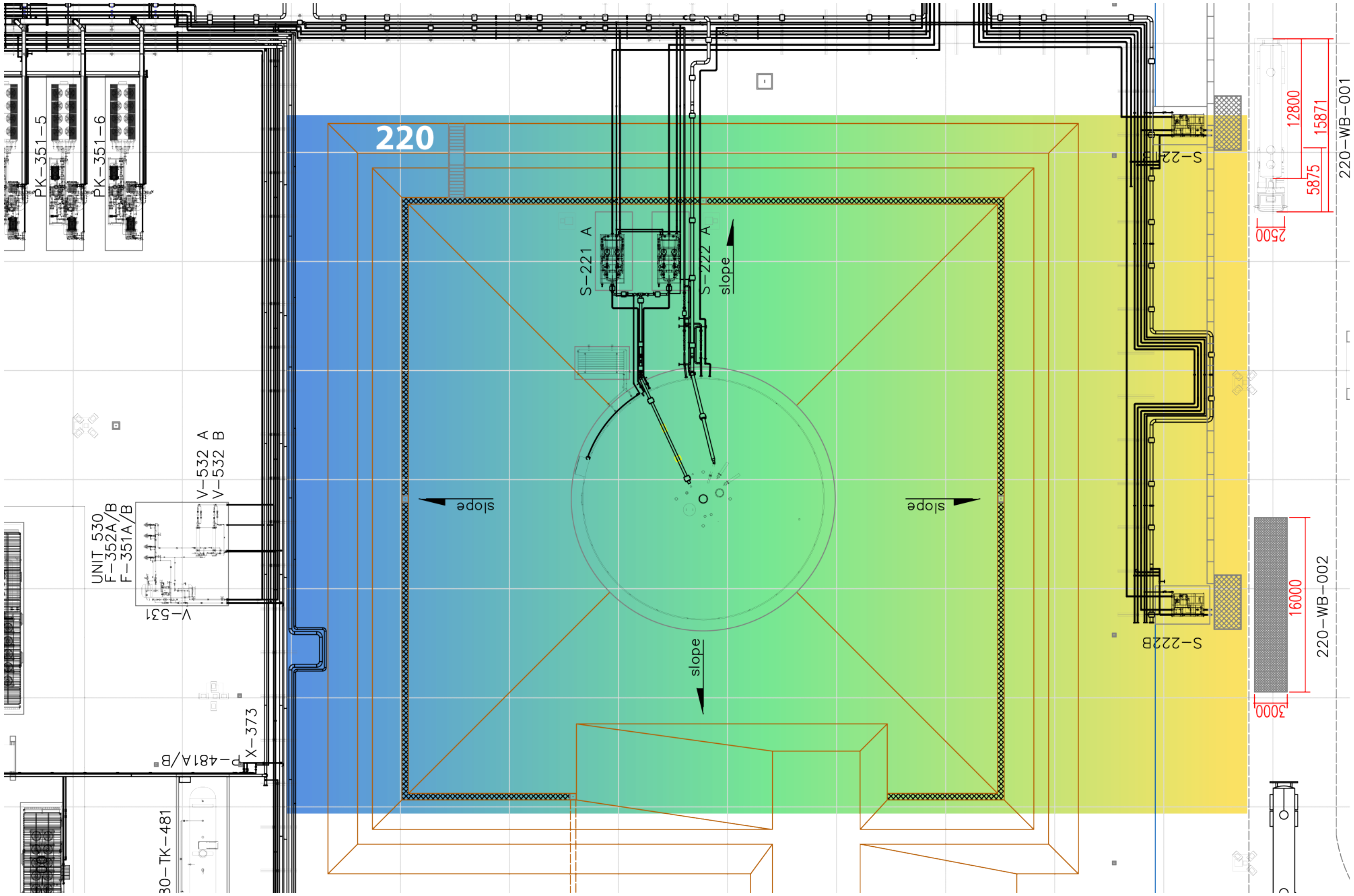


Figure 4.1: Unit 220 GA - Risk location

4.3 Relevant Documents

Table 4.2 — Unit 220 – Relevant Docuemnts

ID	DOC NO	TITLE
1	PRO02-ING-FR02	PROCESS DESCRIPTION
2	PRO10-ING-PR03	PROCESS FLOW DIAGRAMS (GAS LIQUEFACTION)
3	PRO13-ING-PR23	SKID S221-AB
4	PRO13-ING-PR24	SKID S222-AB
5	PRO13-ING-PR25	LNG SEPARATOR V-221
6	PRO13-ING-PR26	LNG STORAGE TANK TA-221
7	PRO13-ING-PR52	WEIGHBRIDGE 221-WB-001/002
8	SAF01-ING-FR01	PROCESS SAFETY AND ISOLATION PHILOSOPHY
9	SAF12-ING-FR01	CAUSE AND EFFECT MATRIX
10	SAF05-ING-FR01	RELIEF AND BLOWDOWN PHILOSOPHY
11	21044-AU-HMI-00002	HMI PHILOSOPHY

4.4 Process Description

Constitutes the final stage of the LNG production process. Its primary function is to receive the liquefied natural gas from the downstream Nitrogen Rejection Unit (Unit 350), provide safe cryogenic storage, perform custody-transfer metering, and manage the controlled loading of LNG into cryogenic road tankers for distribution. This unit serves as the interface between the liquefaction system and the transportation network, ensuring that the LNG is delivered in compliance with quality, safety, and operational standards.

LNG from the NRU outlet expands across LV-2211 into the LNG Separator V-221, where vapor and liquid phases are separated. The stabilized liquid is transferred through the level control valve LV-2232 to the atmospheric cryogenic storage tank TA-221 for temporary storage prior to truck loading, while the separated vapor is directed to the Boil-Off Gas (BOG) recovery header.

The LNG transfer line from the LNG Separator V-221 to the storage and truck loading system is equipped with an online gas chromatograph (AT-3571), as shown in the P&ID. The gas chromatograph continuously analyzes the LNG composition, including nitrogen content, to verify compliance with contractual product quality specifications prior to send LNG to storage.

Within TA-221 tank pressure is monitored by pressure transmitter PT-2201 and indicated locally by PG-2201. The tank pressure controller PIC-2202 is used for low-pressure control and maintains the minimum tank pressure by regulating the operation of the vaporizer (PBUC).

Overpressure protection of the tank is provided by the tank safety valves TSV-2201 and TSV-2202, which discharge to atmosphere only in abnormal or emergency conditions.

The tank vent valve TSV-2202 is provided only as an emergency protection device and shall remain closed during normal operation.

During normal operation, one of the cryogenic loading pumps (P-221A/B or P-222A/B) is started when truck loading is authorized. The selected pump transfers LNG from TA-221 to the corresponding loading skid, where the transferred quantity is measured by the custody transfer metering system (FT-2212 or FT-2222) before being delivered to the loading arms/hoses. The LNG loading flowrate is regulated by the control valve PV-2211 or PV-2221, providing a smooth and controlled filling rate as per the loading procedure. Temperature transmitters TT-2217 and TT-2227 monitor the LNG inlet temperature at each skid to confirm proper cooldown conditions prior to initiating/continuing loading.

In case of overpressure in TA-221, PAHH-2201 initiates a Local Shutdown (LSD) by closing SDV-2201 and SDV-2202 to isolate the tank from incoming LNG streams, and closing XV-3601 (BOG superheater inlet) as per the cause & effect logic. Tank overpressure protection is provided by the dedicated mechanical relief/vent devices installed on TA-221.

The LNG storage tank TA-221 is protected against abnormal liquid level conditions by dedicated level alarms. A high-high liquid level in TA-221 (LAHH-2201) initiates a Process Shutdown (PSD) to prevent tank overfilling. A low-low liquid level in TA-221 (LALL-2201) initiates a Local Shutdown (LSD), resulting in the tripping of all LNG loading pumps (P-221A/B and P-222A/B) to protect the cryogenic pumps against cavitation and dry running.

V-221 – LNG Separator

V-221 receives LNG at low pressure from the upstream liquefaction section and provides phase separation between residual vapor and liquid. The separated liquid is routed from the bottom of the vessel to the LNG storage tank TA-221, while the vapor phase is relieved from the top of the separator in accordance with the pressure control strategy. Pressure in V-221 is monitored by PT-2231 and controlled by PIC-2231 acting on control valve PV-2231, maintaining stable separator pressure and preventing excessive flash vaporization upstream of the storage tank. Liquid level in the separator is measured by LT-2232 and controlled by LIC-2232, which modulates the liquid outlet control valve LV-2232 to ensure continuous and stable transfer of LNG to TA-221. High-high and low-low liquid level conditions in V-221 are detected by LAHH-2233 and LALL-2233 respectively, providing protection against abnormal operating conditions in accordance with the defined PSD/LSD philosophy.

TA-221 – LNG Storage Tank

TA-221 is a single-containment, flat-bottom, atmospheric cryogenic LNG storage tank designed in accordance with API 620 and EN 14620. The tank is insulated with perlite to minimize heat ingress and limit boil-off generation. LNG is stored at near-atmospheric pressure and cryogenic temperature, approximately −160 °C. LNG temperature within the tank is monitored by temperature transmitter TT-2201.

Tank pressure is monitored by pressure transmitter PT-2201 and indicated locally by PG-2201. The tank pressure controller PIC-2202 is used for low-pressure control and maintains the minimum tank pressure by regulating the operation of the vaporizer (PBUC).

Overpressure protection of the tank is provided by the tank safety valves TSV-2201 and TSV-2202, which discharge to atmosphere only in abnormal or emergency conditions.

Nitrogen injection is provided for tank inerting and purging purposes through pressure control valves PCV-2201 and PCV-2202. Nitrogen supply pressure is monitored for supervision and alarm purposes only and does not form part of an automatic pressure control loop.

Liquid level protection in TA-221 is provided by dedicated level alarms. A high-high liquid level (LAHH-2203) initiates a Process Shutdown (PSD) to prevent tank overfilling. A low-low liquid level (LALL-2201) initiates a Local Shutdown (LSD), resulting in the tripping of all LNG loading pumps (P-221A/B and P-222A/B) to protect the pumps against cavitation and dry running.

Cryogenic Loading Pumps P-221A/B and P-222A/B

Two identical LNG truck loading skids, S-221 and S-222, are provided to ensure redundancy and operational flexibility. Each skid is equipped with two cryogenic centrifugal loading pumps (P-221A/B and P-222A/B respectively), arranged in duty/standby configuration. The pumps are installed in cryostats with submerged electric motors and draw LNG from the storage tank TA-221, delivering it to the truck loading arms through vacuum-insulated transfer lines. Pump discharge pressure is monitored by dedicated pressure transmitters (PT-2211 and PT-2213 for skid S-221, and PT-2221 and PT-2223 for skid S-222) for supervision and protection purposes. Additional pressure transmitters (PT-2212/2214 and PT-2222/2224) monitor vent and cryostat pressures to detect abnormal conditions such as leaks or excessive vapor generation. Temperature transmitters installed on the pump cryostats and associated piping provide continuous monitoring of cryogenic conditions, supporting leak detection and ensuring the integrity of the loading system during operation. Pump operation is managed via variable frequency drives (VFDs), allowing controlled start-up, speed adjustment and smooth flow regulation in accordance with the defined loading sequence and flow setpoints.

Metering and Quality Control System

Each LNG loading skid is equipped with mass flowmeters intended for the measurement of the LNG quantity delivered to each truck, as a secondary source of information since the primary measurement will be based on the weight of the trucks. LNG flow is measured by Coriolis flowmeters FT-2212 and FT-2222 installed on loading skids S-221 and S-222, respectively. The metering system is fully integrated into the PCS and records transaction data for accounting purposes.

In addition to quantity measurement, LNG product quality is verified prior to dispatch by an online gas chromatograph AT-2211 installed on the LNG transfer line between TA-221 and the loading skids. AT-2211 has two ports, one connected to each loading bay, and it will perform measurements whenever the PCS sequence requires it to do so. Sampling is initiated by a start signal for each loading batch, and composition data are recorded together with the flow measurement data as part of the LNG loading transaction and archived in the plant historian.

A dedicated nitrogen gas chromatograph, AT-3571, installed at the outlet of the LNG separator V-221, provides continuous monitoring of residual nitrogen content in the LNG. This measurement supports product quality verification and operational monitoring by ensuring that the nitrogen concentration remains within the specified limits prior to storage and loading.

All critical valves, including SDV-2201/2202 and the loading line valves (XV-22XX), are fitted with limit switches (ZSH/ZSL-22XX) providing open/closed position feedback to the Process Control System (PCS). The system enforces all permissive logic and safety interlocks before authorizing loading operations, ensuring safe containment, reliable metering, and efficient LNG transfer to the road tankers while maintaining product integrity and operational safety.

BOG Return Line and Handling Interface.

During LNG truck loading operations, vapor displaced from the receiving road tanker is routed through the vapor return line of the loading skid to maintain pressure equilibrium between the truck and the LNG transfer system. The returned vapor flow is measured by dedicated flow transmitters FT-2213A and FT-2223A installed on loading skids S-221 and S-222 respectively. These measurements are used for mass balance, custody transfer accounting and operational monitoring of the loading process.

4.5 Technological Objects

4.5.1 Main Equipment List

Table 4.3 — Unit 220 – Main Equipment List

ID	TAG	EQUIPMENT
1	MP-221A/B	TRUCK LOADING PUMPS ELECTRIC MOTORS
2	MP-222A/B	TRUCK LOADING PUMPS ELECTRIC MOTORS
3	P-221A/B	TRUCK LOADING PUMPS
4	P-222A/B	TRUCK LOADING PUMPS
5	TA-221	LNG STORAGE TANK
6	V-221	LNG SEPARATOR

4.5.2 Electrical Loads

Table 4.4 — Unit 220 – Electrical Loads

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	P-221A	TRUCK LOADING PUMP	VFD	400	30
2	P-221B	TRUCK LOADING PUMP	VFD	400	30
3	P-222A	TRUCK LOADING PUMP	VFD	400	30
4	P-222B	TRUCK LOADING PUMP	VFD	400	30
5	480-JG-221	TRUCK SUPPLY 1	480-JG-221	230	0,5
6	480-JG-222	TRUCK SUPPLY 2	480-JG-222	230	0,5

4.5.3 Valves

Table 4.5 — Unit 220 – Valves

ID	TAG	SERVICE	SYSTEM	NOTES
1	LEV-2232	V-221 LIQUID LEVEL CONTROL	ESD	-
2	LV-2232	V-221 LIQUID LEVEL CONTROL	-	-
3	PSE-2201	TA-221 INTERSPACE RELIEF	-	SET PRESSURE 5 MMH2O
4	PSV-2201A	TA-221 INNER TANK RELIEF	-	SET PRESSURE 200/-2,2 MMH2O

ID	TAG	SERVICE	SYSTEM	NOTES
5	PSV-2201B	TA-221 INNER TANK RELIEF	-	SET PRESSURE 200/-2,2 MMH2O
6	PSV-2201C	TA-221 INTERSPACE RELIEF	-	SET PRESSURE 4,5/-2,5 MMH2O
7	PSV-2211A	P-221A PRESSURE RELIEF	-	SET PRESSURE 5 barg
8	PSV-2211B	P-221A PRESSURE RELIEF	-	SET PRESSURE 5 barg
9	PSV-2213A	P-221B PRESSURE RELIEF	-	SET PRESSURE 5 barg
10	PSV-2213B	P-221B PRESSURE RELIEF	-	SET PRESSURE 5 barg
11	PSV-2231A	P-222A PRESSURE RELIEF	-	SET PRESSURE 5 barg
12	PSV-2231B	P-222A PRESSURE RELIEF	-	SET PRESSURE 5 barg
13	PSV-2233A	P-222B PRESSURE RELIEF	-	SET PRESSURE 5 barg
14	PSV-2233B	P-222B PRESSURE RELIEF	-	SET PRESSURE 5 barg
15	PSV-2251A	V-221 PRESSURE RELIEF	-	SET PRESSURE 10 barg
16	PSV-2251B	V-221 PRESSURE RELIEF	-	SET PRESSURE 10 barg
17	PV-2211	LNG TO TRUCK TRAILER PRESSURE CONTROL	-	-
18	PV-2215	LNG BY PASS TO TA-221	-	-
19	PV-2221	LNG TO TRUCK TRAILER PRESSURE CONTROL	-	-
20	PV-2225	LNG BY PASS TO TA-221	-	-
21	PV-2231	V-221 PRESSURE CONTROL	-	-
22	SDEV-2201	TA-221 BOTTOM LNG FEED SHUT-DOWN	ESD	-
23	SDEV-2202	TA-221 TOP LNG FEED SHUTDOWN	ESD	-
24	SDV-2201	TA-221 BOTTOM LNG FEED SHUT-DOWN	-	-
25	SDV-2202	TA-221 TOP LNG FEED SHUTDOWN	-	-
26	TSV-2201	VAPORIZER THERMAL RELIEF	-	SET PRESSURE
27	TSV-2202	VAPORIZER THERMAL RELIEF	-	SET PRESSURE
28	TSV-2212	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
29	TSV-2214	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
30	TSV-2215	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
31	TSV-2216	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
32	TSV-2217	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
33	TSV-2218	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
34	TSV-2219	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
35	TSV-2220	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
36	TSV-2221	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
37	TSV-2222	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
38	TSV-2223	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
39	TSV-2224	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
40	TSV-2225	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg

ID	TAG	SERVICE	SYSTEM	NOTES
41	TSV-2232	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
42	TSV-2234	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
43	TSV-2235	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
44	TSV-2236	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
45	TSV-2237	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
46	TSV-2238	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
47	TSV-2239	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
48	TSV-2240	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
49	TSV-2241	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
50	TSV-2242	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
51	TSV-2243	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
52	TSV-2244	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
53	TSV-2245	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 15 barg
54	TSV-2252	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
55	TSV-2253	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
56	TSV-2254	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
57	TSV-2261	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
58	TSV-2262	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
59	TSV-2263	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
60	TSV-2264	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
61	TSV-2265	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
62	TSV-2266	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
63	TSV-2267	LINE THERMAL EXPANSION RELIEF	-	SET PRESSURE 5 barg
64	XEV-2203	TA-221 VAPORIZER SUCTION SHUT DOWN	PCS	-
65	XEV-2204	LNG TO P-221A/B SUCTION SHUT DOWN	PCS	-
66	XEV-2205	LNG TO P-222A/B SUCTION SHUT DOWN	PCS	-
67	XEV-2211	LNG TO P-221A INLET SHUT OFF	PCS	-
68	XEV-2212	P-221A DISCHARGE SHUT OFF	PCS	-
69	XEV-2213	LNG TO P-221B INLET SHUT OFF	PCS	-
70	XEV-2214	P-221B DISCHARGE SHUT OFF	PCS	-
71	XEV-2216	P-221B RETURN TO TA-221 GAS PHASE SHUT OFF	PCS	-
72	XEV-2217	P-221A RETURN TO TA-221 GAS PHASE SHUT OFF	PCS	-
73	XEV-2221	LNG TO P-222A INLET SHUT OFF	PCS	-
74	XEV-2222	P-222A DISCHARGE SHUT OFF	PCS	-
75	XEV-2223	LNG TO P-222B INLET SHUT OFF	PCS	-
76	XEV-2224	P-222B DISCHARGE SHUT OFF	PCS	-

ID	TAG	SERVICE	SYSTEM	NOTES
77	XEV-2226	P-222B RETURN TO TA-221 GAS PHASE SHUT OFF	PCS	-
78	XEV-2227	P-222A RETURN TO TA-221 GAS PHASE SHUT OFF	PCS	-
79	XV-2203	TA-221 VAPORIZER SUCTION SHUT DOWN	-	-
80	XV-2204	LNG TO P-221A/B SUCTION SHUT DOWN	-	-
81	XV-2205	LNG TO P-222A/B SUCTION SHUT DOWN	-	-
82	XV-2211	LNG TO P-221A INLET SHUT OFF	-	-
83	XV-2212	P-221A DISCHARGE SHUT OFF	-	-
84	XV-2213	LNG TO P-221B INLET SHUT OFF	-	-
85	XV-2214	P-221B DISCHARGE SHUT OFF	-	-
86	XV-2216	P-221B RETURN TO TA-221 GAS PHASE SHUT OFF	-	-
87	XV-2217	P-221A RETURN TO TA-221 GAS PHASE SHUT OFF	-	-
88	XV-2221	LNG TO P-222A INLET SHUT OFF	-	-
89	XV-2222	P-222A DISCHARGE SHUT OFF	-	-
90	XV-2223	LNG TO P-222B INLET SHUT OFF	-	-
91	XV-2224	P-222B DISCHARGE SHUT OFF	-	-
92	XV-2226	P-222B RETURN TO TA-221 GAS PHASE SHUT OFF	-	-
93	XV-2227	P-222A RETURN TO TA-221 GAS PHASE SHUT OFF	-	-

4.5.4 Instrumentation

Table 4.6 — Unit 220 – Instrument List

ID	TAG	SERVICE	RANGE	UNITS	SYSTEM
1	PG-2201	TA-221 PRESSURE	0-200	mbarg	-
2	PT-2201	TA-221 PRESSURE	0-200	mbarg	ESD
3	PT-2202	TA-221 LOW PRESSURE CONTROL WITH VAPORIZER	0-200	mbarg	PCS
4	PT-2203	NITROGEN PRESSURE	-50/50	mbarg	PCS
5	PT-2211	P-221A DISCHARGE PRESSURE	0..16	barg	PCS
6	PT-2212	P-221A VENT PRESSURE	0..2,5	barg	PCS
7	PT-2213	P-221B DISCHARGE PRESSURE	0..16	barg	PCS

ID	TAG	SERVICE	RANGE	UNITS	SYSTEM
8	PT-2214	P-221B VENT PRESSURE	0..2,5	barg	PCS
9	PT-2215	LNG TO TRUCK TRAILER PRESSURE	-200/+200	barg	PCS/ESD
10	PT-2216	BY PASS LINE PRESSURE	-200/+200	barg	PCS
11	PT-2221	P-222A DISCHARGE PRESSURE	0..16	barg	PCS
12	PT-2222	P-222A VENT PRESSURE	0..2,5	barg	PCS
13	PT-2223	P-222B DISCHARGE PRESSURE	0..16	barg	PCS
14	PT-2224	P-222B VENT PRESSURE	0..2,5	barg	PCS
15	PT-2225	LNG TO TRUCK TRAILER PRESSURE	-200/+200	barg	PCS/ESD
16	PT-2226	BY PASS LINE PRESSURE	-200/+200	barg	PCS
17	PT-2231	LNG TO V-221 PRESSURE	0..10	barg	PCS
18	TT-2201B	TA-221 TEMPERATURE	-196/100	°C	PCS
19	TT-2205	TA-221 INTERSPACE PERLITE TEMPERATURE	-196/80	°C	PCS
20	TT-2206	TA-221 INTERSPACE PERLITE TEMPERATURE	-196/80	°C	PCS
21	TT-2213	P-221A RETURN TO TA-221 GAS PHASE TEMPERATURE	-200/+200	°C	PCS
22	TT-2214	P-221B RETURN TO TA-221 GAS PHASE TEMPERATURE	-200/+200	°C	PCS
23	TT-2215	P-221A LEAK DETECTION STREAM TEMPERATURE	-200/+200	°C	PCS
24	TT-2216	P-221B LEAK DETECTION STREAM TEMPERATURE	-200/+200	°C	PCS
25	TT-2217	BY PASS SKID S-221B INLET TEMPERATURE	-200/+200	°C	PCS
26	TT-2223	P-222A RETURN TO TA-221 GAS PHASE TEMPERATURE	-200/+200	°C	PCS
27	TT-2224	P-222B RETURN TO TA-221 GAS PHASE TEMPERATURE	-200/+200	°C	PCS
28	TT-2225	P-222A LEAK DETECTION STREAM TEMPERATURE	-200/+200	°C	PCS
29	TT-2226	P-222B LEAK DETECTION STREAM TEMPERATURE	-200/+200	°C	PCS
30	TT-2227	BY PASS SKID S-222B INLET TEMPERATURE	-200/+200	°C	PCS
31	ZSH-2201	SDV-2201 OPEN POSITION INDICATION	-	-	ESD
32	ZSH-2202	SDV-2202 OPEN POSITION INDICATION	-	-	ESD
33	ZSH-2203	XV-2203 OPEN POSITION INDICATION	-	-	PCS
34	ZSH-2204	XV-2204 OPEN POSITION INDICATION	-	-	PCS
35	ZSH-2205	XV-2205 OPEN POSITION INDICATION	-	-	PCS
36	ZSH-2211	XV-2211 OPEN POSITION INDICATION	-	-	PCS
37	ZSH-2212	XV-2212 OPEN POSITION INDICATION	-	-	PCS
38	ZSH-2213	XV-2213 OPEN POSITION INDICATION	-	-	PCS
39	ZSH-2214	XV-2214 OPEN POSITION INDICATION	-	-	PCS
40	ZSH-2216	XV-2216 OPEN POSITION INDICATION	-	-	PCS
41	ZSH-2217	XV-2217 OPEN POSITION INDICATION	-	-	PCS
42	ZSH-2221	XV-2221 OPEN POSITION INDICATION	-	-	PCS
43	ZSH-2222	XV-2222 OPEN POSITION INDICATION	-	-	PCS

ID	TAG	SERVICE	RANGE	UNITS	SYSTEM
44	ZSH-2223	XV-2223 OPEN POSITION INDICATION	-	-	PCS
45	ZSH-2224	XV-2224 OPEN POSITION INDICATION	-	-	PCS
46	ZSH-2226	XV-2226 OPEN POSITION INDICATION	-	-	PCS
47	ZSH-2227	XV-2227 OPEN POSITION INDICATION	-	-	PCS
48	ZSL-2201	SDV-2201 CLOSED POSITION INDICATION	-	-	ESD
49	ZSL-2202	SDV-2202 CLOSED POSITION INDICATION	-	-	ESD
50	ZSL-2203	XV-2203 CLOSED POSITION INDICATION	-	-	PCS
51	ZSL-2204	XV-2204 CLOSED POSITION INDICATION	-	-	PCS
52	ZSL-2205	XV-2205 CLOSED POSITION INDICATION	-	-	PCS
53	ZSL-2211	XV-2211 CLOSED POSITION INDICATION	-	-	PCS
54	ZSL-2212	XV-2212 CLOSED POSITION INDICATION	-	-	PCS
55	ZSL-2213	XV-2213 CLOSED POSITION INDICATION	-	-	PCS
56	ZSL-2214	XV-2214 CLOSED POSITION INDICATION	-	-	PCS
57	ZSL-2216	XV-2216 CLOSED POSITION INDICATION	-	-	PCS
58	ZSL-2217	XV-2217 CLOSED POSITION INDICATION	-	-	PCS
59	ZSL-2221	XV-2221 CLOSED POSITION INDICATION	-	-	PCS
60	ZSL-2222	XV-2222 CLOSED POSITION INDICATION	-	-	PCS
61	ZSL-2223	XV-2223 CLOSED POSITION INDICATION	-	-	PCS
62	ZSL-2224	XV-2224 CLOSED POSITION INDICATION	-	-	PCS
63	ZSL-2226	XV-2226 CLOSED POSITION INDICATION	-	-	PCS
64	ZSL-2227	XV-2227 CLOSED POSITION INDICATION	-	-	PCS

4.5.5 Junction Boxes

Table 4.7 — Unit 220 – Junction Boxes

ID	TAG	SERVICE	CONNECTION	TYPE
1	P-PN-S221A.1	LNG STORAGE TANK	COPPER-FIBER	HA
2	P-PN-S221B.1	LNG LOADING UNITS	COPPER-COPPER	HA
3	P-PN-S222A.1	LNG STORAGE TANK	COPPER-COPPER	HA
4	P-PN-S222B.1	LNG LOADING UNITS	COPPER-FIBER	HA
5	P-PN-TA221.1	LNG STORAGE TANK	COPPER-COPPER	STANDARD

4.5.6 Controllers

Table 4.8 — Unit 220 – Controllers

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
LIC-2232	V-221 LIQUID LEVEL CONTROLLER	LV-2232	0..100 %	50 %
PIC-2202	TA-221 LOW PRESSURE CONTROLLER	TSV-2202	0..200 mbarg	150
PIC-2231	LNG TO V-221 PRESSURE CONTROLLER	PV-2231	0..10 barg	—
XC-2212	P-221A/B TRUCK LOADING FLOW CONTROLLER	PV-2211	0..120 m^3/h	60
XC-2222	P-222A/B TRUCK LOADING FLOW CONTROLLER	PV-2221	0..120 m^3/h	60

4.6 Operating Interface

4.6.1 PCS Operating Screens

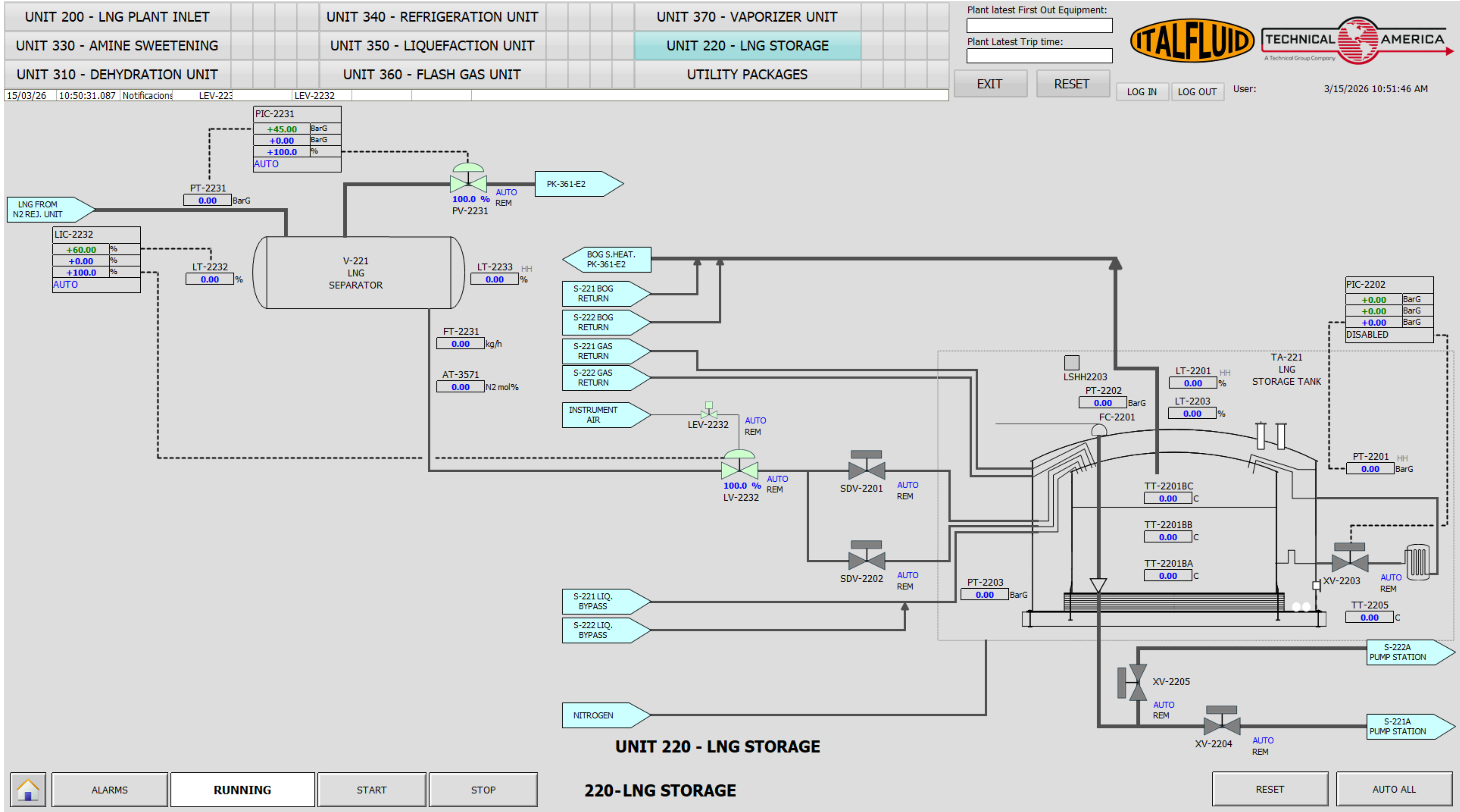


Figure 4.2: Unit 220 - HMI Operating Screen

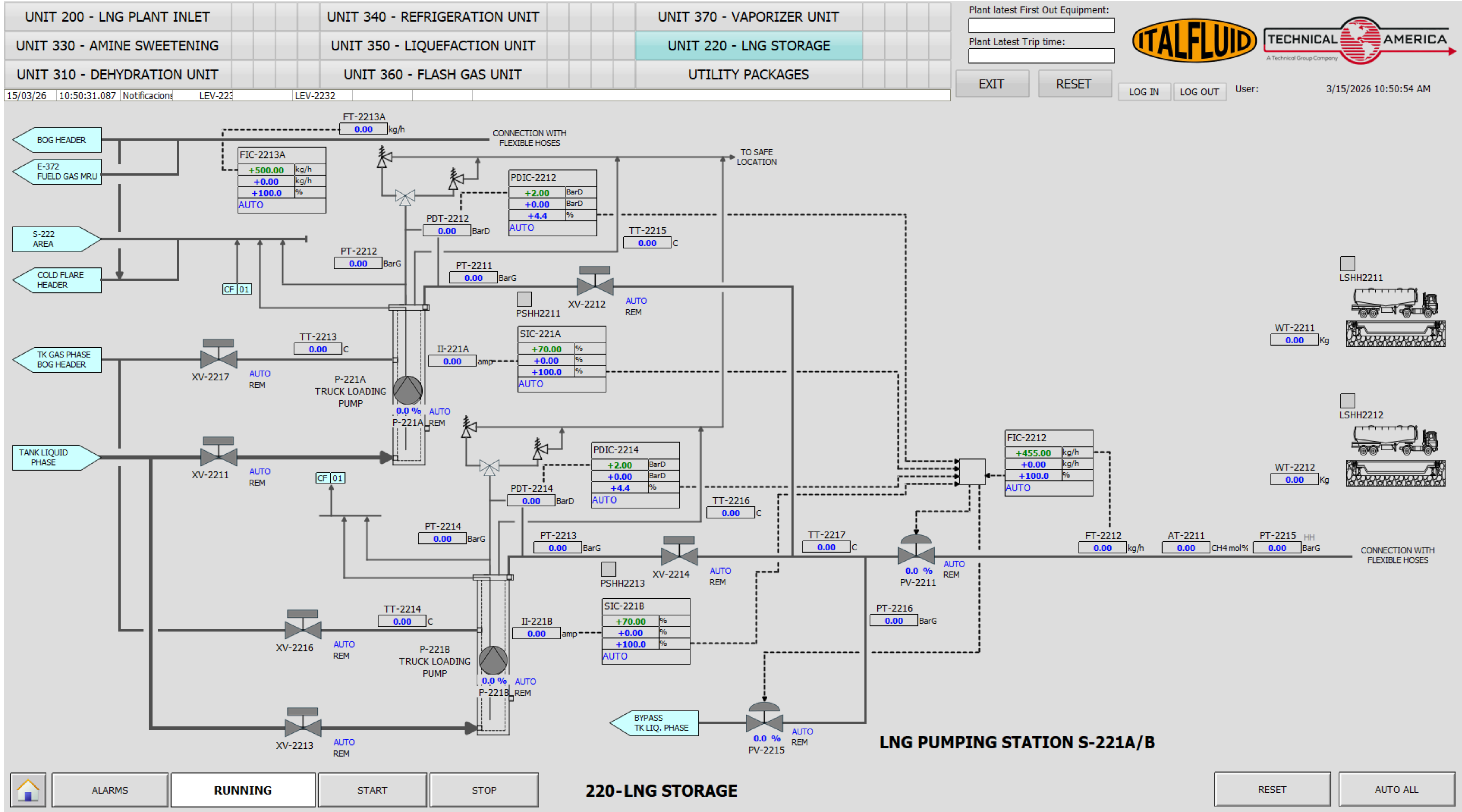


Figure 4.3: Unit 220 - HMI Operating Screen

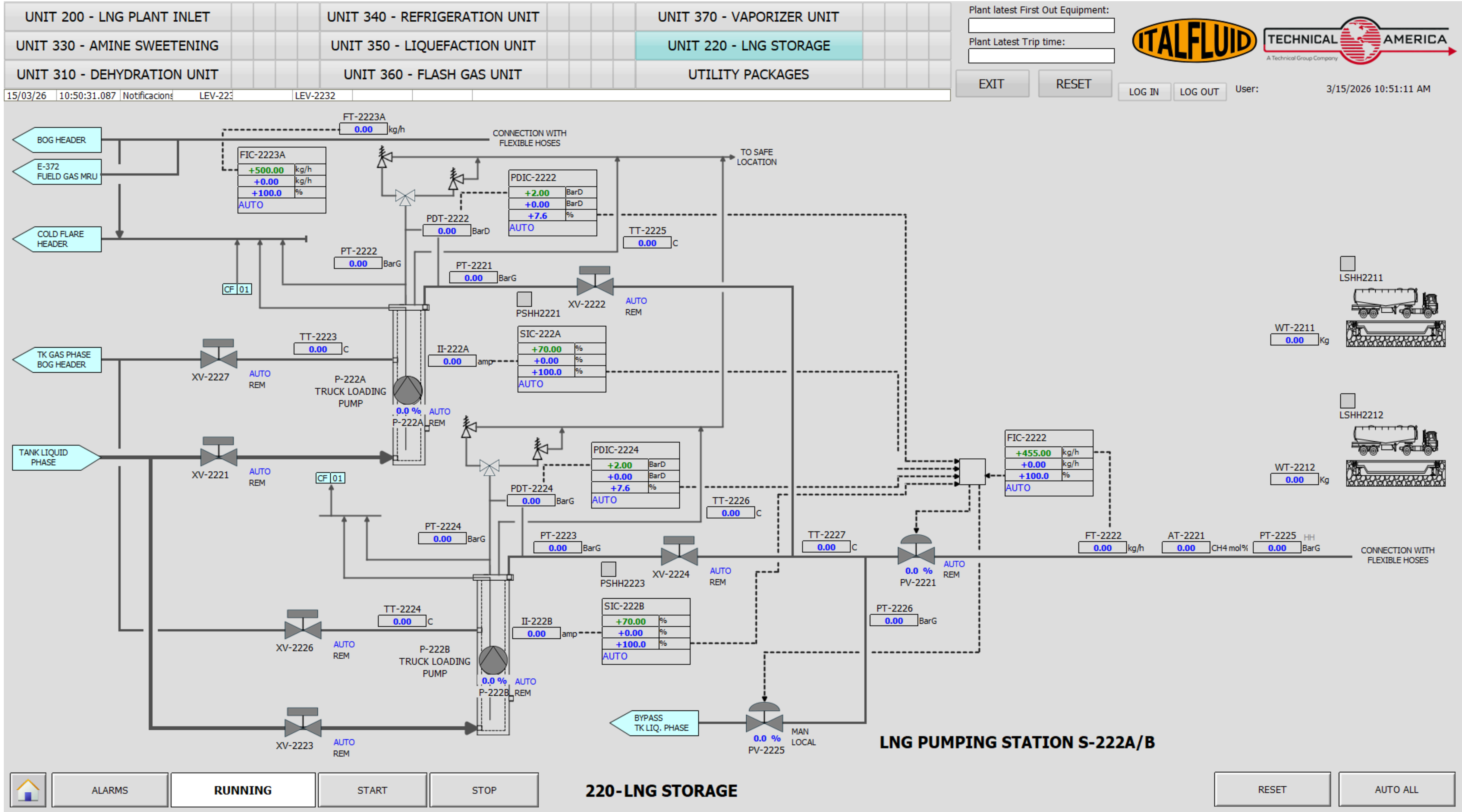


Figure 4.4: Unit 220 - HMI Operating Screen

4.6.2 Critical Alarms and Trips

Table 4.9 — Unit 220 – Critical Alarms

ALARM	INSTRUMENT	TYPE	SP	DESCRIPTION
AAHH-3571	AT-3571	PSD	—	GAS CROMATOGRAPHER HIGH NITROGEN CONTENT
LAHH-2201	LT-2201	PSD	—	HIGH HIGH LIQUID LEVEL ON TA-221
LAHH-2233	LT-2233	PSD	—	HIGH HIGH LIQUID LEVEL ON V-221
LALL-2233	LT-2233	LSD	—	LOW LOW LIQUID LEVEL ON V-221
LALL-2201	LT-2201	LSD	—	LOW LOW LIQUID LEVEL ON TA-221
PAHH-2021	PIT-2021	PSD	—	HIGH HIGH PRESSURE TA-221
PAHH-2201	PT-2201	LSD	—	HIGH HIGH PRESSURE ON TA-221
PAHH-2211	PSHH-2211	LSD	—	HIGH HIGH PRESSURE ON P-221A OUTLET
PAHH-2213	PSHH-2213	LSD	—	HIGH HIGH PRESSURE ON P-221B OUTLET
PAHH-2215	PT-2215	LSD	—	HIGH HIGH PRESSURE ON S-221 OUTLET
PAHH-2221	PSHH-2221	LSD	—	HIGH HIGH PRESSURE ON P-222A OUTLET
PAHH-2223	PSHH-2223	LSD	—	HIGH HIGH PRESSURE ON P-222B OUTLET
PAHH-2225	PT-2225	LSD	—	HIGH HIGH PRESSURE ON S-222 OUTLET
TSL-2213	TT-2213	LSD	—	LOSS OF PERMISSIVE DUE TO HIGH TEMPERATURE
TSL-2214	TT-2214	LSD	—	LOSS OF PERMISSIVE DUE TO HIGH TEMPERATURE
TSL-2223	TT-2223	LSD	—	LOSS OF PERMISSIVE DUE TO HIGH TEMPERATURE
TSL-2224	TT-2224	LSD	—	LOSS OF PERMISSIVE DUE TO HIGH TEMPERATURE
LSD25	—	LSD	—	S-221 PACKAGE EMERGENCY STOP
LSD26	—	LSD	—	S-222 PACKAGE EMERGENCY STOP

4.6.3 Unit Cause and Effect Matrix

Table 4.10 — Unit 220 – Cause & Effect Matrix

ALARM	INSTRUMENT	LEV-2232	SDV-2201	SDV-2202	XV-2211	XV-2212	XV-2213	XV-2214	XV-2216	XV-2217	P-221A	P-221B	XV-2221	XV-2222	XV-2223	XV-2224	XV-2226	XV-2227	P-222A	P-222B	XV-XXXX	XV-YYYY
LALL-2233	LT-2233	C	C	C																		
LALL-2201	LT-2201										T	T							T	T		
PAHH-2201	PT-2201		C	C																		
S-221	Local PB				C	C	C	C	C	C	T	T										
S-222	Local PB												C	C	C	C	C	C	T	T		
PAHH-2211	PSHH-2211										T											
PAHH-2213	PSHH-2213											T										
PAHH-2221	PSHH-2221																		T			
PAHH-2223	PSHH-2223																			T		
PAHH-2215	PT-2215				C		C	C	C	C	T	T										
PAHH-2225	PT-2225												C	C	C	C	C	C	T	T		
TSL-2213	TT-2213										T											
TSL-2214	TT-2214											T										
TSL-2223	TT-2223																		T			
TSL-2224	TT-2224																			T		
FALL-2213A	FT-2213A										T	T										
FALL-2223A	FT-2223A																		T	T		
WAHH-2211	WE-2211										T	T										
WAHH-2212	WE-2212																		T	T		
QAHH-2211	FQT-2211										T	T										
QAHH-2212	FQT-2212																		T	T		
LAHH-2211	LSH-2211																				T	
LAHH-2212	LSH-2212																					T

4.7 Operation and Control

All operations in the LNG transfer system involve the handling of cryogenic liquids at extremely low temperatures (approximately $-160\,°C$). Contact with cryogenic fluids, cold surfaces, or cold vapors can cause severe injuries, including cryogenic burns, material embrittlement, and equipment damage. Strict safety procedures shall be followed at all times when operating equipment in this area. If you are not trained, not authorized, or not familiar with these procedures, do not operate any equipment. Contact your HSE supervisor before performing any operation in this system.

The LNG transfer system is located in a classified hazardous area where flammable gases may be present. All equipment installed in this area is designed and certified for operation in potentially explosive atmospheres. Strict compliance with all safety procedures, operating instructions, and hazardous area regulations is mandatory. Unauthorized equipment, ignition sources, or deviations from approved procedures may result in fire or explosion hazards. Only authorized and trained personnel shall operate or perform work in this area.

4.7.1 Special Controllers XC-2212 and XC-2222

Both controllers refer to dedicated control functions that perform a set of actions to ensure correct sequential operation and protection override functions. These controllers are described in the following table.

Table 4.11 — Unit 220 – Special Controllers XC-2212/2222

ID	TAG	DESCRIPTION
CONTROLLER XC-2212		
1	PV-2215	In case of high-high pressure alarm PAHH-2216, the controller forces the opening of valve PV-2215 in order to divert LNG flow back to TA-221 and relieve excessive pressure in the LNG loading line.

ID	TAG	DESCRIPTION
2	FIC-2212	The controller regulates the LNG loading flow by modulating the discharge control valve PV-2211 according to the setpoint received from the PCS.
3	FIC-2212	The controller receives remote setpoints from the PCS to operate under predefined operating modes such as low-flow and high-flow LNG loading conditions.
4	PDIC-2212	The controller calculates the differential pressure of pump P-221A/B using the difference between discharge and suction pressure measurements in order to monitor pump operating conditions and protect the equipment.
5	PUMP START INTERLOCK	The controller prevents pump start-up unless the corresponding discharge valve PV-2211 is confirmed open, avoiding pump dead-head operation and protecting the pump.
6	GC MEASUREMENT TRIGGER	The controller sends a start measurement signal to the gas chromatographs through Modbus communication in order to initiate LNG composition analysis associated with each loading batch.
7	LOADING DATA MANAGEMENT	The controller stores LNG loading information in the plant relational database including loading date, manually entered truck weight, and associated measurement data for traceability and reporting.
8	COOL DOWN STATUS MANAGEMENT	The controller manages the status bits associated with the LNG line cool-down sequence, ensuring correct sequencing, permissive verification, and operational status monitoring before loading operations can start.
CONTROLLER XC-2222		
9	PV-2225	In case of high-high pressure alarm PAHH-2226, the controller forces the opening of valve PV-2225 in order to divert LNG flow back to TA-221 and relieve excessive pressure in the LNG loading line.
10	FIC-2222	The controller regulates the LNG loading flow by modulating the discharge control valve PV-2221 according to the setpoint received from the PCS.
11	FIC-2222	The controller receives remote setpoints from the PCS to operate under predefined operating modes such as low-flow and high-flow LNG loading conditions.
12	PDIC-2222	The controller calculates the differential pressure of pump P-222A/B using the difference between discharge and suction pressure measurements in order to monitor pump operating conditions and protect the equipment.
13	PUMP START INTERLOCK	The controller prevents pump start-up unless the corresponding discharge valve PV-2221 is confirmed open, avoiding pump dead-head operation and protecting the pump.
14	GC MEASUREMENT TRIGGER	The controller sends a start measurement signal to the gas chromatographs through Modbus communication in order to initiate LNG composition analysis associated with each loading batch.
15	LOADING DATA MANAGEMENT	The controller stores LNG loading information in the plant relational database including loading date, manually entered truck weight, and associated measurement data for traceability and reporting.

ID	TAG	DESCRIPTION
16	COOL DOWN STATUS MANAGEMENT	The controller manages the status bits associated with the LNG line cool-down sequence, ensuring correct sequencing, permissive verification, and operational status monitoring before loading operations can start.

4.7.2 Loading Panel Operation

The LNG loading station is equipped with a local control panel located in the loading bay. This panel allows the operator to perform essential loading functions while remaining close to the truck during the transfer operation. The panel layout and the location of the main control buttons are illustrated in Figure 4.5.

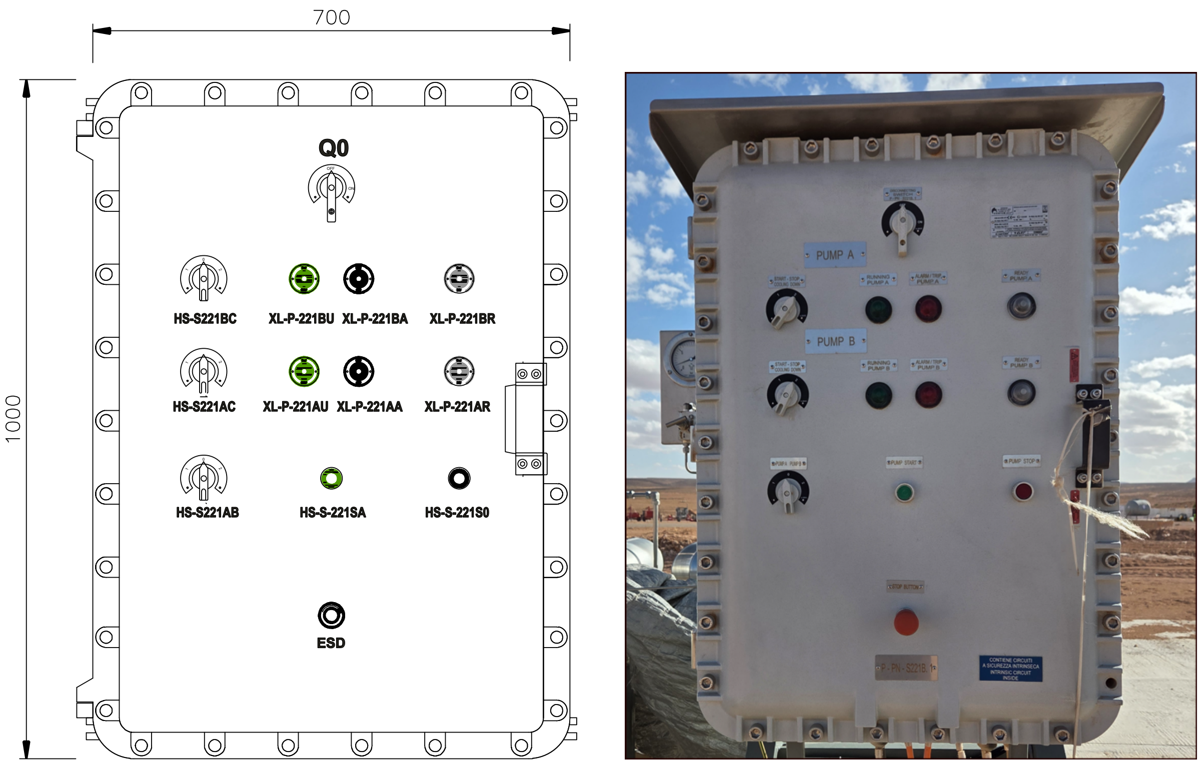


Figure 4.5: LNG loading station field control panel S221B

The loading panel provides the following controls:

Table 4.12 — Loading station field panel components

TAG	SIGNAL	DESCRIPTION
ESD	Emergency Shutdown	Emergency shutdown pushbutton
HS-S-221SA	START PUMP	Start command for selected pump
HS-S-221SO	STOP PUMP	Stop command for selected pump
HS-S221AB	PUMP SELECTION	Select active loading pump (Pump A or Pump B)
HS-S221AC	PRECOOLING PUMP A	Request pre-cooling procedure for Pump A
HS-S221BC	PRECOOLING PUMP B	Request pre-cooling procedure for Pump B
Q0	POWER SWITCH	Main selector to energize or isolate the loading panel.
XL-P-221AA	PUMP A ALARM	Pump A alarm/trip indication
XL-P-221AR	PUMP A READY	Pump A ready indicaton
XL-P-221AU	PUMP A RUNNING	Pump A running indication
XL-P-221BA	PUMP B ALARM	Pump B alarm/trip indication
XL-P-221BR	PUMP B READY	Pump B ready indicaton
XL-P-221BU	PUMP B RUNNING	Pump B running indication

4.7.3 LNG Transfer System

The speed of the pump is controlled by the variable frequency drive (VFD) and is regulated until the required discharge pressure is achieved.

The discharge pressure depends on the following factors:

- Counter-pressure of the receiving tank or downstream utilizer.
- Pressure losses along the pipeline.
- Liquid level in the storage tank.

The flow rate depends on the demand of the utilizer. If the required flow is lower than the minimum continuous flow of the pump, a portion of the pumped liquid must be recirculated through the bypass line in order to maintain the minimum allowable pump flow.

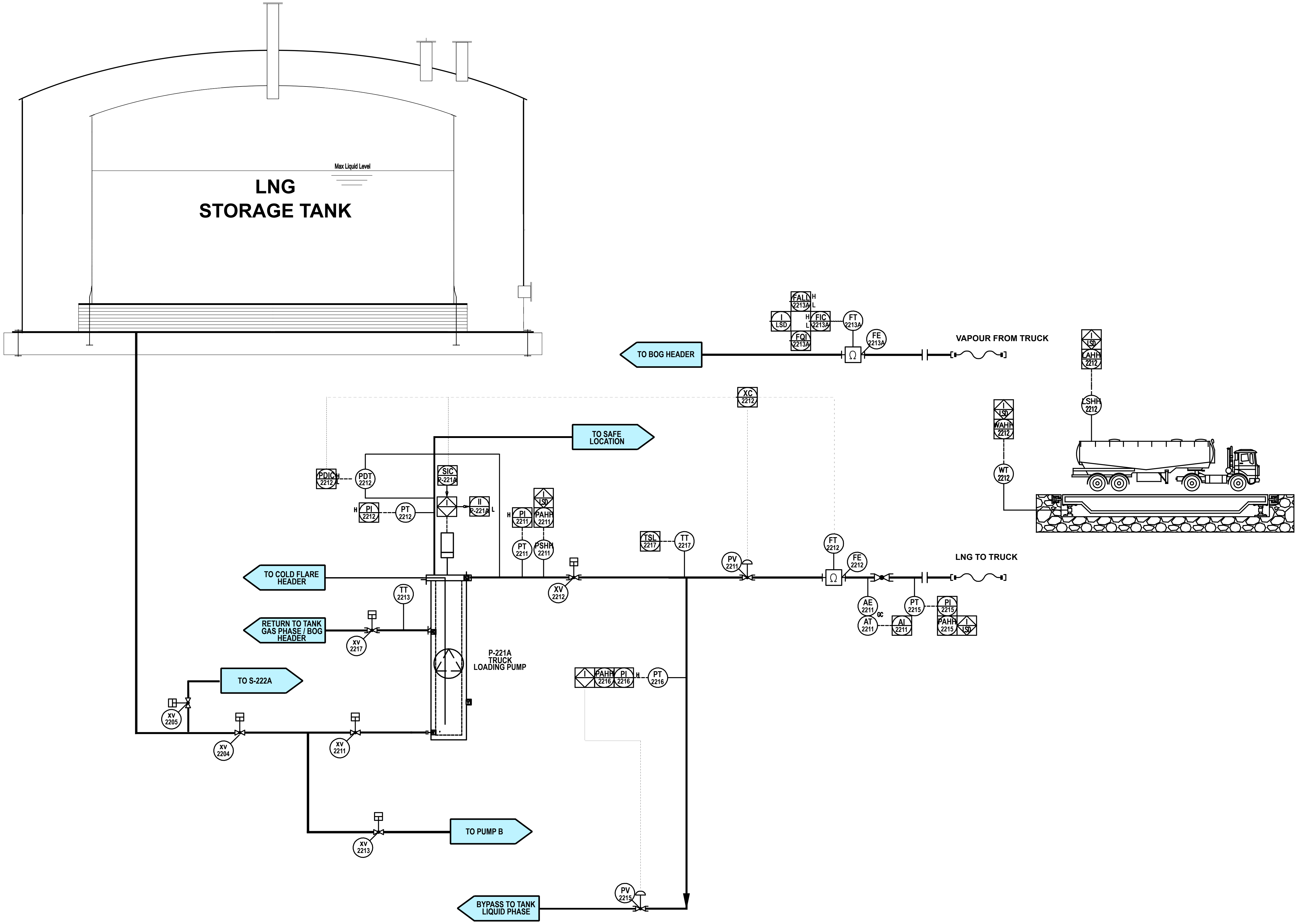


Figure 4.6: Unit 220 - LNG Loading Diagram Pump P-221

4.7.4 Liquid Level

The liquid level in the tank shall ensure that the available Net Positive Suction Head ($NPSH_a$) is greater than the required Net Positive Suction Head ($NPSH_r$). In addition, the liquid level must guarantee proper lubrication of the rotating components (bearings, DU bushings, etc.) and adequate cooling of the motor windings.

The $NPSH_a$ is primarily provided by the static head of the liquid column and can be increased by applying a temporary artificial pressure to the tank through the Pressure Build-Up (PBU) system. A PBU pressure of 200 mbar above the actual vapor pressure, considering a liquid density of 450 kg/m^3, corresponds to an additional contribution of approximately 4.5 m to the $NPSH_a$.

The $NPSH_r$ is determined by the impeller design and can be reduced by lowering the rotational speed to the minimum allowable value and regulating the pump flow.

The NPSH reference datum corresponds to the inducer inlet, located 150 mm from the pump suction nozzle.

To prevent pump damage, the start of the pumps shall be inhibited when the LNG level in tank TA-221 is below the minimum level required to guarantee an available NPSH greater than the pump required NPSH.

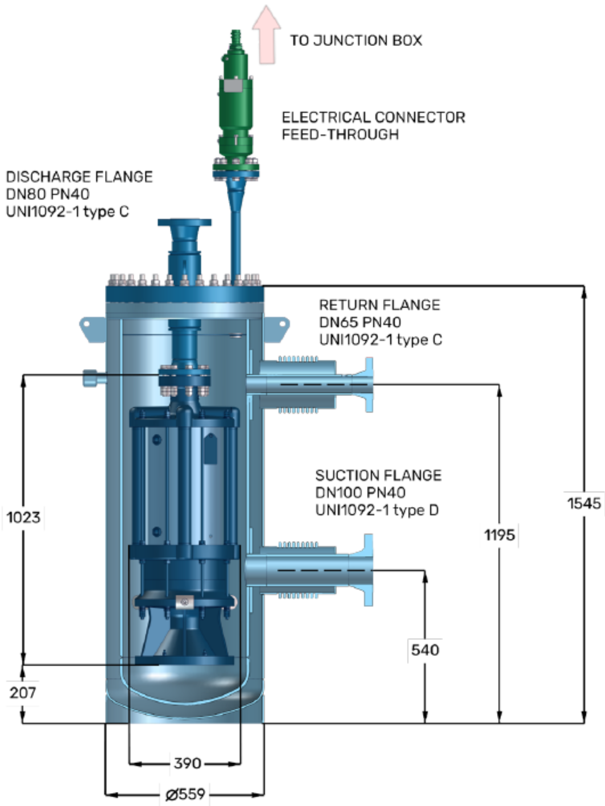
The LNG level is monitored by level transmitter LT-2201. When the level reaches the low-low alarm setpoint, the signal LALL-2201 is generated. This signal acts as an interlock and start permissive, preventing the pumps from starting when the liquid level is below the minimum level required for safe operation. Pump start shall only be permitted once the minimum liquid level condition is restored.

The minimum liquid level for normal operation (see Figure 4.7) corresponds to the lowest LNG level that ensures proper lubrication of the bearings and prevents dry running. This level coincides with the position of the return line connection, located 1,195 mm above ground level.

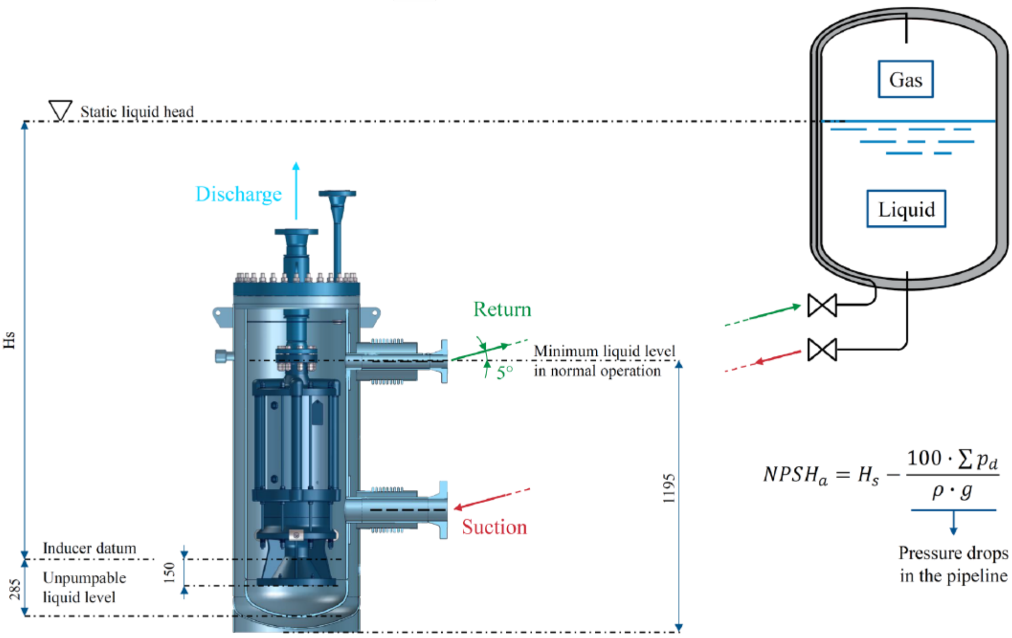
A temperature probe TT-2213 is installed to monitor the temperature conditions required to ensure proper pump cooling prior to restart and during normal operation. The corresponding instrument tag shall be verified against the latest revision of the PID and the tags assigned to the respective pumps.

Cryostat stripping operations of short and limited duration can be performed thanks to the pump self-lubricating system, which ensures a minimum level of cooling and lubrication.

Once the suction valve is closed, set the VFD at the minimum allowable speed (condition of minimum NPSHr) and strip away the LNG from the Cryostat until the prime is lost (unpumbable liquid level). The remaining LNG can be drained opening the liquid vent valve (221-HV-018) on the cover of the Cryostat.



(a) Cryogenic submerged pump installation configuration.



(b) Available NPSH configuration for cryogenic pump operation.

Figure 4.7: Cryogenic submerged pump installation and hydraulic conditions.

4.7.5 Suction Line Cooling Down

This operation consists of the gradual cooling of the suction system, which includes the cryostat, the pump, and the lines connecting the cryostat to the LNG storage tank, followed by the filling of the cryostat with LNG.

This procedure is performed manually, as it requires the operator to carry out several manual valve opening and closing actions during the cooldown process. Therefore, continuous operator attention is required to ensure a controlled and safe precooling of the system.

Table 4.13 — Suction line cooldown procedure

Step	Tag	SP		Action	Description
		Value	Unit		
1	XV-2204	–	–	OC	Open the storage tank valves.
2	XV-2217	–	–	OC	Open the pneumatic valve on the return line.
3	221-HV-021	–	–	OC	Allow the system to cool down gradually by circulating cold gas from the storage tank and venting through valve 221-HV-021
4	221-HV-021	–	–	OC	Once the temperature stabilizes, close the gas vent valve.
5	XV-2211	–	–	OC	Open the pneumatic valve on the suction line. The cryostat begins to fill with liquid and the displaced gas flows through the return line back to the storage tank.
6	TT-2213	-40	°C/h	CK	Control valve opening to avoid excessive thermal stress. The cooling rate measured by TT-2213 shall be approximately -40 °C/h.
7	TT-2213	-165	°C	CK	When the temperature transmitter reaches the imposed set point, start a 15 minute stabilization countdown. If the temperature rises above the threshold the countdown shall restart.

If the pneumatic valves on the suction and return lines (XV-2211, XV-2217) are kept open the pump will be ready to start (Stand-by condition).

Start the pump only if completely submerged into the cryogenic liquid. The cryogenic liquid provides the lubrication of the rotating parts (bearings, motor shaft, etc.) and the cooling of the motor.

4.7.6 Discharge Cooling Down

After the suction line has been cooled down, it is necessary to cool down the discharge line running from skid S-221A/S-222A to S-221B/S-222B, which are approximately 60 m and 110 m long respectively.

Table 4.14 — Discharge line cooldown procedure

Step	Tag	SP		Action	Description
		Value	Unit		
1	HS-S221AC	–	–	OC	Set the selector switch to the ON position to initiate the automatic precooling sequence.
2	XC-2212	–	–	SC	System command CHECK COOLING LINE STATUS BIT
3	PV-2215	20	%	SC	System command SET PV-2215.SP=20%
4	PV-2211	0	%	SC	System command SET PV-2211.SP=0%
5	XV-2212	–	–	SC	System command OPEN valve and confirm its position
6	SIC-P-221A	30	%	SC	System command SET SIC-P-221A/B.SP=30%
7	P-221A/B	–	–	SC	System command START the selected LNG loading pump
8	PI-2211	2	barg	SC	System command WAIT until presure is built in the pump
9	TT-2217	-40	°C/h	CK	Monitor the temperature decrease at TT-2217 and the pressure increase in the bypass return line at PT-2216. Ensure the cooling rate does not exceed -40 °C/h.
10	TT-2217	-165	°C	SC	System command CHECK SP: When the temperature reaches the imposed set point, start a 15 minute stabilization countdown. If the temperature rises above the threshold the countdown shall restart.
11	–	–	–	CK	After completion of the cooldown procedure, the line is ready for normal operation.
12	P-221A/B	–	–	SC	System command STOP the selected LNG loading pump
13	PV-2215	0	%	SC	System command SET PV-2215.SP=0%.
14	TT-2217	-155	°C	SC	System command CHECK temperature and assign STATUS BIT

Voltage during operation of pump-motor set should be within 5% of the rated voltage. Similarly current should be below the rated current as per nameplate on the motor.

4.7.7 Truck Cooling Down

Once the complete transfer line, from the storage tank to the loading station S-222A/B, has been cooled down, the system is ready to deliver product to the trucks.

If truck cooling down is requested, a dedicated procedure shall be initiated in order to account for the amount of LNG used for this purpose. The following steps shall be performed.

Table 4.15 — Truck cooldown operating procedure

Step	Tag	SP		Action	Description
		Value	Unit		
1	–	–	–	CK	Confirm that the truck is correctly positioned in the loading bay and that the parking brake is engaged
2	–	–	–	CK	Verify that the truck grounding cable is properly connected and that the grounding permissive signal is active
3	–	–	–	CK	Confirm that vapor return and liquid loading hoses are properly connected and that all mechanical connections are secure
4	–	–	–	CK	Verify that the truck is ready to receive cryogenic LNG for cooldown purposes.
5	HS-S221AC	–	–	OC	Set the selector switch to the ON position to initiate the automatic precooling sequence.
6	XC-2212	–	–	SC	System command CHECK COOLING LINE STATUS BIT
7	XV-2212	–	–	SC	System command OPEN valve and confirm its position.
8	SIC-P-221A	30	%	SC	System command SET SIC-P-221A/B.SP=30%
9	FQI-2212	0	–	SC	System command RESET the LNG loading flow totalizer
10	FQI-2213A	0	–	SC	System command RESET the LNG return flow totalizer
11	AE-2211	–	–	SC	System command Analyzer countdown for LNG quality monitoring.

Step	Tag	SP		Action	Description
		Value	Unit		
12	P-221A/B	–	–	SC	System command START the selected LNG loading pump
13	PDIC-2212	2	bar	SC	System command WAIT until presure is built in the pump
14	FIC-2212	12	m^3/h	SC	System command SET FIC-2212.SP=12 m^3/h
15	PV-2215	0	%	SC	System command SET PV-2215.SP=0%
16	HS-S221AC	–	–	OC	Once truck cooldown, set the selector switch to the OFF position to stop the automatic precooling sequence.
17	P-221A/B	–	–	SC	System command STOP the selected LNG loading pump
18	PV-2211	0	%	SC	System command SET PV-2211.SP=0%
19	PV-2215	100	%	SC	System command SET PV-2215.SP=100%.
20	PCS	–	–	OC	Press “Save Truck Cooldown Procedure”. The relevant information for invoicing purposes will be automatically stored in the database.

4.7.8 Truck LNG Loading

Once the truck cooldown procedure has been completed and the transfer lines are confirmed to be at cryogenic temperature, the system is ready to start the LNG loading operation.

Truck loading shall be performed under continuous operator supervision. The loading system incorporates a dead-man confirmation philosophy requiring operator presence in the loading bay in order to initiate and continue LNG transfer.

The operator shall remain in the loading area during the complete transfer operation and shall be ready to stop the loading operation in case of abnormal conditions.

Table 4.16 — Truck LNG loading operating procedure

Step	Tag	SP		Action	Description
		Value	Unit		
1	–	–	–	CK	Confirm that the truck is correctly positioned in the loading bay and that the parking brake is engaged
2	–	–	–	CK	Verify that the truck grounding cable is properly connected and that the grounding permissive signal is active

Step	Tag	SP		Action	Description
		Value	Unit		
3	–	–	–	CK	Confirm that vapor return and liquid loading hoses are properly connected and that all mechanical connections are secure
4	–	–	–	CK	Verify that the truck is ready to receive cryogenic LNG, confirming that the tank pressure is within the acceptable range or that the cooldown procedure has been completed.
5	HS-S-221SA	–	–	OC	Press the START button in the panel
6	XV-2212	–	–	SC	System command OPEN valve
7	PV-2211	10	%	SC	System command SET PV-2211.SP=10%
8	PV-2215	50	%	SC	System command SET PV-2215.SP=50%
9	SIC-P-221A/B	50	%	SC	System command SET SIC-P-221A/B.SP=50%
10	FQI-2212	0	–	SC	System command RESET the LNG loading flow totalizer
11	FQI-2213A	0	–	SC	System command RESET the LNG return flow totalizer
12	AE-2211	–	–	SC	System command Analyzer countdown for LNG quality monitoring
13	P-221A/B	–	–	SC	System command START the selected LNG loading pump
14	PDIC-2212	2	bar	SC	System command WAIT until presure is built in the pump
15	FIC-2212	12	m^3/h	SC	System command SET FIC-2212.SP=12 m^3/h
16	PV-2215	0	%	SC	System command SET PV-2215.SP=0%
17	–	–	–	CK	LNG transfer starts at minimum flow rate. Maintain reduced flow for approximately five minutes to stabilize the truck tank conditions
18	–	–	–	SC	System command WAIT 5 minutes, a new start signal is required to continue loading procedure and incrfease flow (DEAD MAN)
19	HS-S-221SA	–	–	OC	Press the START button in the panel
20	FIC-2212	60	m^3/h	SC	System command SET FIC-2212.SP=60 m^3/h
21	SIC-P-221A/B	100	%	SC	System command RAMP-UP pump speed to reach 100%
22	PV-2211	–	–	CK	The LNG loading control valve automatically regulates the flowrate to maintain stable transfer conditions
23	PT-2215	10	bar	CK	Continuously monitor truck tank pressure and confirm that pressure remains within acceptable limits

Step	Tag	SP		Action	Description
		Value	Unit		
24	FQI-2212	–	kg	CK	Monitor the transferred LNG quantity according to the truck loading order
25	HS-S-221SO	–	–	OC	Press the *Stop Loading* pushbutton when the required LNG quantity has been transferred
26	P-221A/B	–	–	SC	System command STOP the selected LNG loading pump
27	XV-2212	–	–	SC	System command CLOSE valve
28	FIC-2212	0	m^3/h	SC	System command SET FIC-2212.SP=0 m^3/h
29	PV-2211	0	%	SC	System command SET PV-2211.SP=0% to fully isolate the transfer line
30	PV-2215	100	%	SC	System command SET PV-2215.SP=100% to depressurize the loading line toward the storage tank
31	PCS	–	–	OC	Press “Save Truck Loading Operation”. The transferred LNG quantity will be automatically stored in the database for invoicing purposes.
32	PT-2211	1	bar	CK	Verify that pressure in the loading line decreases to a safe level before disconnection of hoses
33	–	–	–	CK	Confirm that LNG transfer has stopped and that no flow is detected in the loading line
34	–	–	–	CK	Disconnect the liquid loading hose followed by the vapor return hose and secure all connections
35	–	–	–	CK	Disconnect the truck grounding cable and confirm that the loading bay is clear of equipment
36	–	–	–	CK	Authorize truck departure from the loading bay once all safety checks have been completed

4.8 Troubleshooting

Table 4.17 — Unit 220 – LNG Pump Troubleshooting

TROUBLE	POSSIBLE CAUSE	SOLUTION
Motor doesn't start (buzzing noise or no noise)	Interruption of 1 or 2 phases	Check the power supply
Low insulation	Short circuit in the motor winding	Disassemble the pump and replace the parts (*)
Motor amperage fluctuations	Pump operates below minimum continuous flow	Increase flow
High current absorbed	Ice formation into the bearings	Empty and flush the cryostat with nitrogen gas
	Worn bearings	Disassemble the pump and replace the parts (if necessary) (*)
Rated speed not achieved	Wrong setting of the VFD	Check voltage–frequency law
Pump doesn't prime or cavitates	Insufficient cooling	Extend cool down time
	Insufficient NPSHa	Raise liquid level or pressure into the tank
	Wrong direction of rotation	Reverse polarity from electric control panel
	Discharge flow of the pump too high or too low	Adjust the discharge flow using the throttle regulating valve
	Filter clogged	Remove the conical filter from the suction line. Clean or replace the part
Too low head measured	Wrong direction of rotation	Reverse polarity from electric control panel
No flow measured	Overheating of the discharged fluid	Provide the cooling of the lines by liquid recirculation
Rated head not achieved	System counter pressure too high	Check installation
	Impeller clogged	Disassemble the pump and replace the parts (if necessary) (*)

TROUBLE	POSSIBLE CAUSE	SOLUTION
	Worn bearings	Disassemble the pump and replace the parts (if necessary) (*)
Unusual noise or vibration	Foreign substance into the processed fluid	Install filters on the suction line
	Pump operates below minimum continuous flow	Increase flow
	Rotor unbalanced	Disassemble the pump and provide the rotor balancing (*)
	Ice formation into the vessel	Empty and flush the cryostat with nitrogen gas
Bearings wear out rapidly	Foreign substance into the processed fluid	Install filters on the suction line
	Rotor unbalanced	Disassemble the pump and provide the rotor balancing (*)
Pressure drop into the intermediate part of the feed-through connector	If the pressure < tank pressure, leakages from the detection system (fittings or valves)	Check the sealing using leakage detector
	If the pressure < tank pressure, upper element damaged	Replace the upper sealing element or the O-rings
	If the pressure = tank pressure, lower element damaged	The system can continue to work but schedule the replacement
Low insulation	Moisture in the connector	Purge and drain with N2 gas
Leakages of gas from the connector	Failure of the both seals barrier	Interrupt the power supply. Replace the component
Ice formation on the cryostat shell	Vacuum loss	Connect a vacuum pump to the valve and restore the vacuum

(*) These operations should be performed in TECHNIACL AMERICA facilities and may include replacement of some parts.

5 Unit 310 - Gas Dehydration Unit (molecular sieve)

5.1 Operating Summary

CATEGORY	DESCRIPTION
Inputs	Sweet gas from UNIT 330, pre-cooled in E-311 (NH_3 refrigeration).
	Regeneration gas primarily from BOG recycle compressors PK-361/PK-362.
	Automatic make-up sweet gas to regeneration via FV-3110A when BOG is insufficient.
Outputs	Dry gas to UNIT 350 (moisture specification as per HMB, monitored by AT-3101).
	Condensed water from V-311 and V-312 routed to UNIT 530.
	Regeneration off-gas recycled to E-311; excess diverted to the fuel-gas system via PDV-3112.
Key Operating Specs	Thermal Swing Adsorption cycle: 6 h adsorption / 6 h regeneration per bed.
	Regeneration temperature controlled at approximately 232–235 °C.
	Adsorber differential pressure monitored locally by PD gauges (not connected to PCS).
	Liquid levels in V-311 and V-312 protected by LALL/LAHH interlocks leading to local shutdown actions.
Major Equipment	T-311A/B molecular sieve adsorbers, E-311 natural gas pre-cooler.
	X-312 regeneration gas heater, E-313 regeneration gas air cooler.
	V-311 cold wet gas separator, V-312 regeneration wet gas separator.
	F-311A/B and F-312A/B dust filters, T-312A/B mercury guard beds.

5.2 General Arrangement and Risk Location

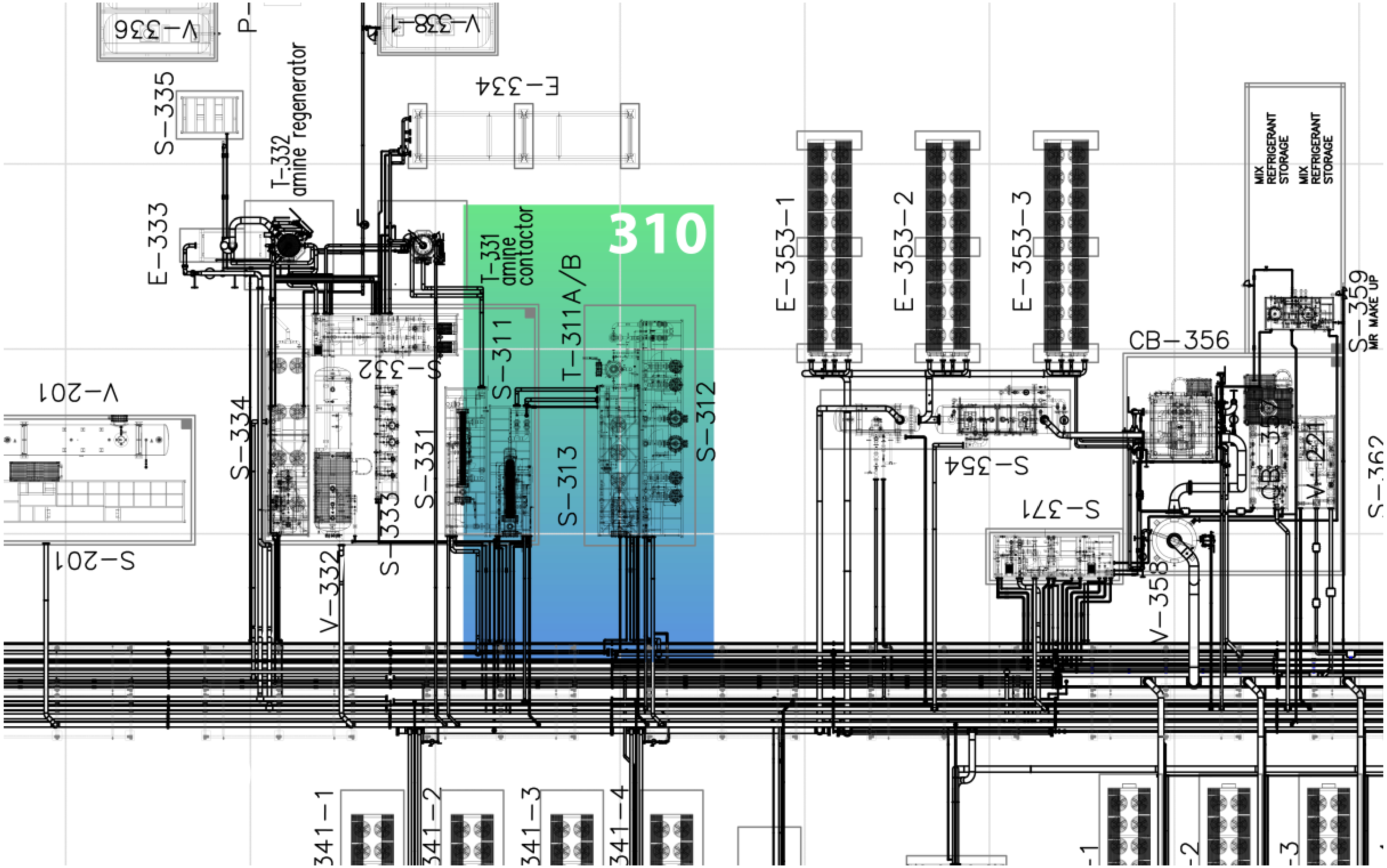


Figure 5.1: Unit 310 GA - Risk location

5.3 Relevant Documents

Table 5.2 — Unit 310 – Relevant Documents

ID	DOC NO	TITLE
1	PRO02-ING-FR02	PROCESS DESCRIPTION
2	PRO10-ING-PR01	PROCESS FLOW DIAGRAMS
3	PRO13-ING-PR08	DEHYDRATION UNIT 310 – S-311 / S-313
4	PRO13-ING-PR09	DEHYDRATION UNIT 310 – S-313 / S-312
5	PRO13-ING-PR10	DEHYDRATION UNIT 310 – S-313 / S-312
6	SAF12-ING-FR01	CAUSE AND EFFECT MATRIX
7	SAF05-ING-FR01	RELIEF AND BLOWDOWN PHILOSOPHY
8	SAF01-ING-FR01	SAFETY AND ISOLATION PHILOSOPHY

5.4 Process Description

The Dehydration Unit (Unit 310) is designed to reduce the residual water content of the sweet natural gas stream exiting the Amine Sweetening Unit (Unit 330) to levels suitable for cryogenic liquefaction in the downstream Liquefaction Unit (Unit 350). Effective moisture removal is required to prevent hydrate formation, ice blockage, and operational instability under low-temperature operating conditions.

The dehydration system operates based on a Thermal Swing Adsorption (TSA) process, utilizing two parallel molecular sieve adsorption beds (T-311A and T-311B). The beds operate in an alternating manner, such that one bed is always in adsorption service while the other is either in regeneration or standby mode, ensuring continuous dehydration of the process gas.

Feed Gas Conditioning

Sweet gas from Unit 330 enters the Natural Gas Pre-Cooler (E-311), a shell-and-tube heat exchanger cooled by an ammonia refrigeration loop. The liquid ammonia inventory within E-311 is controlled by level controller LIC-3111, ensuring stable thermal performance and proper refrigerant management.

The evaporating pressure of the ammonia in the pre-cooler is regulated by PCV-3111, which injects hot ammonia vapor from the compressor discharge header. This pressure control strategy prevents excessive subcooling of the natural gas and ensures stable operation during normal operation, start-up, turndown, and standby conditions.

The outlet temperature of the pre-cooled gas is controlled by TIC-3101, operating in cascade mode to regulate the liquid ammonia flow to E-311.

Downstream of E-311, the gas flows to the Cold Wet Gas Separator V-311, where condensed water is removed. The liquid level in V-311 is controlled by LIC-3101, which modulates LV-3101 to discharge separated water to the Process Water Treatment Unit (Unit 530).

Adsorption Beds

The molecular sieve beds T-311A and T-311B operate alternately in adsorption and regeneration modes. Each bed is equipped with a dedicated set of automated on/off isolation valves, which are sequenced by the Process Control System (PCS) in accordance with a predefined TSA cycle.

A dedicated digital interlock controller KRC-3100 supervises the valve sequencing to ensure that:

- Exactly one adsorption bed is in dehydration service at all times.
- The non-operating bed is either in regeneration or standby mode and fully isolated from the main process gas flow.
- Simultaneous adsorption or regeneration of both beds is prevented.

All bed changeovers are executed automatically by PCS and KRC-3100 logic. No manual valve operation is required for normal TSA cycle transitions.

Differential pressure across each adsorption bed is measured locally using Pressure Differential Gauges (PDGs), which are not connected to the PCS. Operators are required to verify the pressure drop during routine field inspections. If abnormal differential pressure is detected, a manual bed changeover can be initiated from the PCS HMI.

Regeneration System

The regeneration process restores the adsorption capacity of the molecular sieve beds by desorbing accumulated water using a heated regeneration gas stream.

The primary source of regeneration gas is Boil-Off Gas (BOG) supplied by the BOG recycle compressors PK-361 and PK-362. When available BOG flow is insufficient, makeup sweet gas is automatically introduced through FV-3110A downstream of dust filters F-311A/B to maintain the required regeneration gas flow.

The regeneration gas flow is managed as follows:

- FV-3110C modulates the main BOG flow to the Regeneration Gas Heater X-312.
- FV-3110B diverts excess BOG to the fuel gas system to prevent overpressure and allow recycling.

- FV-3110A supplies makeup sweet gas only when BOG flow is insufficient.

The regeneration gas is heated in the electric Regeneration Gas Heater X-312, with a total installed capacity of 120 kW divided into three independent 40 kW stages. The outlet temperature is controlled by TIC-3122 to approximately 232–235 °C. High-temperature protection is provided by TAHH-3122.

The hot regeneration gas flows upward through the saturated molecular sieve bed, desorbing the adsorbed water. The spent gas is then cooled in the Regeneration Gas Air Cooler E-313 and routed to the Regeneration Wet Gas Separator V-312, where condensed water is separated. The separator level is controlled by LIC-3102, which modulates LV-3102 to discharge water to the Process Water Treatment Unit.

Downstream of V-312, the regeneration gas passes through dust filters F-312A/B and Mercury Guard Beds T-312A/B before being routed back to the inlet of the Natural Gas Pre-Cooler E-311. Excess regeneration gas is diverted to the fuel gas system through backpressure valve PDV-3112 to maintain stable regeneration circuit pressure.

Differential pressure across the adsorption beds is monitored by AT-3101. Abnormal conditions generate alarms in the PCS, prompting operator intervention as required to protect downstream equipment and ensure safe operation.

TSA Regeneration Cycle

Each adsorption bed operates on a 12-hour TSA cycle, comprising approximately 6 hours of adsorption followed by 6 hours of regeneration. The regeneration sequence is executed automatically according to the predefined cycle logic summarized in Table 5.3.

TSA Regeneration Cycle

Each adsorption bed operates on a 12-hour TSA cycle, comprising approximately 6 hours of adsorption followed by 6 hours of regeneration. The regeneration sequence is executed automatically according to the predefined cycle logic summarized in Table 5.3.

Table 5.3 — TSA Regeneration Cycle – Typical Sequence and Duration

Step	Description	Typical Duration
1	Bed changeover: switch to fresh bed and isolate spent bed	∼5 min
2	Controlled pressure transition (maximum 3 bar/min)	∼5 min
3	Regeneration gas temperature ramp-up to 232 °C	∼5 min
4	Heating soak period at approximately 235 °C	∼168 min

Continued on next page

Step	Description	Typical Duration
5	Bed cooling to within 16 °C of feed gas temperature	∼66 min
6	Standby mode, ready for next adsorption cycle	∼111 min

5.5 Technological Objects

5.5.1 Main Equipment List

Table 5.4 — Unit 310 – Main Equipment List

ID	TAG	EQUIPMENT
1	E-311	Natural gas precooler – BXU type
2	V-311	Cold wet gas separator
3	T-311A	Water adsorber bed A
4	T-311B	Water adsorber bed B
5	F-311A	Dust filter A
6	F-311B	Dust filter B
7	X-312	Regeneration gas heater
8	E-313	Regeneration gas aircooler
9	V-312	Regeneration wet gas separator
10	T-312A	Mercury bed guard A
11	T-312B	Mercury bed guard B
12	F-312A	Dust filter A
13	F-312B	Dust filter B
14	F-313A/B/C/D	Aircooler fans

5.5.2 Electrical Loads

Table 5.5 — Unit 310 – Electrical Loads

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	X-312_1	Regeneration gas trim heater	DOL	690	40,00
2	X-312_2	Regeneration gas trim heater	DOL	690	40,00
3	X-312_3	Regeneration gas trim heater	DOL	690	40,00

5.5.3 Motors as Remote MCC

Table 5.6 — Unit 310 – Motors as Remote MCC

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	MF-313A	Regeneration gas cooler fan electric motor A	DOL	690	4,00
2	MF-313B	Regeneration gas cooler fan electric motor B	DOL	690	4,00
3	MF-313C	Regeneration gas cooler fan electric motor C	DOL	690	4,00

5.5.4 Valves

Table 5.7 — Unit 310 – Valves

ID	TAG	SERVICE	SYSTEM	NOTES
1	BDEV-3101	T-311A/B blowdown	ESD	-
2	BDV-3101	T-311A/B blowdown	-	-
3	BDEV-3561	Cold box depressurization	ESD	-
4	BDV-3561	Cold box depressurization	-	-
5	FEV-3110A	Dehydrated gas to X-312	PCS	-
6	FV-3110A	Dehydrated gas to X-312	-	-
7	FEV-3110B	BOG return to sweet gas inlet	PCS	-
8	FV-3110B	BOG return to sweet gas inlet	-	-
9	FEV-3110C	BOG to T-311A/B regeneration flow control	PCS	-
10	FV-3110C	BOG to T-311A/B regeneration flow control	-	-
11	LEV-3111	E-311 ammonia liquid level control	PCS	-
12	LV-3111	E-311 ammonia liquid level control	-	-
13	LV-3101	V-311 liquid level control	-	-
14	LV-3102	V-312 liquid level control	-	-
15	PEV-3101	E-311 pressure control	PCS	With 5 m cable
16	PCV-3101	E-311 pressure control	-	With 5 m cable
17	PEV-3111	E-311 pressure control (hot bypass)	PCS	-
18	PCV-3111	E-311 pressure control (hot bypass)	-	-
19	PSV-3101A	E-311 pressure relief	-	Set pressure 28 barg
20	PSV-3101B	E-311 pressure relief	-	Set pressure 28 barg
21	PSV-3102A	V-311 pressure relief	-	Set pressure 50 barg
22	PSV-3102B	V-311 pressure relief	-	Set pressure 50 barg
23	PSV-3103	T-311A pressure relief	-	Set pressure 50 barg
24	PSV-3104	T-311B pressure relief	-	Set pressure 50 barg

ID	TAG	SERVICE	SYSTEM	NOTES
25	PSV-3106A	V-312 pressure relief	-	Set pressure 50 barg
26	PSV-3106B	V-312 pressure relief	-	Set pressure 50 barg
27	PSV-3107A	T-312A pressure relief	-	Set pressure 50 barg
28	PSV-3107B	T-312B pressure relief	-	Set pressure 50 barg
29	PSV-3110A	F-311B pressure relief	-	Set pressure 50 barg
30	PSV-3110B	F-311A pressure relief	-	Set pressure 50 barg
31	PSV-3112A	F-312A pressure relief	-	Set pressure 50 barg
32	PSV-3112B	F-312B pressure relief	-	Set pressure 50 barg
33	SDEV-3110	V-311 liquid outlet to process water package shutdown	ESD	-
34	SDV-3110	V-311 liquid outlet to process water package shutdown	-	-
35	SDEV-3112	V-312 liquid outlet to process water package shutdown	ESD	-
36	SDV-3112	V-312 liquid outlet to process water package shutdown	-	-
37	SDEV-3113	V-312 gas outlet shutdown	ESD	-
38	SDV-3113	V-312 gas outlet shutdown	-	-
39	SDEV-3560	Clean gas to liquefier shutdown	ESD	-
40	SDV-3560	Clean gas to liquefier shutdown	-	-
41	XEV-3103A	T-311A inlet shut off	PCS	-
42	XV-3103A	T-311A inlet shut off	-	-
43	XEV-3103B	T-311B inlet shut off	PCS	-
44	XV-3103B	T-311B inlet shut off	-	-
45	XEV-3104A	T-311A outlet shut off	PCS	-
46	XV-3104A	T-311A outlet shut off	-	-
47	XEV-3104B	T-311B outlet shut off	PCS	-
48	XV-3104B	T-311B outlet shut off	-	-
49	XEV-3105A	T-311A regeneration gas outlet shut off	PCS	-
50	XV-3105A	T-311A regeneration gas outlet shut off	-	-
51	XEV-3105B	T-311B regeneration gas outlet shut off	PCS	-
52	XV-3105B	T-311B regeneration gas outlet shut off	-	-
53	XEV-3106A	T-311A regeneration gas inlet shut off	PCS	-
54	XV-3106A	T-311A regeneration gas inlet shut off	-	-
55	XEV-3106B	T-311B regeneration gas inlet shut off	PCS	-
56	XV-3106B	T-311B regeneration gas inlet shut off	-	-

5.5.5 Instrumentation

Table 5.8 — Unit 310 – Instrument List

ID	TAG	SERVICE	RANGE	OP	UNITS
1	AE-3101	Dehydrated gas moisture content	-	–	-
2	AT-3101	Dehydrated gas moisture content	-	–	ppm
3	FE-3110A	Regeneration gas flow	-	–	-
4	FE-3560A	Clean gas to liquefier flow	-	–	-
5	FT-3110A	Regeneration gas flow	-	–	MMSCFD
6	FT-3560A	Clean gas to liquefier flow	0..25	–	MMSCFD
7	LSHH-3101	E-311 scrubber high high liquid level	-	–	-
8	LT-3101	V-311 liquid level	-	–	mm
9	LT-3102	V-312 liquid level	-	–	mm
10	LT-3103	V-311 liquid level	-	–	mm
11	LT-3105	V-312 liquid level	-	–	mm
12	LT-3111	E-311 ammonia liquid level control	-	–	mm
13	PT-3102	T-311A regeneration gas outlet pressure	0..50	–	barg
14	PT-3103	T-311B regeneration gas outlet pressure	0..50	–	barg
15	PT-3122	Regeneration gas pressure	0..50	–	barg
16	PT-3566	Clean gas to liquefier pressure	0..50	–	barg
17	RO-3101	BDV-3101 blowdown restriction	-	–	-
18	RO-3561	BDV-3561 blowdown restriction	-	–	-
19	TAHH-X-312	X-312 high high temperature switch	285	–	°C
20	TG-3101	Sweet gas to dehydration temperature	-15..150	–	°C
21	TG-3102	E-313 outlet temperature	-	–	-
22	TT-3101	E-311 outlet temperature	-50..200	–	°C
23	TT-3102	T-311A/B regeneration gas outlet temperature	-15..350	–	°C
24	TT-3103	E-313 outlet temperature	-15..350	–	°C
25	TT-3122	Regeneration gas temperature	-15..350	–	°C
26	TT-3560A	Clean gas to liquefier temperature	-50..200	–	°C
27	ZSH-3101	BDV-3101 open position indication	-	–	-
28	ZSH-3103A	XV-3103A open position indication	-	–	-
29	ZSH-3103B	XV-3103B open position indication	-	–	-
30	ZSH-3104A	XV-3104A open position indication	-	–	-
31	ZSH-3104B	XV-3104B open position indication	-	–	-
32	ZSH-3105A	XV-3105A open position indication	-	–	-
33	ZSH-3105B	XV-3105B open position indication	-	–	-
34	ZSH-3106A	XV-3106A open position indication	-	–	-
35	ZSH-3106B	XV-3106B open position indication	-	–	-
36	ZSH-3110	SDV-3110 open position indication	-	–	-

ID	TAG	SERVICE	RANGE	OP	UNITS
37	ZSH-3112	SDV-3112 open position indication	-	–	-
38	ZSH-3113	V-312 gas outlet shutdown position indication	-	–	-
39	ZSH-3560	SDV-3560 open position indication	-	–	-
40	ZSH-3561	BDV-3561 open position indication	-	–	-
41	ZSL-3101	BDV-3101 closed position indication	-	–	-
42	ZSL-3103A	XV-3103A closed position indication	-	–	-
43	ZSL-3103B	XV-3103B closed position indication	-	–	-
44	ZSL-3104A	XV-3104A closed position indication	-	–	-
45	ZSL-3104B	XV-3104B closed position indication	-	–	-
46	ZSL-3105A	XV-3105A closed position indication	-	–	-
47	ZSL-3105B	XV-3105B closed position indication	-	–	-
48	ZSL-3106A	XV-3106A closed position indication	-	–	-
49	ZSL-3106B	XV-3106B closed position indication	-	–	-
50	ZSL-3110	SDV-3110 closed position indication	-	–	-
51	ZSL-3112	SDV-3112 closed position indication	-	–	-
52	ZSL-3113	V-312 gas outlet shutdown position indication	-	–	-
53	ZSL-3560	SDV-3560 closed position indication	-	–	-
54	ZSL-3561	BDV-3561 closed position indication	-	–	-

5.5.6 Junction Boxes

Table 5.9 — Unit 310 – Junction Boxes

ID	TAG	SERVICE	CONNECTION	TYPE
1	JG-313	Dehydration	COPPER-COPPER	STANDARD
2	P-PN-0S311.1	Dehydration	COPPER-COPPER	STANDARD
3	P-PN-0S312.1	Dehydration	COPPER-COPPER	STANDARD

5.5.7 Controllers

Table 5.10 — Unit 310 – Controllers

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
FIC-3110A	REGENERATION GAS FLOW CONTROLLER	FV-3110 A/B/C	0..0,66 MMSCFD	—
FIC-3560A	CLEAN GAS TO LIQUEFIER FLOW CONTROLLER	—	0..25,0 MMSCFD	—

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
LIC-3101	V-311 LIQUID LEVEL CONTROLLER	LV-3101	0..100 %	—
LIC-3102	V-312 LIQUID LEVEL CONTROLLER	LV-3102	0..100 %	—
LIC-3111	E-311 AMMONIA LIQUID LEVEL CONTROLLER	LV-3111	0..100 %	—
TIC-3101	E-311 OUTLET TEMPERATURE CONTROLLER	LV-3111	-50..200 °C	—

5.6 Operating Interface

5.6.1 PCS Operating Screens – Unit 310

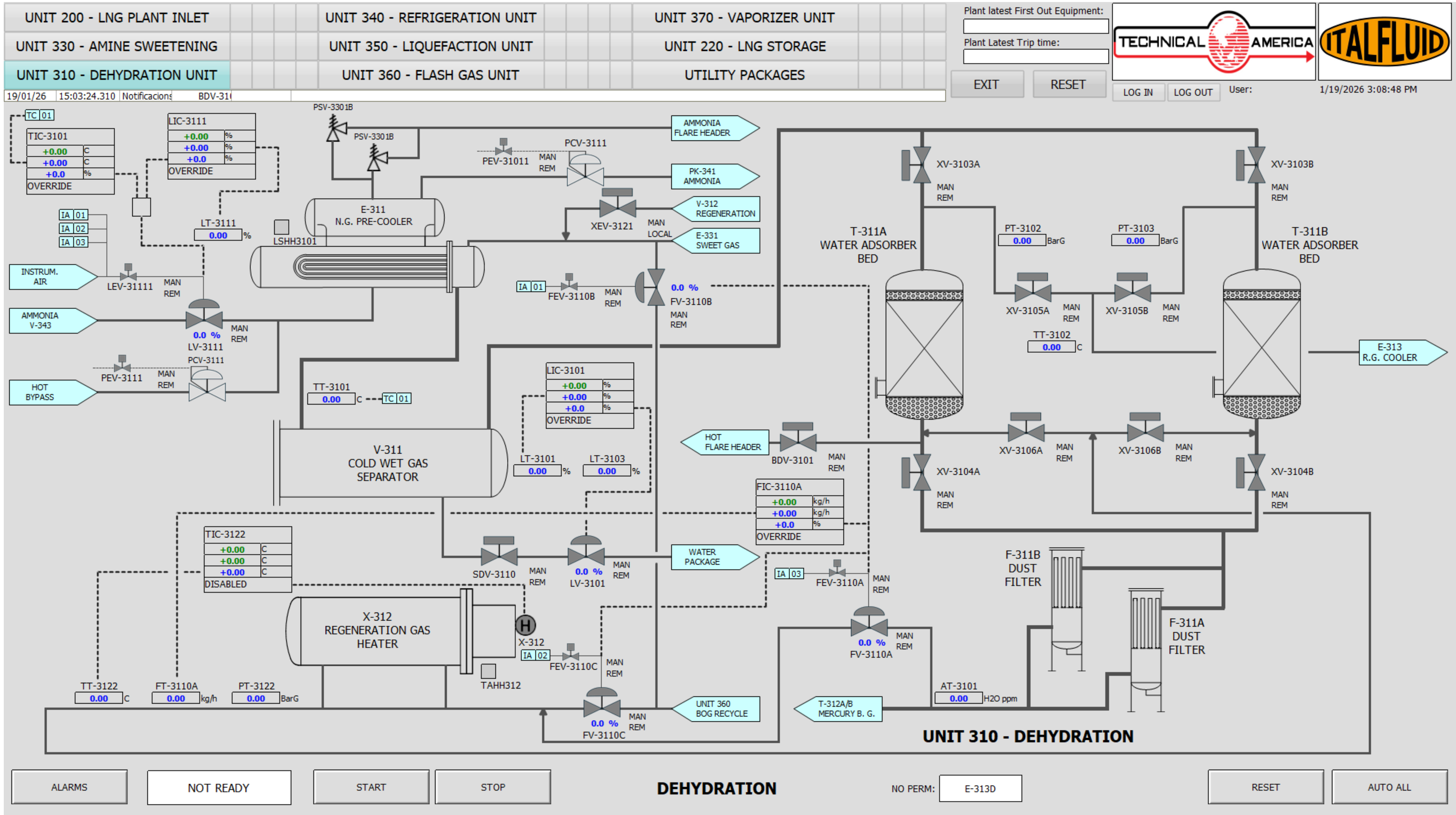


Figure 5.2: UNIT 310 - HMI Operating Screen

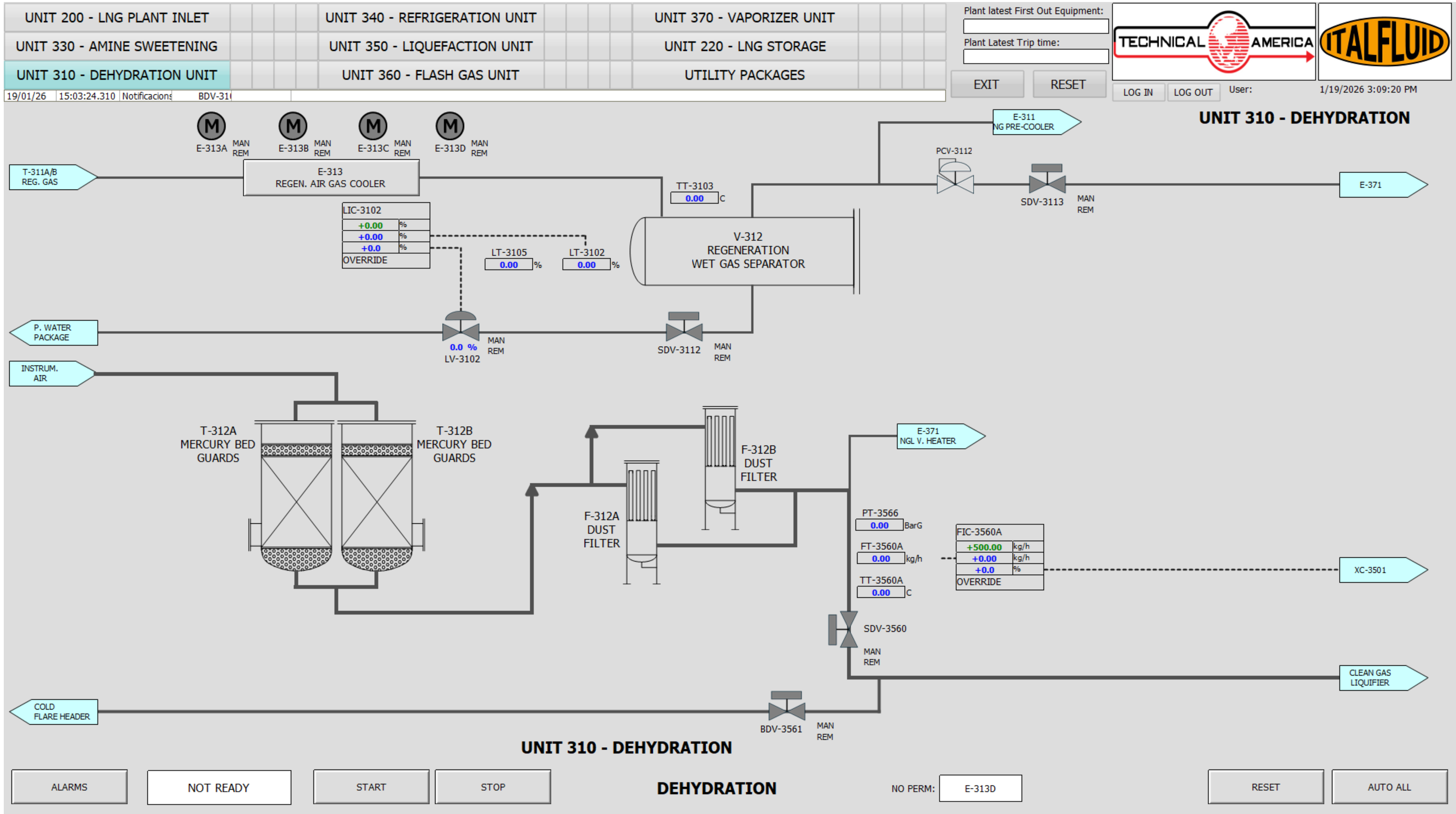


Figure 5.3: UNIT 310 - HMI Operating Screen

5.6.2 Critical Alarms and Trips

Table 5.11 — Unit 310 – Critical Alarms

ALARM	INSTRUMENT	TYPE	SP	DESCRIPTION
AAHH-3101	AT-3101	PSD	—	MOISTURE HIGH LEVEL
LAH-3111	LT-3111	LSD	—	HIGH LIQUID LEVEL ON E-311
LAHH-3101	LS-3101	PSD	—	HIGH HIGH LIQUID LEVEL ON E-311
LAHH-3103	LT-3103	PSD	—	HIGH HIGH LIQUID LEVEL ON V-311
LAHH-3105	LT-3105	LSD	—	HIGH HIGH LIQUID LEVEL ON V-312
LALL-3103	LT-3103	LSD	—	LOW LOW LIQUID LEVEL ON V-311
TAHH-3102	TT-3102	LSD	—	HIGH HIGH TEMPERATURE ON OUTLET OF X-312
TAHH-X-312	TAHH-X-312	LSD	—	HIGH HIGH TEMPERATURE ON X-312

5.6.3 Unit Cause and Effect Matrix

Table 5.12 — Unit 310 – Cause & Effect Matrix

ALARM	INSTRUMENT	PEV-3101	X-312	XV-3105A/B	XV-3106A/B	SDV-3110	SDV-3112	SDV-3113
LAH-3111	LIC-3111	C						
LAHH-3105	LT-3105		T	C	C			C
LALL-3103	LT-3103					C		
LALL-3105	LT-3105						C	
TAHH-3102	TT-3102		T					
TAHH-X-312	TIC-3122		T					

6 Unit 330 - Amine Sweetening Unit (CO2 removal)

6.1 Operating Summary

CATEGORY	DESCRIPTION
Inputs	Sour natural gas from UNIT 200, pre-heated in gas/gas heat exchanger E-331, with CO_2 content up to 10 mol%. Feed gas passes through inlet separator V-331 for free water removal prior to absorption.
	Utilities include hot oil for amine reboiler E-333, electrical power for pumps and air coolers, instrument air, and nitrogen supply for reflux system pressure protection.
Outputs	Sweetened gas, water-saturated, routed to UNIT 310 (Dehydration Unit).
	Rich amine routed to the regeneration section via V-332 (Rich Amine Separator) and T-332 (Amine Regenerator).
	CO_2-rich acid gas discharged via the amine vent header, under controlled pressure conditions and protected against vacuum by nitrogen blanketing.
Key Operating Specs	CO_2 removal is achieved in trayed absorber T-331 operating under counter-current contact between gas and lean amine. The absorption process is exothermic, requiring close monitoring of absorber temperature profile and liquid levels.
	Rich amine flashing pressure in V-332 is controlled by PCV-3303, set at approximately 4.5 barg, to release dissolved hydrocarbons prior to regeneration. Stable operation depends on coordinated control of flow, level, temperature, and pressure throughout the amine loop.
Major Equipment	T-331 amine absorber, V-331 inlet separator, V-332 rich amine separator,
	T-332 amine regenerator, E-332 rich/lean amine heat exchanger, E-333 amine reboiler, E-334 lean amine cooler, E-335 reflux condenser, V-334 reflux accumulator, associated pumps, filters, and air-cooled fans.

6.2 General Arrangement and Risk Location

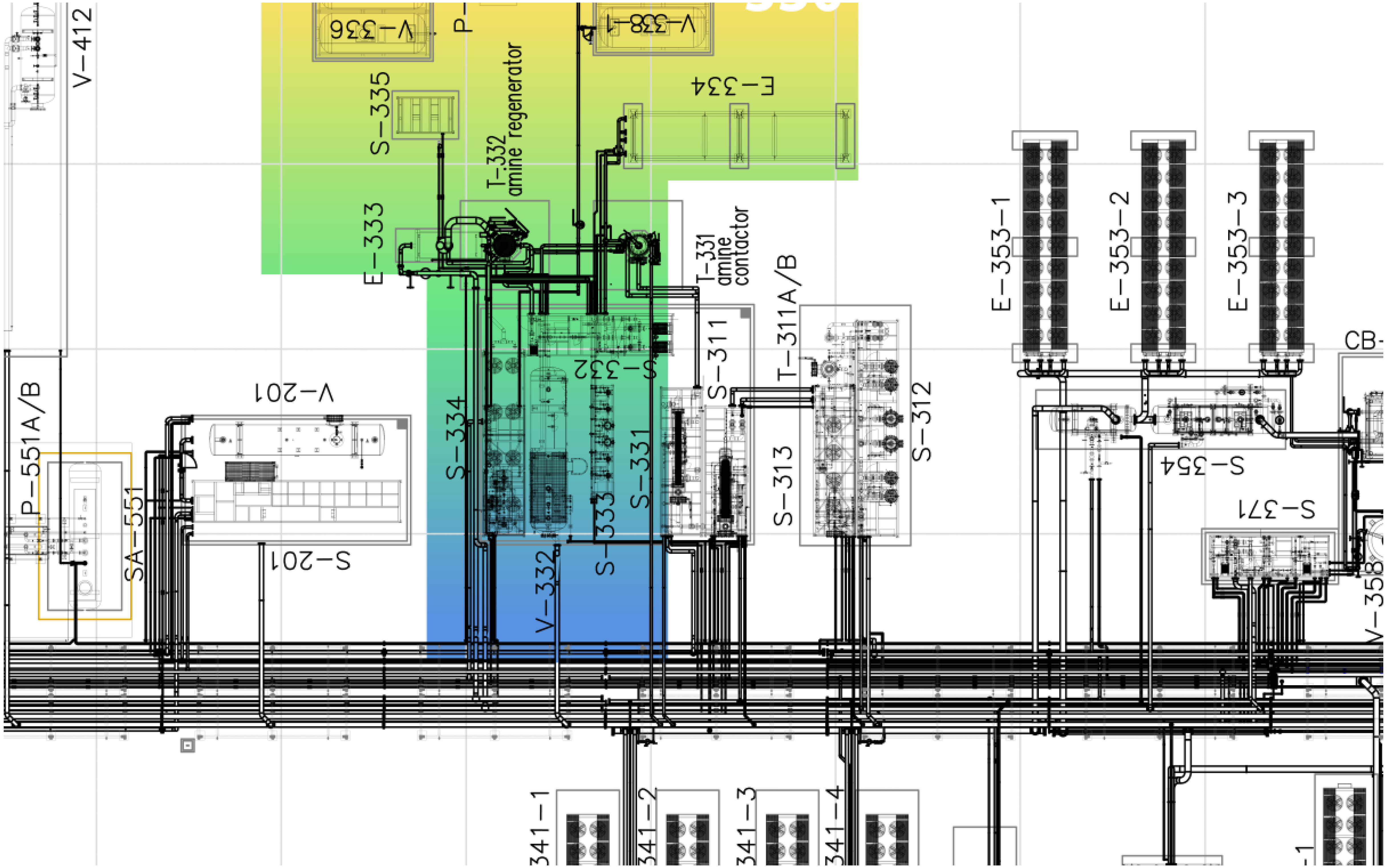


Figure 6.1: Unit 330 GA - Risk location

6.3 Relevant Documents

Table 6.2 — Unit 330 – Relevant Documents

ID	DOC NO	TITLE
1	PRO02-ING-FR02	PROCESS DESCRIPTION
2	PRO10-ING-PR01	PROCESS FLOW DIAGRAMS
3	PRO13-ING-PR02	SWEETENING UNIT 330 – SKID S-331
4	PRO13-ING-PR03	SWEETENING UNIT 330 – T-331 / T-332 / E-333
5	PRO13-ING-PR04	SWEETENING UNIT 330 – SKID S-332
6	PRO13-ING-PR05	SWEETENING UNIT 330 – SKID S-334
7	PRO13-ING-PR06	SWEETENING UNIT 330 – AMINE UTILITIES / S-335
8	PRO13-ING-PR07	SWEETENING UNIT 330 – SKID S-332 / S-333
9	SAF12-ING-FR01	CAUSE AND EFFECT MATRIX
10	SAF05-ING-FR01	RELIEF AND BLOWDOWN PHILOSOPHY

6.4 Process Description

Unit 330 – Amine Gas Sweetening is responsible for removing acid gases, primarily carbon dioxide (CO_2), from the raw natural gas stream. This treatment step is essential to meet specifications for downstream cryogenic processing and LNG production, as well as to prevent CO_2 freeze-out and protect equipment from corrosion. The unit uses a regenerative aqueous MDEA solution to selectively absorb acid gases through a continuous contact and regeneration cycle.

Feed Gas Conditioning and Absorption.

The untreated raw gas enters Unit 330 via the Gas/Gas Heat Exchanger (E-331), where it is preheated by recovering thermal energy from sweetened gas exiting the Amine Contactor (T-331). This heat integration improves overall energy efficiency and stabilizes process conditions.

After preheating, the gas flows into the Inlet Separator (V-331), a two-phase horizontal vessel designed to remove free water. The separated water is directed to the Process Water Package, while the gas, containing any residual hydrocarbon mist, continues to the Amine Contactor (T-331).

Inside T-331, the gas enters from the bottom and rises through a trayed column in counter-current contact with lean amine flowing downward. Carbon dioxide is absorbed into the amine solution, producing a sweet gas that exits the top of the contactor and is routed to the Dehydration Unit (Unit 310).

The absorber operation is monitored by pressure and temperature transmitters (PT-3311A/B/C, TT-3301, TT-3302, TT-3307, TT-3308, TT-3309) and by the CO_2 analyzer AE-3301, which measures the outlet gas CO_2 content. If excessive CO_2 is detected, the high-high alarm AAHH-3301 is triggered, initiating a Process Shutdown (PSD) to protect downstream units and ensure gas quality.

Foaming control in the absorber is handled by a dedicated Antifoam Injection System, comprising the Antifoam Agent Tank (V-337) and Antifoam Pumps P-337A/B. This system is pneumatically actuated and manually operated in the field; it is not connected to the PCS or ESD and must be operated by field personnel.

Rich Amine Flashing and Pre-heating.

The rich amine, now containing absorbed CO_2, flows from the bottom of T-331 to the Rich Amine Separator (V-332), where it undergoes a pressure reduction to flash off any entrained hydrocarbon gases. The flashed gas is vented to the flare via PCV-3303, and the vessel liquid level is controlled by LV-3304, with associated high-high and low-low alarms to protect against abnormal operating conditions.

Any hydrocarbon condensates collected in the dedicated section of separator V-332 are discharged to the closed drain system under level control operated by the on-off valve XV-3302.

The rich amine solution then passes through the Rich/Lean Amine Heat Exchanger (E-332), where heat is recovered from the lean amine stream, increasing overall thermal efficiency and reducing reboiler duty.

Amine Regeneration.

The preheated rich amine enters the upper section of the Amine Regenerator (T-332), where steam generated by the Amine Reboiler (E-333) strips the absorbed acid gases from the solution. The regenerator operates at low pressure and moderate temperature, with key operating parameters monitored by PT-3312A/B, TT-3333, TT-3304, TT-3305, and TT-3306.

Overhead vapors, consisting mainly of water vapor and acid gases, flow to the Amine Reflux Condenser (E-335). The condensed liquid is collected in the Reflux Accumulator (V-334) and pumped back to the top of the regenerator by the Reflux Pumps P-334A/B under level control by LIC-3305. Condenser fans MF-335-1 to MF-335-8 are automatically staged by TIC-3314 to maintain the desired condenser outlet temperature.

To ensure stable operating pressure in the reflux system, PCV-3302, a self-actuated backpressure valve, injects nitrogen into V-334 to maintain a minimum vessel pressure of 0.5 barg, preventing vacuum conditions and protecting pump suction. In addition, PCV-3304 vents excess non-condensable gases, mainly CO_2, to the flare if the condenser outlet pressure exceeds 0.58 barg.

Lean Amine Cooling and Circulation.

The regenerated lean amine is withdrawn from the bottom of T-332 and routed through E-332 for heat recovery before being pumped by Lean Amine Pumps P-332A/B to the Lean Amine Cooler (E-334), where it is cooled to the optimal absorption temperature.

Lean amine circulation back to the absorber is provided by Lean Amine Pumps P-332A/B, arranged in a one-duty and one-standby configuration to ensure uninterrupted operation and high system reliability. The PCS includes a dedicated pump selector to allow operators to switch between the duty and standby pump as required. Manual valves on the suction side must be verified by field operators. Each pump is equipped with a variable frequency drive (SIC-MP-332A/B) to allow precise flow control and improved energy efficiency.

Temperature control at the outlet of the Lean Amine Cooler (E-334) is achieved by the Temperature Differential Indicator Controller TDIC-3301. This controller monitors the temperature difference between the rich gas entering T-331 and the lean amine exiting E-334, and modulates the three-way control valve TV-3301. The valve receives a 4–20 mA control signal and dynamically splits the lean amine flow between the bypass line and the cooler to maintain the target outlet temperature.

Under high cooling demand, TV-3301 closes the bypass line and directs the full lean amine flow through E-334. When cooling demand decreases, the valve proportionally opens the bypass to optimize energy consumption. This operation is coordinated with the air-cooled exchanger fans MF-334-1 to MF-334-18, which are staged automatically but can also be manually controlled from the PCS if required.

The lean amine circulation loop includes comprehensive filtration through Particulate Pre-Filters (F-331A/B), Charcoal Filters (F-332A/B), and After-Filters (F-333A/B) to maintain amine quality and protect the system from fouling and degradation.

Auxiliary Systems and Make-up.

The amine system includes make-up facilities provided by Make-up Pumps P-333A/B/C, which supply fresh amine and demineralized water as required from the Fresh Amine Tank (V-336) and the Demineralized Water Tank (V-335), respectively. Dedicated heaters, HE-V336 for fresh amine and HE-V335 for demineralized water, maintain the make-up streams at suitable temperatures. These heaters are controlled by TIC-3324 and TIC-3323 to ensure proper conditioning of the make-up fluids prior to injection into the system.

6.5 Technological Objects

6.5.1 Main Equipment List

Table 6.3 — Unit 330 – Main Equipment List

ID	TAG	EQUIPMENT
1	E-331	Gas/gas heat exchanger – BEM type
2	V-331	Inlet separator
3	T-331	Amine contactor
4	V-332	Rich amine separator
5	E-332	Rich/lean amine heat exchanger
6	T-332	Amine regenerator
7	E-333	Amine reboiler
8	E-335	Amine reflux condenser
9	V-334	Reflux accumulator
10	P-334A/B	Amine regenerator reflux pumps
11	P-332A/B	Lean amine pumps
12	F-331A/B	Particulate pre filters
13	F-332A/B	Charcoal filters
14	F-333A/B	Particulate after filters
15	E-334	Lean amine cooler
16	P-333A/B/C	Amine and water make up pumps
17	V-335	Demineralized water solution tank
18	V-336	Fresh amine tank
19	V-337	Antifoam agent tank
20	P-337A/B	Antifoam make up pumps
21	V-338-1/2	Exhaust amine storage tanks
22	P-338	Amine transfer pump on trolley
23	F-334-1...18	Reflux condenser fans
24	F-335-1...8	Lean amine cooler fans

6.5.2 Electrical Loads

Table 6.4 — Unit 330 – Electrical Loads

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	P-332A	Lean amine pump A	VFD	690	160,00
2	P-332B	Lean amine pump B	VFD	690	160,00

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
3	P-333A	Amine and water make up pump A	DOL	690	1,10
4	P-333B	Amine and water make up pump B	DOL	690	1,10
5	P-333C	Amine and water make up pump C	DOL	690	1,10
6	P-334A	Amine regenerator reflux pump A	VFD	690	1,50
7	P-334B	Amine regenerator reflux pump B	VFD	690	1,50
8	X-332A	Amine electric motor heater A	INTERLOCKED	230	0,13
9	X-332B	Amine electric motor heater B	INTERLOCKED	230	0,13

6.5.3 Motors as Remote MCC

Table 6.5 — Unit 330 – Motors as Remote MCC

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	MF-334-1	Fan motor 1 – Lean amine cooler E-334	DOL	690	N/A
2	MF-334-2	Fan motor 2 – Lean amine cooler E-334	DOL	690	N/A
3	MF-334-3	Fan motor 3 – Lean amine cooler E-334	DOL	690	N/A
4	MF-334-4	Fan motor 4 – Lean amine cooler E-334	DOL	690	N/A
5	MF-334-5	Fan motor 5 – Lean amine cooler E-334	DOL	690	N/A
6	MF-334-6	Fan motor 6 – Lean amine cooler E-334	DOL	690	N/A
7	MF-334-7	Fan motor 7 – Lean amine cooler E-334	DOL	690	N/A
8	MF-334-8	Fan motor 8 – Lean amine cooler E-334	DOL	690	N/A
9	MF-334-9	Fan motor 9 – Lean amine cooler E-334	DOL	690	N/A
10	MF-334-10	Fan motor 10 – Lean amine cooler E-334	DOL	690	N/A
11	MF-334-11	Fan motor 11 – Lean amine cooler E-334	DOL	690	N/A
12	MF-334-12	Fan motor 12 – Lean amine cooler E-334	DOL	690	N/A
13	MF-334-13	Fan motor 13 – Lean amine cooler E-334	DOL	690	N/A
14	MF-334-14	Fan motor 14 – Lean amine cooler E-334	DOL	690	N/A
15	MF-334-15	Fan motor 15 – Lean amine cooler E-334	DOL	690	N/A
16	MF-334-16	Fan motor 16 – Lean amine cooler E-334	DOL	690	N/A
17	MF-334-17	Fan motor 17 – Lean amine cooler E-334	DOL	690	N/A
18	MF-334-18	Fan motor 18 – Lean amine cooler E-334	DOL	690	N/A
19	MF-335-1	Fan motor 1 – Amine reflux condenser E-335	DOL	690	N/A
20	MF-335-2	Fan motor 2 – Amine reflux condenser E-335	DOL	690	N/A
21	MF-335-3	Fan motor 3 – Amine reflux condenser E-335	DOL	690	N/A
22	MF-335-4	Fan motor 4 – Amine reflux condenser E-335	DOL	690	N/A
23	MF-335-5	Fan motor 5 – Amine reflux condenser E-335	DOL	690	N/A
24	MF-335-6	Fan motor 6 – Amine reflux condenser E-335	DOL	690	N/A
25	MF-335-7	Fan motor 7 – Amine reflux condenser E-335	DOL	690	N/A

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
26	MF-335-8	Fan motor 8 – Amine reflux condenser E-335	DOL	690	N/A
27	MP-334A	Amine regenerator reflux pump A motor	VFD	690	1,50
28	MP-334B	Amine regenerator reflux pump B motor	VFD	690	1,50
29	MP-332A	Lean amine pump A motor	VFD	690	160,00
30	MP-332B	Lean amine pump B motor	VFD	690	160,00
31	MP-333A	Amine and water make up pump A motor	DOL	690	1,10
32	MP-333B	Amine and water make up pump B motor	DOL	690	1,10
33	MP-333C	Amine and water make up pump C motor	DOL	690	1,10

6.5.4 Valves

Table 6.6 — Unit 330 – Valves

ID	TAG	SERVICE	SYSTEM	NOTES
1	FV-3301	Lean amine to T-331 flow control	-	-
2	LEV-3303	T-331 bottom liquid level control	ESD	-
3	LV-3303	T-331 bottom liquid level control	-	-
4	LEV-3304	V-332 level control	PCS	-
5	LV-3304	V-332 level control	-	-
6	PCV-3303	V-332 pressure control	-	Set pressure 4.5 barg
7	PCV-3304	V-334 pressure control	-	Set pressure 0.58 barg
8	PSV-3301A	V-331 pressure relief	-	Set pressure 50 barg
9	PSV-3301B	V-331 pressure relief	-	Set pressure 50 barg
10	PSV-3302A	T-331 pressure relief	-	Set pressure 50 barg
11	PSV-3302B	T-331 pressure relief	-	Set pressure 50 barg
12	PSV-3303A	T-332 pressure relief	-	Set pressure 5 barg
13	PSV-3303B	T-332 pressure relief	-	Set pressure 5 barg
14	PSV-3304A	V-332 pressure relief	-	Set pressure 10 barg
15	PSV-3304B	V-332 pressure relief	-	Set pressure 10 barg
16	PSV-3306A	P-333A pressure relief	-	Set pressure 5 bar diff.
17	PSV-3306B	P-333B pressure relief	-	Set pressure 5 bar diff.
18	PSV-3306C	P-333C pressure relief	-	Set pressure 5 bar diff.
19	PSV-3307A	P-334A pressure relief	-	Set pressure 5 bar diff.
20	PSV-3307B	P-334B pressure relief	-	Set pressure 5 bar diff.
21	PSV-3310A	F-331A pressure relief	-	Set pressure 10 barg
22	PSV-3310B	F-331B pressure relief	-	Set pressure 10 barg
23	PSV-3320A	F-332A pressure relief	-	Set pressure 10 barg
24	PSV-3320B	F-332B pressure relief	-	Set pressure 10 barg

ID	TAG	SERVICE	SYSTEM	NOTES
25	PSV-3330A	F-333A pressure relief	-	Set pressure 10 barg
26	PSV-3330B	F-333B pressure relief	-	Set pressure 10 barg
27	PSV-3371	IA to antifoam pumps	-	Set pressure TBD
28	SDEV-3304	Lean amine filtering system shutdown	ESD	-
29	SDV-3304	Lean amine filtering system shutdown	-	-
30	SV-3371	Antifoam pump A start	PCS	-
31	SV-3372	Antifoam pump B start	PCS	-
32	TSV-3305	E-332 thermal expansion protection	-	Set pressure 10 barg
33	TV-3301	Lean amine to T-331 temperature control	-	-
34	TV-3333	T-332 temperature control	-	-
35	XEV-3301	Antifoam to lean amine to T-331	PCS	-
36	XV-3301	Antifoam to lean amine to T-331	-	-
37	XEV-3302	V-332 condensate level control	PCS	-
38	XV-3302	V-332 condensate level control	-	-
39	XEV-3303	Antifoam injection to T-332	PCS	-
40	XV-3303	Antifoam injection to T-332	-	-
41	XEV-3311	V-331 liquid level control	PCS	-
42	XV-3311	V-331 liquid level control	-	-

6.5.5 Instrumentation

Table 6.7 — Unit 330 – Instrument List

ID	TAG	SERVICE	RANGE	OP	UNITS
1	AE-3301	Sweetening unit outlet CO_2 concentration	-	–	-
2	AT-3301	Sweetening unit outlet CO_2 concentration	-	–	ppm
3	FE-3301	Lean amine to T-331 flow	-	–	-
4	FE-3302	Reflux flow to T-332	-	–	-
5	FI-3301	Amine system purge	-	–	Nm^3/h
6	FT-3301	Lean amine to T-331 flow	0..100	–	m^3/h
7	FT-3302	Reflux flow to T-332	-	–	m^3/h
8	LSLL-3301	V-334 low low liquid level switch	-	–	-
9	LT-3302	V-332 condensate level	-	–	mm
10	LT-3303	T-331 bottom liquid level	-	–	mm
11	LT-3304	V-332 liquid level	-	–	mm
12	LT-3305	V-334 liquid level control	-	–	mm
13	LT-3306	T-332 bottom liquid level	-	–	mm
14	LT-3307	V-332 liquid level	-	–	mm

ID	TAG	SERVICE	RANGE	OP	UNITS
15	LT-3308	T-331 bottom liquid level	-	–	mm
16	LT-3311	V-331 liquid level	-	–	mm
17	PT-3311A	T-331 top pressure	0..50	–	barg
18	PT-3311B	T-331 liquid outlet pressure	0..50	–	barg
19	PT-3311C	T-331 gas inlet pressure	0..50	–	barg
20	PT-3312A	T-332 top pressure	0..10	–	barg
21	PT-3312B	T-332 liquid outlet pressure	0..10	–	barg
22	PT-3313	V-332 pressure	0..10	–	barg
23	PT-3314	P-332A/B suction pressure	0..10	–	barg
24	PT-3322	V-334 pressure	0..10	–	barg
25	PT-3332	P-332A/B discharge pressure	0..55	–	barg
26	PT-3334	FV-3301 outlet pressure	0..15	–	barg
27	RD-3320	V-334 vacuum protection	-	–	mm H_2O
28	TG-3301	Sweet gas to dehydration temperature	-15..150	–	°C
29	TG-3309	Rich amine to T-332 temperature	-15..250	–	°C
30	TG-3310	Lean amine outlet from E-332 temperature	-15..150	–	°C
31	TG-3321	V-336 temperature	-15..150	–	°C
32	TG-3322	V-335 temperature	-15..150	–	°C
33	TT-3301	Lean amine to T-331 temperature	-50..200	–	°C
34	TT-3302	T-331 top temperature	-15..250	–	°C
35	TT-3303	T-331 bottom liquid temperature	-15..250	–	°C
36	TT-3304	T-332 top temperature	-15..250	–	°C
37	TT-3305	T-332 bottom liquid temperature	-15..250	–	°C
38	TT-3306	Amine outlet from E-333 temperature	-15..250	–	°C
39	TT-3307	T-331 tray 11 temperature	-15..250	–	°C
40	TT-3308	T-331 tray 14 temperature	-15..250	–	°C
41	TT-3309	T-331 tray 18 temperature	-15..250	–	°C
42	TT-3310	CO_2 rich gas to T-331 temperature	-50..200	–	°C
43	TT-3314	E-335 outlet temperature	-50..200	–	°C
44	TT-3323	V-336 temperature	-15..150	–	°C
45	TT-3324	V-335 temperature	-15..150	–	°C
46	TT-3333	T-332 gas outlet temperature	-15..250	–	°C
47	UHS-3301A	XV-3301 local open button	-	–	-
48	UHS-3301B	XV-3301 local close button	-	–	-
49	UHS-3303A	XV-3303 local open button	-	–	-
50	UHS-3303B	XV-3303 local close button	-	–	-
51	ZSH-3301	XV-3301 open position indication	-	–	-
52	ZSH-3302	V-332 condensate level open	0..100	–	%
53	ZSH-3303	XV-3303 open position indication	-	–	-
54	ZSH-3304	SDV-3304 open position indication	-	–	-

ID	TAG	SERVICE	RANGE	OP	UNITS
55	ZSH-3311	V-331 liquid level control open position	-	–	-
56	ZSL-3301	XV-3301 closed position indication	-	–	-
57	ZSL-3302	V-332 condensate level closed	0..100	–	%
58	ZSL-3303	XV-3303 closed position indication	-	–	-
59	ZSL-3304	SDV-3304 closed position indication	-	–	-
60	ZSL-3311	V-331 liquid level control closed position	-	–	-

6.5.6 Junction Boxes

Table 6.8 — Unit 330 – Junction Boxes

ID	TAG	SERVICE	CONNECTION	TYPE
1	JG-334	Amine utilities	COPPER-COPPER	STANDARD
2	JG-335	Amine regenerator	COPPER-COPPER	STANDARD
3	P-PN-OS331.1	Amine inlet skid	COPPER-COPPER	STANDARD
4	P-PN-OS332.1	Amine lean/rich skid	COPPER-COPPER	STANDARD
5	P-PN-OS334.1	Amine reflux skid	COPPER-COPPER	STANDARD
6	P-PN-OT331.1	Amine contactor-absorber	COPPER-COPPER	STANDARD

6.5.7 Controllers

Table 6.9 — Unit 330 – Controllers

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
FIC-3301	LEAN AMINE TO T-331 FLOW CONTROLLER	FV-3301	0..100 m^3/H	—
LIC-3302	V-332 CONDENSATE LEVEL CONTROLLER	LV-3302	0..100 %	—
LIC-3303	T-331 BOTTOM LIQUID LEVEL CONTROLLER	LV-3303	0..100 %	—
LIC-3304	V-332 LIQUID LEVEL CONTROLLER	LV-3304	0..100 %	—
LIC-3305	V-334 LIQUID LEVEL CONTROLLER	SIC-MP-334A/B	0..100 %	—
LIC-3351	LEVEL IN V-335	PK-514A/B	0..100 %	—
LIC-3306	T-332 BOTTOM LIQUID LEVEL CONTROLLER	P-333A/B/C	0..100 %	—
TDIC-3301	ΔT TEMPERATURE TT-3310 AND TT-3301	TV-3301	-50..200 °C	—
TIC-3314	E-335 OUTLET TEMPERATURE CONTROLLER	MF-335-1-8	-50..200 °C	—
TIC-3323	V-336 TEMPERATURE CONTROLLER	HEATER (TAG TBD)	-15..150 °C	—
TIC-3324	V-335 TEMPERATURE CONTROLLER	HEATER (TAG TBD)	-15..150 °C	—

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
TIC-3333	T-332 GAS OUTLET TEMPERATURE CONTROLLER	TV-3333	-15..250 °C	—

6.6 Operating Interface

6.6.1 PCS Operating Screens – Unit 330

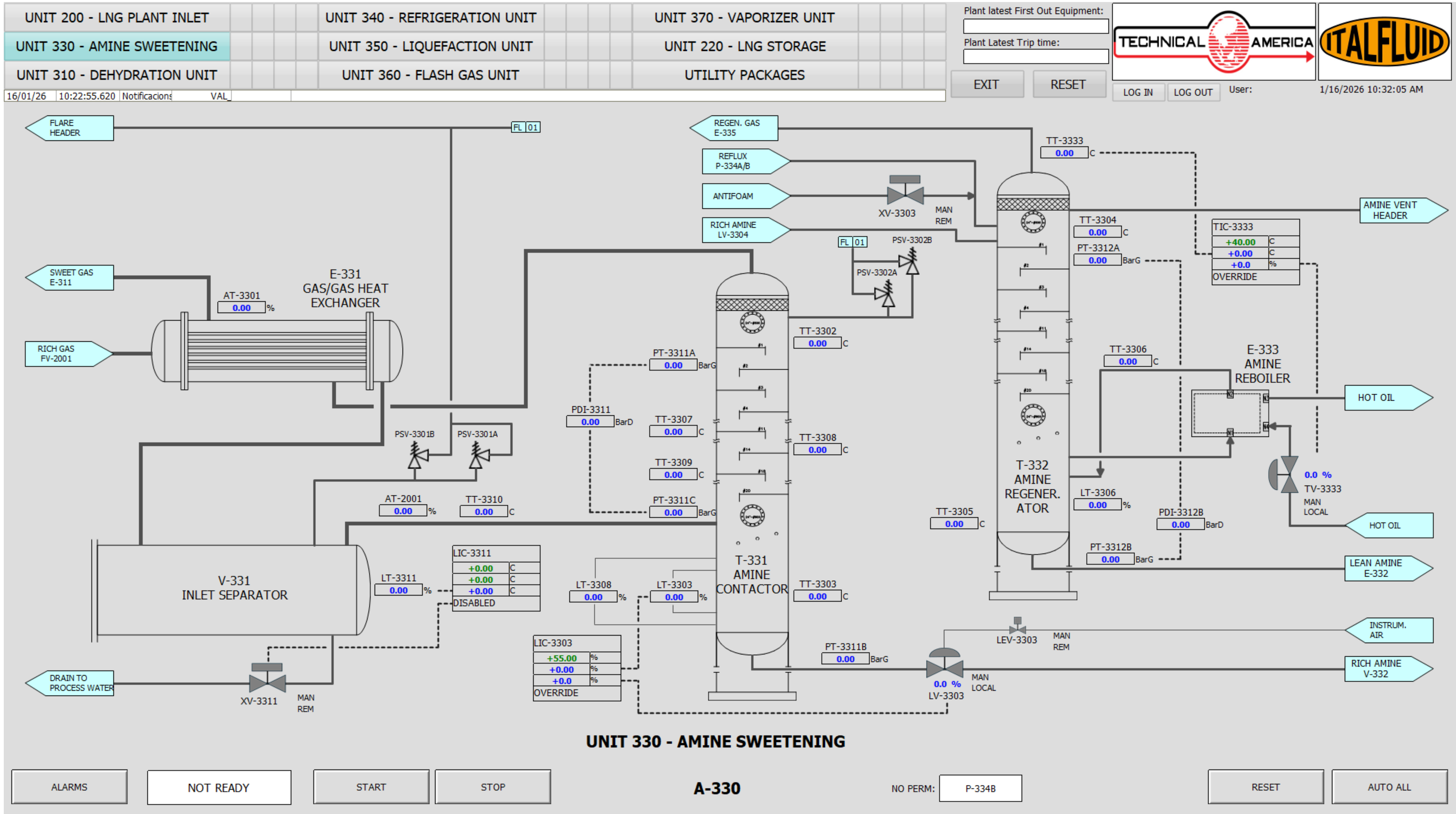


Figure 6.2: UNIT 330 - HMI Operating Screen

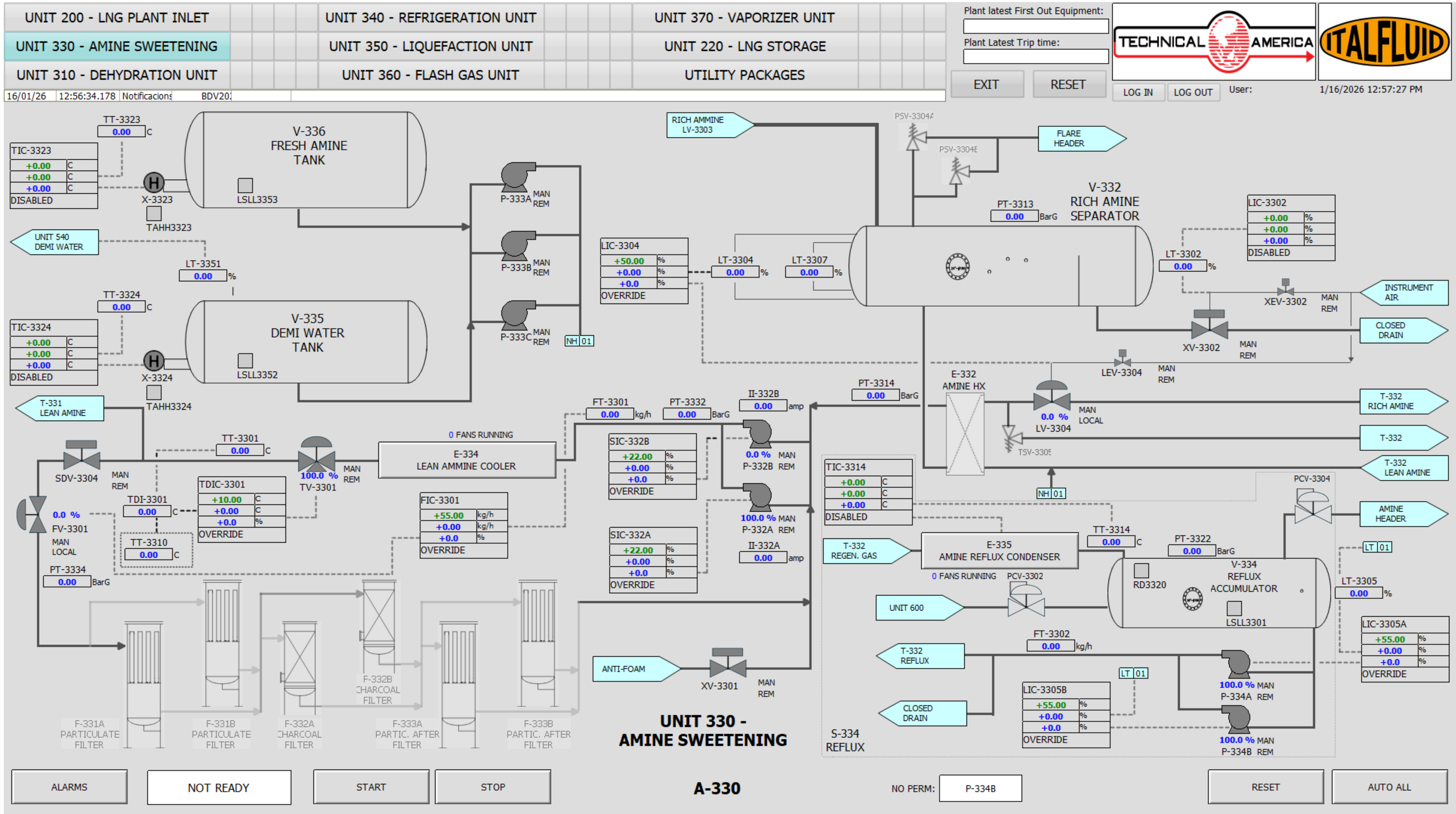


Figure 6.3: UNIT 330 - HMI Operating Screen

6.6.2 Critical Alarms and Trips

Table 6.10 — Unit 330 – Critical Alarms

ALARM	INSTRUMENT	TYPE	SP	DESCRIPTION
AAHH-3301	AT-3301	PSD	—	CO2 HIGH LEVEL
FALL-3301	FT-3301	PSD	—	LOW LOW FLOW OF LEAN AMINE
LAHH-3305	LT-3305	PSD	—	HIGH HIGH LIQUID LEVEL ON V-334
LAHH-3307	LT-3307	PSD	—	HIGH HIGH LIQUID LEVEL ON V-332
LAHH-3308	LT-3308	PSD	—	HIGH HIGH LIQUID LEVEL ON T-331
LALL-3304	LT-3304	LSD	—	LOW LOW LIQUID LEVEL ON V-332
LALL-3305	LS-3301	LSD	—	LOW LOW LIQUID LEVEL ON V-334
LALL-3306	LT-3306	LSD	—	LOW LOW LIQUID LEVEL ON T-332
LALL-3307	LT-3307	PSD	—	LOW LOW LIQUID LEVEL ON V-332
LALL-3308	LT-3308	PSD	—	LOW LOW LIQUID LEVEL ON T-331
LALL-3352	LSLL-3352	LSD	—	LOW LOW LIQUID LEVEL ON V-335
LALL-3353	LSLL-3353	LSD	—	LOW LOW LIQUID LEVEL ON V-336
PAHH-3313	PT-3313	PSD	—	HIGH HIGH PRESSURE ON V-332
PAHH-3322	PT-3322	PSD	—	HIGH HIGH PRESSURE ON V-334
PAHH-3334	PT-3334	LSD	—	HIGH HIGH PRESSURE ON AMINE FILTRATION
PALL-3314	PT-3314	PSD	—	LOW LOW PRESSURE ON P-332A/B SUCTION
PDHH-3311	PT-3311A/C	PSD	—	HIGH HIGH DIFFERENTIAL PRESSURE ON T-331
TAHH-3323	TAHH-3323	LSD	—	HIGH HIGH TEMPERATURE ON V-336
TAHH-3324	TAHH-3324	LSD	—	HIGH HIGH TEMPERATURE ON V-335
PSD05	PSD05	PSD	—	LOSS OF P-332A/B

6.6.3 Unit Cause and Effect Matrix

Table 6.11 — Unit 330 – Cause & Effect Matrix

ALARM	INSTRUMENT	LEV-3304	P-334A/B	P-332A/B	P-333A/B/C	X-V336	X-V335	SDV-3304
LALL-3304	LT-3304	C						
LALL-3305	LSLL-3301		T					
LALL-3306	LT-3306			T				
LALL-3352	LALL-3352				T			

ALARM	INSTRUMENT	LEV-3304	P-334A/B	P-332A/B	P-333A/B/C	X-V336	X-V335	SDV-3304
LALL-3353	LALL-3353				T			
PAHH-3334	PT-3334							C
TAHH-3323	TT-3323					T		
TAHH-3324	TT-3324						T	

7 Unit 340 - Refrigeration Unit (ammonia R1 loop)

7.1 Operating Summary

CATEGORY	DESCRIPTION
Inputs	Closed NH_3 cycle; compressors 560 kW each (3W+1S). Cooling duty for E-311 and E-354.
Outputs	Refrigeration supplied to precoolers E-311 and E-354; condensed NH_3 collected in receiver V-343, with heat rejected to atmosphere via air-cooled condensers E-342.
Key Operating Specs	Single-stage mechanical refrigeration system per EN 378. Common suction pressure controlled by PIC-3552 via coordinated slide-valve modulation of operating compressors. Evaporator liquid levels controlled by LIC-3111 (E-311) and LIC-3540 (E-354). Local back-pressure control PCV-3111 on E-311 to maintain evaporating temperature hierarchy. Receiver pressure controlled by PIC-3401 through on/off control of air condenser fans E-342.
Major Equipment	PK-341 ammonia compressor packages (C-341-1…4), V-341 oil separators, E-341 oil coolers, E-342-1…6 air-cooled condensers, V-343 ammonia receiver.

7.2 General Arrangement and Risk Location

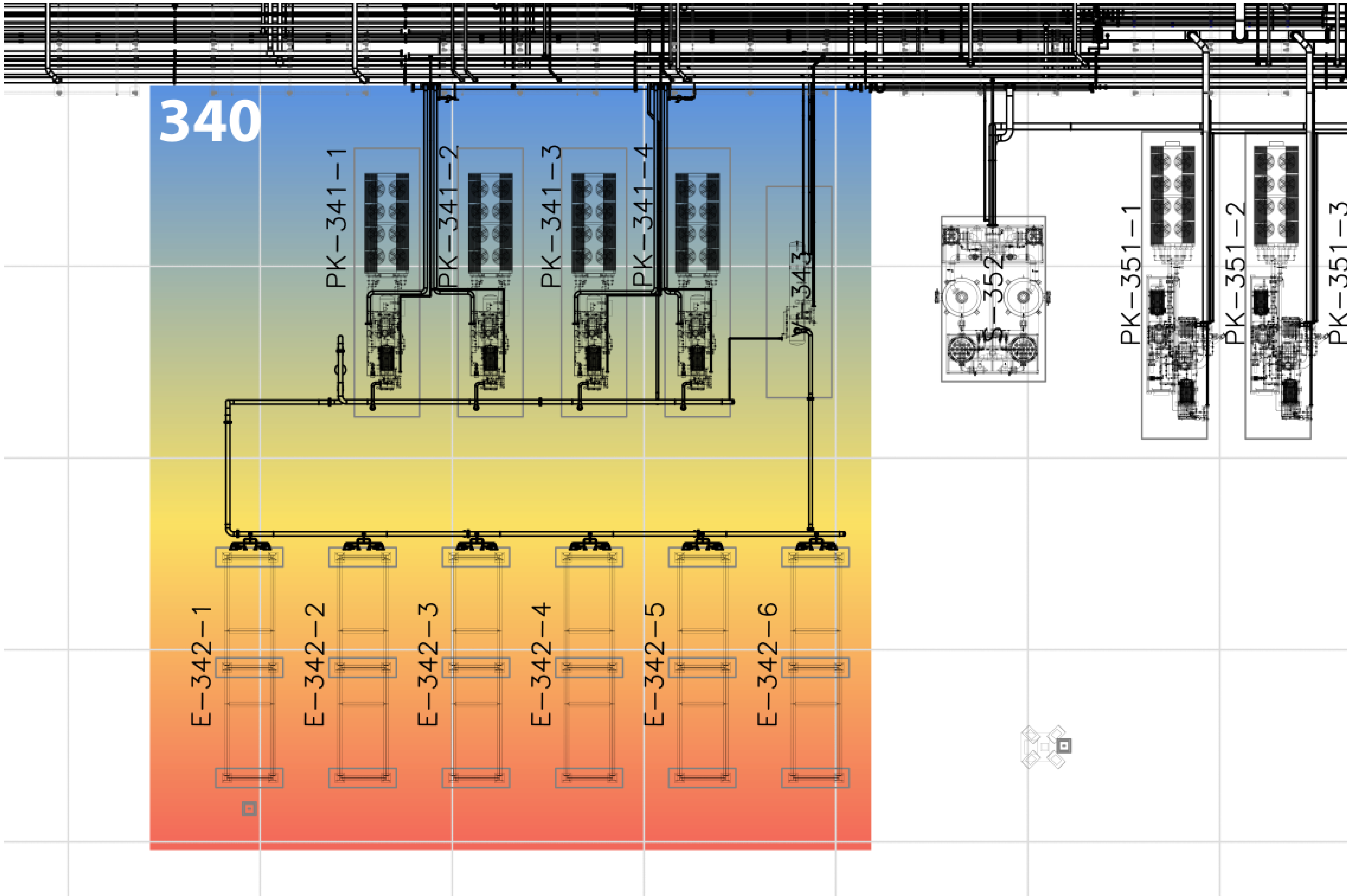


Figure 7.1: Unit 340 GA - Risk location

7.3 Relevant Documents

Table 7.2 — Unit 340 – Relevant Docuemnts

ID	DOC NO	TITLE
1	PRO02-ING-FR02	PROCESS DESCRIPTION
2	PRO10-ING-PR07	UTILITY FLOW DIAGRAM (R1 REFRIGERANT LOOP)
3	PRO13-ING-PR34	PACKAGE PK-341-1 (UNIT 340)
4	PRO13-ING-PR38	REFRIGERATION UNIT 340
5	SAF01-ING-FR01	PROCESS SAFETY AND ISOLATION PHILOSOPHY
6	SAF12-ING-FR01	CAUSE AND EFFECT MATRIX
7	SAF05-ING-FR01	RELIEF AND BLOWDOWN PHILOSOPHY
8	21044-AU-HMI-00002	HMI PHILOSOPHY

7.4 Process Description

Area 340 provides the single-stage mechanical refrigeration required to supply ammonia refrigerant (R1 loop) to the natural gas precooler E-311 in Unit 310 (Dehydration) and to the ammonia mixed-refrigerant cooler E-354 in Unit 350 (Liquefaction). The refrigeration system operates as a closed ammonia loop designed in accordance with EN 378.

The system consists of four oil-flooded screw compressor trains PK-341-1/2/3/4 arranged in a 3 Working + 1 Standby configuration, six parallel air-cooled ammonia condensers E-342-1 to E-342-6, a common horizontal liquid receiver V-343, and the associated oil systems, piping, instrumentation, and safety devices.

Each compressor train includes a single-stage oil-flooded screw compressor C-341-x driven by a 560 kW, 690 V, 50 Hz electric motor. The compressor package incorporates a primary oil separator V-341-x, an air-cooled oil cooler E-341-x, and a closed lubrication circuit equipped with a positive-displacement oil pump P-341-x and duplex oil filters arranged as one working and one standby.

The condenser bank E-342 rejects the heat of compression to atmosphere and condenses the ammonia vapor. The condensed liquid refrigerant drains by gravity into the common liquid receiver V-343, which provides refrigerant inventory and hydraulic separation between the condensing and evaporating sections of the refrigeration loop.

Process Flow and Refrigeration Users

Low-pressure ammonia vapor generated in E-311, the natural gas precooler in Unit 310, and E-354, the ammonia mixed-refrigerant cooler in Unit 350, is collected in a common suction header and routed to the refrigeration compressor package PK-341.

The number of compressors in service is selected manually by the operator according to plant operating requirements. No automatic compressor start, stop, or sequencing logic is provided. Each compressor is equipped with an integral slide valve used for continuous capacity control.

The evaporating pressure of the R1 refrigeration loop is controlled by pressure indicator controller PIC-3552, which measures the common compressor suction pressure. The capacity of each operating compressor is automatically adjusted through its slide valve based on the deviation between the measured suction pressure and the PIC-3552 setpoint. This control strategy matches the refrigeration capacity to the combined cooling demand of E-311 and E-354 while maintaining operator control of compressor availability.

Compressed ammonia vapor is discharged to the air-cooled condenser bank E-342, where it is condensed before flowing to liquid receiver V-343. High-pressure liquid ammonia from V-343 is then distributed to the refrigeration users E-311 and E-354.

At each exchanger, liquid ammonia expands through a dedicated level control valve into the shell side of the exchanger, generating a two-phase refrigerant mixture. The liquid refrigerant inventory is maintained independently in each evaporator by level control. LIC-3111 controls the liquid level in E-311, while LIC-3540 controls the liquid level in E-354. Refrigerant vapor leaving each evaporator returns to the common suction header, completing the refrigeration cycle.

A back-pressure control valve PCV-3111 is installed on the ammonia vapor outlet of E-311. This valve maintains a minimum evaporating pressure in the dehydration precooler and ensures that the evaporating temperature of E-311 remains equal to or higher than that of E-354. This arrangement prevents excessive cooling of the natural gas upstream of the dehydration adsorbers and reduces the risk of hydrate formation.

The refrigeration system therefore operates with manual compressor selection, automatic compressor capacity control based on common suction pressure, independent evaporator level control, and controlled evaporating-pressure hierarchy between the refrigeration users.

7.5 Technological Objects

7.5.1 Main Equipment List

Table 7.3 — Unit 340 – Main Equipment List

ID	TAG	EQUIPMENT
1	**PK-341-1**	**AMMONIA COMPRESSOR PACKAGE 1**
1.1	C-341-1	Ammonia compressor 1
1.2	MC-341-1	Electric motor
1.3	V-341-1	Oil separator
1.4	E-341-1	Oil cooler
1.4.1	F-341-1-1..10	Oil cooler fans (10)
1.4.2	MF-341-1-1..10	Oil cooler fan motors (10)
1.5	P-341-1	Oil pump
1.6	F-341-1A	Main oil filter A
1.7	F-341-1B	Main oil filter B
1.8	X-341-1	Oil heater 1
2	**PK-341-2**	**AMMONIA COMPRESSOR PACKAGE 2**
2.1	C-341-2	Ammonia compressor 2
2.2	MC-341-2	Electric motor
2.3	V-341-2	Oil separator
2.4	E-341-2	Oil cooler
2.4.1	F-341-2-1..10	Oil cooler fans (10)
2.4.2	MF-341-2-1..10	Oil cooler fan motors (10)
2.5	P-341-2	Oil pump
2.6	F-341-2A	Main oil filter A
2.7	F-341-2B	Main oil filter B
2.8	X-341-2	Oil heater 2
3	**PK-341-3**	**AMMONIA COMPRESSOR PACKAGE 3**
3.1	C-341-3	Ammonia compressor 3
3.2	MC-341-3	Electric motor
3.3	V-341-3	Oil separator
3.4	E-341-3	Oil cooler
3.4.1	F-341-3-1..10	Oil cooler fans (10)
3.4.2	MF-341-3-1..10	Oil cooler fan motors (10)
3.5	P-341-3	Oil pump
3.6	F-341-3A	Main oil filter A
3.7	F-341-3B	Main oil filter B
3.8	X-341-3	Oil heater 3
4	**PK-341-4**	**AMMONIA COMPRESSOR PACKAGE 4**
4.1	C-341-4	Ammonia compressor 4

ID	TAG	EQUIPMENT
4.2	MC-341-4	Electric motor
4.3	V-341-4	Oil separator
4.4	E-341-4	Oil cooler
4.4.1	F-341-4-1..10	Oil cooler fans (10)
4.4.2	MF-341-4-1..10	Oil cooler fan motors (10)
4.5	P-341-4	Oil pump
4.6	F-341-4A	Main oil filter A
4.7	F-341-4B	Main oil filter B
4.8	X-341-4	Oil heater 4
5	**E-342-1**	**AMMONIA CONDENSER 1**
5.1	F-342-1-1..12	Condenser fans (12)
5.2	MF-342-1-1..12	Condenser fan motors (12)
6	**E-342-2**	**AMMONIA CONDENSER 2**
6.1	F-342-2-1..12	Condenser fans (12)
6.2	MF-342-2-1..12	Condenser fan motors (12)
7	**E-342-3**	**AMMONIA CONDENSER 3**
7.1	F-342-3-1..12	Condenser fans (12)
7.2	MF-342-3-1..12	Condenser fan motors (12)
8	**E-342-4**	**AMMONIA CONDENSER 4**
8.1	F-342-4-1..12	Condenser fans (12)
8.2	MF-342-4-1..12	Condenser fan motors (12)
9	**E-342-5**	**AMMONIA CONDENSER 5**
9.1	F-342-5-1..12	Condenser fans (12)
9.2	MF-342-5-1..12	Condenser fan motors (12)
10	**E-342-6**	**AMMONIA CONDENSER 6**
10.1	F-342-6-1..12	Condenser fans (12)
10.2	MF-342-6-1..12	Condenser fan motors (12)
11	**V-343**	**AMMONIA RECEIVER**

7.5.2 Electrical Loads

Table 7.4 — Unit 340 – Electrical Loads

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	C-341-1	AMMONIA COMPRESSOR	VFD	690	560,00
2	X-341-1	OIL HEATER (OIL SEPARATOR)	DOL	690	5,30
3	P-341-1	OIL PUMP	DOL	690	4,00
4	C-341-2	AMMONIA COMPRESSOR	VFD	690	560,00
5	X-341-2	OIL HEATER (OIL SEPARATOR)	DOL	690	5,30

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
6	P-341-2	OIL PUMP	DOL	690	4,00
7	C-341-3	AMMONIA COMPRESSOR	VFD	690	560,00
8	X-341-3	OIL HEATER (OIL SEPARATOR)	DOL	690	5,30
9	P-341-3	OIL PUMP	DOL	690	4,00
10	C-341-4	AMMONIA COMPRESSOR	VFD	690	560,00
11	X-341-4	OIL HEATER (OIL SEPARATOR)	DOL	690	5,30
12	P-341-4	OIL PUMP	DOL	690	4,00

7.5.3 Motors as Remote MCC

Table 7.5 — Unit 340 – Motors as Remote MCC

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	MF-341-1-1	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
2	MF-341-1-2	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
3	MF-341-1-3	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
4	MF-341-1-4	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
5	MF-341-1-5	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
6	MF-341-1-6	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
7	MF-341-1-7	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
8	MF-341-1-8	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
9	MF-341-1-9	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
10	MF-341-1-10	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
11	MF-341-2-1	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
12	MF-341-2-2	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
13	MF-341-2-3	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
14	MF-341-2-4	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
15	MF-341-2-5	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
16	MF-341-2-6	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
17	MF-341-2-7	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
18	MF-341-2-8	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
19	MF-341-2-9	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
20	MF-341-2-10	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
21	MF-341-3-1	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
22	MF-341-3-2	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
23	MF-341-3-3	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
24	MF-341-3-4	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
25	MF-341-3-5	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
26	MF-341-3-6	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
27	MF-341-3-7	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
28	MF-341-3-8	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
29	MF-341-3-9	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
30	MF-341-3-10	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
31	MF-341-4-1	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
32	MF-341-4-2	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
33	MF-341-4-3	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
34	MF-341-4-4	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
35	MF-341-4-5	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
36	MF-341-4-6	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
37	MF-341-4-7	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
38	MF-341-4-8	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
39	MF-341-4-9	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
40	MF-341-4-10	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
41	MF-342-1-1	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
42	MF-342-1-2	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
43	MF-342-1-3	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
44	MF-342-1-4	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
45	MF-342-1-5	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
46	MF-342-1-6	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
47	MF-342-1-7	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
48	MF-342-1-8	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
49	MF-342-1-9	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
50	MF-342-1-10	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
51	MF-342-1-11	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
52	MF-342-1-12	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
53	MF-342-2-1	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
54	MF-342-2-2	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
55	MF-342-2-3	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
56	MF-342-2-4	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
57	MF-342-2-5	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
58	MF-342-2-6	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
59	MF-342-2-7	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
60	MF-342-2-8	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
61	MF-342-2-9	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
62	MF-342-2-10	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
63	MF-342-2-11	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
64	MF-342-2-12	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
65	MF-342-3-1	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
66	MF-342-3-2	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
67	MF-342-3-3	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
68	MF-342-3-4	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
69	MF-342-3-5	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
70	MF-342-3-6	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
71	MF-342-3-7	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
72	MF-342-3-8	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
73	MF-342-3-9	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
74	MF-342-3-10	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
75	MF-342-3-11	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
76	MF-342-3-12	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
77	MF-342-4-1	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
78	MF-342-4-2	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
79	MF-342-4-3	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
80	MF-342-4-4	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
81	MF-342-4-5	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
82	MF-342-4-6	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
83	MF-342-4-7	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
84	MF-342-4-8	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
85	MF-342-4-9	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
86	MF-342-4-10	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
87	MF-342-4-11	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
88	MF-342-4-12	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
89	MF-342-5-1	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
90	MF-342-5-2	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
91	MF-342-5-3	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
92	MF-342-5-4	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
93	MF-342-5-5	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
94	MF-342-5-6	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
95	MF-342-5-7	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
96	MF-342-5-8	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
97	MF-342-5-9	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
98	MF-342-5-10	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
99	MF-342-5-11	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
100	MF-342-5-12	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
101	MF-342-6-1	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
102	MF-342-6-2	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
103	MF-342-6-3	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
104	MF-342-6-4	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
105	MF-342-6-5	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
106	MF-342-6-6	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
107	MF-342-6-7	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
108	MF-342-6-8	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
109	MF-342-6-9	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
110	MF-342-6-10	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
111	MF-342-6-11	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00
112	MF-342-6-12	AMMONIA CONDENSER FAN ELECTRIC MOTOR	DOL	690	4,00

7.5.4 Valves

Table 7.6 — Unit 340 – Valves

ID	TAG	SERVICE	SYSTEM	NOTES
1	PCV-3401	AMMONIA LOOP PRESSURE CONTROL	-	SET PRESSURE
2	PEV-3401	AMMONIA LOOP PRESSURE CONTROL	PCS	WITH 5 MT CABLE
3	PEV-3402	AMMONIA LOOP PRESSURE CONTROL	PCS	WITH 5 MT CABLE
4	PSV-3401A	V-343 PRESSURE RELIEF	-	SET PRESSURE
5	PSV-3401B	V-343 PRESSURE RELIEF	-	SET PRESSURE
6	PSV-3411.1	C-341-1 BLOCKED OUTLET RELIEF	-	SET PRESSURE 28 barg
7	PSV-3411.2	C-341-2 BLOCKED OUTLET RELIEF	-	SET PRESSURE 28 barg
8	PSV-3411.3	C-341-3 BLOCKED OUTLET RELIEF	-	SET PRESSURE 28 barg
9	PSV-3411.4	C-341-4 BLOCKED OUTLET RELIEF	-	SET PRESSURE 28 barg
10	PSV-3417.1A	V-341-1 PRESSURE RELIEF	-	SET PRESSURE 28 barg
11	PSV-3417.1B	V-341-1 PRESSURE RELIEF	-	SET PRESSURE 28 barg
12	PSV-3417.2A	V-341-2 PRESSURE RELIEF	-	SET PRESSURE 28 barg
13	PSV-3417.2B	V-341-2 PRESSURE RELIEF	-	SET PRESSURE 28 barg
14	PSV-3417.3A	V-341-3 PRESSURE RELIEF	-	SET PRESSURE 28 barg
15	PSV-3417.3B	V-341-3 PRESSURE RELIEF	-	SET PRESSURE 28 barg
16	PSV-3417.4A	V-341-4 PRESSURE RELIEF	-	SET PRESSURE 28 barg
17	PSV-3417.4B	V-341-4 PRESSURE RELIEF	-	SET PRESSURE 28 barg
18	PSV-3418.1	P-341-1 PRESSURE RELIEF	-	SET PRESSURE 6 bar diff.
19	PSV-3418.2	P-341-2 PRESSURE RELIEF	-	SET PRESSURE 6 bar diff.
20	PSV-3418.3	P-341-3 PRESSURE RELIEF	-	SET PRESSURE 6 bar diff.
21	PSV-3418.4	P-441-4 PRESSURE RELIEF	-	SET PRESSURE 6 bar diff.
22	ZEV-3411.1	C-341-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
23	ZEV-3411.2	C-341-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
24	ZEV-3411.3	C-341-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
25	ZEV-3411.4	C-341-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704

ID	TAG	SERVICE	SYSTEM	NOTES
26	ZEV-3412.1	C-341-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
27	ZEV-3412.2	C-341-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
28	ZEV-3412.3	C-341-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
29	ZEV-3412.4	C-341-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
30	ZEV-3413.1	C-341-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
31	ZEV-3413.2	C-341-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
32	ZEV-3413.3	C-341-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
33	ZEV-3413.4	C-341-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
34	ZEV-3414.1	C-341-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
35	ZEV-3414.2	C-341-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
36	ZEV-3414.3	C-341-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
37	ZEV-3414.4	C-341-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704

7.5.5 Instrumentation

Table 7.7 — Unit 340 – Instrument List

ID	TAG	SERVICE	RANGE	OP	UNITS
1	LSL-3417.1	V-341-1 OIL LEVEL	-	–	mm
2	LSL-3417.2	V-341-2 OIL LEVEL	-	–	mm
3	LSL-3417.3	V-341-3 OIL LEVEL	-	–	mm
4	LSL-3417.4	V-341-4 OIL LEVEL	-	–	mm
5	LT-3401	V-343 LIQUID LEVEL	0..100	50.00	%
6	PT-3401	V-343 PRESSURE	0..28	25.81	barg
7	PT-3411.1	C-341-1 SUCTION PRESSURE	0..13.80	3.69	bara
8	PT-3411.2	C-341-2 SUCTION PRESSURE	0..13.80	3.69	bara
9	PT-3411.3	C-341-3 SUCTION PRESSURE	0..13.80	3.69	bara
10	PT-3411.4	C-341-4 SUCTION PRESSURE	0..13.80	3.69	bara
11	PT-3412.1	V-341-1 PRESSURE	0..34.47	25.93	barg
12	PT-3412.2	V-341-2 PRESSURE	0..34.47	25.93	barg
13	PT-3412.3	V-341-3 PRESSURE	0..34.47	25.93	barg
14	PT-3412.4	V-341-4 PRESSURE	0..34.47	25.93	barg
15	PT-3413.1	C-341-1 OIL HEADER PRESSURE	0..34.47	28.93	barg
16	PT-3413.2	C-341-2 OIL HEADER PRESSURE	0..34.47	28.93	barg
17	PT-3413.3	C-341-3 OIL HEADER PRESSURE	0..34.47	28.93	barg
18	PT-3413.4	C-341-4 OIL HEADER PRESSURE	0..34.47	28.93	barg
19	TT-3412.1	C-341-1 DISCHARGE TEMPERATURE	-50..200	88.00	°C
20	TT-3412.2	C-341-2 DISCHARGE TEMPERATURE	-50..200	88.00	°C

ID	TAG	SERVICE	RANGE	OP	UNITS
21	TT-3412.3	C-341-3 DISCHARGE TEMPERATURE	-50..200	88.00	°C
22	TT-3412.4	C-341-4 DISCHARGE TEMPERATURE	-50..200	88.00	°C
23	TT-3417.1	V-341-1 OIL TEMPERATURE	-50..200	88.00	°C
24	TT-3417.2	V-341-2 OIL TEMPERATURE	-50..200	88.00	°C
25	TT-3417.3	V-341-3 OIL TEMPERATURE	-50..200	88.00	°C
26	TT-3417.4	V-341-4 OIL TEMPERATURE	-50..200	88.00	°C
27	TT-3418.1	OIL TO C-341-1 TEMPERATURE	-50..200	57.00	°C
28	TT-3418.2	OIL TO C-341-2 TEMPERATURE	-50..200	57.00	°C
29	TT-3418.3	OIL TO C-341-3 TEMPERATURE	-50..200	57.00	°C
30	TT-3418.4	OIL TO C-341-4 TEMPERATURE	-50..200	57.00	°C
31	TT-3419.1	C-341-1 SUCTION TEMPERATURE	-50..200	-1.25	°C
32	TT-3419.2	C-341-2 SUCTION TEMPERATURE	-50..200	-1.25	°C
33	TT-3419.3	C-341-3 SUCTION TEMPERATURE	-50..200	-1.25	°C
34	TT-3419.4	C-341-4 SUCTION TEMPERATURE	-50..200	-1.25	°C

7.5.6 Junction Boxes

Table 7.8 — Unit 340 – Junction Boxes

ID	TAG	SERVICE	CONNECTION	TYPE
1	P-PN-OV351.1	AMMONIA RECEIVER	COPPER-COPPER	STANDARD
2	P-PN-PK-341-1	AMMONIA COMPRESSOR	COPPER-COPPER	STANDARD
3	P-PN-PK-341-2	AMMONIA COMPRESSOR	COPPER-COPPER	STANDARD
4	P-PN-PK-341-3	AMMONIA COMPRESSOR	COPPER-COPPER	STANDARD
5	P-PN-PK-341-4	AMMONIA COMPRESSOR	COPPER-COPPER	STANDARD
6	JG-341-1C	AMMONIA COMPRESSOR OIL COOLER	COPPER-COPPER	STANDARD
7	JG-341-2C	AMMONIA COMPRESSOR OIL COOLER	COPPER-COPPER	STANDARD
8	JG-341-3C	AMMONIA COMPRESSOR OIL COOLER	COPPER-COPPER	STANDARD
9	JG-341-4C	AMMONIA COMPRESSOR OIL COOLER	COPPER-COPPER	STANDARD
10	JG-342-1	AIR COOLED AMMONIA CONDENSER FANS	COPPER-FIBER	STANDARD
11	JG-342-2	AIR COOLED AMMONIA CONDENSER FANS	COPPER-COPPER	STANDARD
12	JG-342-3	AIR COOLED AMMONIA CONDENSER FANS	COPPER-COPPER	STANDARD
13	JG-342-4	AIR COOLED AMMONIA CONDENSER FANS	COPPER-COPPER	STANDARD
14	JG-342-5	AIR COOLED AMMONIA CONDENSER FANS	COPPER-COPPER	STANDARD
15	JG-342-6	AIR COOLED AMMONIA CONDENSER FANS	COPPER-COPPER	STANDARD

7.5.7 Controllers

Table 7.9 — Unit 340 – Controllers

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
PIC-3401	V-343 AMMONIA RECEIVER PRESSURE CONTROLLER	E-342	0..28 barg	25 barg

7.6 Operating Interface

7.6.1 PCS Operating Screens – Unit 340

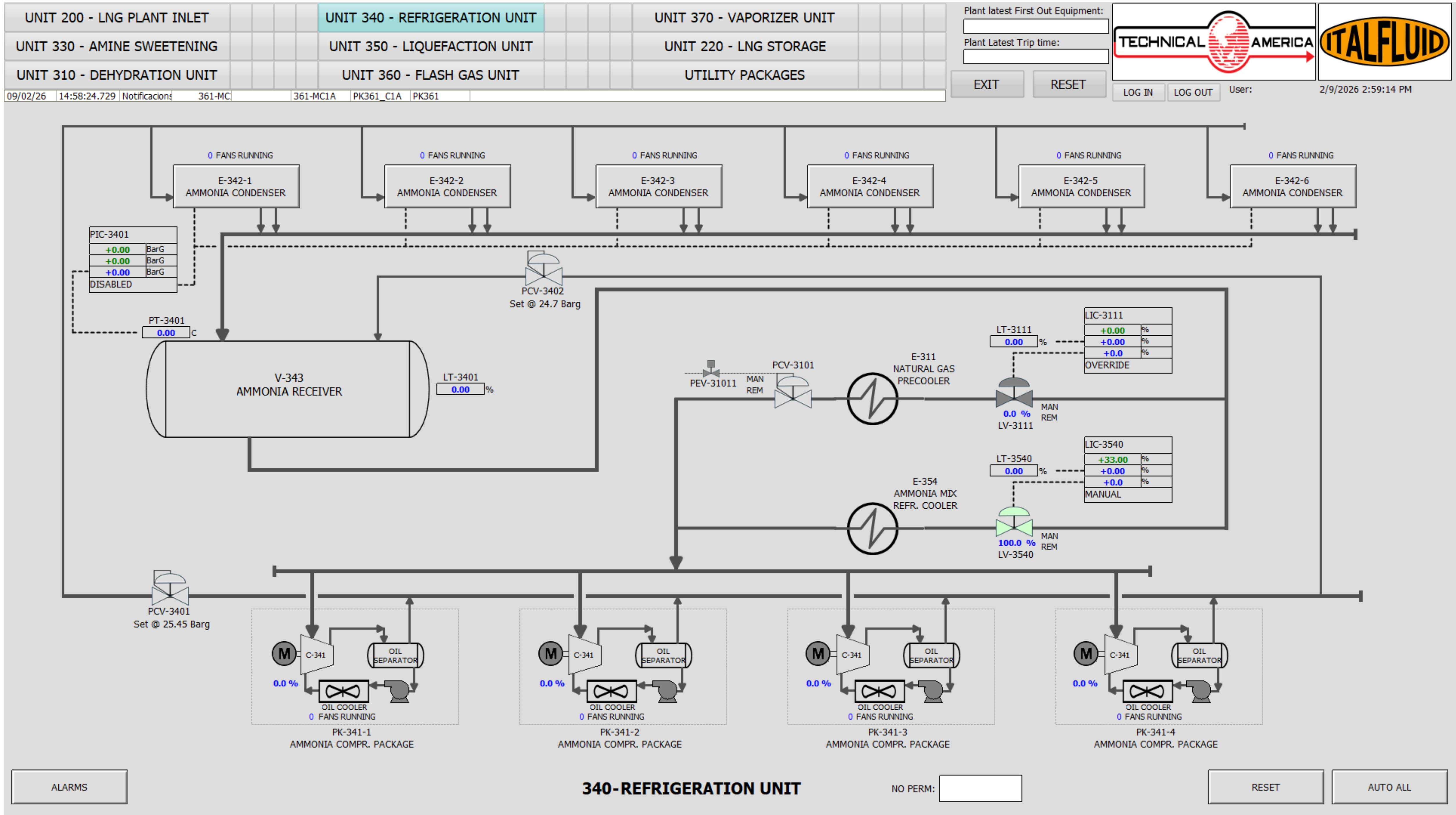


Figure 7.2: Unit 340 - HMI Operating Screen (Overview)

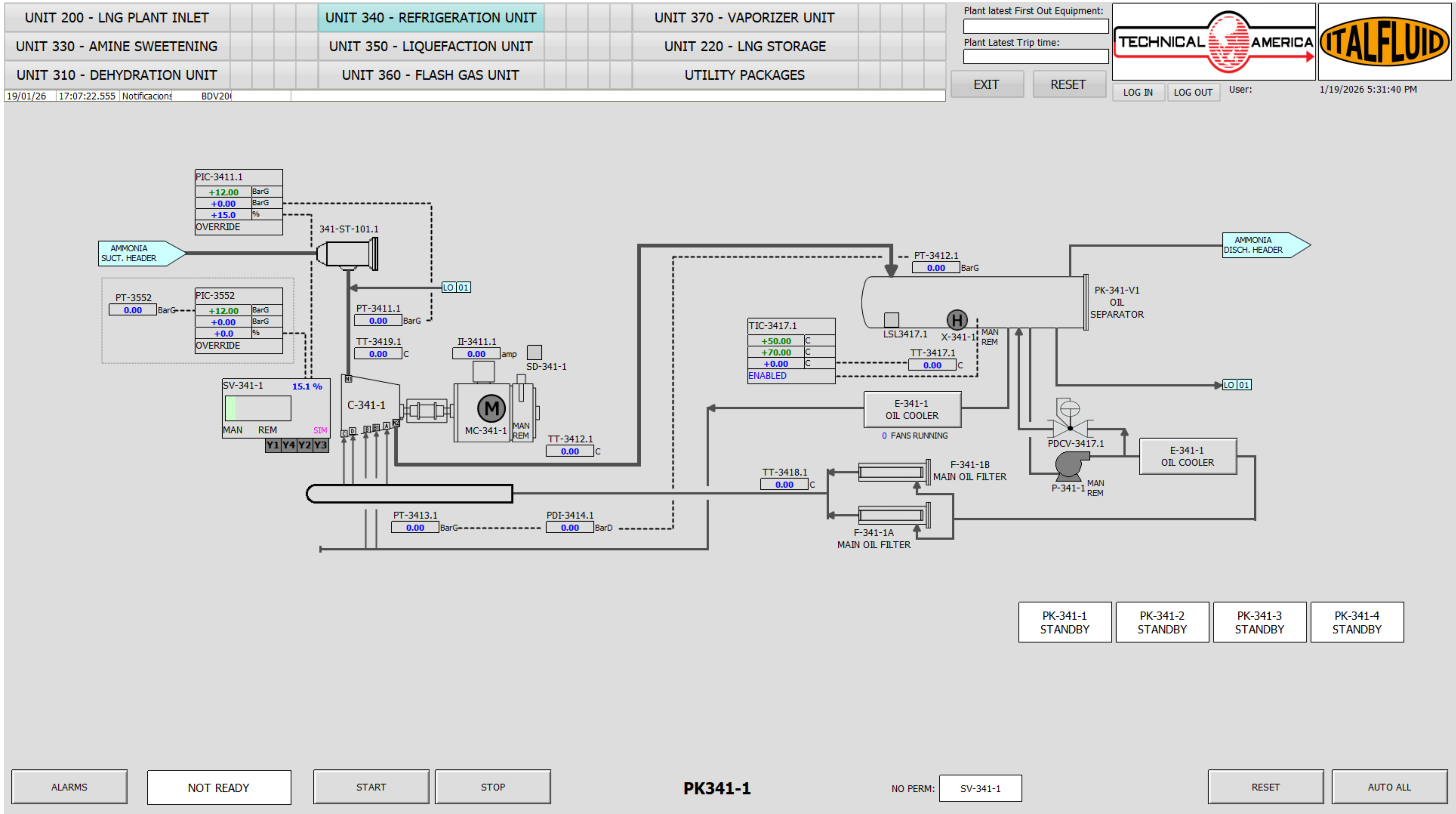


Figure 7.3: Unit 340 - HMI Operating Screen (Compressor Package)

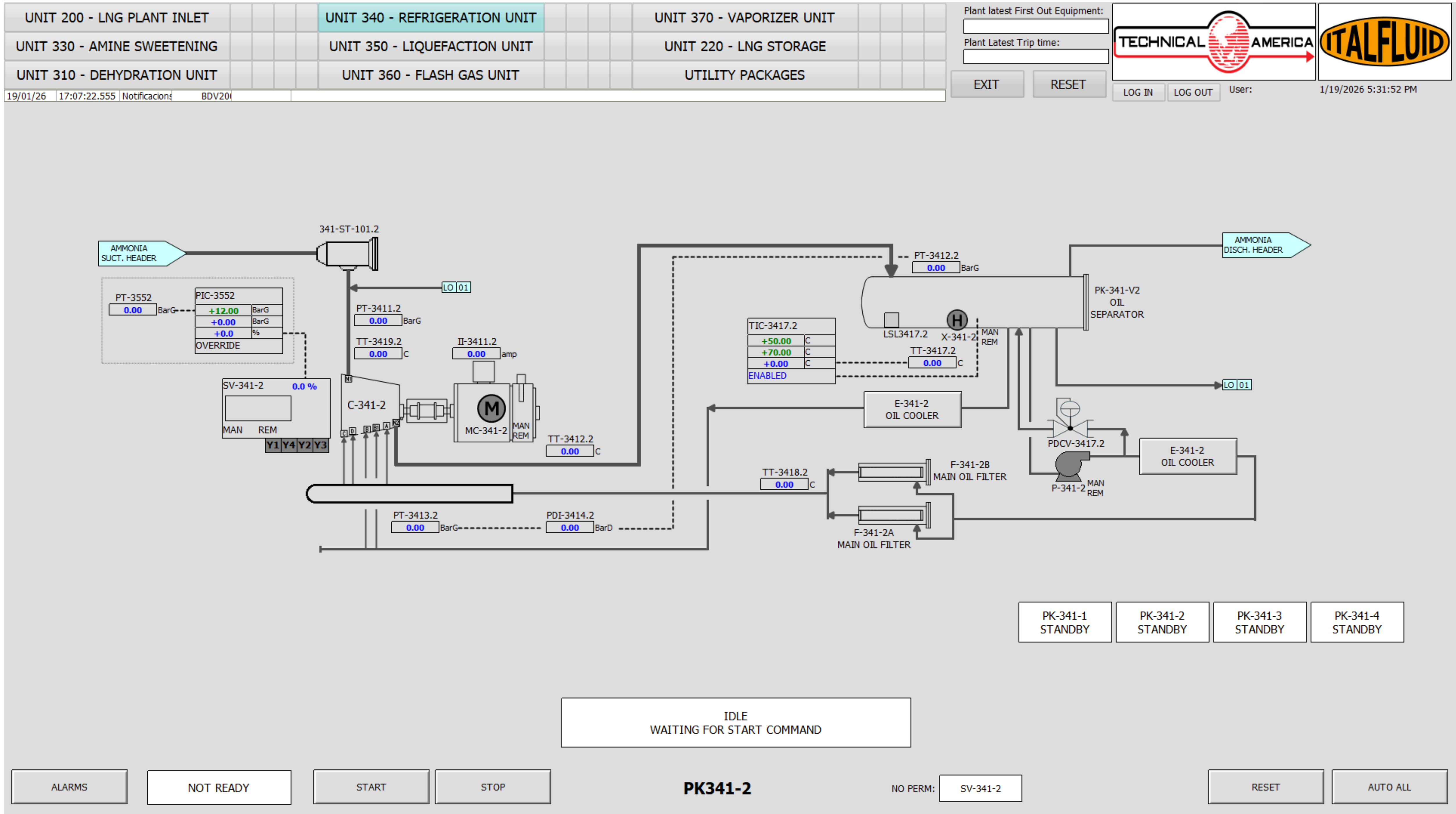


Figure 7.4: Unit 340 - HMI Operating Screen (Oil System)

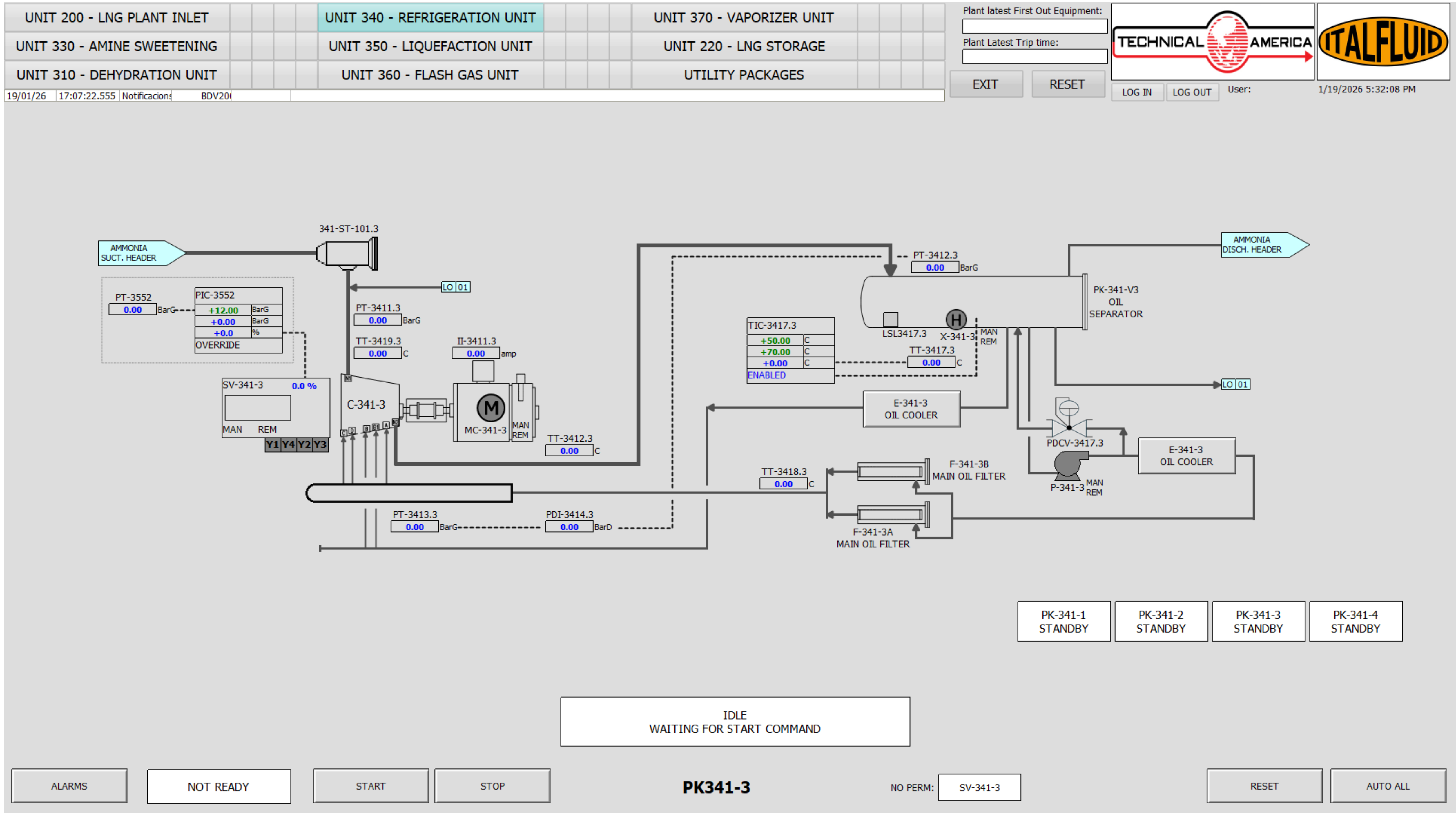


Figure 7.5: Unit 340 - HMI Operating Screen (Condensers)

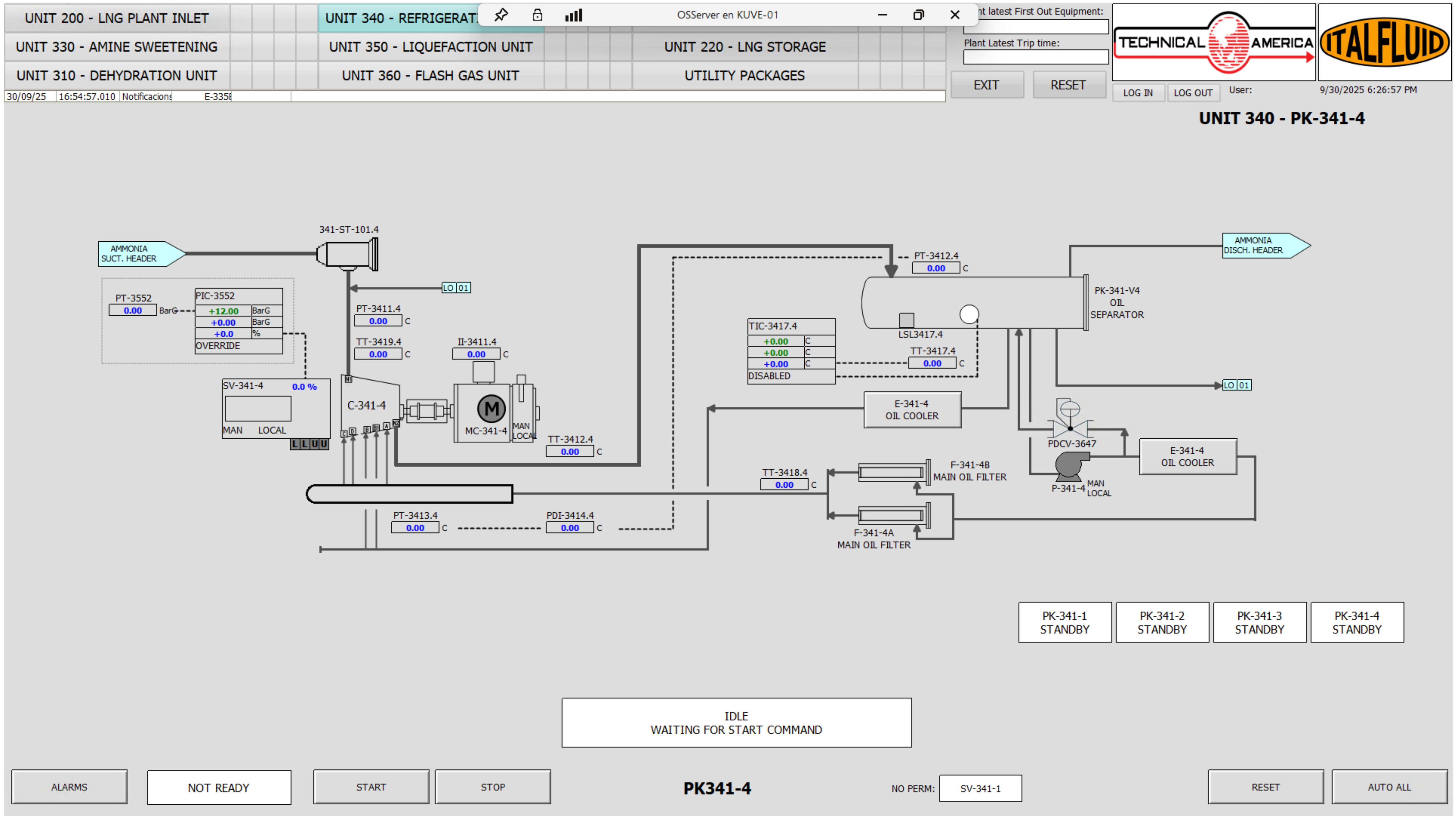


Figure 7.6: Unit 340 - HMI Operating Screen (Auxiliary Systems)

7.6.2 Critical Alarms and Trips

Table 7.10 — Unit 340 – Critical Alarms

ALARM	INSTRUMENT	TYPE	SP	DESCRIPTION
LSD01	LSD01	LSD	—	PK-341-1 EMERGENCY STOP PUSH BUTTON
LSD02	LSD02	LSD	—	PK-341-1 EMERGENCY STOP INTERLOCK
LSD03	LSD03	LSD	—	PK-341-2 EMERGENCY STOP PUSH BUTTON
LSD04	LSD04	LSD	—	PK-341-2 EMERGENCY STOP INTERLOCK
LSD05	LSD05	LSD	—	PK-341-3 EMERGENCY STOP PUSH BUTTON
LSD06	LSD06	LSD	—	PK-341-3 EMERGENCY STOP INTERLOCK
LSD07	LSD07	LSD	—	PK-341-4 EMERGENCY STOP PUSH BUTTON
LSD08	LSD08	LSD	—	PK-341-4 EMERGENCY STOP INTERLOCK
PAHH-3412.1	PT-3412.1	LSD	27.5	C-341-1 HIGH HIGH DISCHARGE PRESSURE
PAHH-3412.2	PT-3412.2	LSD	27.5	C-341-2 HIGH HIGH DISCHARGE PRESSURE
PAHH-3412.3	PT-3412.3	LSD	27.5	C-341-3 HIGH HIGH DISCHARGE PRESSURE
PAHH-3412.4	PT-3412.4	LSD	27.5	C-341-4 HIGH HIGH DISCHARGE PRESSURE
PALL-3411.1	PT-3411.1	LSD	1.5	C-341-1 LOW LOW SUCTION PRESSURE
PALL-3411.2	PT-3411.2	LSD	1.5	C-341-2 LOW LOW SUCTION PRESSURE
PALL-3411.3	PT-3411.3	LSD	1.5	C-341-3 LOW LOW SUCTION PRESSURE
PALL-3411.4	PT-3411.4	LSD	1.5	C-341-4 LOW LOW SUCTION PRESSURE
PDALL-3414.1	PDT-3414.1	LSD	1.5	C-341-1 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3414.2	PDT-3414.2	LSD	1.5	C-341-2 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3414.3	PDT-3414.3	LSD	1.5	C-341-3 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3414.4	PDT-3414.4	LSD	1.5	C-341-4 LOW LOW OIL DIFFERENTIAL PRESSURE
PSD01	PSD01	PSD	—	LOSS OF AMMONIA LOOP
TAHH-3412.1	TT-3412.1	LSD	110	C-341-1 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3412.2	TT-3412.2	LSD	110	C-341-2 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3412.3	TT-3412.3	LSD	110	C-341-3 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3412.4	TT-3412.4	LSD	110	C-341-4 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3418.1	TT-3418.1	LSD	—	C-341-1 HIGH HIGH OIL HEADER TEMPERATURE
TAHH-3418.2	TT-3418.2	LSD	—	C-341-2 HIGH HIGH OIL HEADER TEMPERATURE
TAHH-3418.3	TT-3418.3	LSD	—	C-341-3 HIGH HIGH OIL HEADER TEMPERATURE
TAHH-3418.4	TT-3418.4	LSD	—	C-341-4 HIGH HIGH OIL HEADER TEMPERATURE
TAHH-3419.1	TT-3419.1	LSD	—	C-341-1 HIGH HIGH SUCTION TEMPERATURE
TAHH-3419.2	TT-3419.2	LSD	—	C-341-2 HIGH HIGH SUCTION TEMPERATURE
TAHH-3419.3	TT-3419.3	LSD	—	C-341-3 HIGH HIGH SUCTION TEMPERATURE
TAHH-3419.4	TT-3419.4	LSD	—	C-341-4 HIGH HIGH SUCTION TEMPERATURE

7.6.3 Unit Cause and Effect Matrix

Table 7.11 — Unit 340 – Cause & Effect Matrix

ALARM	INSTRUMENT	C-341-1	V-341-1	P-341-1	E-341-1	C-341-2	V-341-2	P-341-2	E-341-2	C-341-3	V-341-3	P-341-3	E-341-3	C-341-4	V-341-4	P-341-4	E-341-4
LSD01	LSD01	T	T	T	T												
LSD02	LSD02	T	T	T	T												
PAHH-3412.1	PT-3412.1	T	T	T	T												
PALL-3411.1	PT-3411.1	T	T	T	T												
TAHH-3412.1	TT-3412.1	T	T	T	T												
LSD03	LSD03					T	T	T	T								
LSD04	LSD04					T	T	T	T								
PAHH-3412.2	PT-3412.2					T	T	T	T								
PALL-3411.2	PT-3411.2					T	T	T	T								
TAHH-3412.2	TT-3412.2					T	T	T	T								
LSD05	LSD05									T	T	T	T				
LSD06	LSD06									T	T	T	T				
PAHH-3412.3	PT-3412.3									T	T	T	T				
PALL-3411.3	PT-3411.3									T	T	T	T				
TAHH-3412.3	TT-3412.3									T	T	T	T				
LSD07	LSD07													T	T	T	T
LSD08	LSD08													T	T	T	T
PAHH-3412.4	PT-3412.4													T	T	T	T
PALL-3411.4	PT-3411.4													T	T	T	T
TAHH-3412.4	TT-3412.4													T	T	T	T

7.7 Operation and Control

The ammonia refrigeration system operates with pressurized ammonia in a closed loop. Any loss of containment represents a significant hazard.

Ammonia is toxic and corrosive. Exposure to vapors may cause severe irritation or injury to eyes, skin, and respiratory system.

In case of leakage, immediately evacuate the area and follow the emergency response procedure.

7.7.1 Compressor Capacity Control

The capacity control of the ammonia screw compressors (C-341-1/2/3/4) is achieved by means of an internal slide valve mechanism, which provides continuous modulation of the compressor capacity.

The slide valve moves axially along the rotor casing and modifies the effective compression length. By shifting its position, part of the gas trapped between the rotors is internally returned to the suction side before compression is completed. This results in a reduction of the effective compression volume and, consequently, the compressor capacity.

The slide valve is hydraulically actuated using the compressor lubrication oil system. The movement of the valve is controlled by a set of solenoid valves, which direct oil flow to either side of the actuator piston, allowing precise positioning of the slide valve.

The capacity control system operates as follows:

- When capacity increase is required, the slide valve moves towards the discharge side, increasing the effective compression volume.
- When capacity reduction is required, the slide valve moves towards the suction side, allowing a portion of the gas to bypass compression.
- The compressor can typically operate within a capacity range of approximately 100% down to 15–30% of its nominal capacity.

The slide valve position is controlled by the Process Control System (PCS) based on process demand, typically using suction pressure as the main control variable.

Under normal standalone operation, each compressor operates with its dedicated suction pressure control loop using the local pressure transmitter PI-3411.x. In this mode, the controller

(PIC-3411.x) operates in AUTO and directly modulates the slide valve position, allowing independent control of each compressor.

For parallel operation of multiple compressors, the control strategy can be switched to CASCADE mode. In this configuration, the compressor capacity controller receives its setpoint from a master suction pressure controller based on the common suction header pressure (PT-3552 – Ammonia Suction Pressure). This ensures coordinated load sharing between compressors and stable control of the overall refrigeration system.

This dual control philosophy provides operational flexibility, allowing both independent compressor operation and integrated multi-compressor load management.

This method of capacity control offers the following advantages:

- High energy efficiency compared to external recycle or bypass systems.
- Smooth and continuous capacity modulation without step changes.

During start-up, the compressor is maintained in a fully unloaded condition (minimum capacity) to reduce starting torque and mechanical stress. After start-up, the slide valve is gradually loaded according to process requirements.

The control of the slide valve is interlocked with the lubrication system and compressor permissives to ensure safe operation. Loss of oil pressure, solenoid malfunction, or abnormal operating conditions will result in the slide valve moving to a safe position and/or triggering compressor shutdown as defined in the Cause & Effect matrix.

7.7.2 Ammonia Receiver Pressure Control

The pressure in the ammonia receiver (V-343) is controlled by the pressure control loop PIC-3401, which regulates the condensation capacity of the ammonia refrigeration system.

PIC-3401 maintains the receiver pressure within the defined operating range by automatically starting and stopping the air-cooled condenser (E-342) fan motors. As the receiver pressure increases due to higher refrigerant load or ambient conditions, additional condenser fans are started to increase heat rejection capacity. Conversely, when the pressure decreases, condenser fans are stopped in a controlled sequence.

The controller operates in on/off (step control) mode with predefined pressure thresholds and hysteresis. This control philosophy avoids frequent start/stop cycling of the condenser fans and ensures stable operation of the refrigeration loop.

Maintaining proper receiver pressure is critical for the overall system performance:

- It ensures adequate liquid refrigerant supply to downstream users (E-311, E-354).

- It directly influences compressor discharge pressure and, therefore, compressor power consumption.
- It prevents excessive condensing pressure, which could lead to high compressor load or trips.
- It avoids excessively low pressure conditions that may result in insufficient refrigerant flow or unstable evaporation control.

In case of abnormal conditions (e.g., high-high pressure), protective actions are implemented via the shutdown system as defined in the Cause & Effect matrix.

7.7.3 Compressor Panel Operation

The lubrication system is critical for the operation of oil-flooded screw compressors. Failure or inadequate performance of the lubrication system may result in severe compressor damage.

The refrigeration unit (UNIT 340) is equipped with local control panels associated with each ammonia compressor package (PK-341-1/2/3/4).

Each local control panel allows the operator to select the compressor operating mode (Local or Remote) by means of a selector switch:

- **Remote mode**: the compressor is started from the Process Control System (PCS).
- **Local mode**: the compressor can be started directly from the local panel.

When Local mode is selected, the compressor start-up sequence can be initiated by pressing the **START** pushbutton, while the normal shutdown sequence is initiated by pressing the **STOP** pushbutton.

A dedicated Emergency Shutdown (ESD/LSD) pushbutton is provided on the panel. Its activation initiates a local shutdown of the compressor package in accordance with the Cause & Effect Matrix.

The panel is also equipped with status indications to support safe operation, including:

- Compressor **Running** status
- **Ready to Start** indication
- **Alarm / Trip** indication

These indications provide the operator with real-time feedback on compressor availability and operating condition, ensuring proper interaction between local operation and PCS supervision.

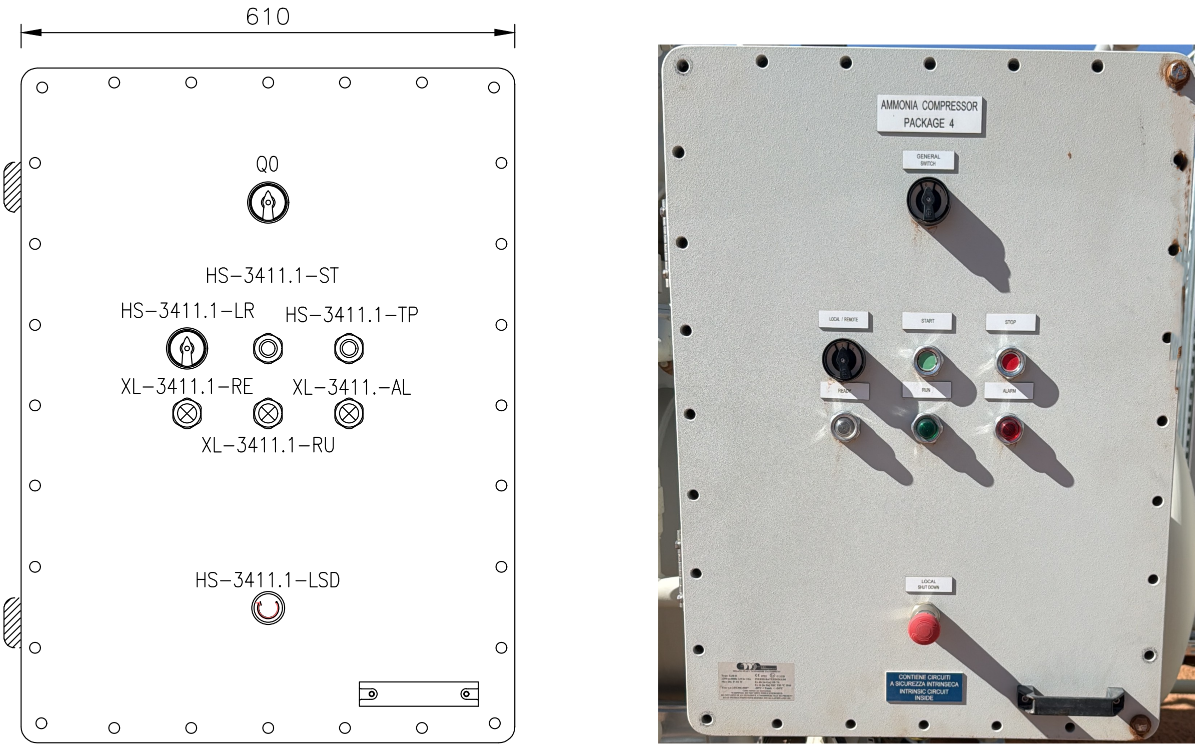


Figure 7.7: Ammonia compressor local control panel

Table 7.12 — Compressor package local panel components

TAG	SIGNAL	DESCRIPTION
HS-3411.x-LR	LOCAL REMOTE SELECTOR	Select local remote control
HS-3411.x-ST	LOCAL START	Local start command
HS-3411.x-TP	LOCAL STOP	Local stop command
HS-3411.x-LSD	LOCAL SHUTDOWN	Local emergency shutdown
XL-3411.x-RU	RUN	Compressor running
XL-3411.x-RE	READY	Compressor ready to run
XL-3411.x-AL	ALARM	Compressor alarm or trip active

7.7.4 Ammonia Refrigeration Loop R1

The ammonia refrigeration system (R1 loop – UNIT 340) operates as a closed-loop single-stage mechanical refrigeration cycle providing cooling duty to process users.

The refrigerant vapor returning from the cooling users is routed to the suction of the ammonia compressor packages (PK-341-1/2/3/4), where it is compressed to the required condensing pressure.

The compressed ammonia is then directed to the air-cooled condensers (E-342), where heat is rejected to ambient air and the refrigerant is condensed. The condensed liquid ammonia is collected in the liquid receiver (V-343), which ensures stable liquid inventory and distribution to the refrigeration users.

From the receiver, liquid ammonia is supplied to the following cooling consumers:

- E-311 – Natural Gas Precooler (UNIT 310)
- E-354 – Mixed Refrigerant Cooler (UNIT 350)

At each user, the refrigerant flow is controlled by a level control valve (e.g. LV-3111, LV-3540), where the liquid ammonia expands from condensing pressure to evaporating pressure, generating the refrigeration effect.

The two-phase ammonia mixture is accumulated in the shell side of each exchanger, where evaporation occurs. The refrigerant level is controlled to ensure proper heat transfer and to avoid liquid carryover to the compressor suction.

A back-pressure control valve (PCV-3101) is installed on the E-311 circuit to maintain a higher evaporating pressure compared to E-354, preventing excessively low temperatures and minimizing the risk of hydrate formation in the natural gas stream.

The vaporized ammonia from all users is collected in a common suction header and returned to the compressors, completing the refrigeration cycle.

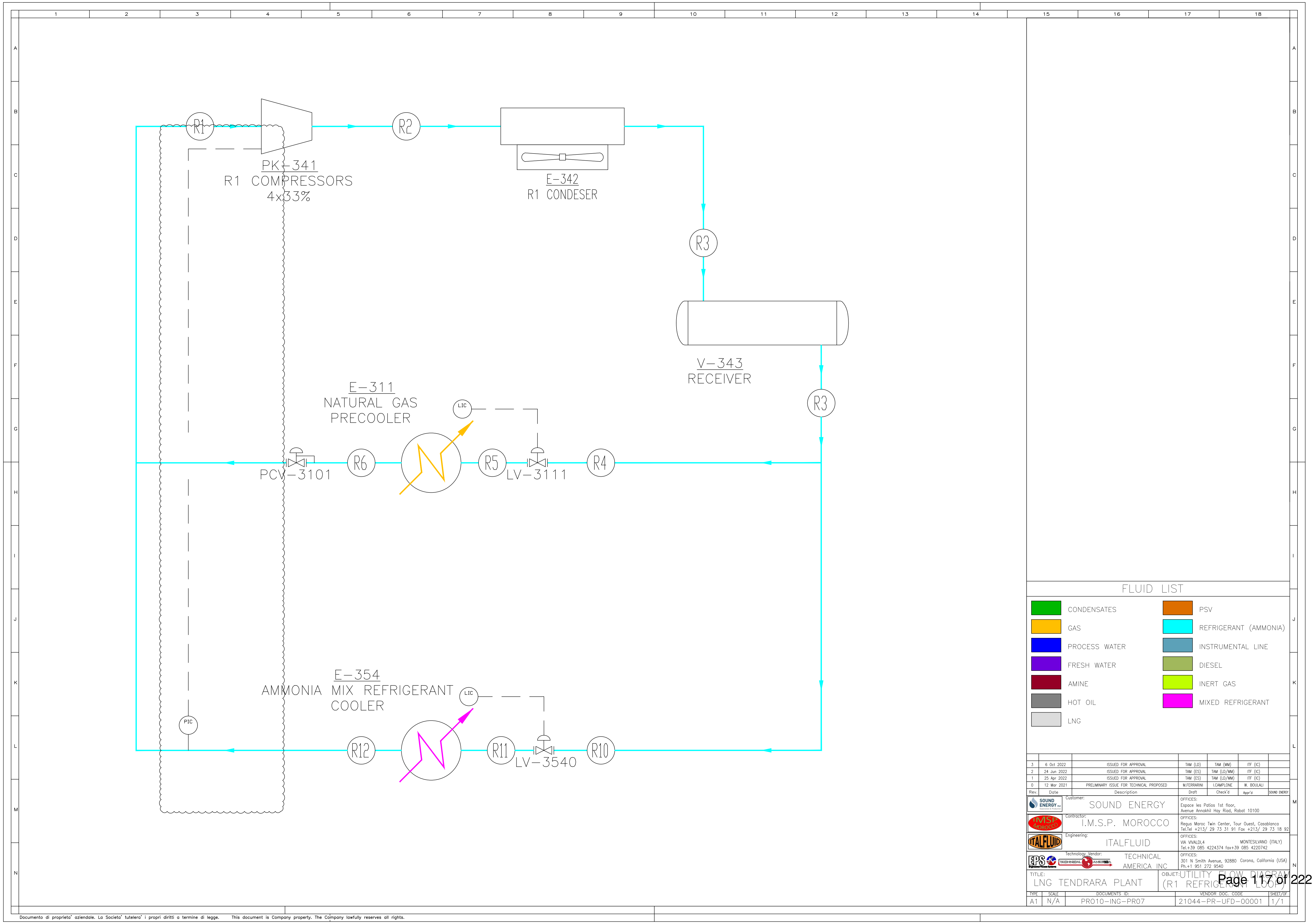


Figure 7.8: Unit 340 – Ammonia Refrigeration Loop (R1)

7.7.5 Natural Gas Precooler Level Control LIC-3111

The natural gas precooler (E-311) operates as a flooded shell-and-tube exchanger using ammonia as refrigerant on the shell side. The liquid level is controlled by LIC-3111 through the ammonia inlet control valve (LV-3111).

The control objective is to maintain sufficient liquid inventory to ensure proper heat transfer while preventing liquid carryover to the compressor suction.

Normal operating level is 35%, with a high level alarm at 41%.

Low level may reduce cooling efficiency, while high level increases the risk of liquid entrainment, which can lead to compressor damage.

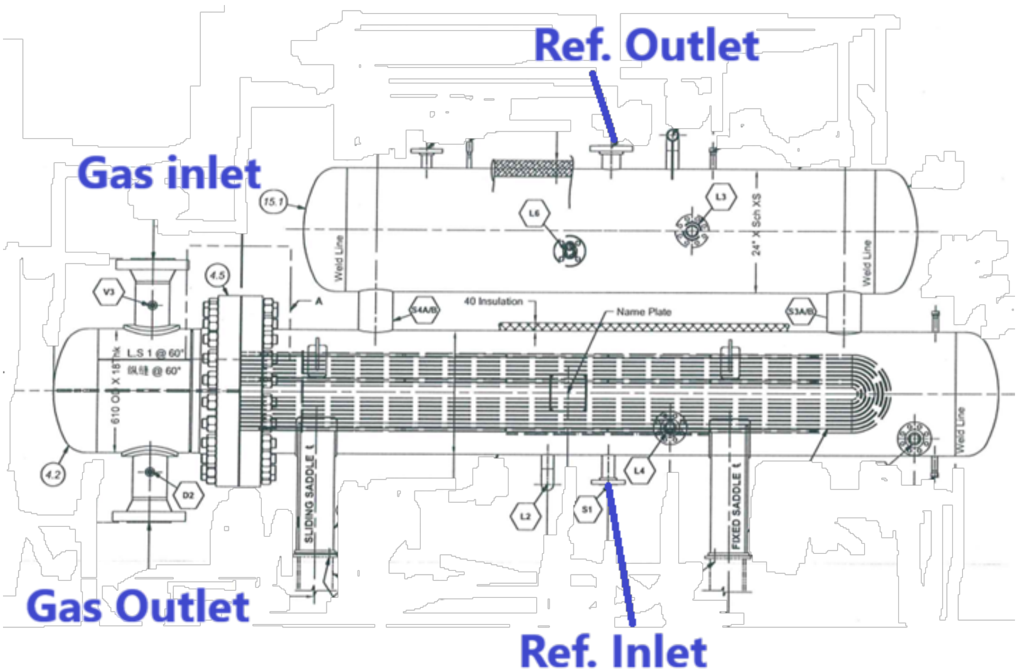


Figure 7.9: Natural Gas Precooler E-311

7.7.6 Ammonia MR Cooler Level Control LIC-3540

The ammonia MR cooler operates as a flooded shell-and-tube heat exchanger, where liquid ammonia evaporates on the shell side to provide cooling to the mixed refrigerant.

The refrigerant level is controlled by LIC-3540 through the ammonia inlet control valve (LV-3540), maintaining adequate liquid inventory for stable heat transfer.

Normal operating level is 90%.

Low level may reduce cooling performance, while excessive level increases the risk of liquid carryover toward the compressor suction.

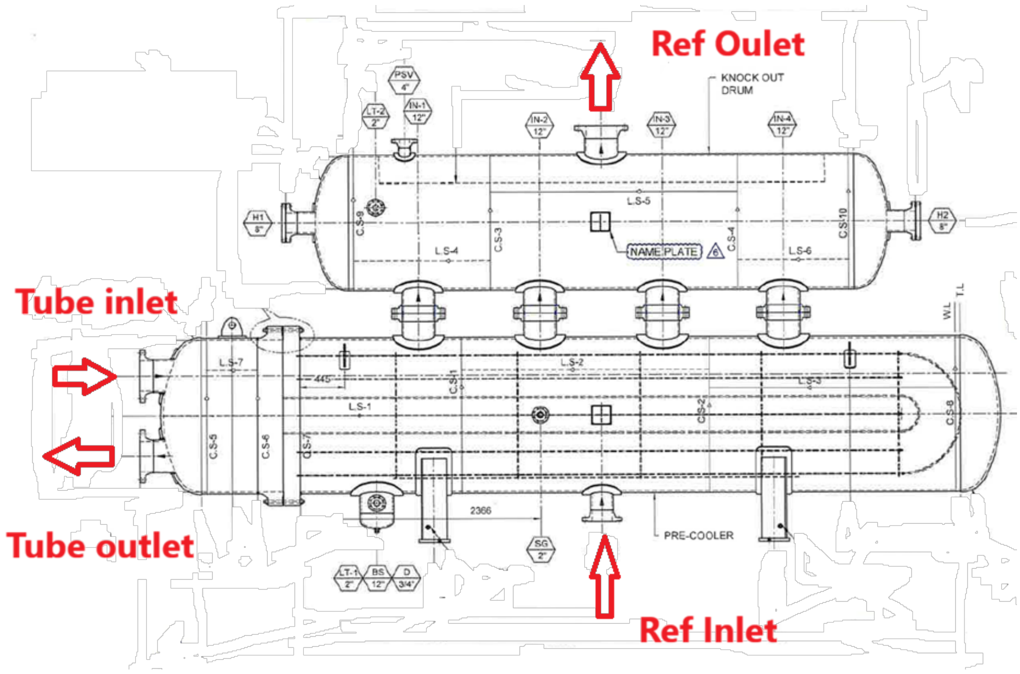


Figure 7.10: Ammonia MR Cooler

7.7.7 Lube Oil System

The lubrication oil system of the ammonia compressor packages (PK-341-1/2/3/4) is a critical subsystem designed to ensure reliable compressor operation by providing lubrication, cooling, and sealing functions.

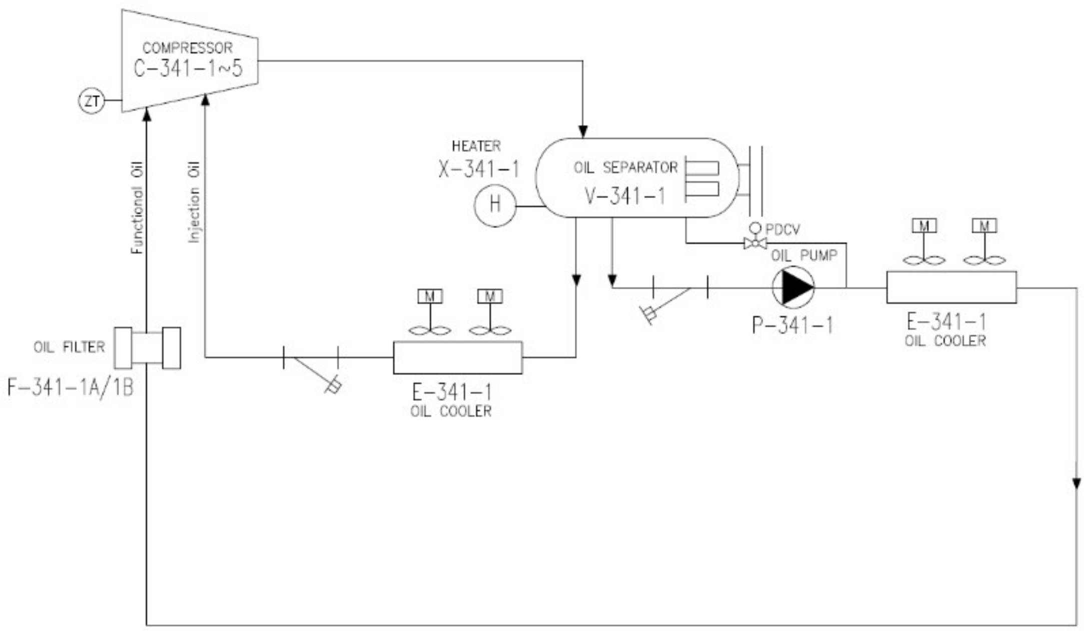


Figure 7.11: Ammonia compressor lube oil system

System Description

The lube oil system performs the following main functions:

- Removal of heat generated during the compression process.
- Lubrication of bearings and moving components.
- Sealing between compressor rotors to improve compression efficiency.

During operation, the lube oil is injected into the compressor together with the refrigerant gas. The oil-gas mixture is discharged from the compressor and routed to the oil separator (V-341-1), where the oil is separated and collected.

The oil separator also serves as the oil reservoir for the system, ensuring adequate oil inventory during operation.

From the separator, the oil is circulated back to the compressor through the lube oil circuit.

Oil Circulation and Conditioning

The oil circulation system includes the following main equipment:

- **Oil Pump (P-341-1):** Positive displacement gear type pump (1 x 100%), supplying pressurized oil to the compressor.

- **Oil Cooler (E-341-1):** Removes heat from the oil to maintain proper viscosity and cooling performance.
- **Oil Filters (F-341-1A/B):** Duplex configuration (1 operating + 1 standby) allowing continuous operation during filter replacement and ensuring clean oil supply.

Oil is pumped from the separator through the cooler and filters before being injected into the compressor for lubrication and capacity control functions.

Control and Protection

The lubrication system is monitored and protected by the control system through key parameters:

- Oil differential pressure across the compressor.
- Oil temperature downstream of the cooler.
- Oil level in the separator.

A minimum oil differential pressure is required to ensure proper lubrication. If the oil differential pressure drops below 1.5 bar, the compressor start sequence is inhibited or the running compressor is tripped.

This protection is critical to avoid damage to bearings, rotors, and internal components.

Operational Considerations

- The oil pump must be in operation prior to compressor start-up.
- Oil temperature shall be within the specified range before loading the compressor.
- Filters shall be monitored and replaced when differential pressure increases.
- Stable oil pressure and temperature are required for proper slide valve operation.

The oil pump is equipped with a magnetic coupling generating a strong magnetic field.

Personnel with implanted medical devices (e.g. pacemakers) shall not approach the equipment.

Maintain a safe distance from the magnetic coupling and avoid placing magnetic or sensitive electronic devices nearby.

7.7.8 Magnetic Coupling

The lube oil pump (P-341-1) is equipped with a magnetic coupling, which transmits torque from the motor to the pump without a direct mechanical shaft connection. This design eliminates the need for dynamic seals and ensures leak-free operation, which is particularly suitable for ammonia service.

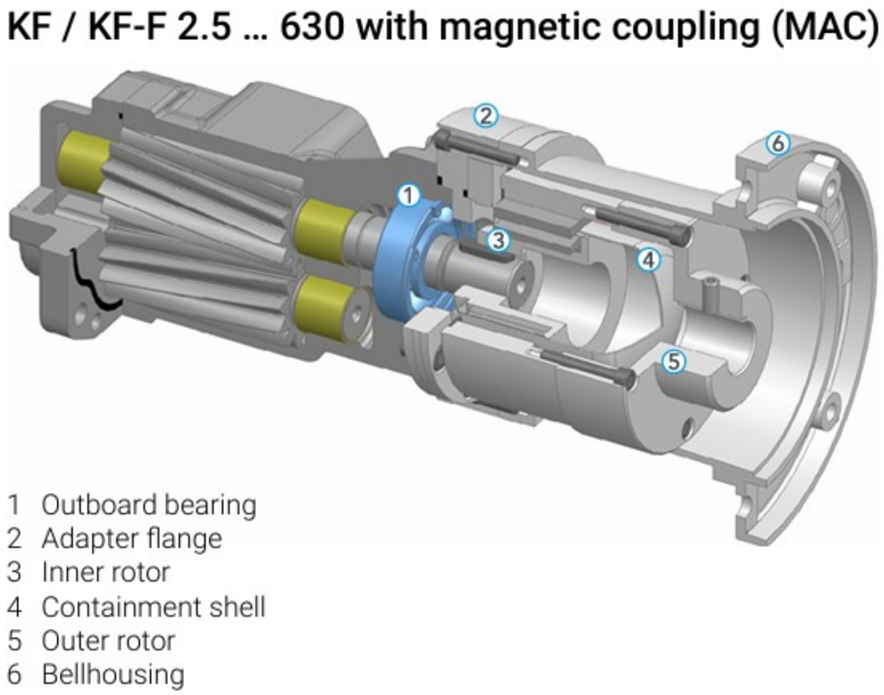


Figure 7.12: Oil pump magnetic coupling

The magnetic coupling consists of an outer rotor connected to the motor shaft and an inner rotor connected to the pump shaft. Torque is transmitted through magnetic forces across a containment shell, allowing complete separation between the driven and driving components.

This configuration provides several advantages, including improved reliability, reduced maintenance requirements, and elimination of potential leakage points. It is particularly suitable for hazardous fluids, as it minimizes the risk of external release.

Operationally, the magnetic coupling has a maximum torque transmission limit. In case of overload, magnetic decoupling may occur, resulting in the pump rotating without effective torque transmission. This condition may lead to loss of oil circulation and must be detected promptly. Dry running conditions shall be strictly avoided, as they can damage internal pump components.

7.7.9 Compressor Start-Up Permissives

Before initiating the automatic start-up sequence of any ammonia compressor package in Unit 340, the following start-up permissives shall be confirmed:

- All compressor motors to be available in **AUTO MODE** (typically three duty compressors available, with one package in standby).
- All lube oil pumps to be available in **AUTO MODE**.
- All oil cooler fans to be available in **AUTO MODE**.
- Compressor capacity control / slide valve to be in **AUTO MODE**.
- Compressor suction and discharge isolation valves to be confirmed **open**.
- Lube oil temperature to be within the normal start-up range.
- Compressor to be in **unloaded condition** before start.
- Associated solenoid valves to be in their normal **AUTO MODE**.
- No active shutdowns, trips, or critical alarms present on the selected package.
- Motor restart inhibit / cool-down timer to be expired.

These permissives are intended to ensure that the compressor starts under safe mechanical and process conditions, with the lubrication system, cooling system, and capacity control system available before the motor is energized. This is consistent with the Unit 340 refrigeration package configuration and with the plant shutdown and safety philosophy for package operation.

7.7.10 Compressor Start-Up Sequence

After completion of the refrigeration system preparation and verification of permissive conditions, the ammonia compressor package can be started following the automatic sequence controlled by the PCS.

Table 7.13 — Ammonia compressor start-up procedure

Step	Tag	SP		Action	Description
		Value	Unit		
1	HS-3411.x-ST	–	–	OC	Start automatic sequence by local panel or PCS

Step	Tag	SP		Action	Description
		Value	Unit		
2	P-341-x	–	–	SC	System command START auxiliary oil pump to establish lubrication flow.
3	PDI-3414.x	1.5	barg	SC	System command CHECK oil differential pressure $\geq$ 1.5 barg before start permissive.
4	ZT-3411.x	<5	%	SC	System command SET slide valve to fully unloaded position and confirm feedback.
5	MC-341-x	–	–	SC	System command START compressor main motor and accelerate to nominal speed.
6	–	30	s	CK	Maintain compressor at minimum load for stabilization period.
7	E-341-x	–	–	SC	System command START oil cooler fans to control oil temperature.
8	PIC-3433.x	–	–	SC	System command ENABLE suction pressure and capacity control loops.
9	–	–	–	CK	Compressor reaches steady-state operation under automatic control.

7.7.11 Compressor Shutdown Sequence

After confirmation of stable operating conditions and load reduction of the refrigeration system, the ammonia compressor package can be stopped following the automatic sequence controlled by the PCS.

Table 7.14 — Ammonia compressor shutdown procedure

Step	Tag	SP		Action	Description
		Value	Unit		
1	HS-3411.x-ST	–	–	OC	Start automatic shutdown sequence by local panel or PCS.
2	ZT-3411.x	<5	%	SC	System command SET slide valve to fully unloaded position and confirm feedback.
3	MC-341-x	–	–	SC	System command STOP compressor main motor and confirm de-energization.
4	P-341-x	–	–	CK	Auxiliary oil pump remains running to ensure post-lubrication during coast-down.
5	–	30	s	CK	Maintain post-lubrication period to protect bearings and seals.
6	PDI-3414.x	–	–	CK	Monitor oil differential pressure decay and confirm safe lubrication conditions.

Step	Tag	SP		Action	Description
		Value	Unit		
7	P-341-x	–	–	SC	System command STOP auxiliary oil pump after post-lubrication period.
8	MC-341-x	–	–	CK	Confirm compressor fully stopped and no active start command present.
9	PIC-3433.x	–	–	SC	System command DISABLE capacity control loops.

7.8 Troubleshooting

Table 7.15 — Unit 340 – Ammonia Compressor Troubleshooting

TROUBLE	POSSIBLE CAUSE	SOLUTION
Complete loss of power	Main power switch disconnected	Check all power wiring
	Main fuse failure	Replace the burned-out fuse
Partial loss of power	Auxiliary fuse failure	Replace the burned-out fuse
	Power disconnected to auxiliary equipment	Check wiring to auxiliary equipment
Oil pump will not run in HAND mode	Overload open	Reset overload
	Compressor is running	Oil pumps must be in AUTO while compressor is operating
	Main power off	Check starter disconnect and power supply
	Panel control power off	Turn panel control power on
	Motor burned out	Inspect and repair or replace motor
	Pump rotation reversed	Check motor rotation and correct if required
Oil pump will not run in AUTO mode	Compressor start has not been initiated	Initiate compressor start sequence
	Pump not developing sufficient pressure	Adjust pressure differential control valve to 30–60 psid
	Oil too cold	Warm oil to approximately 100°F

TROUBLE	POSSIBLE CAUSE	SOLUTION
	Pump rotation reversed	Check motor rotation and correct if required
Oil pump runs in AUTO mode but compressor will not start	Lube oil supply pressure too low	Adjust pressure differential control valve to 30–60 psid
	Compressor motor wired incorrectly	Check and correct motor wiring
	Slide valve not at less than 5% position	Confirm compressor is fully unloaded and hydraulic oil block valves are open
Oil pump runs in AUTO mode, compressor starts, but stops immediately	Motor starter or auxiliary contact wired incorrectly	Check and correct wiring; auxiliary contact must seal within 3 seconds
	Error in compressor start logic	Check and correct control logic
High or low lube oil temperature	High suction temperature	Check operating conditions and correct
	Thermal expansion valve not functioning properly	Check and correct valve operation
Low lube oil pressure	Oil pump not developing enough pressure	Adjust PDCV to 50–60 psid at regulator inlet; isolate PDCV and open bypass to verify pressure build-up
	Oil too cold	Run pump in HAND mode to warm oil to at least 90°F; close compressor suction valve while warming
	Oil pressure switch defective or incorrectly set	Check switch setting
	Control panel contact not closing properly or loose wiring	Check and correct
	Lube oil filter clogged	Check and replace filter if necessary
	Low oil level in oil reservoir	Add oil to vessel
Bad oil quality	Solids in oil	Drain oil and refill with clean oil; flush system if necessary; analyze oil and eliminate source of contamination
High oil carryover	Coalescer element loose	Tighten and mount coalescer properly; verify differential pressure across coalescer is not less than 0.25 psid

TROUBLE	POSSIBLE CAUSE	SOLUTION
High differential pressure across oil filter	Coalescer drain line choked	Check for oil accumulation in sight glass and open globe valve further to restore oil flow
	Oil filter plugged	Check and replace oil filter if necessary
Decrease in oil level in oil reservoir	Oil leakage from system	Check and correct leaks
	Coalescers not tight enough to seal	Tighten coalescers
	Oil return line from oil separator blocked	Adjust or clear block valve / return line
	Excess leakage from compressor seal	Check and correct
	Oil leaving system through a relief valve	Check all relief valve discharge piping for oil presence
Abnormal noise	Bearings damaged	Replace bearings
	Loose connections	Stop compressor and tighten connections
	Foreign material entered compressor	Stop compressor; check suction strainer and compressor internals
	Misalignment or damaged shaft coupling	Stop compressor; check alignment and coupling condition
	Rotors rubbing against casing inner surface	Stop and disassemble compressor; replace damaged parts and correct internal clearances

8 Unit 350 - Liquefaction Unit (MR compressors, cold box, NRU).

8.1 Operating Summary

CATEGORY	DESCRIPTION
Inputs	Dehydrated and CO_2-free natural gas from UNIT 310.
	Mixed Refrigerant (MR) two-stage compression loop (PK-351).
	Ammonia refrigeration (E-354) for MR pre-cooling.
Outputs	Liquefied Natural Gas (LNG) to V-221 and then to UNIT 220 storage.
	Nitrogen-rich overhead stream from NRU (T-357) to fuel gas system.
Key Operating Specs	Two-stage MR compression: 5W + 1S screw compressors (PK-351-1/6).
	Multi-stream cryogenic PFHEs (E-356A/B) for precooling and liquefaction.
	Heavy hydrocarbon separation in V-356A/B at $-75°$C.
	LNG flashing through LV-2211 into V-221 prior to storage.
	NRU distillation column (T-357) achieving LNG $N_2 \lesssim 1.1$ mol%.
	All MR streams conditioned through finite oil removal (S-352).
Major Equipment	PK-351 MR compression package: LP/HP screw compressors, V-351 oil separators,
	E-351 oil coolers, P-351 lubrication circuits, V-355A/B MR separators.
	E-353 air MR precoolers, ammonia refrigeration system providing cooling duty to MR pre-cooler E-354.
	CB-356 cold box with PFHE E-356A/B and separators V-356A/B.
	T-357 NRU cryogenic distillation column with electric reboiler X-357.
	MR storage tanks V-357-3/4; MR dryers V-359A/B; particulate filters F-359.

8.2 General Arrangement and Risk Location

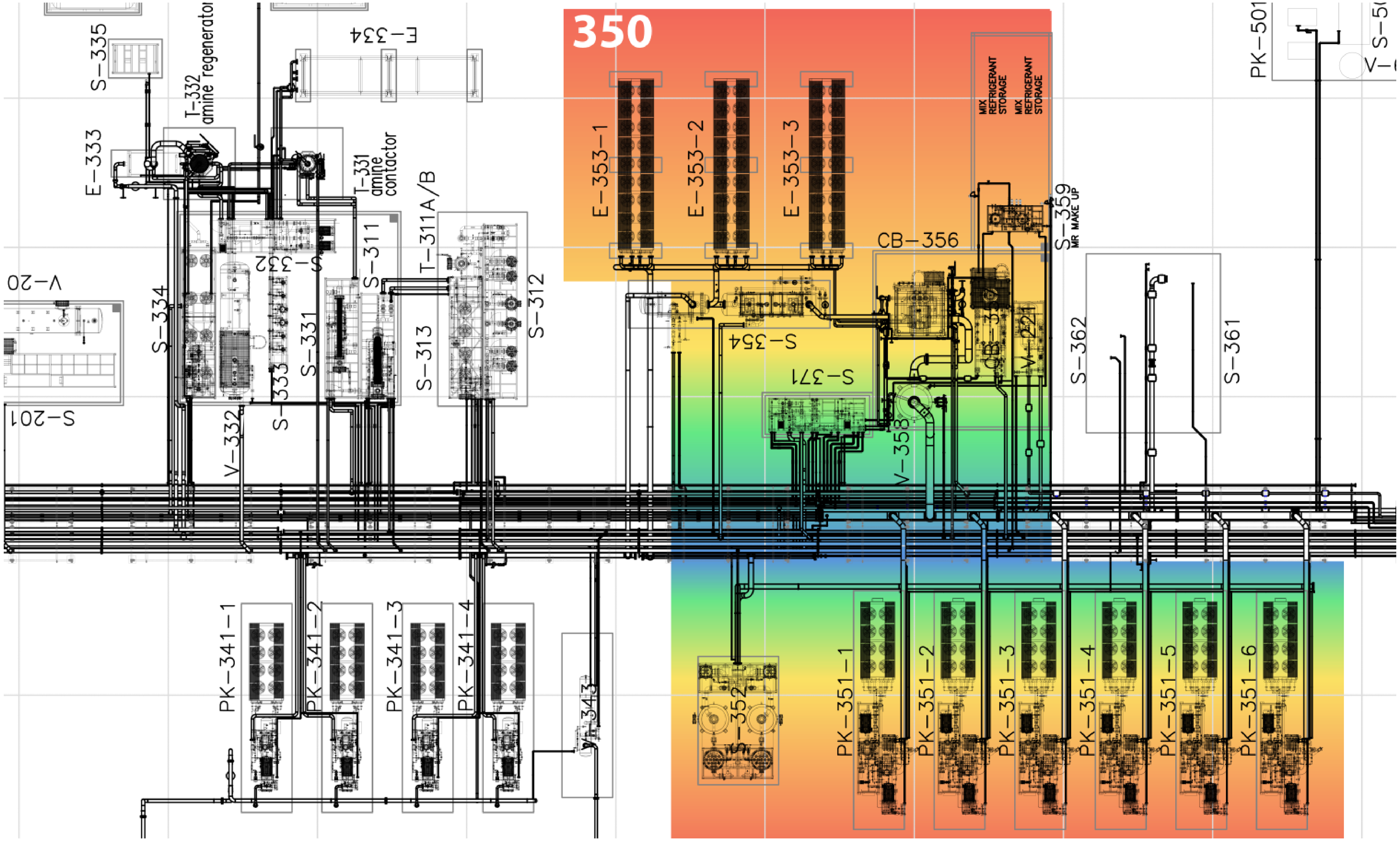


Figure 8.1: Unit 350 GA - Risk location

8.3 Relevant Documents

Table 8.2 — Unit 350 – Relevant Docuemnts

ID	DOC NO	TITLE
1	PRO02-ING-FR02	PROCESS DESCRIPTION
2	PRO10-ING-PR02	PROCESS FLOW DIAGRAM (GAS DEHYDRATION)
3	PRO10-ING-PR03	PROCESS FLOW DIAGRAM (GAS LIQUEFACTION)
4	PRO10-ING-PR06	UTILITY FLOW DIAGRAM (MIXED REFRIGERANT LOOP)
5	PRO10-ING-PR10	UNIT 310 SKID S-312 / S-313
6	PRO13-ING-PR11	COLD BOX CB-356
7	PRO13-ING-PR12	COLD BOX CB-357
8	21044-AU-HMI-00002	HMI PHILOSOPHY
9	PRO13-ING-PR13	PACKAGE PK-351-1
10	PRO13-ING-PR14	PACKAGE PK-351-2
11	PRO13-ING-PR15	PACKAGE PK-351-3
12	PRO13-ING-PR16	PACKAGE PK-351-4
13	PRO13-ING-PR17	PACKAGE PK-351-5
14	PRO13-ING-PR18	PACKAGE PK-351-6
15	PRO13-ING-PR19	S-357 MR MAKEUP
16	PRO13-ING-PR20	UNIT 350 – COMPRESSOR & KO DRUM
17	PRO13-ING-PR21	UNIT 350 – SKID S-352 – ORS
18	PRO13-ING-PR22	S-354 – UNIT 350
19	SAF12-ING-FR01	CAUSE AND EFFECT MATRIX
20	SAF05-ING-FR01	RELIEF AND BLOWDOWN PHILOSOPHY
21	SAF01-ING-FR01	SAFETY AND ISOLATION PHILOSOPHY

8.4 Process Description

The Liquefaction Unit (Unit 350) constitutes the core cryogenic section of the LNG plant, where pretreated natural gas is cooled and liquefied using a closed-loop Mixed Refrigerant (MR) system and cryogenic plate-fin heat exchangers installed within Cold Box CB-356.

The unit integrates feed gas precooling, staged cryogenic cooling, mixed refrigerant compression and expansion, oil removal and suction conditioning systems, and supervisory control logic to balance process cooling demand with available refrigeration capacity. Unit 350 is designed for continuous operation with high availability, incorporating multiple parallel MR compressor trains, mechanical redundancy, and flexible operator-selectable control modes.

Feed Gas Reception and Precooling

Pretreated natural gas from the upstream dehydration unit enters Unit 350 through shutdown valve SDV-3560 and is routed to the cryogenic cold box CB-356.

The feed gas is first precooled in the multistream plate-fin heat exchanger E-356A. Precooling is achieved through indirect heat exchange against colder process streams and expanded natural gas generated by controlled pressure reduction across valve PV-3561A.

Valve PV-3561A is installed on the natural gas process line and operates under pressure control to reduce gas pressure and temperature. The expanded gas stream provides refrigeration duty to E-356A, improving the thermal efficiency of the liquefaction process while maintaining stable upstream and downstream operating conditions.

Following precooling, the partially cooled gas is routed to the vertical separator V-356A, where any condensed heavy hydrocarbons are removed. Although the design basis feed gas is not expected to contain significant heavy hydrocarbon content, this separator is provided as a safeguard to accommodate potential variations in gas composition and to prevent freezing or fouling of cryogenic heat exchangers.

Separated condensates are routed to a dedicated vaporization system for controlled heating and recovery to the fuel gas system.

Cryogenic Cooling and Gas Liquefaction

Downstream of the first separation stage, the natural gas is returned to the cold box and further cooled in the cryogenic plate-fin heat exchanger E-356B, reaching temperatures of approximately $-75\,^{\circ}$C.

A second vertical separator, V-356B, is installed downstream of this cooling stage to remove any residual condensates prior to deep cryogenic cooling, providing additional protection for the cold box heat exchangers.

Deep liquefaction of the natural gas is achieved through staged heat exchange within the cold box using the mixed refrigerant loop as the refrigeration medium. The mixed refrigerant undergoes compression, cooling, phase separation, and controlled expansion, enabling efficient heat absorption across the full temperature profile required for LNG production.

The resulting liquefied natural gas exits the cold box at cryogenic conditions and is routed downstream to the LNG handling system.

Compressor Control and Coordination

Refrigeration for Unit 350 is provided by the PK-351 mixed refrigerant compressor package, consisting of six identical oil-flooded screw compressor trains arranged in parallel. Five trains operate under normal conditions, while one train is maintained in standby to ensure high availability.

Each compressor train includes low-pressure and high-pressure screw compressors, dedicated oil separation, oil cooling, lubrication systems, and redundant oil filtration. A common mechanical oil removal system, S-352, located downstream of the compressor package, provides cryogenic-grade oil separation to protect the plate-fin heat exchangers.

The refrigeration capacity of Unit 350 is coordinated with the LNG production rate through a cascade control philosophy. The operator manually starts the required number of compressor trains according to the desired plant throughput.

Two operator-selectable master control modes are available via the PCS HMI.

Flow-Based Control Mode

In this mode, the master flow controller FIC-3560A, receiving the signal from FT-3560A, determines the required refrigeration capacity based on the natural gas feed rate. The output of FIC-3560A generates a suction pressure setpoint for the compressor system.

Of the running compressor trains, $N - 1$ trains operate at full capacity, while one train, designated as the swing train, modulates its slide valve to maintain the suction pressure setpoint.

Superheat-Based Control Mode

In this mode, a calculated superheat parameter is used as the master control variable. The superheat controller generates the suction pressure setpoint, while the same $N - 1$ plus one swing train philosophy is maintained.

Slide valve modulation of the swing train is achieved through individual compressor suction pressure control loops. Selection of the active control mode is performed by the operator through the PCS HMI.

In the event of a compressor train trip, the available refrigeration capacity is reduced. To re-establish thermal balance, the operator may start an additional compressor train or reduce the natural gas feed rate to the liquefaction unit.

Cold Box and Nitrogen Rejection

Cold Box CB-356 integrates the cryogenic heat exchangers and separators required for final cooling, liquefaction, and mixed refrigerant circulation. The cold box provides the physical and thermal interface between the natural gas process streams and the mixed refrigerant refrigeration loop, enabling staged heat exchange and phase separation under controlled cryogenic conditions.

Overall refrigeration balance within the cold box is governed by valve PV-3561B, installed on the mixed refrigerant process line. Valve PV-3561B is controlled by PIC-3561B, whose setpoint is generated by the supervisory multivariable controller XC-3561B.

Controller XC-3561B receives multiple process inputs, including the high-pressure mixed refrigerant separator pressure at V-355A (PT-3550), the mixed refrigerant suction pressure at the compressor inlet V-358 (PT-3501), and the differential pressure across PV-3561B (PDI-3569).

Based on these inputs, controller XC-3561B establishes balanced operating conditions for the refrigeration circuit, ensuring stable cold box operation and proper matching between process cooling demand and available refrigeration capacity. The controller can operate under different operator-selectable modes, including differential pressure control and discharge pressure control. Detailed internal control algorithms are implemented within the PCS and are outside the scope of this process description.

Downstream of the cold box, the liquefied natural gas stream is routed to the Nitrogen Rejection Unit (NRU), consisting of the cryogenic distillation column T-357. The NRU is designed to remove excess nitrogen from the LNG stream in order to meet specified product quality requirements.

The nitrogen rejection system is equipped with an electric reboiler X-357, which provides the thermal duty required for nitrogen separation. Nitrogen-rich vapor is withdrawn from the column overhead under pressure control, while nitrogen-lean LNG is collected at the column bottom under level control.

The feed to the NRU and the associated reboiler system are supervised by controller XC-3551, which acts on control valve CV-3551. Controller XC-3551 receives process inputs including the cold box outlet gas temperature (TT-3551) and the gas flow rate (FT-3551).

Based on these inputs, XC-3551 provides a supervisory limiting function to ensure that gas is admitted to the NRU only when cold box outlet conditions are within the required operating range. If the temperature is not sufficiently low, indicating inadequate cold box performance, control valve CV-3551 is driven toward a closed position to prevent the passage of gas outside optimal process conditions and to protect downstream systems.

8.5 Technological Objects

8.5.1 Main Equipment List

Table 8.3 — Unit 350 – Main Equipment List

ID	TAG	EQUIPMENT
1	**PK-351-1**	**MR COMPRESSOR PACKAGE 1**
1.1	C-351L-1	MR compressor LP 1
1.2	C-351H-1	MR compressor HP 1
1.3	V-351-1	Oil separator
1.4	E-351-1	Oil cooler
1.4.1	F-351-1-1..8	Oil cooler fans (8)
1.5	P-351-1	Oil pump
1.6	F-351-1A/B	Main oil filters A/B
1.7	X-351-1	Oil heater
2	**PK-351-2**	**MR COMPRESSOR PACKAGE 2**
2.1	C-351L-2	MR compressor LP 2
2.2	C-351H-2	MR compressor HP 2
2.3	V-351-2	Oil separator
2.4	E-351-2	Oil cooler
2.4.1	F-351-2-1..8	Oil cooler fans (8)
2.5	P-351-2	Oil pump
2.6	F-351-2A/B	Main oil filters A/B
2.7	X-351-2	Oil heater
3	**PK-351-3**	**MR COMPRESSOR PACKAGE 3**
3.1	C-351L-3	MR compressor LP 3
3.2	C-351H-3	MR compressor HP 3
3.3	V-351-3	Oil separator
3.4	E-351-3	Oil cooler
3.4.1	F-351-3-1..8	Oil cooler fans (8)
3.5	P-351-3	Oil pump
3.6	F-351-3A/B	Main oil filters A/B
3.7	X-351-3	Oil heater
4	**PK-351-4**	**MR COMPRESSOR PACKAGE 4**
4.1	C-351L-4	MR compressor LP 4
4.2	C-351H-4	MR compressor HP 4
4.3	V-351-4	Oil separator
4.4	E-351-4	Oil cooler
4.4.1	F-351-4-1..8	Oil cooler fans (8)
4.5	P-351-4	Oil pump
4.6	F-351-4A/B	Main oil filters A/B

ID	TAG	EQUIPMENT
4.7	X-351-4	Oil heater
5	**PK-351-5**	**MR COMPRESSOR PACKAGE 5**
5.1	C-351L-5	MR compressor LP 5
5.2	C-351H-5	MR compressor HP 5
5.3	V-351-5	Oil separator
5.4	E-351-5	Oil cooler
5.4.1	F-351-5-1..8	Oil cooler fans (8)
5.5	P-351-5	Oil pump
5.6	F-351-5A/B	Main oil filters A/B
5.7	X-351-5	Oil heater
6	**PK-351-6**	**MR COMPRESSOR PACKAGE 6**
6.1	C-351L-6	MR compressor LP 6
6.2	C-351H-6	MR compressor HP 6
6.3	V-351-6	Oil separator
6.4	E-351-6	Oil cooler
6.4.1	F-351-6-1..8	Oil cooler fans (8)
6.5	P-351-6	Oil pump
6.6	F-351-6A/B	Main oil filters A/B
6.7	X-351-6	Oil heater
7	V-352-1/2	COALESCING SEPARATORS
8	V-352-3/4	CHARCOAL ADSORBERS
9	F-352A/B	ADSORBER AFTER FILTERS
10	**E-353-1**	**AIR COOLED MR PRECOOLER 1**
10.1	F-353-1-1..18	Precooler 1 fans (18)
11	**E-353-2**	**AIR COOLED MR PRECOOLER 2**
11.1	F-353-2-1..18	Precooler 2 fans (18)
12	**E-353-3**	**AIR COOLED MR PRECOOLER 3**
12.1	F-353-3-1..18	Precooler 3 fans (18)
13	E-354	AMMONIA MR COOLER
14	V-355A	HP MIX REFRIGERANT SEPARATOR
15	V-355B	LP MIX REFRIGERANT SEPARATOR
16	P-355A/B	MR PUMPS
17	E-356A	GAS-GAS PFHE
18	E-356B	LIQUIFIER PFHE
19	V-356A	NGL SEPARATOR
20	V-356B	NGL SEPARATOR
21	T-357	N2 STRIPPER
22	X-357	N2 REJECTION REBOILER
23	V-357-3	C2 STORAGE VESSEL
24	V-357-4	nC4 STORAGE VESSEL
25	V-358	MR SUCTION KO DRUM
26	V-359A/B	MR MAKEUP DRYERS

ID	TAG	EQUIPMENT
27	F-359	PARTICULATE AFTER FILTER
28	X-359	MR MAKEUP REGENERATION HEATER

8.5.2 Electrical Loads

Table 8.4 — Unit 350 – Electrical Loads

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	C-351L-1	MR COMPRESSOR	VFD	690	560,00
2	C-351H-1	MR COMPRESSOR	VFD	690	560,00
3	X-351-1	OIL HEATER (OIL SEPARATOR)	DOL	690	5,43
4	P-351-1	OIL PUMP	DOL	690	4,00
5	C-351L-2	MR COMPRESSOR	VFD	690	560,00
6	C-351H-2	MR COMPRESSOR	VFD	690	560,00
7	X-351-2	OIL HEATER (OIL SEPARATOR)	DOL	690	5,43
8	P-351-2	OIL PUMP	DOL	690	4,00
9	C-351L-3	MR COMPRESSOR	VFD	690	560,00
10	C-351H-3	MR COMPRESSOR	VFD	690	560,00
11	X-351-3	OIL HEATER (OIL SEPARATOR)	DOL	690	5,43
12	P-351-3	OIL PUMP	DOL	690	4,00
13	C-351L-4	MR COMPRESSOR	VFD	690	560,00
14	C-351H-4	MR COMPRESSOR	VFD	690	560,00
15	X-351-4	OIL HEATER (OIL SEPARATOR)	DOL	690	5,43
16	P-351-4	OIL PUMP	DOL	690	4,00
17	C-351L-5	MR COMPRESSOR	VFD	690	560,00
18	C-351H-5	MR COMPRESSOR	VFD	690	560,00
19	X-351-5	OIL HEATER (OIL SEPARATOR)	DOL	690	5,43
20	P-351-5	OIL PUMP	DOL	690	4,00
21	C-351L-6	MR COMPRESSOR	VFD	690	560,00
22	C-351H-6	MR COMPRESSOR	VFD	690	560,00
23	X-351-6	OIL HEATER (OIL SEPARATOR)	DOL	690	5,43
24	P-351-6	OIL PUMP	DOL	690	4,00
25	P-355A	MR PUMP	DOL	690	15,00
26	P-355B	MR PUMP	DOL	690	15,00
27	X-357	N2 REJECTION REBOILER	SCR	690	90,00
28	X-359	MR MAKE UP REGENERATION HEATER	DOL	690	15,00

8.5.3 Motors as Remote MCC

Table 8.5 — Unit 350 – Motors as Remote MCC

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	MF-351-1-1	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
2	MF-351-1-2	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
3	MF-351-1-3	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
4	MF-351-1-4	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
5	MF-351-1-5	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
6	MF-351-1-6	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
7	MF-351-1-7	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
8	MF-351-1-8	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
9	MF-351-1-9	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
10	MF-351-1-10	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
11	MF-351-2-1	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
12	MF-351-2-2	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
13	MF-351-2-3	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
14	MF-351-2-4	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
15	MF-351-2-5	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
16	MF-351-2-6	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
17	MF-351-2-7	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
18	MF-351-2-8	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
19	MF-351-2-9	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
20	MF-351-2-10	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
21	MF-351-3-1	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
22	MF-351-3-2	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
23	MF-351-3-3	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
24	MF-351-3-4	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
25	MF-351-3-5	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
26	MF-351-3-6	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
27	MF-351-3-7	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
28	MF-351-3-8	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
29	MF-351-3-9	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
30	MF-351-3-10	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
31	MF-351-4-1	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
32	MF-351-4-2	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
33	MF-351-4-3	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
34	MF-351-4-4	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
35	MF-351-4-5	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
36	MF-351-4-6	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
37	MF-351-4-7	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
38	MF-351-4-8	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
39	MF-351-4-9	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
40	MF-351-4-10	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
41	MF-351-5-1	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
42	MF-351-5-2	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
43	MF-351-5-3	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
44	MF-351-5-4	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
45	MF-351-5-5	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
46	MF-351-5-6	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
47	MF-351-5-7	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
48	MF-351-5-8	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
49	MF-351-5-9	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
50	MF-351-5-10	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
51	MF-351-6-1	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
52	MF-351-6-2	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
53	MF-351-6-3	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
54	MF-351-6-4	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
55	MF-351-6-5	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
56	MF-351-6-6	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
57	MF-351-6-7	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
58	MF-351-6-8	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
59	MF-351-6-9	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
60	MF-351-6-10	OIL COOLER FAN ELECTRIC MOTOR	DOL	690	4,00
61	MF-353-1-1	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
62	MF-353-1-2	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
63	MF-353-1-3	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
64	MF-353-1-4	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
65	MF-353-1-5	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
66	MF-353-1-6	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
67	MF-353-1-7	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
68	MF-353-1-8	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
69	MF-353-1-9	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
70	MF-353-1-10	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
71	MF-353-2-1	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
72	MF-353-2-2	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
73	MF-353-2-3	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
74	MF-353-2-4	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
75	MF-353-2-5	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
76	MF-353-2-6	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
77	MF-353-2-7	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
78	MF-353-2-8	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
79	MF-353-2-9	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
80	MF-353-2-10	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
81	MF-353-3-1	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
82	MF-353-3-2	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
83	MF-353-3-3	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
84	MF-353-3-4	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
85	MF-353-3-5	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
86	MF-353-3-6	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
87	MF-353-3-7	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
88	MF-353-3-8	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
89	MF-353-3-9	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00
90	MF-353-3-10	AIR COOLED MR PRECOOLER FAN MOTOR	DOL	690	4.00

8.5.4 Valves

Table 8.6 — Unit 350 – Valves

ID	TAG	SERVICE	SYSTEM	NOTES
1	CEV-3551	LNG TO NRU FLOW CONTROL	ESD	-
2	CV-3551	LNG TO NRU FLOW CONTROL	-	-
3	LEV-2211	X-357 LIQUID LEVEL CONTROL	ESD	-
4	LEV-3540	E-354 LIQUID LEVEL CONTROL	ESD	-
5	LV-2211	X-357 LIQUID LEVEL CONTROL	-	-
6	LV-3540	E-354 LIQUID LEVEL CONTROL	-	-
7	LV-3550	V-355A LIQUID LEVEL CONTROL	-	-
8	LV-3551	V-355A LIQUID LEVEL CONTROL	-	-
9	PCV-3561	N2 TO CB-356 PRESSURE CONTROL	-	
10	PEV-3511.1	C-351H-1 OIL HEADER PRESSURE SHUT OFF	PCS	WITH 5 MT CABLE
11	PEV-3511.2	C-351H-2 OIL HEADER PRESSURE SHUT OFF	PCS	WITH 5 MT CABLE
12	PEV-3511.3	C-351H-3 OIL HEADER PRESSURE SHUT OFF	PCS	WITH 5 MT CABLE
13	PEV-3511.4	C-351H-4 OIL HEADER PRESSURE SHUT OFF	PCS	WITH 5 MT CABLE
14	PEV-3511.5	C-351H-5 OIL HEADER PRESSURE SHUT OFF	PCS	WITH 5 MT CABLE
15	PEV-3511.6	C-351H-6 OIL HEADER PRESSURE SHUT OFF	PCS	WITH 5 MT CABLE
16	PEV-3540	E-354 PRESSURE CONTROL (HOT BY PASS)	ESD	WITH 5 MT CABLE
17	PEV-3561A	V-356A INLET PRESSURE CONTROL	ESD	-
18	PEV-3561B	V-355B INLET PRESSURE CONTROL	ESD	-

ID	TAG	SERVICE	SYSTEM	NOTES
19	PSV-3503A	E-354 PRESSURE RELIEF	-	-
20	PSV-3503B	E-354 PRESSURE RELIEF	-	-
21	PSV-3511.1H	C-351H-1 BLOCKED OUTLET RELIEF	-	SET 28 barg
22	PSV-3511.2H	C-351H-2 BLOCKED OUTLET RELIEF	-	SET 28 barg
23	PSV-3511.3H	C-351H-3 BLOCKED OUTLET RELIEF	-	SET 28 barg
24	PSV-3511.4H	C-351H-4 BLOCKED OUTLET RELIEF	-	SET 28 barg
25	PSV-3511.5H	C-351H-5 BLOCKED OUTLET RELIEF	-	SET 28 barg
26	PSV-3511.6H	C-351H-6 BLOCKED OUTLET RELIEF	-	SET 28 barg
27	PSV-3517.1A	V-351-1 PRESSURE RELIEF	-	SET 28 barg
28	PSV-3517.1B	V-351-1 PRESSURE RELIEF	-	SET 28 barg
29	PSV-3517.2A	V-351-2 PRESSURE RELIEF	-	SET 28 barg
30	PSV-3517.2B	V-351-2 PRESSURE RELIEF	-	SET 28 barg
31	PSV-3517.3A	V-351-3 PRESSURE RELIEF	-	SET 28 barg
32	PSV-3517.3B	V-351-3 PRESSURE RELIEF	-	SET 28 barg
33	PSV-3517.4A	V-351-4 PRESSURE RELIEF	-	SET 28 barg
34	PSV-3517.4B	V-351-4 PRESSURE RELIEF	-	SET 28 barg
35	PSV-3517.5A	V-351-5 PRESSURE RELIEF	-	SET 28 barg
36	PSV-3517.5B	V-351-5 PRESSURE RELIEF	-	SET 28 barg
37	PSV-3517.6A	V-351-6 PRESSURE RELIEF	-	SET 28 barg
38	PSV-3517.6B	V-351-6 PRESSURE RELIEF	-	SET 28 barg
39	PSV-3518.1	P-351-1 PRESSURE RELIEF	-	SET 6 BAR DIFF.
40	PSV-3518.2	P-351-2 PRESSURE RELIEF	-	SET 6 BAR DIFF.
41	PSV-3518.3	P-351-3 PRESSURE RELIEF	-	SET 6 BAR DIFF.
42	PSV-3518.4	P-351-4 PRESSURE RELIEF	-	SET 6 BAR DIFF.
43	PSV-3518.5	P-351-5 PRESSURE RELIEF	-	SET 6 BAR DIFF.
44	PSV-3518.6	P-351-6 PRESSURE RELIEF	-	SET 6 BAR DIFF.
45	PSV-3521A	V-352-1 AND V-352-3 PRESSURE SAFETY VALVE	-	SET 28 barg
46	PSV-3521B	V-352-1 AND V-352-3 PRESSURE SAFETY VALVE	-	SET 28 barg
47	PSV-3522A	V-352-2 AND V-352-4 PRESSURE SAFETY VALVE	-	SET 28 barg
48	PSV-3522B	V-352-2 AND V-352-4 PRESSURE SAFETY VALVE	-	SET 28 barg
49	PSV-3523A	F-352A PRESSURE SAFETY VALVE	-	SET 28 barg
50	PSV-3523B	F-352B PRESSURE SAFETY VALVE	-	SET 28 barg
51	PSV-3550A	V-355A PRESSURE RELIEF	-	SET 28 barg
52	PSV-3550B	V-355A PRESSURE RELIEF	-	SET 28 barg
53	PSV-3561A	V-355B PRESSURE RELIEF	-	SET 28 barg
54	PSV-3561B	V-355B PRESSURE RELIEF	-	SET 28 barg
55	PSV-3562A	V-356B PRESSURE RELIEF	-	SET 50 barg
56	PSV-3562B	V-356B PRESSURE RELIEF	-	SET 50 barg
57	PSV-3571A	X-357 PRESSURE RELIEF	-	SET 10 barg
58	PSV-3571B	X-357 PRESSURE RELIEF	-	SET 10 barg

ID	TAG	SERVICE	SYSTEM	NOTES
59	PSV-3580A	V-358 PRESSURE RELIEF	-	SET 28 barg
60	PSV-3580B	V-358 PRESSURE RELIEF	-	SET 28 barg
61	PSV-3591A	V-359A PRESSURE RELIEF	-	SET 15 barg
62	PSV-3591B	V-359B PRESSURE RELIEF	-	SET 15 barg
63	PSV-3592	F-359 PRESSURE SAFETY VALVE	-	SET 15 barg
64	PV-3551	T-357 PRESSURE CONTROL	-	-
65	PV-3561A	V-356A INLET PRESSURE CONTROL	-	-
66	PV-3561B	V-355B INLET PRESSURE CONTROL	-	-
67	PV-3581	V-358 PRESSURE CONTROL (HOT BY PASS)	-	-
68	TSV-3572	LNG TO V-221 THERMAL EXPANSION PROTECTION	-	-
69	TSV-3573	LNG TO V-221 THERMAL EXPANSION PROTECTION	-	-
70	XEV-3511.1	C-351L-1 SUCTION SHUT DOWN	PCS	-
71	XEV-3511.2	C-351L-2 SUCTION SHUT DOWN	PCS	-
72	XEV-3511.3	C-351L-3 SUCTION SHUT DOWN	PCS	-
73	XEV-3511.4	C-351L-4 SUCTION SHUT DOWN	PCS	-
74	XEV-3511.5	C-351L-5 SUCTION SHUT DOWN	PCS	-
75	XEV-3511.6	C-351L-6 SUCTION SHUT DOWN	PCS	-
76	XEV-3512.1	C-351L-1 SUCTION DEPRESSURIZATION BYPASS	PCS	WITH 5 MT CABLE
77	XEV-3512.2	C-351L-2 SUCTION DEPRESSURIZATION BYPASS	PCS	WITH 5 MT CABLE
78	XEV-3512.3	C-351L-3 SUCTION DEPRESSURIZATION BYPASS	PCS	WITH 5 MT CABLE
79	XEV-3512.4	C-351L-4 SUCTION DEPRESSURIZATION BYPASS	PCS	WITH 5 MT CABLE
80	XEV-3512.5	C-351L-5 SUCTION DEPRESSURIZATION BYPASS	PCS	WITH 5 MT CABLE
81	XEV-3512.6	C-351L-6 SUCTION DEPRESSURIZATION BYPASS	PCS	WITH 5 MT CABLE
82	XEV-3513.1	MR FROM OIL SEPARATOR TO EJ1-351.1 SHUTDOWN	PCS	WITH 5 MT CABLE
83	XEV-3513.2	MR FROM OIL SEPARATOR TO EJ1-351.2 SHUTDOWN	PCS	WITH 5 MT CABLE
84	XEV-3513.3	MR FROM OIL SEPARATOR TO EJ1-351.3 SHUTDOWN	PCS	WITH 5 MT CABLE
85	XEV-3513.4	MR FROM OIL SEPARATOR TO EJ1-351.4 SHUTDOWN	PCS	WITH 5 MT CABLE
86	XEV-3513.5	MR FROM OIL SEPARATOR TO EJ1-351.5 SHUTDOWN	PCS	WITH 5 MT CABLE
87	XEV-3513.6	MR FROM OIL SEPARATOR TO EJ1-351.6 SHUTDOWN	PCS	WITH 5 MT CABLE
88	XEV-3514.1	MR FROM SUCTION HEADER TO EJ1-351.1 SHUTDOWN	PCS	-
89	XEV-3514.2	MR FROM SUCTION HEADER TO EJ1-351.2 SHUTDOWN	PCS	-

ID	TAG	SERVICE	SYSTEM	NOTES
90	XEV-3514.3	MR FROM SUCTION HEADER TO EJ1-351.3 SHUTDOWN	PCS	-
91	XEV-3514.4	MR FROM SUCTION HEADER TO EJ1-351.4 SHUTDOWN	PCS	-
92	XEV-3514.5	MR FROM SUCTION HEADER TO EJ1-351.5 SHUTDOWN	PCS	-
93	XEV-3514.6	MR FROM SUCTION HEADER TO EJ1-351.6 SHUTDOWN	PCS	-
94	XEV-3521.1	OIL FROM OIL SEPARATOR TO C-351H-1 SHUTDOWN	PCS	WITH 5 MT CABLE
95	XEV-3521.2	OIL FROM OIL SEPARATOR TO C-351H-2 SHUTDOWN	PCS	WITH 5 MT CABLE
96	XEV-3521.3	OIL FROM OIL SEPARATOR TO C-351H-3 SHUTDOWN	PCS	WITH 5 MT CABLE
97	XEV-3521.4	OIL FROM OIL SEPARATOR TO C-351H-4 SHUTDOWN	PCS	WITH 5 MT CABLE
98	XEV-3521.5	OIL FROM OIL SEPARATOR TO C-351H-5 SHUTDOWN	PCS	WITH 5 MT CABLE
99	XEV-3521.6	OIL FROM OIL SEPARATOR TO C-351H-6 SHUTDOWN	PCS	WITH 5 MT CABLE
100	XEV-3522.1	OIL FROM OIL SEPARATOR TO C-351H-1 SHUTDOWN	PCS	WITH 5 MT CABLE
101	XEV-3522.2	OIL FROM OIL SEPARATOR TO C-351H-2 SHUTDOWN	PCS	WITH 5 MT CABLE
102	XEV-3522.3	OIL FROM OIL SEPARATOR TO C-351H-3 SHUTDOWN	PCS	WITH 5 MT CABLE
103	XEV-3522.4	OIL FROM OIL SEPARATOR TO C-351H-4 SHUTDOWN	PCS	WITH 5 MT CABLE
104	XEV-3522.5	OIL FROM OIL SEPARATOR TO C-351H-5 SHUTDOWN	PCS	WITH 5 MT CABLE
105	XEV-3522.6	OIL FROM OIL SEPARATOR TO C-351H-6 SHUTDOWN	PCS	WITH 5 MT CABLE
106	XEV-3591	LNG TO MR MAKE UP ON-OFF VALVE	PCS	WITH 5 MT CABLE
107	XEV-3592	C2 TO MR MAKE UP ON-OFF VALVE	PCS	WITH 5 MT CABLE
108	XEV-3593	C4 TO MR MAKE UP ON-OFF VALVE	PCS	WITH 5 MT CABLE
109	XEV-3594	N2 TO MR MAKE UP ON-OFF VALVE	PCS	WITH 5 MT CABLE
110	XV-3511.1	C-351L-1 SUCTION SHUT DOWN	-	-
111	XV-3511.2	C-351L-2 SUCTION SHUT DOWN	-	-
112	XV-3511.3	C-351L-3 SUCTION SHUT DOWN	-	-
113	XV-3511.4	C-351L-4 SUCTION SHUT DOWN	-	-
114	XV-3511.5	C-351L-5 SUCTION SHUT DOWN	-	-
115	XV-3511.6	C-351L-6 SUCTION SHUT DOWN	-	-

ID	TAG	SERVICE	SYSTEM	NOTES
116	XV-3514.1	MR FROM SUCTION HEADER TO EJ1-351.1 SHUTDOWN	-	-
117	XV-3514.2	MR FROM SUCTION HEADER TO EJ1-351.2 SHUTDOWN	-	-
118	XV-3514.3	MR FROM SUCTION HEADER TO EJ1-351.3 SHUTDOWN	-	-
119	XV-3514.4	MR FROM SUCTION HEADER TO EJ1-351.4 SHUTDOWN	-	-
120	XV-3514.5	MR FROM SUCTION HEADER TO EJ1-351.5 SHUTDOWN	-	-
121	XV-3514.6	MR FROM SUCTION HEADER TO EJ1-351.6 SHUTDOWN	-	-
122	ZEV-3511.H1	C-351H-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
123	ZEV-3511.H2	C-351H-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
124	ZEV-3511.H3	C-351H-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
125	ZEV-3511.H4	C-351H-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
126	ZEV-3511.H5	C-351H-5 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
127	ZEV-3511.H6	C-351H-6 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
128	ZEV-3511.L1	C-351L-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
129	ZEV-3511.L2	C-351L-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
130	ZEV-3511.L3	C-351L-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
131	ZEV-3511.L4	C-351L-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
132	ZEV-3511.L5	C-351L-5 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
133	ZEV-3511.L6	C-351L-6 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
134	ZEV-3512.H1	C-351H-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
135	ZEV-3512.H2	C-351H-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
136	ZEV-3512.H3	C-351H-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
137	ZEV-3512.H4	C-351H-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
138	ZEV-3512.H5	C-351H-5 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
139	ZEV-3512.H6	C-351H-6 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
140	ZEV-3512.L1	C-351L-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
141	ZEV-3512.L2	C-351L-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
142	ZEV-3512.L3	C-351L-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
143	ZEV-3512.L4	C-351L-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
144	ZEV-3512.L5	C-351L-5 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
145	ZEV-3512.L6	C-351L-6 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
146	ZEV-3513.H1	C-351H-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
147	ZEV-3513.H2	C-351H-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
148	ZEV-3513.H3	C-351H-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
149	ZEV-3513.H4	C-351H-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
150	ZEV-3513.H5	C-351H-5 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704

ID	TAG	SERVICE	SYSTEM	NOTES
151	ZEV-3513.H6	C-351H-6 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
152	ZEV-3513.L1	C-351L-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
153	ZEV-3513.L2	C-351L-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
154	ZEV-3513.L3	C-351L-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
155	ZEV-3513.L4	C-351L-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
156	ZEV-3513.L5	C-351L-5 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
157	ZEV-3513.L6	C-351L-6 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
158	ZEV-3514.H1	C-351H-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
159	ZEV-3514.H2	C-351H-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
160	ZEV-3514.H3	C-351H-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
161	ZEV-3514.H4	C-351H-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
162	ZEV-3514.H5	C-351H-5 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
163	ZEV-3514.H6	C-351H-6 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
164	ZEV-3514.L1	C-351L-1 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
165	ZEV-3514.L2	C-351L-2 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
166	ZEV-3514.L3	C-351L-3 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
167	ZEV-3514.L4	C-351L-4 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
168	ZEV-3514.L5	C-351L-5 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
169	ZEV-3514.L6	C-351L-6 COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704

8.5.5 Instrumentation

Table 8.7 — Unit 350 – Instrument List

ID	TAG	SERVICE	RANGE	OP	UNITS
1	FE-3551	LNG TO NRU FLOW	-	-	-
2	FI-3591	NITROGEN TO X-359 FLOW	-	-	Nm3/h
3	FT-3551	LNG TO NRU FLOW	0..20	-	MMSCFD
4	LSHH-3541	E-354 HIGH HIGH LEVEL SWITCH	-	-	-
5	LSL-3517.1	V-351-1 OIL LEVEL	-	-	-
6	LSL-3517.2	V-351-2 OIL LEVEL	-	-	-
7	LSL-3517.3	V-351-3 OIL LEVEL	-	-	-
8	LSL-3517.4	V-351-4 OIL LEVEL	-	-	-
9	LSL-3517.5	V-351-5 OIL LEVEL	-	-	-
10	LSL-3517.6	V-351-6 OIL LEVEL	-	-	-
11	LSLL-3550	V-355A LOW LOW LEVEL SWITCH	-	-	-
12	LT-3540	E-354 LIQUID LEVEL	-	-	mm
13	LT-3550	V-355A LIQUID LEVEL	-	-	mm

ID	TAG	SERVICE	RANGE	OP	UNITS
14	LT-3580	V-358 LIQUID LEVEL	-	-	mm
15	PT-3501	MR TO COMPRESSOR PACKAGE INLET PRESSURE	0..30	-	barg
16	PT-3511.H1	C-351H-1 SUCTION PRESSURE	0..34.47	-	barg
17	PT-3511.H2	C-351H-2 SUCTION PRESSURE	0..34.47	-	barg
18	PT-3511.H3	C-351H-3 SUCTION PRESSURE	0..34.47	-	barg
19	PT-3511.H4	C-351H-4 SUCTION PRESSURE	0..34.47	-	barg
20	PT-3511.H5	C-351H-5 SUCTION PRESSURE	0..34.47	-	barg
21	PT-3511.H6	C-351H-6 SUCTION PRESSURE	0..34.47	-	barg
22	PT-3511.L1	C-351L-1 SUCTION PRESSURE	0..13.7895	-	bara
23	PT-3511.L2	C-351L-2 SUCTION PRESSURE	0..13.7895	-	bara
24	PT-3511.L3	C-351L-3 SUCTION PRESSURE	0..13.7895	-	bara
25	PT-3511.L4	C-351L-4 SUCTION PRESSURE	0..13.7895	-	bara
26	PT-3511.L5	C-351L-5 SUCTION PRESSURE	0..13.7895	-	bara
27	PT-3511.L6	C-351L-6 SUCTION PRESSURE	0..13.7895	-	bara
28	PT-3512.H1	C-351H-1 DISCHARGE PRESSURE	0..34.47	-	barg
29	PT-3512.H2	C-351H-2 DISCHARGE PRESSURE	0..34.47	-	barg
30	PT-3512.H3	C-351H-3 DISCHARGE PRESSURE	0..34.47	-	barg
31	PT-3512.H4	C-351H-4 DISCHARGE PRESSURE	0..34.47	-	barg
32	PT-3512.H5	C-351H-5 DISCHARGE PRESSURE	0..34.47	-	barg
33	PT-3512.H6	C-351H-6 DISCHARGE PRESSURE	0..34.47	-	barg
34	PT-3512.L1	C-351L-1 DISCHARGE PRESSURE	0..34.47	-	barg
35	PT-3512.L2	C-351L-2 DISCHARGE PRESSURE	0..34.47	-	barg
36	PT-3512.L3	C-351L-3 DISCHARGE PRESSURE	0..34.47	-	barg
37	PT-3512.L4	C-351L-4 DISCHARGE PRESSURE	0..34.47	-	barg
38	PT-3512.L5	C-351L-5 DISCHARGE PRESSURE	0..34.47	-	barg
39	PT-3512.L6	C-351L-6 DISCHARGE PRESSURE	0..34.47	-	barg
40	PT-3513.H1	C-351H-1 OIL HEADER PRESSURE	0..34.47	-	barg
41	PT-3513.H2	C-351H-2 OIL HEADER PRESSURE	0..34.47	-	barg
42	PT-3513.H3	C-351H-3 OIL HEADER PRESSURE	0..34.47	-	barg
43	PT-3513.H4	C-351H-4 OIL HEADER PRESSURE	0..34.47	-	barg
44	PT-3513.H5	C-351H-5 OIL HEADER PRESSURE	0..34.47	-	barg
45	PT-3513.H6	C-351H-6 OIL HEADER PRESSURE	0..34.47	-	barg
46	PT-3513.L1	C-351L-1 OIL HEADER PRESSURE	0..34.47	-	barg
47	PT-3513.L2	C-351L-2 OIL HEADER PRESSURE	0..34.47	-	barg
48	PT-3513.L3	C-351L-3 OIL HEADER PRESSURE	0..34.47	-	barg
49	PT-3513.L4	C-351L-4 OIL HEADER PRESSURE	0..34.47	-	barg
50	PT-3513.L5	C-351L-5 OIL HEADER PRESSURE	0..34.47	-	barg
51	PT-3513.L6	C-351L-6 OIL HEADER PRESSURE	0..34.47	-	barg
52	PT-3550	V-355A PRESSURE	0..34.47	-	barg
53	PT-3551	LIQUID MR TO E-356B PRESSURE	0..34.47	-	barg

ID	TAG	SERVICE	RANGE	OP	UNITS
54	PT-3552	AMMONIA TO PK-341-1/2/3/4 PRESSURE	0..34.47	-	barg
55	PT-3560	CB-356 COLD BOX INLET PRESSURE	0..50	-	barg
56	PT-3560A	E-356A A-OUTLET PRESSURE	0..50	-	barg
57	PT-3561A	V-356A PRESSURE	0..50	-	barg
58	PT-3561B	E-356B C-OUTLET PRESSURE	0..50	-	barg
59	PT-3562	V-356B PRESSURE	0..50	-	barg
60	PT-3563	E-356B B2-INLET PRESSURE	0..50	-	barg
61	PT-3564	E-356B B3-OUTLET TO T-357 PRESSURE	0..50	-	barg
62	PT-3565	V-355B D-INLET PRESSURE	0..50	-	barg
63	PT-3571	X-357 PRESSURE	0..10	-	barg
64	PT-3572	X-357 PRESSURE	0..10	-	barg
65	TG-3591	V-357-3 TEMPERATURE	-15..100	-	°C
66	TG-3592	NITROGEN TO MR MAKE UP REGENERATION TEMPERATURE	-15..100	-	°C
67	TG-3593	V-357-4 TEMPERATURE	-15..100	-	°C
68	TT-3501	MR TO COMPRESSOR PACKAGE INLET TEMPERATURE	-50..120	-	°C
69	TT-3512.H1	C-351H-1 DISCHARGE TEMPERATURE	-50..200	-	°C
70	TT-3512.H2	C-351H-2 DISCHARGE TEMPERATURE	-50..200	-	°C
71	TT-3512.H3	C-351H-3 DISCHARGE TEMPERATURE	-50..200	-	°C
72	TT-3512.H4	C-351H-4 DISCHARGE TEMPERATURE	-50..200	-	°C
73	TT-3512.H5	C-351H-5 DISCHARGE TEMPERATURE	-50..200	-	°C
74	TT-3512.H6	C-351H-6 DISCHARGE TEMPERATURE	-50..200	-	°C
75	TT-3512.L1	C-351L-1 DISCHARGE TEMPERATURE	-50..200	-	°C
76	TT-3512.L2	C-351L-2 DISCHARGE TEMPERATURE	-50..200	-	°C
77	TT-3512.L3	C-351L-3 DISCHARGE TEMPERATURE	-50..200	-	°C
78	TT-3512.L4	C-351L-4 DISCHARGE TEMPERATURE	-50..200	-	°C
79	TT-3512.L5	C-351L-5 DISCHARGE TEMPERATURE	-50..200	-	°C
80	TT-3512.L6	C-351L-6 DISCHARGE TEMPERATURE	-50..200	-	°C
81	TT-3517.1	V-351-1 TEMPERATURE	-50..200	-	°C
82	TT-3517.2	V-351-2 TEMPERATURE	-50..200	-	°C
83	TT-3517.3	V-351-3 TEMPERATURE	-50..200	-	°C
84	TT-3517.4	V-351-4 TEMPERATURE	-50..200	-	°C
85	TT-3517.5	V-351-5 TEMPERATURE	-50..200	-	°C
86	TT-3517.6	V-351-6 TEMPERATURE	-50..200	-	°C
87	TT-3518.1	OIL TO C-351H-1 TEMPERATURE	-50..200	-	°C
88	TT-3518.2	OIL TO C-351H-2 TEMPERATURE	-50..200	-	°C
89	TT-3518.3	OIL TO C-351H-3 TEMPERATURE	-50..200	-	°C
90	TT-3518.4	OIL TO C-351H-4 TEMPERATURE	-50..200	-	°C
91	TT-3518.5	OIL TO C-351H-5 TEMPERATURE	-50..200	-	°C

ID	TAG	SERVICE	RANGE	OP	UNITS
92	TT-3518.6	OIL TO C-351H-6 TEMPERATURE	-50..200	-	°C
93	TT-3518.H1	OIL TO C-351H/L-1 TEMPERATURE	-50..200	-	°C
94	TT-3518.H2	OIL TO C-351H/L-2 TEMPERATURE	-50..200	-	°C
95	TT-3518.H3	OIL TO C-351H/L-3 TEMPERATURE	-50..200	-	°C
96	TT-3518.H4	OIL TO C-351H/L-4 TEMPERATURE	-50..200	-	°C
97	TT-3518.H5	OIL TO C-351H/L-5 TEMPERATURE	-50..200	-	°C
98	TT-3518.H6	OIL TO C-351H/L-6 TEMPERATURE	-50..200	-	°C
99	TT-3519.L1	C-351L-1 SUCTION TEMPERATURE	-50..200	-	°C
100	TT-3519.L2	C-351L-2 SUCTION TEMPERATURE	-50..200	-	°C
101	TT-3519.L3	C-351L-3 SUCTION TEMPERATURE	-50..200	-	°C
102	TT-3519.L4	C-351L-4 SUCTION TEMPERATURE	-50..200	-	°C
103	TT-3519.L5	C-351L-5 SUCTION TEMPERATURE	-50..200	-	°C
104	TT-3519.L6	C-351L-6 SUCTION TEMPERATURE	-50..200	-	°C
105	TT-3551	LNG TO NRU INLET TEMPERATURE	-196..65	-	°C
106	TT-3552	MR TO E-354 OUTLET TEMPERATURE	-50..120	-	°C
107	TT-3553A	MOTOR TEMPERATURE P-355A	-50..120	-	°C
108	TT-3553B	MOTOR TEMPERATURE P-355B	-50..120	-	°C
109	TT-3554	MR TO E-354 INLET TEMPERATURE	-50..120	-	°C
110	TT-3561A	NATURAL GAS TO V-356A TEMPERATURE	-50..120	-	°C
111	TT-3562.01	CB-356 CORE T STREAM B2 INLET (CB TE-01)	-196..65	-	°C
112	TT-3562.02	CB-356 CORE T STREAM B3 OUTLET (CB TE-02)	-196..65	-	°C
113	TT-3562.03	CB-356 CORE T STREAM C INLET (CB TE-03)	-196..65	-	°C
114	TT-3562.04	CB-356 CORE T STREAM C OUTLET (CB TE-04)	-196..65	-	°C
115	TT-3562.05	CB-356 CORE T STREAM D OUTLET (CB TE-05)	-196..65	-	°C
116	TT-3562.06	CB-356 CORE T STREAM D (CB TE-06)	-196..65	-	°C
117	TT-3562.07	CB-356 CORE T STREAM D (CB TE-07)	-196..65	-	°C
118	TT-3562.08	CB-356 CORE T STREAM D (CB TE-08)	-196..65	-	°C
119	TT-3562.09	CB-356 CORE T STREAM D INLET (CB TE-09)	-196..65	-	°C
120	TT-3563	E-356B B2-INLET TEMPERATURE (CB TE-10)	-196..65	-	°C
121	TT-3564	E-356A A-OUTLET TEMPERATURE	-50..120	-	°C
122	TT-3571	X-357 TEMPERATURE	-196..65	-	°C
123	TT-3592	NITROGEN TO MR MAKE UP REGENERATION TEMPERATURE	-50..300	-	°C
124	ZSH-3511.1	XEV-3511.1 OPEN POSITION INDICATION	-	-	-
125	ZSH-3511.2	XEV-3511.2 OPEN POSITION INDICATION	-	-	-
126	ZSH-3511.3	XEV-3511.3 OPEN POSITION INDICATION	-	-	-
127	ZSH-3511.4	XEV-3511.4 OPEN POSITION INDICATION	-	-	-
128	ZSH-3511.5	XEV-3511.5 OPEN POSITION INDICATION	-	-	-
129	ZSH-3511.6	XEV-3511.6 OPEN POSITION INDICATION	-	-	-
130	ZSH-3514.1	XEV-3514.1 OPEN POSITION INDICATION	-	-	-

ID	TAG	SERVICE	RANGE	OP	UNITS
131	ZSH-3514.2	XEV-3514.2 OPEN POSITION INDICATION	-	-	-
132	ZSH-3514.3	XEV-3514.3 OPEN POSITION INDICATION	-	-	-
133	ZSH-3514.4	XEV-3514.4 OPEN POSITION INDICATION	-	-	-
134	ZSH-3514.5	XEV-3514.5 OPEN POSITION INDICATION	-	-	-
135	ZSH-3514.6	XEV-3514.6 OPEN POSITION INDICATION	-	-	-
136	ZSL-3511.1	XEV-3511.1 CLOSED POSITION INDICATION	-	-	-
137	ZSL-3511.2	XEV-3511.2 CLOSED POSITION INDICATION	-	-	-
138	ZSL-3511.3	XEV-3511.3 CLOSED POSITION INDICATION	-	-	-
139	ZSL-3511.4	XEV-3511.4 CLOSED POSITION INDICATION	-	-	-
140	ZSL-3511.5	XEV-3511.5 CLOSED POSITION INDICATION	-	-	-
141	ZSL-3511.6	XEV-3511.6 CLOSED POSITION INDICATION	-	-	-
142	ZSL-3514.1	XEV-3514.1 CLOSED POSITION INDICATION	-	-	-
143	ZSL-3514.2	XEV-3514.2 CLOSED POSITION INDICATION	-	-	-
144	ZSL-3514.3	XEV-3514.3 CLOSED POSITION INDICATION	-	-	-
145	ZSL-3514.4	XEV-3514.4 CLOSED POSITION INDICATION	-	-	-
146	ZSL-3514.5	XEV-3514.5 CLOSED POSITION INDICATION	-	-	-
147	ZSL-3514.6	XEV-3514.6 CLOSED POSITION INDICATION	-	-	-

8.5.6 Junction Boxes

Table 8.8 — Unit 350 – Junction Boxes

ID	TAG	SERVICE	CONNECTION	TYPE
1	P-PN-CB356.1	LIQUEFACTION	COPPER-COPPER	SIPLUS
2	P-PN-CB357.1	LIQUEFACTION	COPPER-COPPER	SIPLUS
3	JG-351-1C	AMMONIA COMPRESSOR OIL COOLER	COPPER-COPPER	STANDARD
4	JG-351-2C	AMMONIA COMPRESSOR OIL COOLER	COPPER-COPPER	STANDARD
5	JG-351-3C	AMMONIA COMPRESSOR OIL COOLER	COPPER-COPPER	STANDARD
6	JG-351-4C	AMMONIA COMPRESSOR OIL COOLER	COPPER-COPPER	STANDARD
7	JG-351-5C	AMMONIA COMPRESSOR OIL COOLER	COPPER-COPPER	STANDARD
8	JG-351-6C	AMMONIA COMPRESSOR OIL COOLER	COPPER-COPPER	STANDARD
9	JG-353-1	AIR COOLED MR PRECOOLER FANS	COPPER-COPPER	STANDARD
10	JG-353-2	AIR COOLED MR PRECOOLER FANS	COPPER-COPPER	STANDARD
11	JG-353-3	AIR COOLED MR PRECOOLER FANS	COPPER-COPPER	STANDARD
12	P-PN-OS354.1	LIQUEFACTION	COPPER-COPPER	SIPLUS
13	P-PN-PK-351-1	MR COMPRESSOR	COPPER-COPPER	SIPLUS
14	P-PN-PK-351-2	MR COMPRESSOR	COPPER-COPPER	SIPLUS
15	P-PN-PK-351-3	MR COMPRESSOR	COPPER-COPPER	SIPLUS

ID	TAG	SERVICE	CONNECTION	TYPE
16	P-PN-PK-351-4	MR COMPRESSOR	COPPER-COPPER	SIPLUS
17	P-PN-PK-351-5	MR COMPRESSOR	COPPER-COPPER	SIPLUS
18	P-PN-PK-351-6	MR COMPRESSOR	COPPER-COPPER	SIPLUS

8.5.7 Controllers

Table 8.9 — UNIT 350 – Controllers

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
LIC-2211	LIQUID LEVEL IN REBOILER X-357	LV-2211	0..100 %	—
LIC-3540	REFRIGERANT (NH3) LIQUID LEVEL IN E-354	LV-3540	0..100 %	—
LIC-3550	LIQUID LEVEL IN HP MR SEPARATOR V-355A	LV-3550	0..100 %	—
PIC-3501	MR COMPRESSOR SUCTION PRESSURE	—	0..30 barg	—
PIC-3511.H1	HP MR COMPRESSOR 1 SUCTION PRESSURE	ZEV3511.H1	0..34 barg	—
PIC-3511.H2	HP MR COMPRESSOR 2 SUCTION PRESSURE	ZEV3511.H2	0..34 barg	—
PIC-3511.H3	HP MR COMPRESSOR 3 SUCTION PRESSURE	ZEV3511.H3	0..34 barg	—
PIC-3511.H4	HP MR COMPRESSOR SUCTION PRESSURE	ZEV3511.H4	0..34 barg	—
PIC-3511.H5	HP MR COMPRESSOR 5 SUCTION PRESSURE	ZEV3511.H5	0..34 barg	—
PIC-3511.H6	HP MR COMPRESSOR 6 SUCTION PRESSURE	ZEV3511.H6	0..34 barg	—
PIC-3550	MR LOOP PRESSURE AT V-355A	—	0..34 barg	—
PIC-3552	AMMONIA TO PK-341-1/2/3/4 PRESSSURE	—	0..34 barg	—
PIC-3561B	GAS PRESSURE AT E-356B OUTLET	PV-3561B	0..50 barg	—
PIC-3572	NRU REBOILER-SECTION PRESSURE	PV-3551	0..10 barg	—
TIC-3501	LP MR COMPRESSOR SUCTION TEMPERATURE	ZEV3511	-50..120 °C	—
TIC-3517.1	LUBRICATING-OIL TEMPERATURE (X-351-1)	ZEV3511	-50..200 °C	—
TIC-3517.2	LUBRICATING-OIL TEMPERATURE (X-351-2)	ZEV3511	-50..200 °C	—
TIC-3517.3	LUBRICATING-OIL TEMPERATURE (X-351-3)	ZEV3511	-50..200 °C	—
TIC-3517.4	LUBRICATING-OIL TEMPERATURE (X-351-4)	ZEV3511	-50..200 °C	—
TIC-3517.5	LUBRICATING-OIL TEMPERATURE (X-351-5)	ZEV3511	-50..200 °C	—
TIC-3517.6	LUBRICATING-OIL TEMPERATURE (X-351-6)	ZEV3511	-50..200 °C	—
TIC-3551	LNG TEMPERATURE (FOR FLOW COMPENSATION)	CV-3551	-196..65 °C	—
TIC-3571	REBOILER X-357 OPERATING TEMPERATURE	—	-196..65 °C	—
XC-3501	MR COMPRESSOR SUCTION PRESSURE CONTROL	ZEV3511	0..30 barg	—
XC-3551	LNG FLOWRATE TO T-357	CV-3551	—	—
XC-3561B	MR REFRIGERATION BALANCE CONTROL	—	0..30 barg	—

8.6 Operating Interface

8.6.1 PCS Operating Screens – Unit 350

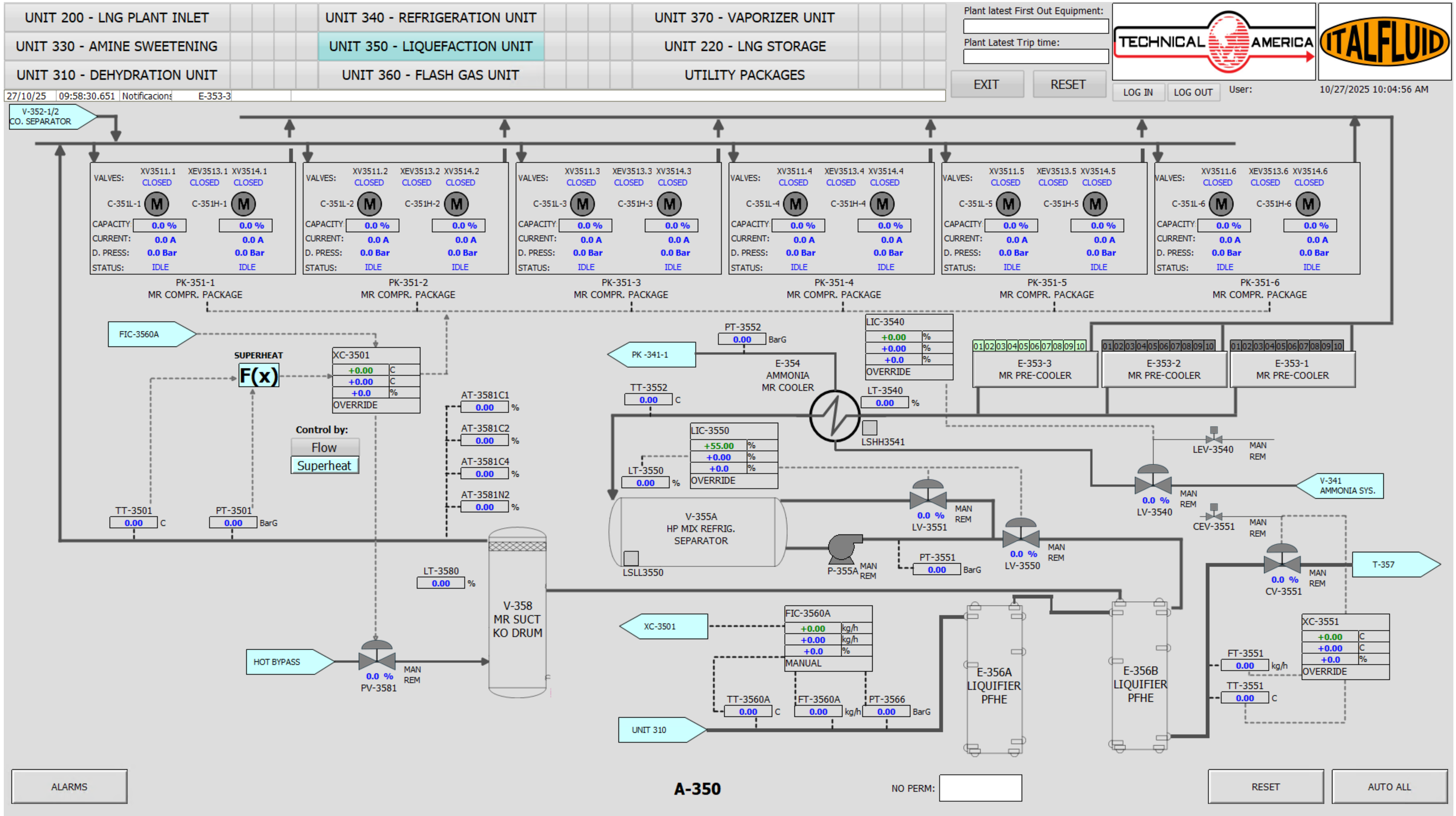


Figure 8.2: UNIT 350 - HMI Operating Screen

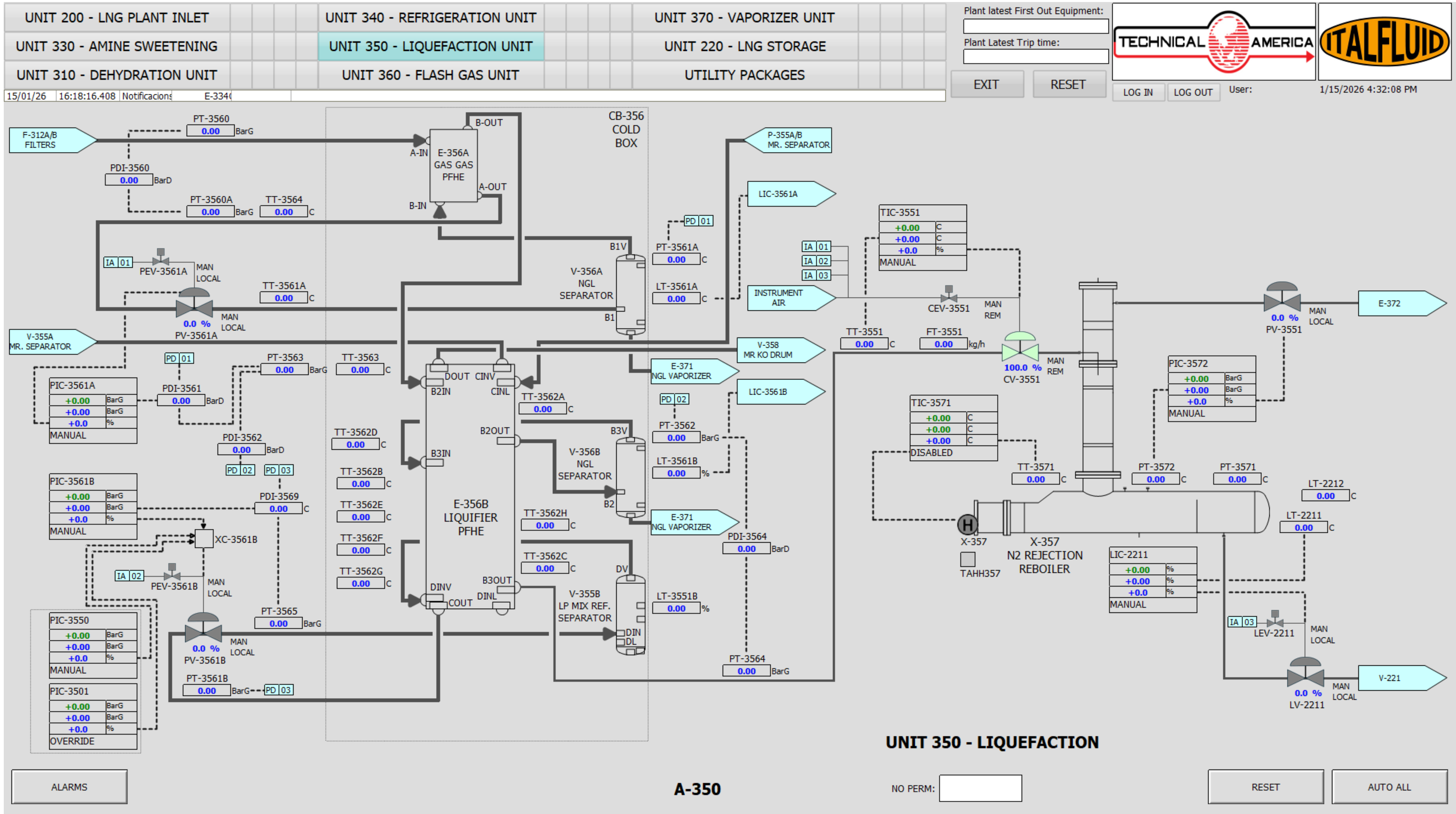


Figure 8.3: UNIT 350 - HMI Operating Screen

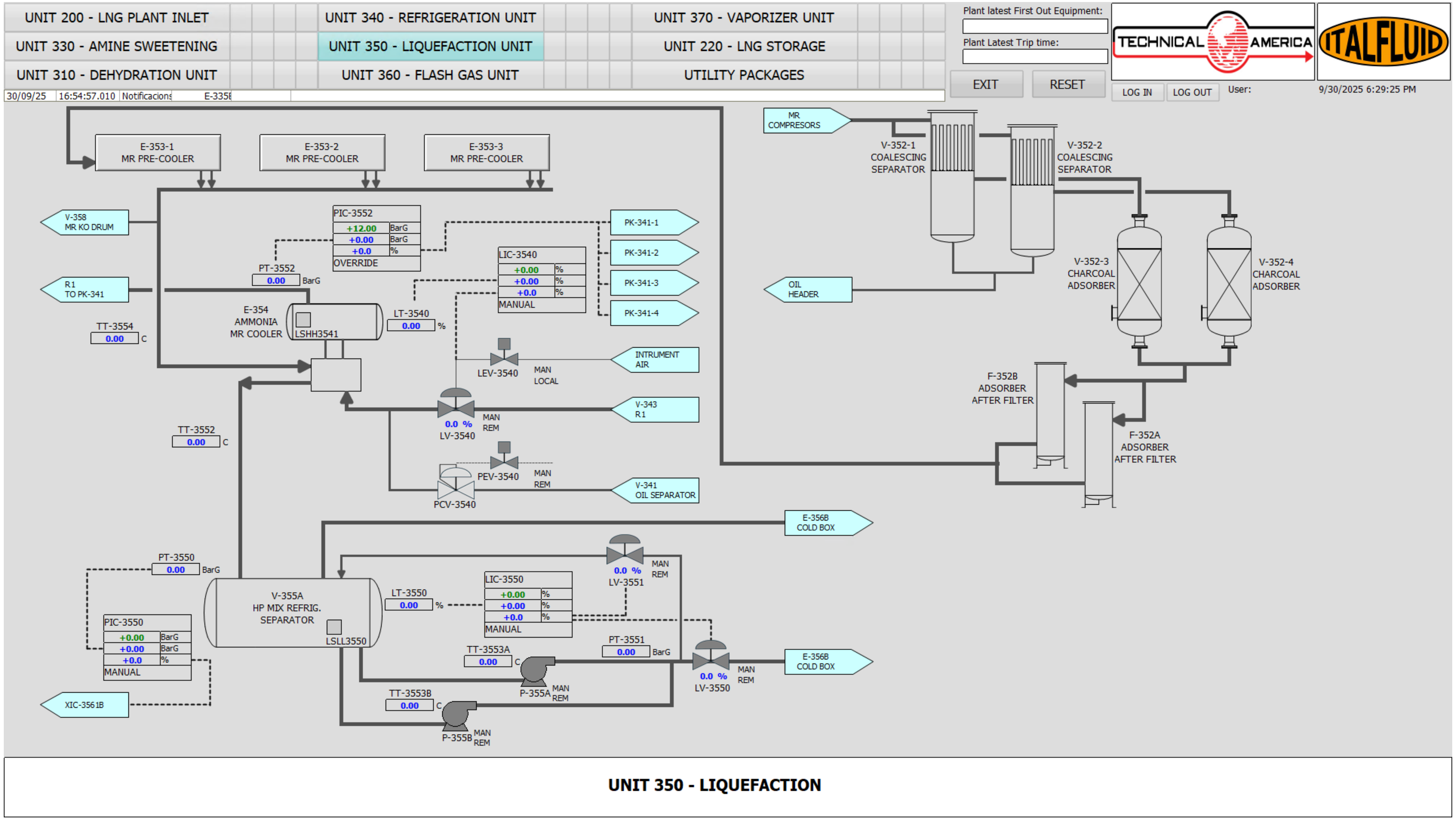


Figure 8.4: UNIT 350 - HMI Operating Screen

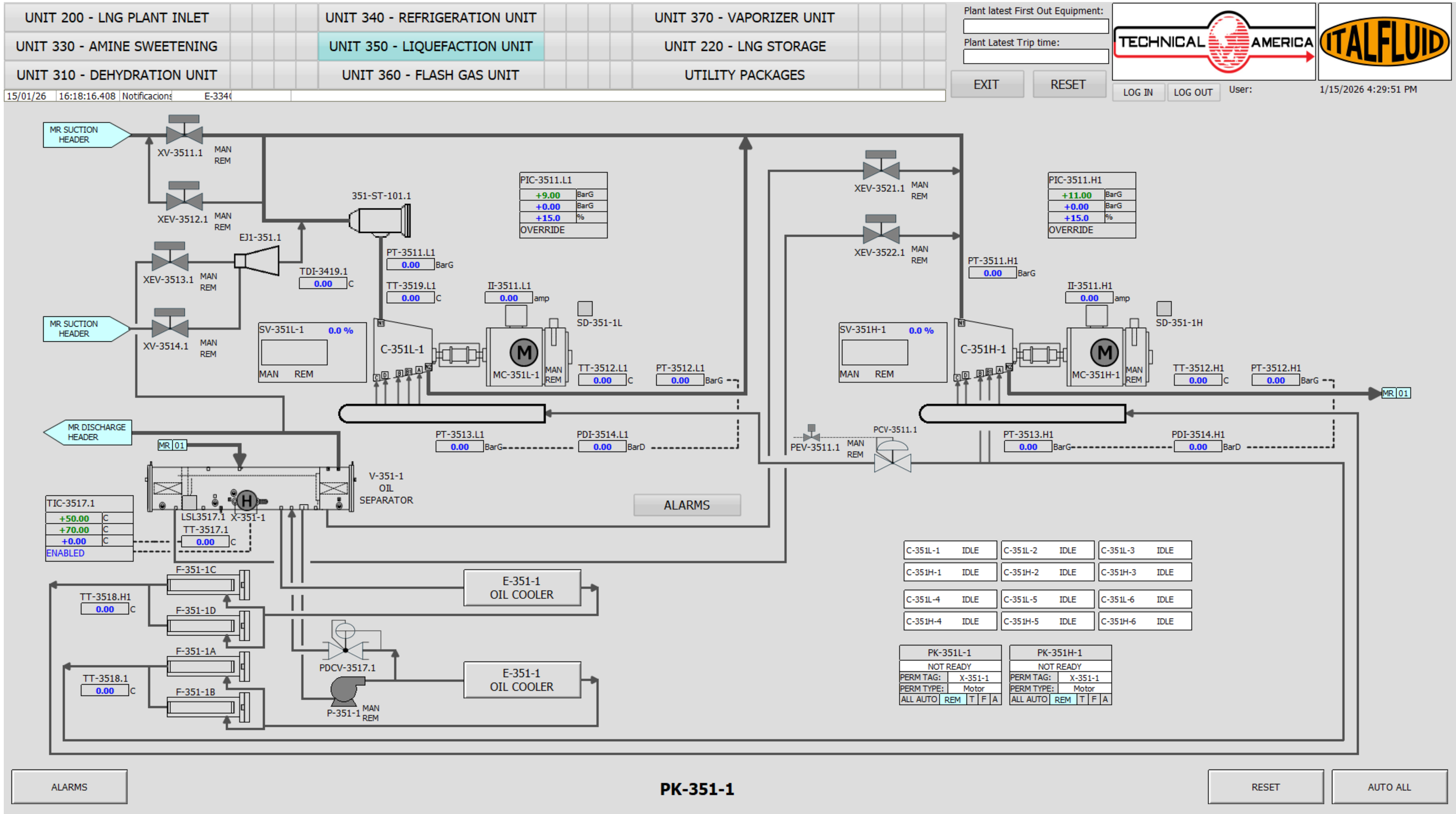


Figure 8.5: UNIT 350 - HMI Operating Screen

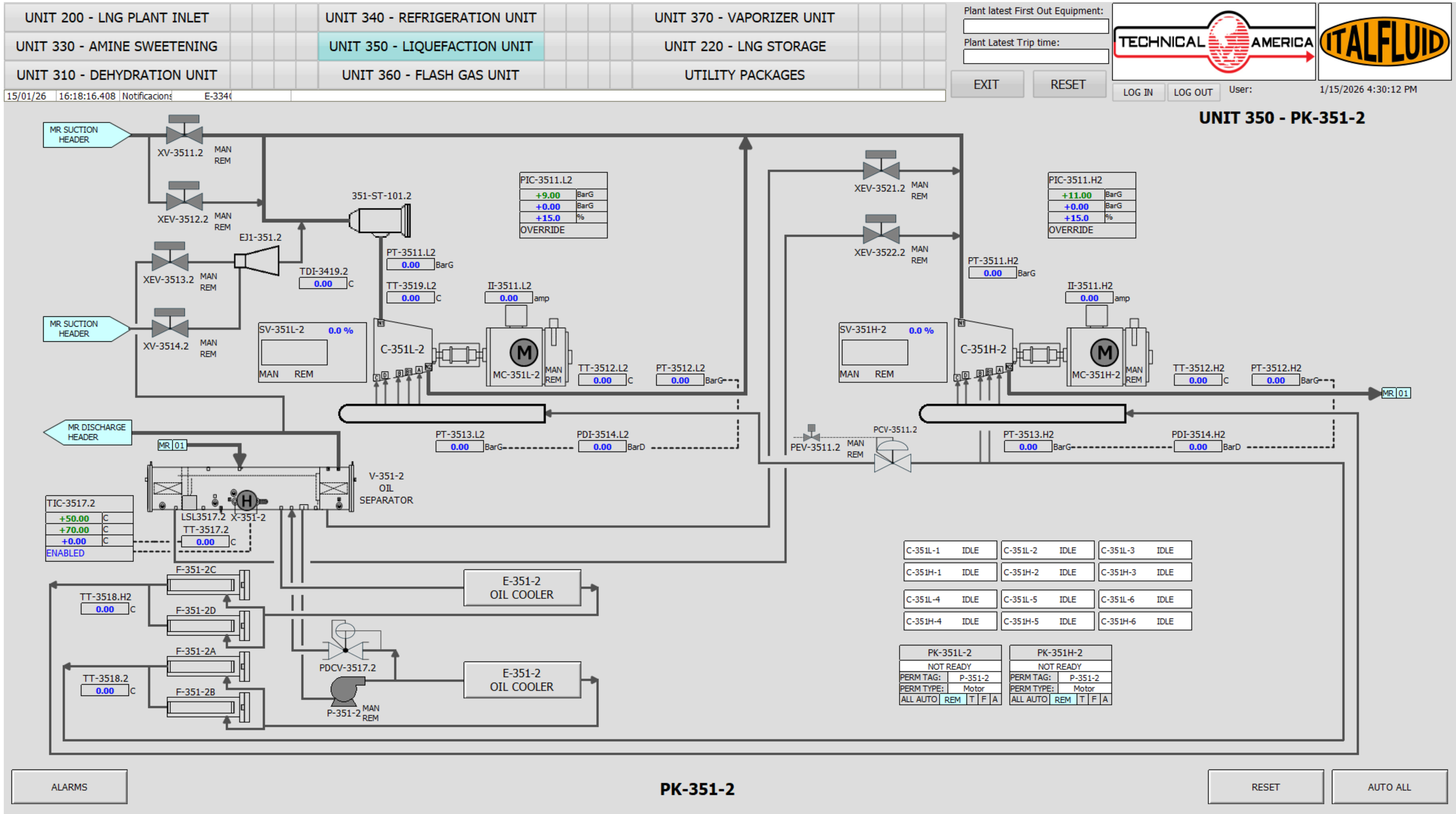


Figure 8.6: UNIT 350 - HMI Operating Screen

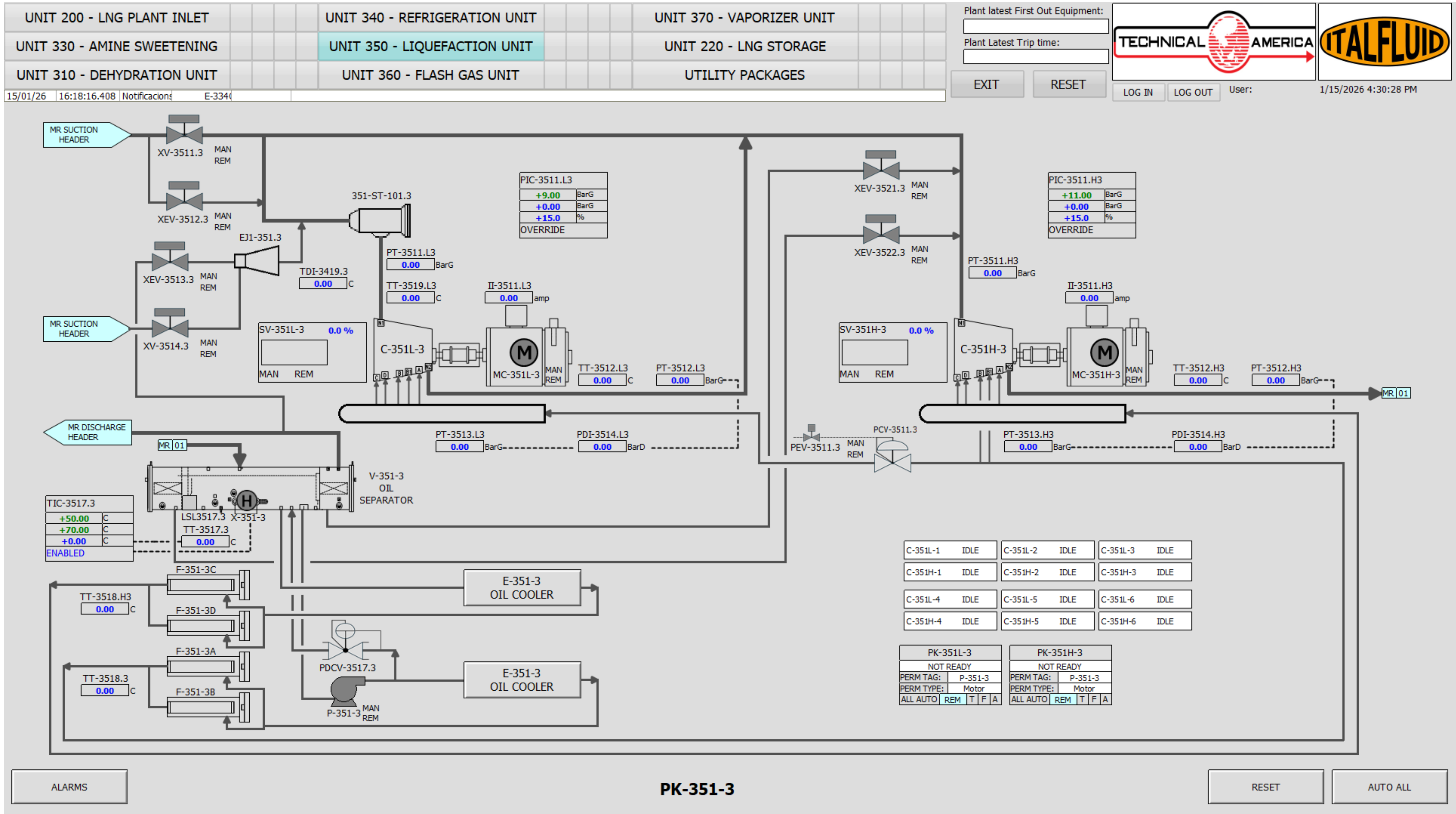


Figure 8.7: UNIT 350 - HMI Operating Screen

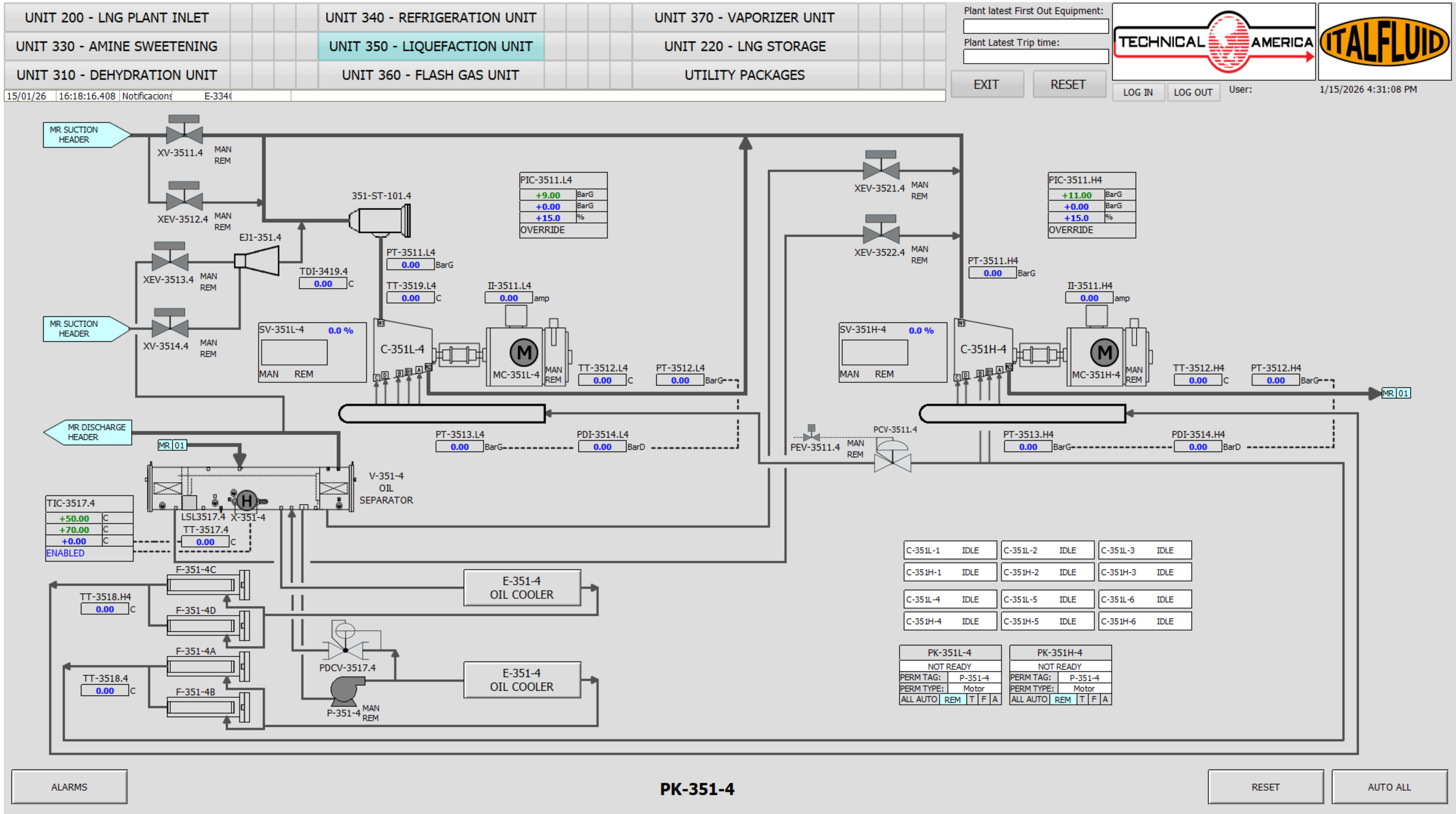


Figure 8.8: UNIT 350 - HMI Operating Screen

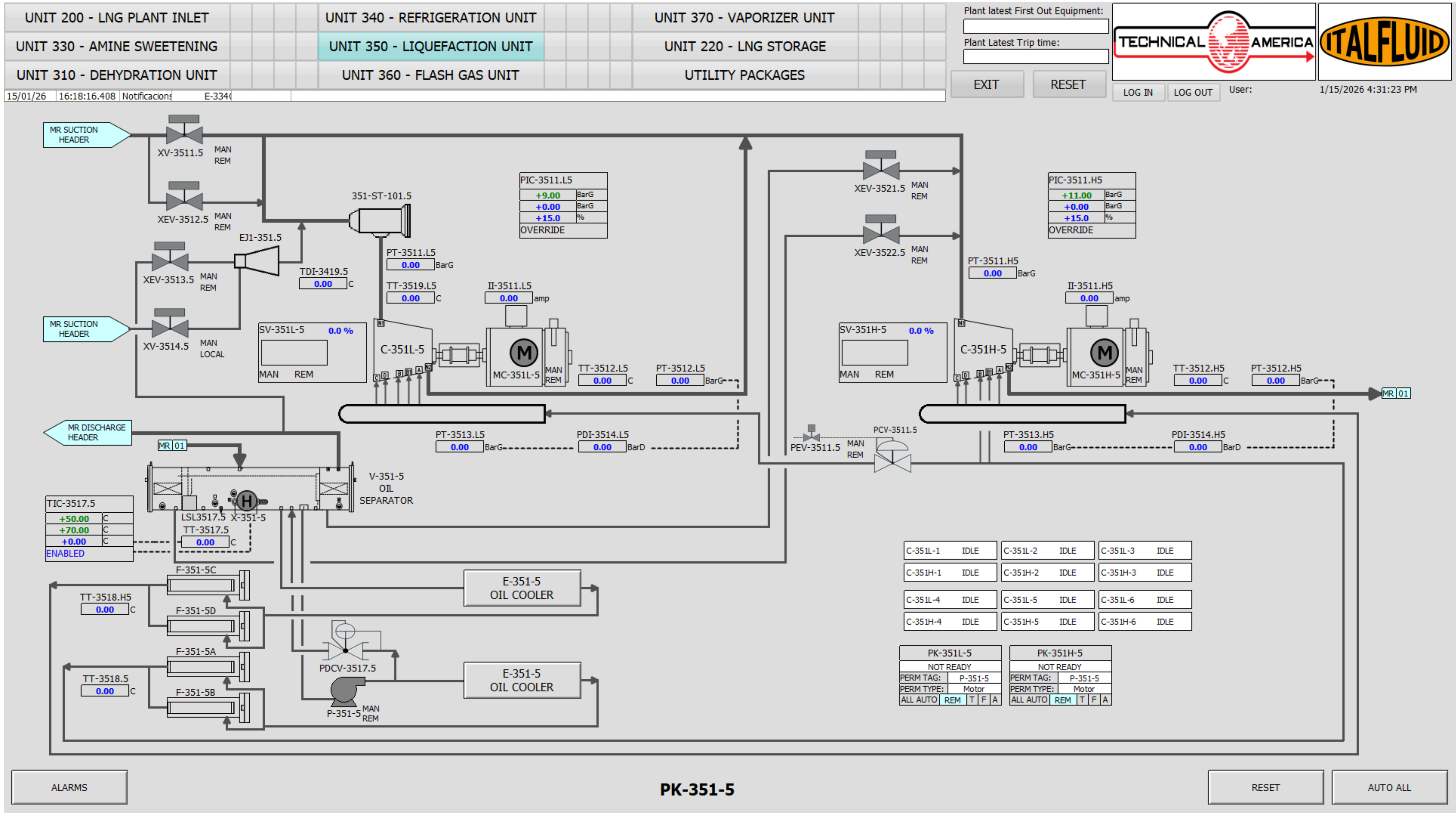


Figure 8.9: UNIT 350 - HMI Operating Screen

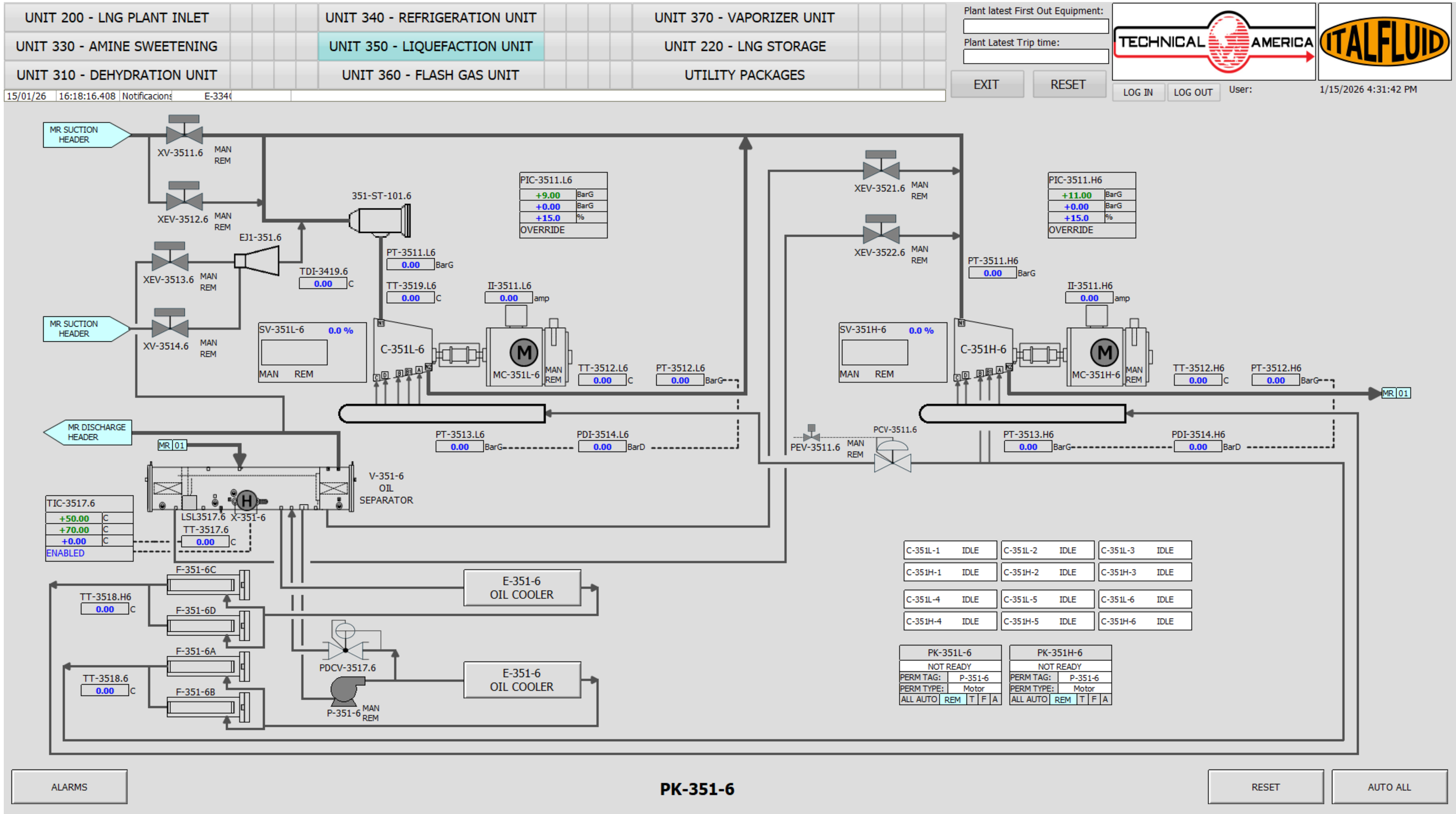


Figure 8.10: UNIT 350 - HMI Operating Screen

8.6.2 Critical Alarms and Trips

Table 8.10 — Unit 350 – Critical Alarms

ALARM	INSTRUMENT	TYPE	SP	DESCRIPTION
LAHH-2212	LT-2212	PSD	—	HIGH HIGH LIQUID LEVEL ON NRU
LAHH-3541	LS-3541	PSD	—	HIGH HIGH LIQUID LEVEL ON E-354
LAHH-3561A	LT-3561A	PSD	—	HIGH HIGH LIQUID LEVEL ON V-356A
LAHH-3561C	LT-3561C	PSD	—	HIGH HIGH LIQUID LEVEL ON V-356B
LAHH-3580	LT-3580	PSD	—	HIGH HIGH LIQUID LEVEL ON V-358
LALL-2212	LT-2212	LSD	—	LOW LOW LIQUID LEVEL ON NRU
LALL-3105	LT-3105	LSD	—	LOW LOW LIQUID LEVEL ON V-312
LALL-3550	LS-3550	LSD	—	LOW LOW LIQUID LEVEL ON V-355A
PAHH-3571	PT-3571	PSD	—	HIGH HIGH PRESSURE ON NRU
PAHH-3512.H1	PT-3512.H1	LSD	—	C-351H-1 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.H2	PT-3512.H2	LSD	—	C-351H-2 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.H3	PT-3512.H3	LSD	—	C-351H-3 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.H4	PT-3512.H4	LSD	—	C-351H-4 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.H5	PT-3512.H5	LSD	—	C-351H-5 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.H6	PT-3512.H6	LSD	—	C-351H-6 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.L1	PT-3512.L1	LSD	—	C-351L-1 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.L2	PT-3512.L2	LSD	—	C-351L-2 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.L3	PT-3512.L3	LSD	—	C-351L-3 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.L4	PT-3512.L4	LSD	—	C-351L-4 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.L5	PT-3512.L5	LSD	—	C-351L-5 HIGH HIGH DISCHARGE PRESSURE
PAHH-3512.L6	PT-3512.L6	LSD	—	C-351L-6 HIGH HIGH DISCHARGE PRESSURE
PALL-3571	PT-3571	PSD	—	LOW LOW PRESSURE ON NRU
PALL-3511.L1	PT-3511.L1	LSD	—	C-351L-1 LOW LOW SUCTION PRESSURE
PALL-3511.L2	PT-3511.L2	LSD	—	C-351L-2 LOW LOW SUCTION PRESSURE
PALL-3511.L3	PT-3511.L3	LSD	—	C-351L-3 LOW LOW SUCTION PRESSURE
PALL-3511.L4	PT-3511.L4	LSD	—	C-351L-4 LOW LOW SUCTION PRESSURE
PALL-3511.L5	PT-3511.L5	LSD	—	C-351L-5 LOW LOW SUCTION PRESSURE
PALL-3511.L6	PT-3511.L6	LSD	—	C-351L-6 LOW LOW SUCTION PRESSURE
PDALL-3514.H1	PDT-3514.H1	LSD	—	C-351H-1 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3514.H2	PDT-3514.H2	LSD	—	C-351H-2 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3514.H3	PDT-3514.H3	LSD	—	C-351H-3 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3514.H4	PDT-3514.H4	LSD	—	C-351H-4 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3514.H5	PDT-3514.H5	LSD	—	C-351H-5 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3514.H6	PDT-3514.H6	LSD	—	C-351H-6 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3514.L1	PDT-3514.L1	LSD	—	C-351L-1 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3514.L2	PDT-3514.L2	LSD	—	C-351L-2 LOW LOW OIL DIFFERENTIAL PRESSURE

ALARM	INSTRUMENT	TYPE	SP	DESCRIPTION
PDALL-3514.L3	PDT-3514.L3	LSD	—	C-351L-3 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3514.L4	PDT-3514.L4	LSD	—	C-351L-4 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3514.L5	PDT-3514.L5	LSD	—	C-351L-5 LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3514.L6	PDT-3514.L6	LSD	—	C-351L-6 LOW LOW OIL DIFFERENTIAL PRESSURE
TAHH-3512.H1	TT-3512.H1	LSD	—	C-351H-1 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.H2	TT-3512.H2	LSD	—	C-351H-2 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.H3	TT-3512.H3	LSD	—	C-351H-3 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.H4	TT-3512.H4	LSD	—	C-351H-4 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.H5	TT-3512.H5	LSD	—	C-351H-5 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.H6	TT-3512.H6	LSD	—	C-351H-6 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.L1	TT-3512.L1	LSD	—	C-351L-1 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.L2	TT-3512.L2	LSD	—	C-351L-2 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.L3	TT-3512.L3	LSD	—	C-351L-3 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.L4	TT-3512.L4	LSD	—	C-351L-4 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.L5	TT-3512.L5	LSD	—	C-351L-5 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3512.L6	TT-3512.L6	LSD	—	C-351L-6 HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3518.H1	TT-3518.H1	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351H-1
TAHH-3518.H2	TT-3518.H2	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351H-2
TAHH-3518.H3	TT-3518.H3	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351H-3
TAHH-3518.H4	TT-3518.H4	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351H-4
TAHH-3518.H5	TT-3518.H5	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351H-5
TAHH-3518.H6	TT-3518.H6	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351H-6
TAHH-3518.L1	TT-3518.L1	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351L-1
TAHH-3518.L2	TT-3518.L2	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351L-2
TAHH-3518.L3	TT-3518.L3	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351L-3
TAHH-3518.L4	TT-3518.L4	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351L-4
TAHH-3518.L5	TT-3518.L5	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351L-5
TAHH-3518.L6	TT-3518.L6	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER C-351L-6
TAHH-3519.L1	TT-3519.L1	LSD	—	HIGH HIGH TEMPERATURE ON C-351L-1 SUCTION
TAHH-3519.L2	TT-3519.L2	LSD	—	HIGH HIGH TEMPERATURE ON C-351L-2 SUCTION
TAHH-3519.L3	TT-3519.L3	LSD	—	HIGH HIGH TEMPERATURE ON C-351L-3 SUCTION
TAHH-3519.L4	TT-3519.L4	LSD	—	HIGH HIGH TEMPERATURE ON C-351L-4 SUCTION
TAHH-3519.L5	TT-3519.L5	LSD	—	HIGH HIGH TEMPERATURE ON C-351L-5 SUCTION
TAHH-3519.L6	TT-3519.L6	LSD	—	HIGH HIGH TEMPERATURE ON C-351L-6 SUCTION
TAHH-X357	TAHH-X-357	LSD	—	HIGH HIGH TEMPERATURE ON X-357
TAHH-X-359	TAHH-X-359	LSD	—	HIGH HIGH TEMPERATURE ON X-359
TALL-3561	TE-3501	PSD	—	SPILL DETECTION - COLD BOX BASIN
LSD09	LSD09	LSD	—	PK-351-1 EMERGENCY STOP PUSH BUTTON
LSD10	LSD10	LSD	—	PK-351-1 EMERGENCY STOP INTERLOCK
LSD11	LSD11	LSD	—	PK-351-2 EMERGENCY STOP PUSH BUTTON

ALARM	INSTRUMENT	TYPE	SP	DESCRIPTION
LSD12	LSD12	LSD	—	PK-351-2 EMERGENCY STOP INTERLOCK
LSD13	LSD13	LSD	—	PK-351-3 EMERGENCY STOP PUSH BUTTON
LSD14	LSD14	LSD	—	PK-351-3 EMERGENCY STOP INTERLOCK
LSD15	LSD15	LSD	—	PK-351-4 EMERGENCY STOP PUSH BUTTON
LSD16	LSD16	LSD	—	PK-351-4 EMERGENCY STOP INTERLOCK
LSD17	LSD17	LSD	—	PK-351-5 EMERGENCY STOP PUSH BUTTON
LSD18	LSD18	LSD	—	PK-351-5 EMERGENCY STOP INTERLOCK
LSD19	LSD19	LSD	—	PK-351-6 EMERGENCY STOP PUSH BUTTON
LSD20	LSD20	LSD	—	PK-351-6 EMERGENCY STOP INTERLOCK
PSD02	PSD02	PSD	—	LOSS OF MIX REFRIGERANT LOOP

8.6.3 Unit Cause and Effect Matrix

Table 8.11 — UNIT 350 – Cause & Effect Matrix

ALARM	INSTRUMENT	LEV-2211	LEV-2232	P-355A	P-355B	X-359	X-373
LALL-2212	LALL-2212	C	C				
LALL-3550	LALL-3550			**T**	**T**		
TAHH-X-357	TAHH-X-357						**T**
TAHH-X-359	TAHH-X-359					**T**	

Table 8.12 — PK-351-1 Cause & Effect Matrix

ALARM	INSTRUMENT	C-351H-1	C-351L-1	XEV-3511-1	XEV-3512-1	XEV-3513-1	XEV-3514-1	LSD-351-1
LSD09	LSD09	**T**	**T**	C	C	C	C	**T**
LSD10	LSD10	**T**	**T**	C	C	C	C	**T**
PALL-3511.L1	PT-3511.L1		**T**					**T**
PAHH-3512.L1	PT-3512.L1		**T**					**T**
TAHH-3512.L1	TT-3512.L1		**T**					**T**
TAHH-3512.H1	TT-3512.H1	**T**						**T**

ALARM	INSTRUMENT	C-351H-1	C-351L-1	XEV-3511-1	XEV-3512-1	XEV-3513-1	XEV-3514-1	LSD-351-1
PDALL-3514.L1	PDT-3514.L1		T					T
PDALL-3514.H1	PDT-3514.H1	T						T
TAHH-3518.L1	TT-3518.L1		T					T
TAHH-3518.H1	TT-3518.H1	T						T
TAHH-3519.L1	TT-3519.L1		T					T

Table 8.13 — PK-351-2 Cause & Effect Matrix

ALARM	INSTRUMENT	C-351H-2	C-351L-2	XEV-3511-2	XEV-3512-2	XEV-3513-2	XEV-3514-2	LSD-351-2
LSD11	LSD11	T	T	C	C	C	C	T
LSD12	LSD12	T	T	C	C	C	C	T
PALL-3511.L2	PT-3511.L2		T					T
PAHH-3512.L2	PT-3512.L2		T					T
TAHH-3512.L2	TT-3512.L2		T					T
TAHH-3512.H2	TT-3512.H2	T						T
PDALL-3514.L2	PDT-3514.L2		T					T
PDALL-3514.H2	PDT-3514.H2	T						T
TAHH-3518.L2	TT-3518.L2		T					T
TAHH-3518.H2	TT-3518.H2	T						T
TAHH-3519.L2	TT-3519.L2		T					T

Table 8.14 — PK-351-3 Cause & Effect Matrix

ALARM	INSTRUMENT	C-351H-3	C-351L-3	XEV-3511-3	XEV-3512-3	XEV-3513-3	XEV-3514-3	LSD-351-3
LSD13	LSD13	T	T	C	C	C	C	T
LSD14	LSD14	T	T	C	C	C	C	T
PALL-3511.L3	PT-3511.L3		T					T

ALARM	INSTRUMENT	C-351H-3	C-351L-3	XEV-3511-3	XEV-3512-3	XEV-3513-3	XEV-3514-3	LSD-351-3
PAHH-3512.L3	PT-3512.L3		T					T
TAHH-3512.L3	TT-3512.L3		T					T
TAHH-3512.H3	TT-3512.H3	T						T
PDALL-3514.L3	PDT-3514.L3		T					T
PDALL-3514.H3	PDT-3514.H3	T						T
TAHH-3518.L3	TT-3518.L3		T					T
TAHH-3518.H3	TT-3518.H3	T						T
TAHH-3519.L3	TT-3519.L3		T					T

Table 8.15 — PK-351-4 Cause & Effect Matrix

ALARM	INSTRUMENT	C-351H-4	C-351L-4	XEV-3511-4	XEV-3512-4	XEV-3513-4	XEV-3514-4	LSD-351-4
LSD15	LSD15	T	T	C	C	C	C	T
LSD16	LSD16	T	T	C	C	C	C	T
PALL-3511.L4	PT-3511.L4		T					T
PAHH-3512.L4	PT-3512.L4		T					T
TAHH-3512.L4	TT-3512.L4		T					T
TAHH-3512.H4	TT-3512.H4	T						T
PDALL-3514.L4	PDT-3514.L4		T					T
PDALL-3514.H4	PDT-3514.H4	T						T
TAHH-3518.L4	TT-3518.L4		T					T
TAHH-3518.H4	TT-3518.H4	T						T
TAHH-3519.L4	TT-3519.L4		T					T

Table 8.16 — PK-351-5 Cause & Effect Matrix

ALARM	INSTRUMENT	C-351H-5	C-351L-5	XEV-3511-5	XEV-3512-5	XEV-3513-5	XEV-3514-5	LSD-351-5
LSD17	LSD17	T	T	C	C	C	C	T
LSD18	LSD18	T	T	C	C	C	C	T
PALL-3511.L5	PT-3511.L5		T					T
PAHH-3512.L5	PT-3512.L5		T					T
TAHH-3512.L5	TT-3512.L5		T					T
TAHH-3512.H5	TT-3512.H5	T						T
PDALL-3514.L5	PDT-3514.L5		T					T
PDALL-3514.H5	PDT-3514.H5	T						T
TAHH-3518.L5	TT-3518.L5		T					T
TAHH-3518.H5	TT-3518.H5	T						T
TAHH-3519.L5	TT-3519.L5		T					T

Table 8.17 — PK-351-6 Cause & Effect Matrix

ALARM	INSTRUMENT	C-351H-6	C-351L-6	XEV-3511-6	XEV-3512-6	XEV-3513-6	XEV-3514-6	LSD-351-6
LSD19	LSD19	T	T	C	C	C	C	T
LSD20	LSD20	T	T	C	C	C	C	T
PALL-3511.L6	PT-3511.L6		T					T
PAHH-3512.L6	PT-3512.L6		T					T
TAHH-3512.L6	TT-3512.L6		T					T
TAHH-3512.H6	TT-3512.H6	T						T
PDALL-3514.L6	PDT-3514.L6		T					T
PDALL-3514.H6	PDT-3514.H6	T						T
TAHH-3518.L6	TT-3518.L6		T					T
TAHH-3518.H6	TT-3518.H6	T						T
TAHH-3519.L6	TT-3519.L6		T					T

8.7 Operation and Control

The Mixed Refrigerant (MR) system operates with hydrocarbons under high pressure and at extremely low temperatures. Loss of containment may result in the release of flammable gas or cryogenic liquid. Contact with cryogenic fluids or cold surfaces can cause severe frostbite and cold burns. Vapor releases may displace oxygen and create an asphyxiation hazard in confined or poorly ventilated areas. Mixed Refrigerant vapors are highly flammable and may form explosive atmospheres if exposed to ignition sources. In case of leakage, immediately evacuate the area, eliminate potential ignition sources, and follow the plant emergency response procedure.

8.7.1 Compressor Capacity Control

The capacity control of the mixed refrigerant screw compressors (C-351L-1/2/3/4/5/6 and C-351H-1/2/3/4/5/6) is achieved by means of an internal slide valve mechanism, which provides continuous modulation of compressor capacity over a wide operating range.

The slide valve moves axially along the rotor casing and modifies the effective compression length. By shifting its position, part of the refrigerant gas trapped between the rotors is internally returned to the suction side before compression is completed. This reduces the effective compression volume and consequently decreases compressor capacity.

The slide valve is hydraulically actuated using the compressor lubrication oil system. The movement of the valve is controlled by dedicated solenoid valves that direct oil flow to either side of the hydraulic actuator, allowing accurate positioning of the slide valve.

The capacity control system operates as follows:

- When an increase in refrigeration capacity is required, the slide valve moves towards the discharge side, increasing the effective compression volume.
- When a reduction in refrigeration capacity is required, the slide valve moves towards the suction side, allowing part of the refrigerant gas to be internally recycled before compression is completed.
- The compressor can typically operate over a capacity range from 100% down to approximately 15–30% of its rated capacity.

The slide valve position is automatically adjusted according to the refrigeration demand of the liquefaction process. Capacity modulation maintains stable operating conditions throughout the mixed refrigerant loop while minimizing energy consumption at part-load operation.

For normal operation, each compressor package regulates its capacity by continuously adjusting the slide valve position in response to process conditions. When multiple compressor packages are operating in parallel, the individual capacity control systems collectively maintain the required mixed refrigerant circulation and suction pressure within the operating limits of the refrigeration system.

This method of capacity control offers several advantages:

- High energy efficiency compared with external recycle or hot-gas bypass control methods.
- Smooth and continuous capacity modulation without step changes.
- Stable operation over a wide range of process loads.
- Reduced power consumption during part-load operation.

During start-up, the compressor is maintained in a fully unloaded condition to minimize starting torque and motor current requirements. After start-up, the slide valve is progressively loaded until the required refrigeration duty is achieved.

The slide valve control system is interlocked with the compressor lubrication and protection systems to ensure safe operation. Abnormal operating conditions such as low oil pressure, compressor trips, or other package protection actions will automatically unload and/or stop the compressor in accordance with the package safeguarding philosophy and the Cause & Effect Matrix.

8.7.2 MR Compression System Capacity Control

The Mixed Refrigerant (MR) compression system is designed to provide the refrigeration duty required by the liquefaction process while maintaining stable operating conditions throughout the MR loop.

The system consists of six independent MR compressor packages (PK-351-1 to PK-351-6), each including a Low Pressure (LP) screw compressor and a High Pressure (HP) screw compressor operating as a single compression train. Compressor capacity is regulated through the internal hydraulic slide valve installed in each compressor.

To accommodate varying plant loads and operating conditions, the MR compression system can operate under two alternative capacity control strategies:

Pressure and Temperature Based Capacity Control (XC-3501A)

In this operating mode, the compressor capacity is determined from the MR suction pressure and suction temperature conditions.

The controller XC-3501A receives the following process variables:

- PT-3501 – MR Suction Pressure
- TT-3501 – MR Suction Temperature

Based on these measurements, XC-3501A calculates the required compressor loading and modulates the compressor slide valve position through the hydraulic actuator and associated solenoid valve assembly (Y1–Y4).

When the refrigeration demand falls below the minimum compressor slide valve operating range, controller XC-3501A modulates hot gas bypass valve PV-3581. The valve recirculates a portion of the mixed refrigerant from the high-pressure side of the cycle back to the MR Suction KO Drum V-358, allowing further capacity reduction while maintaining stable compressor operation, adequate refrigerant circulation, and the desired MR suction pressure and temperature conditions.

Feed Flow Based Capacity Control (XC-3501B)

In this operating mode, the required MR compressor capacity is determined directly from the natural gas feed flow entering the liquefaction process.

The controller XC-3501B receives the feed gas flow signal from FIC-3560A and calculates the refrigeration demand required by the liquefaction unit. Based on this demand, the controller determines the overall compressor capacity required to support the process operating conditions.

8.7.3 Compressor Panel Operation

The lubrication system is critical for the operation of oil-flooded screw compressors. Failure or inadequate performance of the lubrication system may result in severe compressor damage.

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The liquefaction unit (UNIT 350) is equipped with local control panels associated with each mixed refrigerant compressor package (PK-351-1/2/3/4/5/6).

Each local control panel allows the operator to select the operating mode (Local or Remote) for both the LP compressor (C-351L-x) and the HP compressor (C-351H-x) by means of selector switches:

- **Remote mode**: start and stop commands are issued from the Process Control System (PCS).
- **Local mode**: the LP and HP compressors can be operated directly from the local control panel.

When Local mode is selected, dedicated **START** and **STOP** pushbuttons are provided for both the LP and HP compressors. These pushbuttons allow the operator to independently initiate the normal start-up and shutdown sequences of each compressor. A dedicated Emergency Shutdown (ESD/LSD) pushbutton is also provided on the panel. Activation of this pushbutton initiates a local shutdown of the compressor package in accordance with the Cause & Effect Matrix.

The panel includes status indications for both compressors, including:

- **Running** status
- **Ready to Start** indication
- **Alarm / Trip** indication

These indications provide the operator with real-time feedback on the availability and operating condition of the LP and HP compressors, ensuring safe interaction between local operation and PCS supervision.

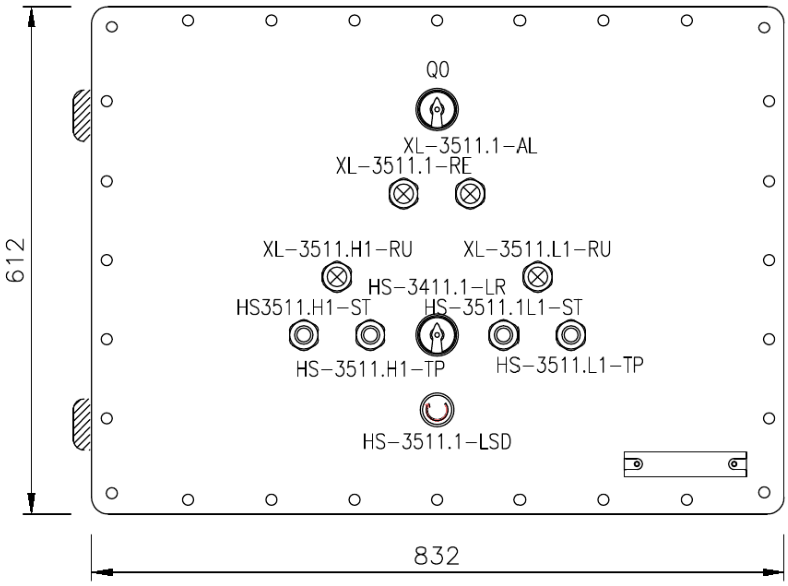


Figure 8.11: MR compressor local control panel

Table 8.18 — MR Compressor package local panel components

TAG	SIGNAL	DESCRIPTION
HS-3511.x-LR	LOCAL REMOTE SELECTOR	Select local remote control
HS-3511.x.L1-ST	LOCAL START	Local start command compressor LP
HS-3511.x.H1-ST	LOCAL START	Local start command compressor HP
HS-3411.x.L1-TP	LOCAL STOP	Local stop command compressor LP
HS-3411.x.H1-TP	LOCAL STOP	Local stop command compressor HP
HS-3411.x-LSD	LOCAL SHUTDOWN	Local emergency shutdown
XL-3411.x.L1-RU	RUN	Compressor LP running
XL-3411.x.H1-RU	RUN	Compressor HP running
XL-3411.x-RE	READY	Package ready to run
XL-3411.x-AL	ALARM	Package alarm or trip active

8.7.4 Mixed Refrigerant Refrigeration Loop

The mixed refrigerant (MR) refrigeration system (UNIT 350) operates as a closed-loop refrigeration cycle providing the cooling and liquefaction duty required in the cryogenic heat exchangers of Cold Box CB-356.

The refrigerant consists of a proprietary mixture of hydrocarbons optimized to provide refrigeration over a wide temperature range. Low-pressure MR vapor returning from the cold boxes is compressed by the mixed refrigerant compressor packages (PK-351-1/2/3/4/5/6), cooled in the air-cooled precoolers (E-353), and further condensed in the ammonia-cooled exchanger (E-354) using refrigeration supplied by the R1 ammonia system.

The condensed refrigerant is separated and distributed through the high-pressure and low-pressure MR separators (V-355A/B) before entering the cryogenic plate-fin heat exchangers (E-356A/B), where it undergoes expansion and evaporation, generating the refrigeration required to cool, condense, and liquefy the natural gas stream.

After providing refrigeration duty in the cold boxes, the vaporized mixed refrigerant is collected in the suction knockout drum (V-358) and returned to the compressor suction, completing the refrigeration cycle.

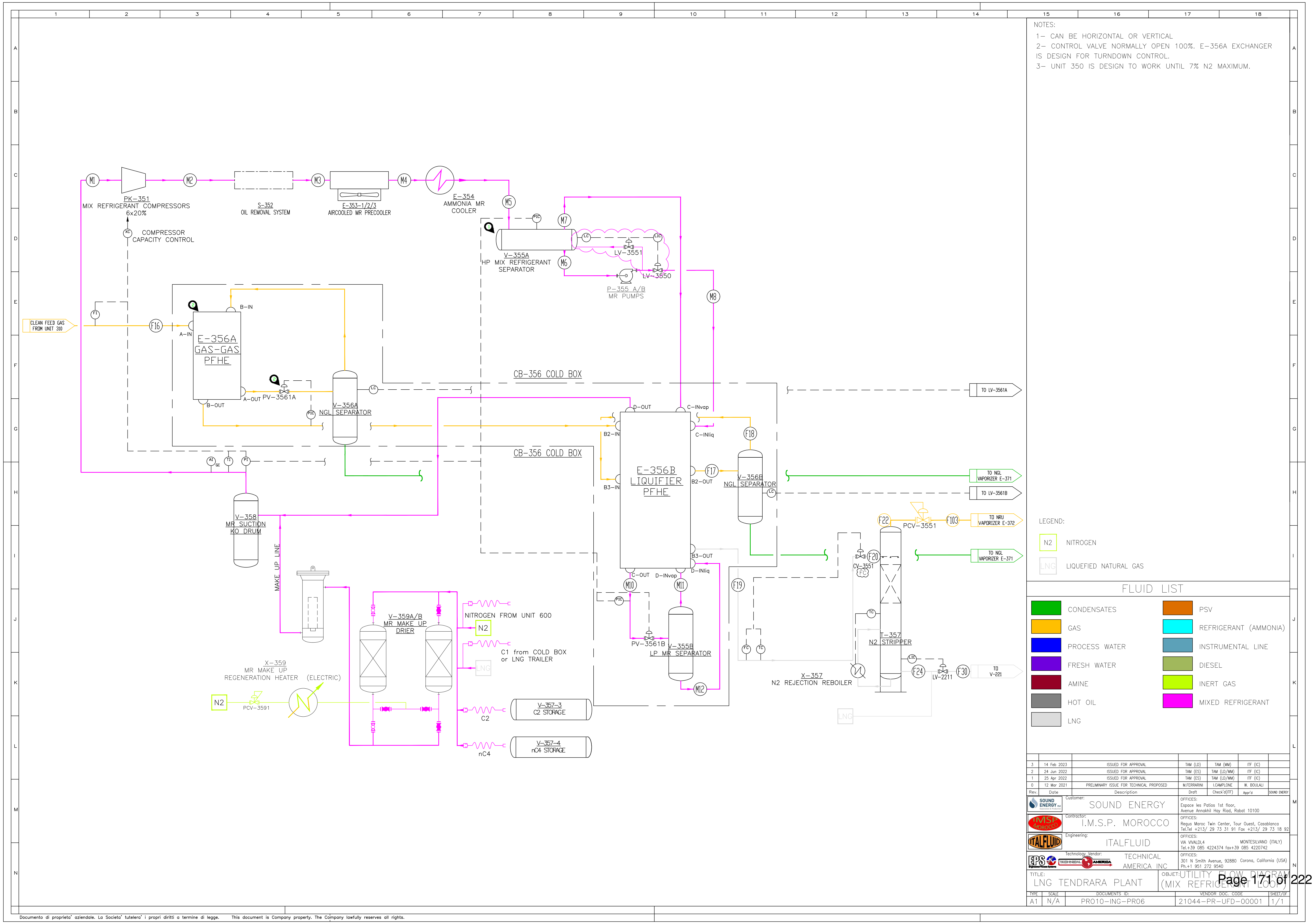


Figure 8.12: Unit 350 – Mixed Refrigerant Loop

8.7.5 Lube Oil System

The lubrication oil system of the mixed refrigerant compressor packages (PK-351-1/2/3/4/5/6) is a critical subsystem designed to ensure reliable compressor operation by providing lubrication, cooling, sealing, and hydraulic actuation of the compressor slide valves.

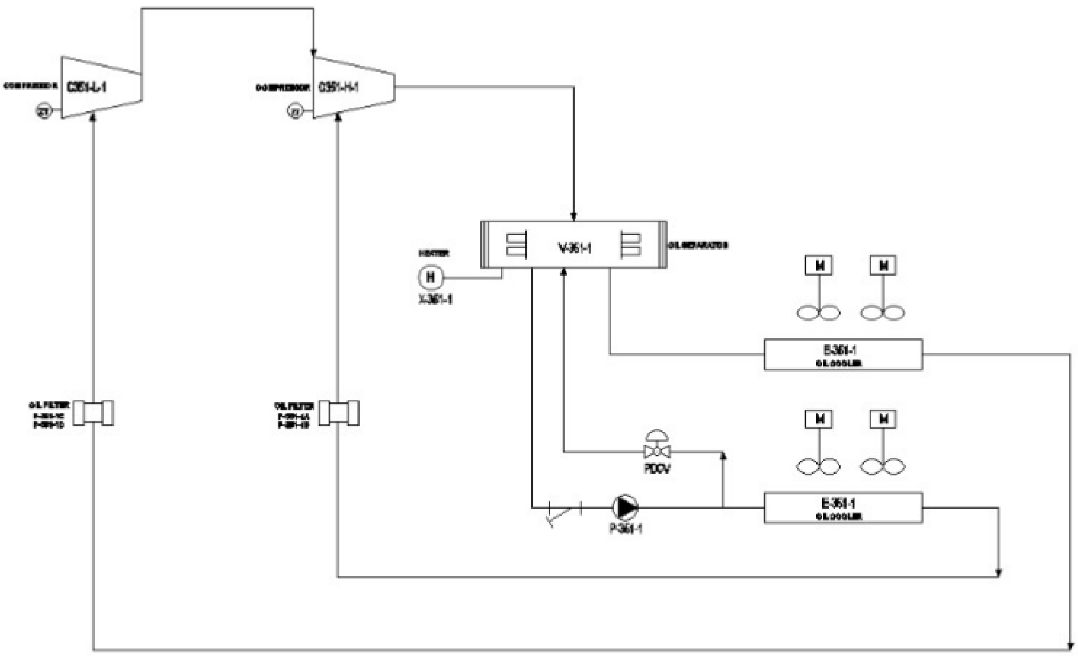


Figure 8.13: MR compressor lube oil system

System Description

The lube oil system performs the following main functions:

- Removal of heat generated during the compression process.
- Lubrication of bearings and moving components.
- Sealing between compressor rotors to improve compression efficiency.

During operation, the lube oil is injected into the compressor together with the refrigerant gas. The oil-gas mixture is discharged from the compressor and routed to the oil separator (V-351-1), where the oil is separated and collected.

The oil separator also serves as the oil reservoir for the system, ensuring adequate oil inventory during operation.

From the separator, the oil is circulated back to the compressor through the lube oil circuit.

Oil Circulation and Conditioning

The oil circulation system includes the following main equipment:

- **Oil Pump (P-351-1):** Positive displacement gear type pump (1 x 100%), supplying pressurized oil to both LP and HP compressors.
- **Oil Cooler (E-351-1):** Removes heat from the oil to maintain proper viscosity and cooling performance.
- **Oil Filters (F-351-1A/B):** Duplex configuration (1 operating + 1 standby) allowing continuous operation during filter replacement and ensuring clean oil supply.
- **Oil Separator (V-351-1):** Separates lubricating oil from the compressed mixed refrigerant and provides oil inventory for the lubrication system.
- **Oil Heater (X-351-1):** Maintains oil temperature during shutdown periods and prevents refrigerant dilution in the lubricant.

Oil is separated from the compressed mixed refrigerant in the oil separator, circulated through the oil cooler and duplex filters, and then supplied to both LP and HP compressors for lubrication, sealing, cooling, and slide valve actuation.

Control and Protection

The lubrication system is monitored and protected by the control system through key parameters:

- Oil differential pressure across the compressor.
- Oil temperature downstream of the cooler.
- Oil level in the separator.

A minimum oil differential pressure is required to ensure proper lubrication. If the oil differential pressure drops below 1.5 bar, the compressor start sequence is inhibited or the running compressor is tripped.

This protection is critical to avoid damage to the bearings, rotors, slide valves, and internal components of both the LP and HP compressors.

Operational Considerations

- The oil pump must be in operation prior to compressor start-up.

- Oil temperature shall be within the specified range before loading the compressor.
- Filters shall be monitored and replaced when differential pressure increases.
- Stable oil pressure and temperature are required for proper operation of the LP and HP compressor slide valves.

The oil pump is equipped with a magnetic coupling generating a strong magnetic field.

Personnel with implanted medical devices (e.g. pacemakers) shall not approach the equipment.

Maintain a safe distance from the magnetic coupling and avoid placing magnetic or sensitive electronic devices nearby.

8.7.6 Magnetic Coupling

The lube oil pump (P-341-1) is equipped with a magnetic coupling, which transmits torque from the motor to the pump without a direct mechanical shaft connection. This design eliminates the need for dynamic seals and ensures leak-free operation, which is particularly suitable for ammonia service.

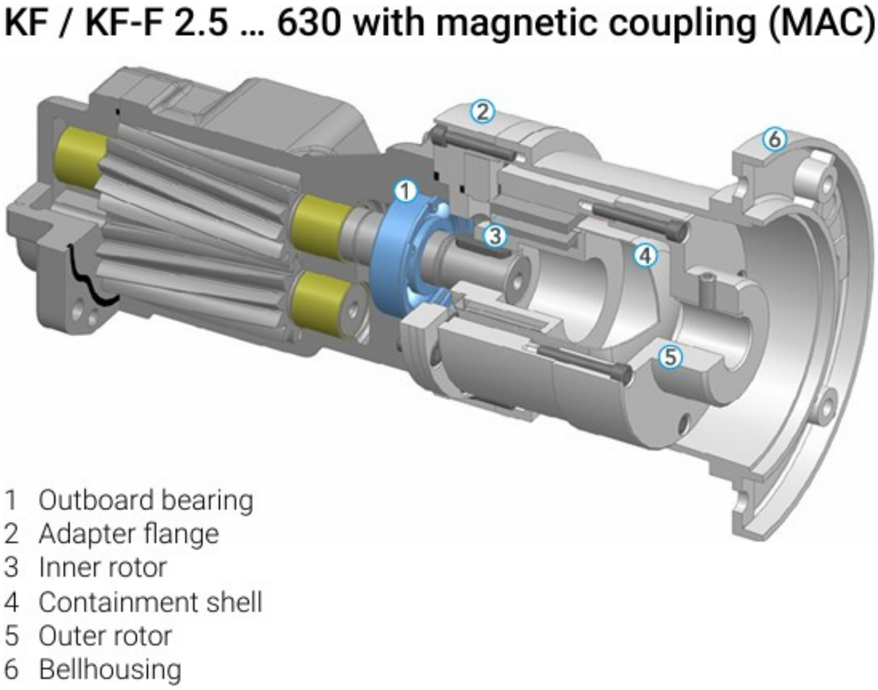


Figure 8.14: Oil pump magnetic coupling

The magnetic coupling consists of an outer rotor connected to the motor shaft and an inner rotor connected to the pump shaft. Torque is transmitted through magnetic forces across a containment shell, allowing complete separation between the driven and driving components.

This configuration provides several advantages, including improved reliability, reduced maintenance requirements, and elimination of potential leakage points. It is particularly suitable for hazardous fluids, as it minimizes the risk of external release.

Operationally, the magnetic coupling has a maximum torque transmission limit. In case of overload, magnetic decoupling may occur, resulting in the pump rotating without effective torque transmission. This condition may lead to loss of oil circulation and must be detected promptly. Dry running conditions shall be strictly avoided, as they can damage internal pump components.

8.7.7 Compressor Start-Up Permissives

Before initiating the automatic start-up sequence of any mixed refrigerant compressor package in Unit 350, the following start-up permissives shall be confirmed:

- The selected LP and HP compressor motors to be available in **AUTO MODE**.
- The lube oil pump to be available in **AUTO MODE**.
- All oil cooler fans to be available in **AUTO MODE**.
- The LP and HP compressor capacity control / slide valves to be in **AUTO MODE**.
- Compressor suction and discharge isolation valves to be confirmed **open**.
- Lube oil temperature to be within the normal start-up range.
- Both compressors to be in **unloaded condition** before start.
- Associated solenoid valves to be in their normal **AUTO MODE**.
- No active shutdowns, trips, or critical alarms present on the selected package.
- Motor restart inhibit / cool-down timers to be expired.

These permissives are intended to ensure that the compressor package starts under safe mechanical and process conditions, with the lubrication system, cooling system, and capacity control system available before the compressor motors are energized. This is consistent with the Unit 350 mixed refrigerant refrigeration package configuration and with the plant shutdown and safety philosophy for package operation.

8.7.8 Compressor Start-Up Sequence

After completion of the mixed refrigerant system preparation and verification of all start-up permissives, the selected MR compressor package (PK-351-x) can be started following the automatic sequence executed by the PCS.

The start-up sequence is designed to establish proper lubrication, start the HP compressor first, stabilize its operation at minimum load, and subsequently start the LP compressor. Once both compressor stages are running and stabilized, the automatic capacity control system is enabled and the package transitions to normal operating mode.

Table 8.19 — Unit 350 – MR compressor automatic start-up sequence

Step	Tag	SP		Action	Description
		Value	Unit		
1	HS-351.x-ST	–	–	OC	Start automatic sequence from local panel or PCS after all start-up permissives are validated.
2	P-351-x	–	–	SC	System command START auxiliary lube oil pump to establish oil circulation for the selected MR compressor package.

Step	Tag	SP		Action	Description
		Value	Unit		
3	PDI-351.x	1.5	bar	SC	System command CHECK lube oil differential pressure $\geq$ 1.5 bar before allowing compressor start.
4	ZT-351H.x	<5	%	SC	System command SET HP compressor slide valve to fully unloaded position and confirm feedback.
5	E-351-x	–	–	SC	System command START oil cooler fans to maintain lube oil temperature within operating range.
6	MC-351H.x	–	–	SC	System command START HP compressor main motor and accelerate to nominal speed.
7	ZT-351H.x	15	%	SC	System command MOVE HP compressor slide valve to minimum load position.
8	C-351H.x	30	s	CK	Maintain HP compressor at minimum load during idling / stabilization time.
9	PIC-351.x	–	–	SC	Enable HP compressor automatic capacity control; system waits for LP compressor start sequence.
10	PDI-351L.x	1.5	bar	SC	System command CHECK LP compressor lube oil differential pressure $\geq$ 1.5 bar.
11	ZT-351L.x	<5	%	SC	System command SET LP compressor slide valve to fully unloaded position and confirm feedback.
12	MC-351L.x	–	–	SC	System command START LP compressor main motor and accelerate to nominal speed.
13	ZT-351L.x	15	%	SC	System command MOVE LP compressor slide valve to minimum load position.
14	C-351L.x	30	s	CK	Maintain LP compressor at minimum load during idling / stabilization time.
15	PIC-351.x	–	–	SC	Enable LP compressor automatic capacity control. Both LP and HP compressors operate under automatic capacity modulation.
16	PK-351-x	–	–	CK	MR compressor package reaches steady-state operation with LP and HP compressors running, lube oil system in service, and control loops enabled.

8.7.9 Compressor Shutdown Sequence

After confirmation of stable operating conditions and reduction of refrigeration load, the MR compressor package (PK-351-x) can be stopped following the automatic shutdown sequence executed by the PCS.

The shutdown sequence is performed in reverse order of the start-up sequence. The LP compressor is unloaded and stopped first, followed by the HP compressor. Post-lubrication is maintained after each compressor stop to ensure adequate bearing and rotor protection before the lubrication system is isolated.

Table 8.20 — Unit 350 – MR compressor automatic shutdown sequence

Step	Tag	SP		Action	Description
		Value	Unit		
1	HS-351.x-SP	–	–	OC	Start automatic shutdown sequence from local panel or PCS.
2	PIC-351.x	–	–	SC	System command DISABLE LP compressor capacity control and unload compressor.
3	ZT-351L.x	<5	%	SC	System command SET LP compressor slide valve to fully unloaded position and confirm feedback.
4	MC-351L.x	–	–	SC	System command STOP LP compressor main motor and confirm de-energization.
5	P-351-x	–	–	CK	Lube oil pump remains in service to provide post-lubrication to the LP compressor.
6	–	30	s	CK	Maintain post-lubrication period for LP compressor.
7	PEV-3511.x	–	–	SC	System command CLOSE oil pressure equalization valve and isolate LP lubrication circuit.
8	MC-351L.x	–	–	CK	Confirm LP compressor fully stopped.
9	PIC-351.x	–	–	SC	System command DISABLE HP compressor capacity control and unload compressor.
10	ZT-351H.x	<5	%	SC	System command SET HP compressor slide valve to fully unloaded position and confirm feedback.
11	MC-351H.x	–	–	SC	System command STOP HP compressor main motor and confirm de-energization.
12	P-351-x	–	–	CK	Lube oil pump remains in service to provide post-lubrication to the HP compressor.
13	–	30	s	CK	Maintain post-lubrication period for HP compressor.
14	PEV-3511.x	–	–	SC	System command CLOSE oil pressure equalization valve and isolate HP lubrication circuit.
15	MC-351H.x	–	–	CK	Confirm HP compressor fully stopped.
16	PK-351-x	–	–	CK	MR compressor package shutdown completed. LP and HP compressors are stopped and all shutdown permissives are reset.

8.8 Troubleshooting

Table 8.21 — Unit 350 – MR Compressor Troubleshooting

TROUBLE	POSSIBLE CAUSE	SOLUTION
Complete loss of power	Main power switch disconnected	Check all power wiring
	Main fuse failure	Replace the burned-out fuse
Partial loss of power	Auxiliary fuse failure	Replace the burned-out fuse
	Power disconnected to auxiliary equipment	Check wiring to auxiliary equipment
Oil pump will not run in HAND mode	Overload open	Reset overload
	Compressor is running	Oil pumps must be in AUTO while compressor is operating
	Main power off	Check starter disconnect and power supply
	Panel control power off	Turn panel control power on
	Motor burned out	Inspect and repair or replace motor
	Pump rotation reversed	Check motor rotation and correct if required
Oil pump will not run in AUTO mode	Compressor start has not been initiated	Initiate compressor start sequence
	Pump not developing sufficient pressure	Adjust pressure differential control valve to 30–60 psid
	Oil too cold	Warm oil to approximately 100°F
	Pump rotation reversed	Check motor rotation and correct if required
Oil pump runs in AUTO mode but compressor will not start	Lube oil supply pressure too low	Adjust pressure differential control valve to 30–60 psid
	Compressor motor wired incorrectly	Check and correct motor wiring
	Slide valve not at less than 5% position	Confirm compressor is fully unloaded and hydraulic oil block valves are open
Oil pump runs in AUTO mode, compressor starts, but stops immediately	Motor starter or auxiliary contact wired incorrectly	Check and correct wiring; auxiliary contact must seal within 3 seconds

TROUBLE	POSSIBLE CAUSE	SOLUTION
	Error in compressor start logic	Check and correct control logic
High or low lube oil temperature	High suction temperature	Check operating conditions and correct
	Thermal expansion valve not functioning properly	Check and correct valve operation
Low lube oil pressure	Oil pump not developing enough pressure	Adjust PDCV to 50–60 psid at regulator inlet; isolate PDCV and open bypass to verify pressure build-up
	Oil too cold	Run pump in HAND mode to warm oil to at least 90°F; close compressor suction valve while warming
	Oil pressure switch defective or incorrectly set	Check switch setting
	Control panel contact not closing properly or loose wiring	Check and correct
	Lube oil filter clogged	Check and replace filter if necessary
	Low oil level in oil reservoir	Add oil to vessel
Bad oil quality	Solids in oil	Drain oil and refill with clean oil; flush system if necessary; analyze oil and eliminate source of contamination
High oil carryover	Coalescer element loose	Tighten and mount coalescer properly; verify differential pressure across coalescer is not less than 0.25 psid
High differential pressure across oil filter	Coalescer drain line choked	Check for oil accumulation in sight glass and open globe valve further to restore oil flow
	Oil filter plugged	Check and replace oil filter if necessary
Decrease in oil level in oil reservoir	Oil leakage from system	Check and correct leaks
	Coalescers not tight enough to seal	Tighten coalescers
	Oil return line from oil separator blocked	Adjust or clear block valve / return line
	Excess leakage from compressor seal	Check and correct

TROUBLE	POSSIBLE CAUSE	SOLUTION
	Oil leaving system through a relief valve	Check all relief valve discharge piping for oil presence
Abnormal noise	Bearings damaged	Replace bearings
	Loose connections	Stop compressor and tighten connections
	Foreign material entered compressor	Stop compressor; check suction strainer and compressor internals
	Misalignment or damaged shaft coupling	Stop compressor; check alignment and coupling condition
	Rotors rubbing against casing inner surface	Stop and disassemble compressor; replace damaged parts and correct internal clearances

9 Unit 360 - Flash Gas / BOG Recovery Unit (PK-361 and PK-362).

9.1 Operating Summary

CATEGORY	DESCRIPTION
Inputs	Boil-Off Gas (BOG) from LNG storage tank TA-221 and LNG separator V-221.
	Low-pressure cryogenic vapor returned from LNG truck loading operations.
Outputs	Conditioned and superheated BOG routed to the plant fuel gas system.
	Excess BOG safely vented or recycled according to operating mode and availability.
Key Operating Specs	Continuous handling of LNG boil-off gas generated by heat ingress and loading operations.
	Controlled pressure stabilization of LNG storage and transfer systems.
	BOG temperature conditioning via electric superheater prior to fuel gas utilization.
	Automatic isolation and shutdown on high pressure or abnormal temperature conditions.
	Integration with LNG storage pressure control and plant-wide fuel gas balance.
Major Equipment	Flash Gas and Boil-Off Gas (BOG) compression packages PK-361 and PK-362, comprising duty/standby screw compressors and complete auxiliary systems.
	Flash/BOG compressors PK-361-C1A/B and BOG compressors PK-362-C1A/B for recovery and pressurization of low-pressure cryogenic vapor streams.
	BOG superheater PK-361-E2 (BEU type) for controlled heating of compressed vapor prior to routing to the plant fuel gas system.
	Lubrication and oil management systems including oil separators PK-361-V1 and PK-362-V1, oil pumps PK-361-P1A/B and PK-362-P1A/B, main oil filters PK-361-F1A/B and PK-362-F1A/B, coalescing separator PK-362-V2, charcoal adsorber PK-362-V3, and oil heaters X-361 and X-362.
	Air-cooled oil cooling systems including oil coolers PK-361-E1 and PK-362-E1 with associated fan assemblies FPK-361-E1 and FPK-362-E1 series.

9.2 General Arrangement and Risk Location

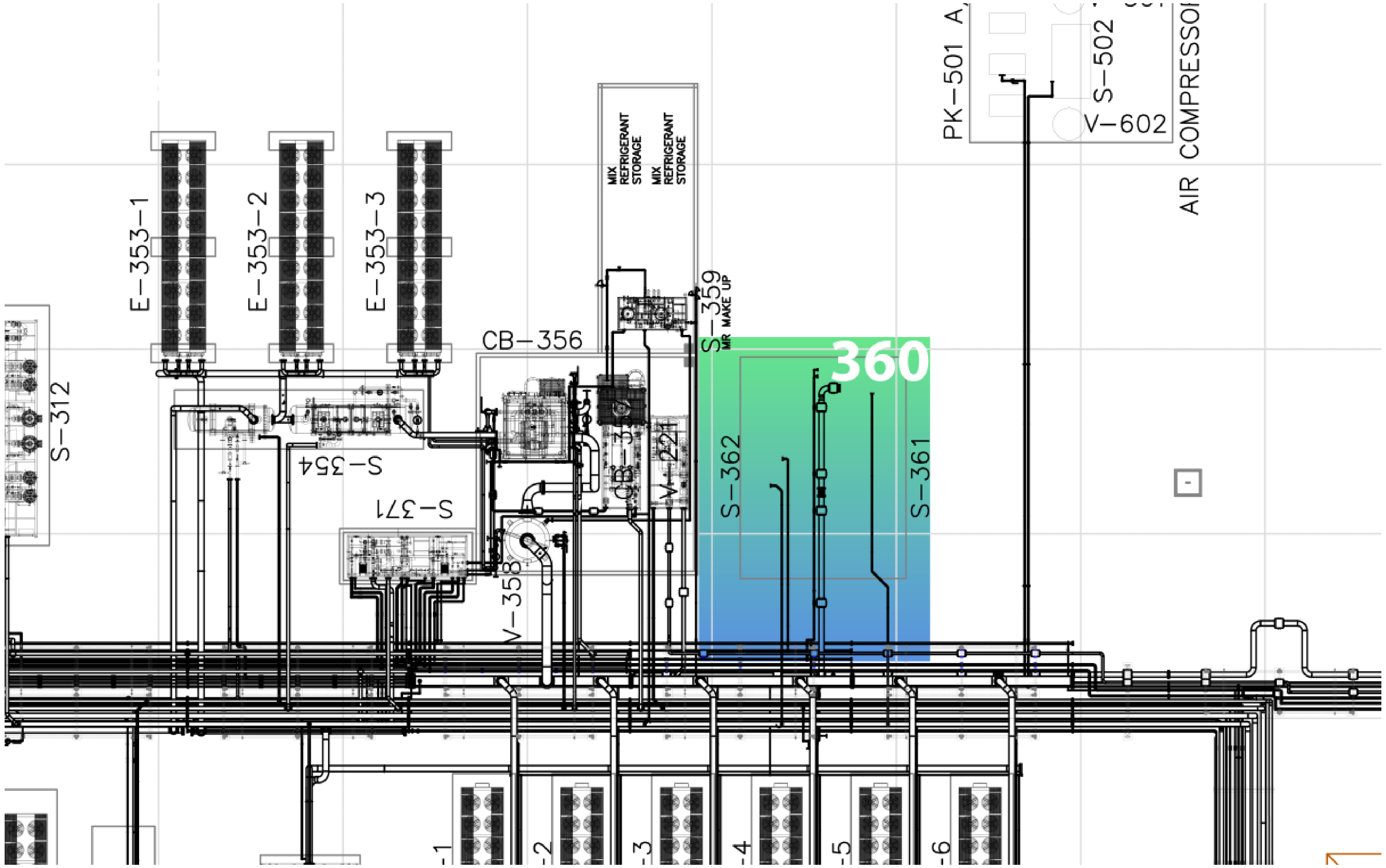


Figure 9.1: Unit 360 GA - Risk location

9.3 Relevant Documents

Table 9.2 — Unit 360 – Relevant Documents

ID	DOC NO	TITLE
1	PRO02-ING-FR02	PROCESS DESCRIPTION
2	PRO10-ING-PR34	PROCESS FLOW DIAGRAMS
3	PRO13-ING-PR29	PK-361-E2
4	PRO13-ING-PR30	SKID S-361
5	PRO13-ING-PR31	OIL SEPARATOR S-361
6	PRO13-ING-PR32	SKID S-362
7	SAF12-ING-FR01	CAUSE AND EFFECT MATRIX
8	SAF05-ING-FR01	RELIEF AND BLOWDOWN PHILOSOPHY
9	SAF01-ING-FR01	SAFETY AND ISOLATION PHILOSOPHY

9.4 Process Description

Area 360 handles all flash gas and boil-off gas (BOG) generated downstream of liquefaction and LNG storage, recovering these streams as useful fuel gas or recycling them upstream of dehydration in order to minimize flaring and stabilize overall plant operation.

Flash gas from the LNG separator V-221 and boil-off gas generated from LNG storage and truck loading operations are routed to the BOG compression system. These low-temperature gas streams first enter the suction superheater PK-361-E2, where controlled superheating ensures suitable compressor suction conditions, with a design minimum temperature of $-60\,°$C. The gas is then compressed in the BOG compressor package PK-361-C1A/B, arranged as one working and one standby unit, to a discharge pressure in the range of approximately 6–11 barg.

Fuel Gas Recovery and Routing

The compressed BOG stream is automatically routed based on fuel-gas network conditions. When the fuel-gas header pressure is within the required operating range, the gas is directed to the fuel-gas network through the discharge pressure control valve PCV-3602, supplying downstream consumers such as the power generation system and the hot-oil unit. Under all operating scenarios, fuel-gas supply has operational priority.

When fuel-gas demand is satisfied, header pressure tends to increase, or upstream recycle is required for process balancing, the BOG recycle compressor package PK-362-C1A/B, also referred to as S-362 in the Process Description, is started.

Recycle Compression and Dehydration Regeneration

In recycle mode, the PK-362 compressor package takes suction from the compressed BOG stream downstream of PK-361 and further compresses the gas before reinjecting it upstream of Unit 310 (Dehydration). This recycled dry gas provides an intermittent regeneration gas source for the molecular sieve dehydration system via exchangers X-312 and E-313 and separator V-312.

This operating mode supports self-balancing between fuel-gas demand and process recycle requirements, allowing dehydration regeneration without routine flaring and enhancing overall plant efficiency.

Compressor Package Design and Control

The PK-361 compressor package comprises an oil-flooded screw compressor train equipped with multistage oil separation in vessel PK-361-V1, an air-cooled oil cooler PK-361-E1, dual oil pumps and oil filters, and a BEU-type suction superheater PK-361-E2. The package also includes associated discharge desuperheating to ensure acceptable downstream gas temperatures.

A hot-gas recycle and bypass control valve CV-3601 is provided to enable start-up recirculation, compressor suction protection, and stable operation under low-load conditions.

The PK-362 recycle compressor package is of similar oil-flooded screw design and includes high-efficiency oil removal and duplex gas filtration to ensure cryogenic-grade gas cleanliness prior to reinjection into the process gas system.

Both compressor packages are equipped with integral slide-valve capacity control, providing an approximate turndown range from 100% to 15% of rated capacity. Operation at very low loads is further supported by hot-gas bypass control, allowing stable compressor operation across the full operating envelope.

Overall Operating Philosophy

This configuration allows the plant to accommodate variable flash gas generated in the LNG separator V-221, together with boil-off gas from the LNG storage tank and vapor return streams from truck loading operations. The system maintains a stable fuel-gas supply as the first operational priority, enables intermittent regeneration of the dehydration system when required, and minimizes flaring under all normal operating conditions.

9.5 Technological Objects

9.5.1 Main Equipment List

Table 9.3 — Unit 360 – Main Equipment List

ID	TAG	EQUIPMENT
1	**PK-361**	**FLASH / BOG COMPRESSOR PACKAGE**
1.1	PK-361-C1A	Flash / BOG compressor A
1.2	PK-361-C1B	Flash / BOG compressor B
1.3	PK-361-V1	Oil separator
1.4	PK-361-E1	Oil cooler
1.4.1	FPK-361-E1-1..4	Oil cooler fans (4)
1.5	PK-361-E2	BOG superheater (BEU type)
1.6	PK-361-P1A/B	Oil pumps
1.7	PK-361-F1A/B	Main oil filters A/B
1.8	X-361	Oil heater
2	**PK-362**	**BOG RECYCLE COMPRESSOR PACKAGE**
2.1	PK-362-C1A	BOG recycle compressor A
2.2	PK-362-C1B	BOG recycle compressor B
2.3	PK-362-V1	Oil separator
2.4	PK-362-E1	Oil cooler
2.4.1	FPK-362-E1-1..n	Oil cooler fans
2.5	PK-362-P1A/B	Oil pumps
2.6	PK-362-F1A/B	Main oil filters A/B
2.7	PK-362-V2	Coalescing separator
2.8	PK-362-V3	Charcoal adsorber
2.9	PK-362-F4A/B	Adsorber after filters A/B
2.10	X-362	Oil heater

9.5.2 Electrical Loads

Table 9.4 — Unit 360 – Electrical Loads

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	PK-361-C1A	FLASH COMPRESSOR A	VFD	690	315,00
2	X-361A	OIL HEATER (OIL SEPARATOR)	DOL	690	3,70
3	PK-361-P1A	OIL PUMP	DOL	690	4,00
4	PK-361-C1B	FLASH COMPRESSOR B	VFD	690	315,00

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
5	PK-361-P1B	OIL PUMP	DOL	690	4,00
6	PK-362-C1A	BOG RECYCLE COMPRESSOR A	SOFT START	690	160,00
7	X-362A	OIL HEATER (OIL SEPARATOR)	DOL	690	3,70
8	PK-362-P1A	OIL PUMP	DOL	690	3,00
9	PK-362-C1B	BOG RECYCLE COMPRESSOR B	SOFT START	690	160,00
10	PK-362-P1B	OIL PUMP	DOL	690	3,00

9.5.3 Motors as Remote MCC

Table 9.5 — Unit 360 – Motors as Remote MCC

ID	TAG	SERVICE	CONNECTED TO PLC	FAIL POSITION
1	MPK-361-E1	OIL COOLER FAN ELECTRIC MOTOR 1	PCS	Stop
2	MPK-361-E2	OIL COOLER FAN ELECTRIC MOTOR 2	PCS	Stop
3	MPK-361-E3	OIL COOLER FAN ELECTRIC MOTOR 3	PCS	Stop
4	MPK-361-E4	OIL COOLER FAN ELECTRIC MOTOR 4	PCS	Stop
5	MPK-362-E1	OIL COOLER FAN ELECTRIC MOTOR 1	PCS	Stop
6	MPK-362-E2	OIL COOLER FAN ELECTRIC MOTOR 2	PCS	Stop
7	MPK-362-E3	OIL COOLER FAN ELECTRIC MOTOR 3	PCS	Stop
8	MPK-362-E4	OIL COOLER FAN ELECTRIC MOTOR 4	PCS	Stop

9.5.4 Valves

Table 9.6 — Unit 360 – Valves

ID	TAG	SERVICE	SYSTEM	NOTES
1	CV-3601	PK-361-C1A/B PRESSURE CONTROL	-	-
2	PSV-3611A	PK-361-C1A BLOCKED OUTLET RELIEF	-	SET 10 barg
3	PSV-3611B	PK-361-C1B BLOCKED OUTLET RELIEF	-	SET 10 barg
4	PSV-3612A	PK-361-P1A PRESSURE RELIEF	-	SET 6 bar diff.
5	PSV-3612B	PK-361-P1B PRESSURE RELIEF	-	SET 6 bar diff.
6	PSV-3617A	PK-361-V1 PRESSURE RELIEF	-	SET 10 barg
7	PSV-3617B	PK-361-V1 PRESSURE RELIEF	-	SET 10 barg
8	PSV-3621A	PK-362-C1A BLOCKED OUTLET RELIEF	-	SET 50 barg
9	PSV-3621B	PK-362-C1B BLOCKED OUTLET RELIEF	-	SET 50 barg

ID	TAG	SERVICE	SYSTEM	NOTES
10	PSV-3622A	PK-362-P1A RELIEF	-	SET 6 bar diff.
11	PSV-3622B	PK-362-P1B RELIEF	-	SET 6 bar diff.
12	PSV-3623A	PK-362-F4A RELIEF	-	SET 50 barg
13	PSV-3623B	PK-362-F4B RELIEF	-	SET 50 barg
14	PSV-3627A	PK-362-V2 RELIEF	-	SET 50 barg
15	PSV-3627B	PK-362-V2 RELIEF	-	SET 50 barg
16	XEV-3601	UNIT-360 INLET SHUT DOWN	PCS	-
17	XEV-3612A	OIL TO PK-361-C1A INLET SHUTDOWN	PCS	WITH 5 MT CABLE
18	XEV-3612B	OIL TO PK-361-C1B INLET SHUTDOWN	PCS	WITH 5 MT CABLE
19	XEV-3613A	PK-361-C1A OIL HEADER SHUTDOWN	PCS	WITH 5 MT CABLE
20	XEV-3613B	PK-361-C1B OIL HEADER SHUTDOWN	PCS	WITH 5 MT CABLE
21	XEV-3614A	LP OIL TO PK-361-C1A SHUTDOWN	PCS	WITH 5 MT CABLE
22	XEV-3614B	LP OIL TO PK-361-C1B SHUTDOWN	PCS	WITH 5 MT CABLE
23	XEV-3622A	OIL TO PK-362-C1A INLET SHUTDOWN	PCS	WITH 5 MT CABLE
24	XEV-3622B	OIL TO PK-362-C1B INLET SHUTDOWN	PCS	WITH 5 MT CABLE
25	XEV-3623A	PK-362-C1A OIL HEADER SHUTDOWN	PCS	WITH 5 MT CABLE
26	XEV-3623B	PK-362-C1B OIL HEADER SHUTDOWN	PCS	WITH 5 MT CABLE
27	XV-3601	UNIT-360 INLET SHUT DOWN	-	-
28	ZEV-3611A	PK-361-C1A COMPRESSOR CAPACITY CONTROL	PCS	-
29	ZEV-3611B	PK-361-C1B COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
30	ZEV-3612A	PK-361-C1A COMPRESSOR CAPACITY CONTROL	PCS	-
31	ZEV-3612B	PK-361-C1B COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
32	ZEV-3613A	PK-361-C1A COMPRESSOR CAPACITY CONTROL	PCS	-
33	ZEV-3613B	PK-361-C1B COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
34	ZEV-3614A	PK-361-C1A COMPRESSOR CAPACITY CONTROL	PCS	-
35	ZEV-3614B	PK-361-C1B COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
36	ZEV-3622A	PK-362-C1A COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
37	ZEV-3622B	PK-362-C1B COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
38	ZEV-3624A	PK-362-C1A COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704
39	ZEV-3624B	PK-362-C1B COMPRESSOR CAPACITY CONTROL	PCS	P/N 018F4704

9.5.5 Instrumentation

Table 9.7 — Unit 360 – Instrument List

ID	TAG	SERVICE	RANGE	OP	UNITS
1	LSL-3617	PK-361-V1 OIL LEVEL	-	-	-
2	LSL-3627	PK-362-V1 OIL LEVEL	-	-	-

ID	TAG	SERVICE	RANGE	OP	UNITS
3	PT-3601	PK-361-C1A/B INLET PRESSURE	0..10	-	barg
4	PT-3611A	PK-361-C1A INLET PRESSURE	0..13.7895	-	bara
5	PT-3611B	PK-361-C1B INLET PRESSURE	0..13.7895	-	bara
6	PT-3612A	PK-361-C1A DISCHARGE PRESSURE	0..34.47	-	barg
7	PT-3612B	PK-361-C1B DISCHARGE PRESSURE	0..34.47	-	barg
8	PT-3613A	PK-361-C1A OIL HEADER PRESSURE	0..34.47	-	barg
9	PT-3613B	PK-361-C1B OIL HEADER PRESSURE	0..34.47	-	barg
10	PT-3621A	PK-362-C1A INLET PRESSURE	0..34.47	-	barg
11	PT-3621B	PK-362-C1B INLET PRESSURE	0..34.47	-	barg
12	PT-3622A	PK-362-C1A DISCHARGE PRESSURE	0..60	-	bara
13	PT-3622B	PK-362-C1B DISCHARGE PRESSURE	0..60	-	bara
14	PT-3623A	PK-362-C1A OIL HEADER PRESSURE	0..60	-	bara
15	PT-3623B	PK-362-C1B OIL HEADER PRESSURE	0..60	-	bara
16	TG-3602	PK-361-E2 TUBE OUTLET TEMPERATURE	-60..60	-	°C
17	TT-3601	PK-361-C1A/B INLET TEMPERATURE	-196..65	-	°C
18	TT-3612A	PK-361-C1A DISCHARGE TEMPERATURE	-50..200	-	°C
19	TT-3612B	PK-361-C1B DISCHARGE TEMPERATURE	-50..200	-	°C
20	TT-3617	PK-361-V1 TEMPERATURE	-50..200	-	°C
21	TT-3618	OIL TO PK-361-C1A/B TEMPERATURE	-50..200	-	°C
22	TT-3622A	PK-362-C1A DISCHARGE TEMPERATURE	-50..200	-	°C
23	TT-3622B	PK-362-C1B DISCHARGE TEMPERATURE	-50..200	-	°C
24	TT-3627	PK-362-V1 OIL TEMPERATURE	-50..200	-	°C
25	TT-3628	OIL TO PK-362-C1A/B TEMPERATURE	-50..200	-	°C
26	ZSH-3601	XEV-3601 OPEN POSITION INDICATION	-	-	-
27	ZSL-3601	XEV-3601 CLOSED POSITION INDICATION	-	-	-
28	PT-3602	FUEL GAS TO LP MANIFOLD PRESSURE	0..10	-	barg
29	PT-3702	FUEL GAS TO LP MANIFOLD PRESSURE	0..10	-	barg
30	TT-3701	FUEL GAS TO LP MANIFOLD TEMPERATURE	-50..120	-	°C
31	TT-3702	FUEL GAS OUTLET FROM E-372 TEMPERATURE	-50..120	-	°C
32	TT-3710	FUEL GAS OUTLET FROM E-371 TEMPERATURE	-50..200	-	°C
33	TT-3720	FUEL GAS OUTLET FROM E-372 TEMPERATURE	-50..200	-	°C
34	TT-3721	HP FUEL GAS TO POWER GENERATORS TEMPERATURE	-15..120	-	°C
35	ZSH-3701	SDV-3701 OPEN POSITION INDICATION	-	-	-
36	ZSH-3702	SDV-3702 OPEN POSITION INDICATION	-	-	-
37	ZSL-3701	SDV-3701 CLOSED POSITION INDICATION	-	-	-
38	ZSL-3702	SDV-3702 CLOSED POSITION INDICATION	-	-	-

9.5.6 Junction Boxes

Table 9.8 — Unit 360 – Junction Boxes

ID	TAG	SERVICE	CONNECTION	TYPE
1	JG-361-E1	OIL COOLER	COPPER-COPPER	STANDARD
2	JG-362-E1	OIL COOLER	COPPER-COPPER	STANDARD
3	P-PN-OS361.1	FLASH GAS	COPPER-COPPER	STANDARD
4	P-PN-OS362.1	FLASH GAS	COPPER-COPPER	STANDARD

9.5.7 Controllers

Table 9.9 — Unit 360 – Controllers

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
PIC-3601	PK-361-C1A/B INLET PRESSURE CONTROL	CV-3601	0..10 barg	—

9.6 Operating Interface

9.6.1 PCS Operating Screens – Unit 360

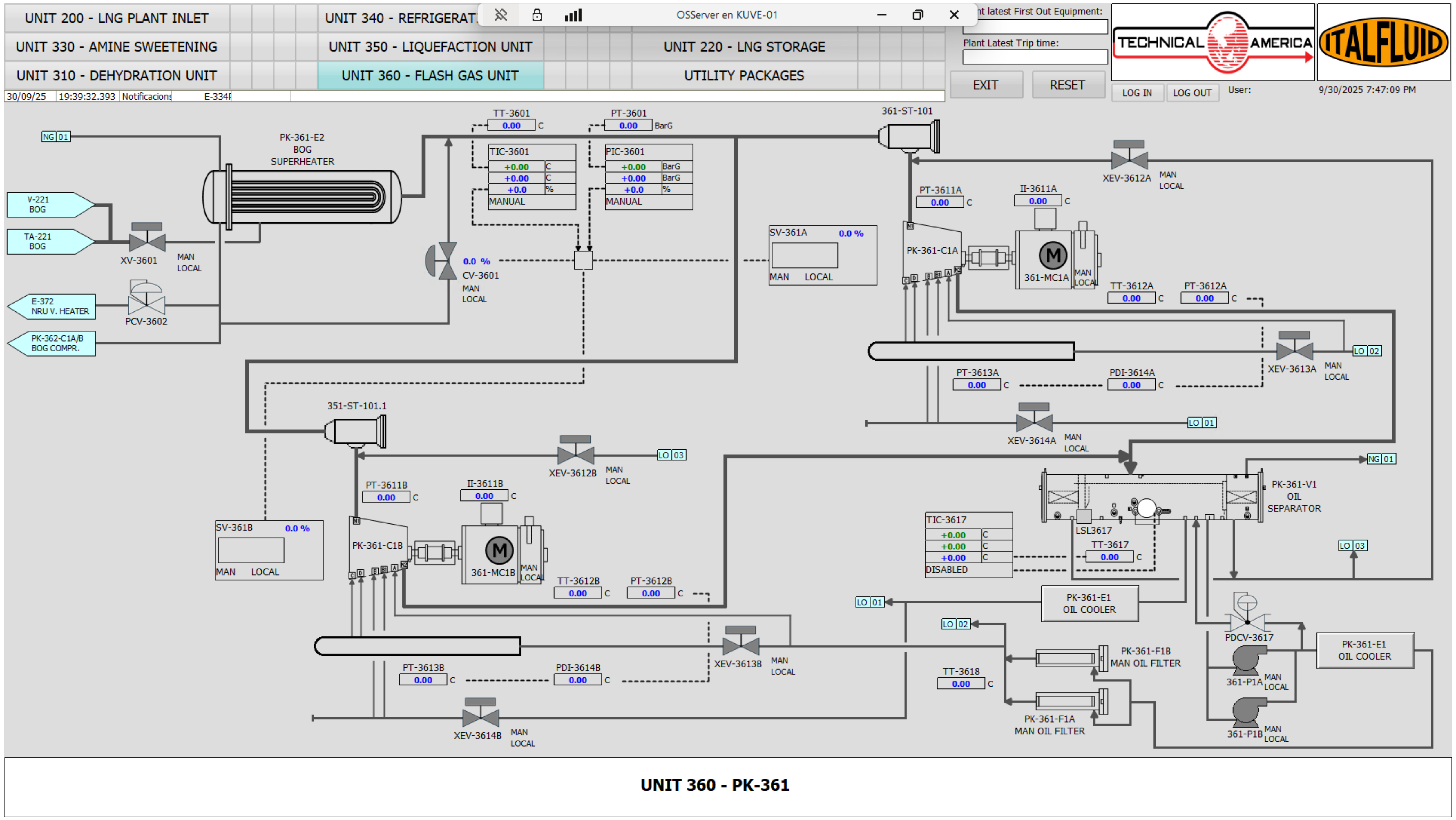


Figure 9.2: UNIT 360 - HMI Operating Screen

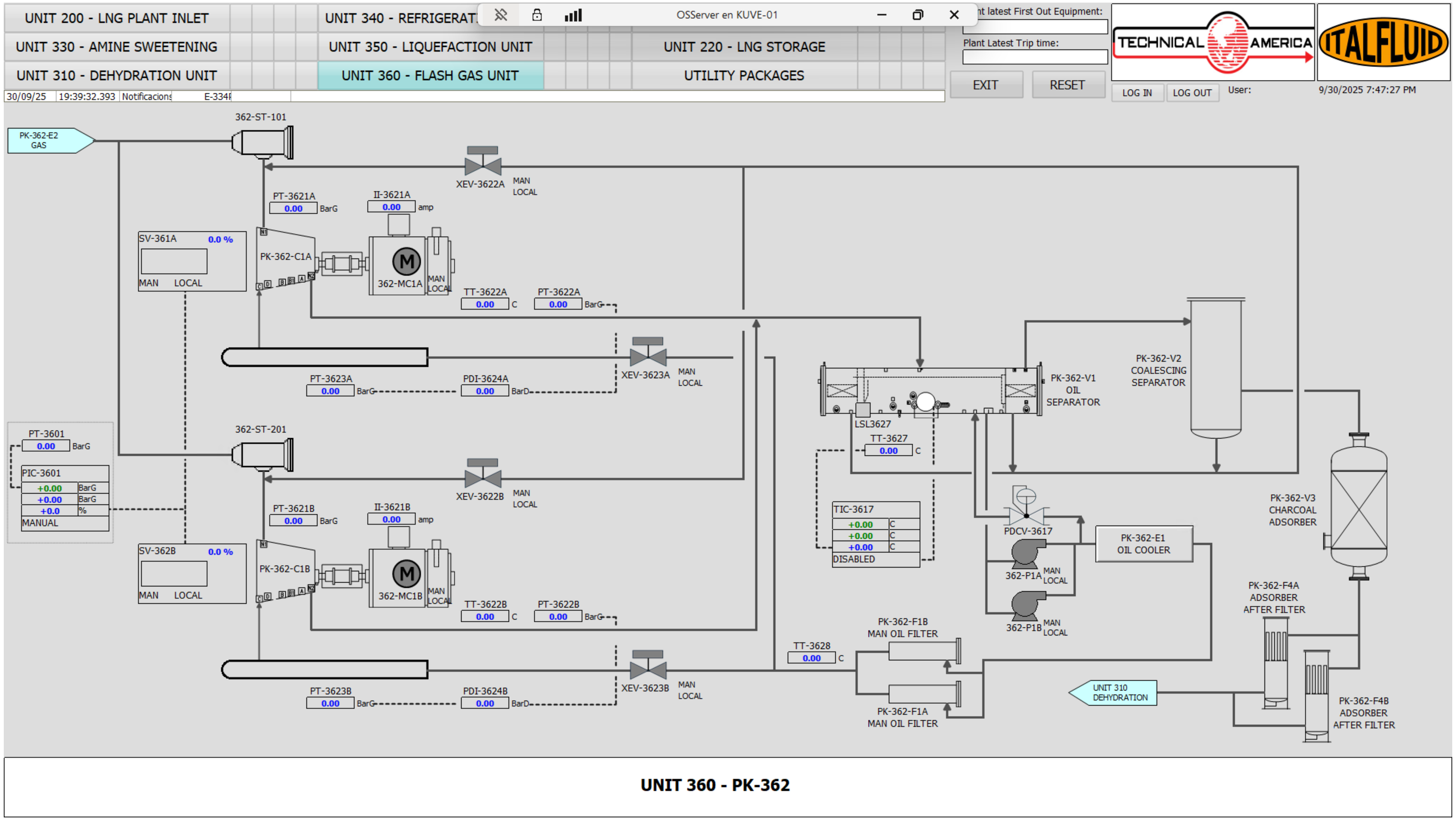


Figure 9.3: UNIT 360 - HMI Operating Screen

9.6.2 Critical Alarms and Trips

Table 9.10 — Unit 360 – Critical Alarms

ALARM	INSTRUMENT	TYPE	SP	DESCRIPTION
LSD21	LSD21	LSD	—	PK-361 EMERGENCY STOP PUSH BUTTON
LSD22	LSD22	LSD	—	PK-361 EMERGENCY STOP INTERLOCK
LSD23	LSD23	LSD	—	PK-362 EMERGENCY STOP PUSH BUTTON
LSD24	LSD24	LSD	—	PK-362 EMERGENCY STOP INTERLOCK
PSD04	PSD04	PSD	—	LOSS OF UNIT 360
PAHH-3612A	PT-3612A	LSD	—	PK-361-C1A HIGH HIGH DISCHARGE PRESSURE
PAHH-3612B	PT-3612B	LSD	—	PK-361-C1B HIGH HIGH DISCHARGE PRESSURE
PAHH-3622A	PT-3622A	LSD	—	PK-362-C1A HIGH HIGH DISCHARGE PRESSURE
PAHH-3622B	PT-3622B	LSD	—	PK-362-C1B HIGH HIGH DISCHARGE PRESSURE
PALL-3611A	PT-3611A	LSD	—	PK-361-C1A LOW LOW SUCTION PRESSURE
PALL-3611B	PT-3611B	LSD	—	PK-361-C1B LOW LOW SUCTION PRESSURE
PALL-3621A	PT-3621A	LSD	—	PK-362-C1A LOW LOW SUCTION PRESSURE
PALL-3621B	PT-3621B	LSD	—	PK-362-C1B LOW LOW SUCTION PRESSURE
PDALL-3614A	PDT-3614A	LSD	—	PK-361-C1A LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3614B	PDT-3614B	LSD	—	PK-361-C1B LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3624A	PDT-3624A	LSD	—	PK-362-C1A LOW LOW OIL DIFFERENTIAL PRESSURE
PDALL-3624B	PDT-3624B	LSD	—	PK-362-C1B LOW LOW OIL DIFFERENTIAL PRESSURE
TALL-3601	TT-3601	LSD	—	LOW LOW SUCTION TEMPERATURE
TAHH-3612A	TT-3612A	LSD	—	PK-361-C1A HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3612B	TT-3612B	LSD	—	PK-361-C1B HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3618	TT-3618	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER
TAHH-3622A	TT-3622A	LSD	—	PK-362-C1A HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3622B	TT-3622B	LSD	—	PK-362-C1B HIGH HIGH DISCHARGE TEMPERATURE
TAHH-3628	TT-3628	LSD	—	HIGH HIGH TEMPERATURE ON OIL HEADER

9.6.3 Unit Cause and Effect Matrix

Table 9.11 — Unit 360 – Cause & Effect Matrix

ALARM	INSTRUMENT	PK-361-C1A	PK-361-C1B	X-361	PK-361-P1A/B	XEV-3612A	XEV-3613A	XEV-3614A	XEV-3612B	XEV-3613B	XEV-3614B	PK-361-E1
LSD21	LSD21	T	T	T	T	C	C	C	C	C	C	T
LSD22	LSD22	T	T	T	T	C	C	C	C	C	C	T
PAHH-3612A	PT-3612A	T			T	C	C	C				T
PAHH-3612B	PT-3612B		T		T				C	C	C	T
PALL-3611A	PT-3611A	T			T	C	C	C				T
PALL-3611B	PT-3611B		T		T				C	C	C	T
PDALL-3614A	PDT-3614A	T			T	C	C	C				T
PDALL-3614B	PDT-3614B		T		T				C	C	C	T
TAHH-3612A	TT-3612A	T			T	C	C	C				T
TAHH-3612B	TT-3612B		T		T				C	C	C	T
TAHH-3618	TT-3618	T	T		T	C	C	C	C	C	C	T
TALL-3601	TT-3601	T	T		T	C	C	C	C	C	C	T

Table 9.12 — PK-362 Cause & Effect Matrix

ALARM	INSTRUMENT	PK-362-C1A	PK-362-C1B	X-362	PK-362-P1A/B	XEV-3622A	XEV-3623A	XEV-3622B	XEV-3623B	PK-362-E1
LSD23	LSD23	T	T	T	T	C	C	C	C	T
LSD24	LSD24	T	T	T	T	C	C	C	C	T
PAHH-3622A	PT-3622A	T				C	C			
PAHH-3622B	PT-3622B		T					C	C	
PALL-3621A	PT-3621A	T				C	C			
PALL-3621B	PT-3621B		T					C	C	
PDALL-3624A	PDT-3624A	T				C	C			
PDALL-3624B	PDT-3624B		T					C	C	
TAHH-3622A	TT-3622A	T				C	C			
TAHH-3622B	TT-3622B		T					C	C	

ALARM	INSTRUMENT	PK-362-C1A	PK-362-C1B	X-362	PK-362-P1A/B	XEV-3622A	XEV-3623A	XEV-3622B	XEV-3623B	PK-362-E1
TAHH-3628	TT-3628	**T**	**T**		**T**	C	C	C	C	**T**

9.7 Operation and Control

The Flash Gas Compression System (UNIT 360) handles flammable natural gas under pressure. Loss of containment may result in fire, explosion, or asphyxiation hazards. Compressor packages contain rotating equipment, pressurized oil systems, and hot surfaces. In case of gas leakage, abnormal vibration, fire, or emergency shutdown activation, immediately follow plant emergency procedures and evacuate the area if required.

9.7.1 Compressor Capacity Control

The capacity control of the flash gas compressors (PK-361-C1A/B) and BOG recycle compressors (PK-362-C1A/B) is achieved by means of an internal slide valve mechanism, which provides continuous modulation of compressor capacity over a wide operating range.

The slide valve moves axially along the rotor casing and modifies the effective compression length. By shifting its position, part of the gas trapped between the rotors is internally returned to the suction side before compression is completed. This reduces the effective compression volume and consequently decreases compressor capacity.

The slide valve is hydraulically actuated using the compressor lubrication oil system. Its movement is controlled by dedicated solenoid valves that direct oil flow to either side of the hydraulic actuator, allowing accurate positioning of the slide valve.

The capacity control system operates as follows:

- When an increase in gas handling capacity is required, the slide valve moves towards the discharge side, increasing the effective compression volume.

- When a reduction in capacity is required, the slide valve moves towards the suction side, allowing part of the gas to be internally recycled before compression is completed.
- The compressor can typically operate over a capacity range from 100

During normal operation, compressor capacity is automatically adjusted to maintain the required suction pressure and gas flow conditions within the flash gas recovery and recycle systems.

This method of capacity control offers several advantages:

- High energy efficiency compared with external recycle or hot-gas bypass control methods.
- Smooth and continuous capacity modulation without step changes.
- Stable operation over a wide range of process loads.
- Reduced power consumption during part-load operation.

During start-up, the compressor is maintained in a fully unloaded condition to minimize starting torque and motor current requirements. After start-up, the slide valve is progressively loaded until the required operating conditions are achieved.

The slide valve control system is interlocked with the compressor lubrication and protection systems to ensure safe operation. Abnormal operating conditions such as low oil pressure, compressor trips, or other package protection actions will automatically unload and/or stop the compressor in accordance with the package safeguarding philosophy and the Cause & Effect Matrix.

9.7.2 BOG Compression System Capacity Control

The capacity of the BOG compressor packages (PK-361-C1A/B) is controlled by the compressor internal hydraulic slide valve.

The compressor capacity control system receives the signals from PIC-3601 and TIC-3601 and automatically modulates the slide valve position to maintain the required BOG system operating conditions.

As the BOG generation rate increases, the compressor is progressively loaded. As the BOG generation rate decreases, the compressor is unloaded accordingly.

The compressor is started in the minimum capacity position and is progressively loaded after start-up.

Under low-load conditions, recycle valve CV-3601 may open to maintain stable compressor operation and prevent insufficient flow through the package.

Compressor protection functions remain independent from the capacity control system and will unload and/or stop the compressor in accordance with the package safeguarding philosophy and Cause & Effect Matrix.

9.7.3 BOG Recycle Compressor Capacity Control

The capacity of the BOG recycle compressor packages (PK-362-C1A/B) is controlled by the compressor internal hydraulic slide valve.

The compressor capacity control system automatically modulates the slide valve position based on the fuel gas manifold pressure controller PIC-3602.

As the fuel gas manifold pressure increases above its target value, the compressor is progressively loaded to recycle excess BOG back to the process. As the fuel gas demand increases and manifold pressure decreases, the compressor is unloaded accordingly.

The compressor is started in the minimum capacity position and is progressively loaded after start-up.

This control strategy allows automatic balancing between fuel gas consumption and BOG generation while minimizing venting and maintaining stable fuel gas header pressure.

Compressor protection functions remain independent from the capacity control system and will unload and/or stop the compressor in accordance with the package safeguarding philosophy and Cause & Effect Matrix.

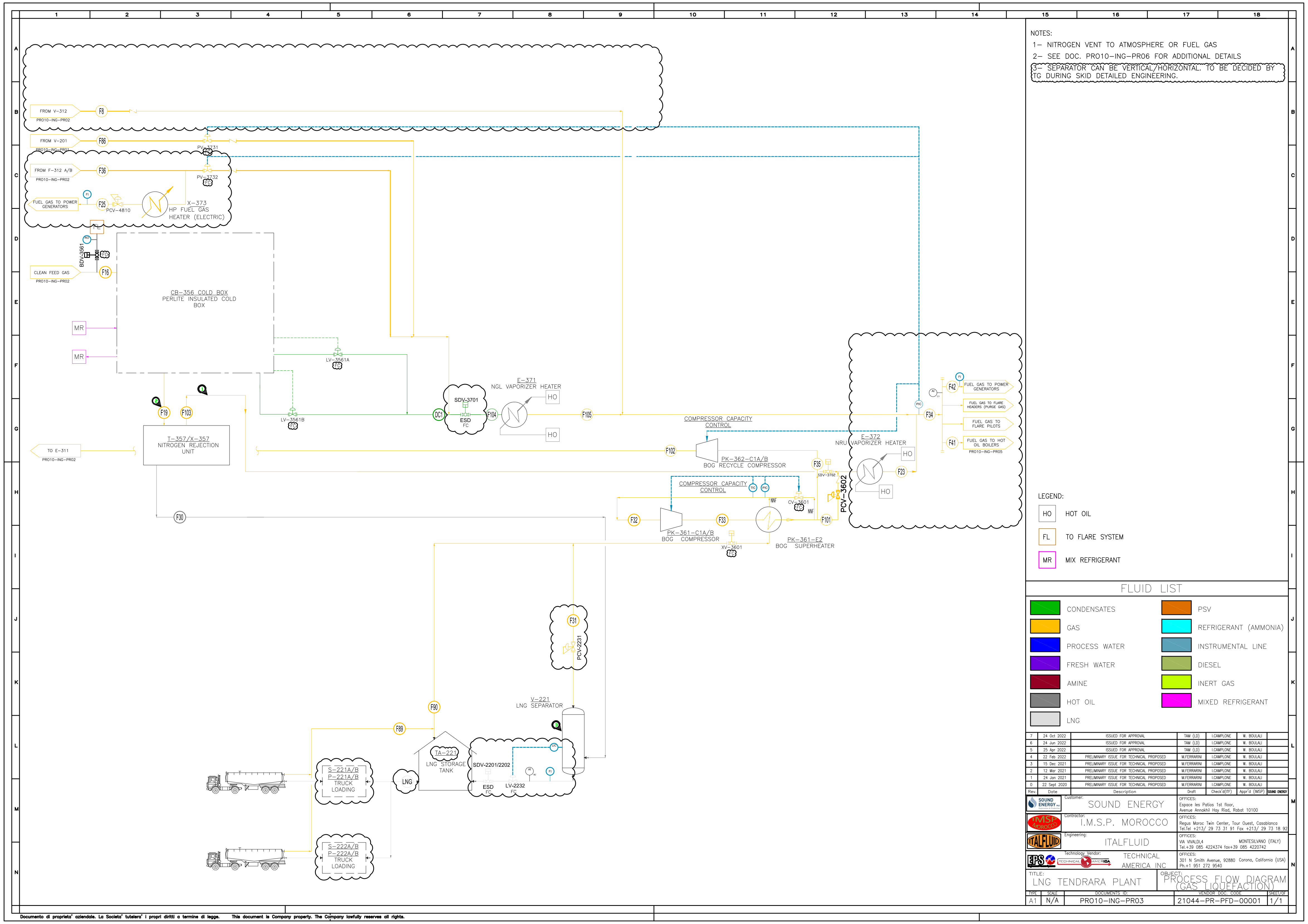


Figure 9.4: Process Gas Liquefaction

9.7.4 Compressor Panel Operation

> The lubrication system is critical for the operation of oil-flooded screw compressors. Failure or inadequate performance of the lubrication system may result in severe compressor damage.

The flash gas compression unit (UNIT 360) is equipped with local package control panels associated with the BOG compressor package (PK-361) and the recycle gas compressor package (PK-362).

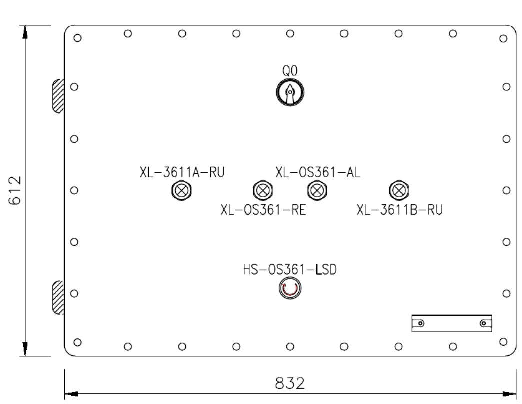
The compressor start-up and shutdown sequences are normally initiated from the Process Control System (PCS). The package controller executes all compressor protection, lubrication system management, capacity control, alarm handling, and shutdown functions required for safe compressor operation.

A dedicated Local Shutdown (LSD) pushbutton is provided on each package panel. Activation of this pushbutton initiates shutdown of the corresponding compressor package in accordance with the Unit 360 Cause & Effect Matrix.

The local panels provide the operator with basic package status indications, allowing rapid verification of compressor operation and package availability. The following indications are provided:

- **Compressor A Running**
- **Compressor B Running**
- **Package Ready**
- **Package Alarm / Trip**

These indications provide real-time feedback regarding compressor operating status, package readiness for start-up, and active alarm or shutdown conditions.



(a) PK-36x local control panel

(b) PK-36x local control panel

Figure 9.5: Unit 360 compressor package local control panels

Table 9.13 — Unit 360 compressor package local panel indications

TAG	SIGNAL	DESCRIPTION
XL-36x1A-RU	RUNNING	Compressor A running indication.
XL-36x1B-RU	RUNNING	Compressor B running indication.
XL-OS36x-RE	READY	Package ready for compressor start indication.
XL-OS36x-AL	ALARM / TRIP	Package alarm or shutdown condition active.

The local panel indications are intended to provide a simple visual summary of compressor package status and availability while detailed operating information, alarms, and shutdown diagnostics remain available through the PCS operator interface.

9.7.5 Lub Oil Systems (PK-361 and PK-362)

The BOG compressor package (PK-361) and the BOG recycle compressor package (PK-362) are each provided with an independent closed-loop lubrication oil system designed to supply lubrication, cooling, sealing, and hydraulic power for compressor capacity control.

The lubrication oil system is an integral part of each screw compressor package and is required for reliable operation of the oil-flooded compressors. Oil is injected into the compressor during the compression process, where it performs several essential functions:

- Lubrication of bearings and rotating components.

- Sealing between the screw rotors to improve volumetric efficiency.
- Removal of compression heat.
- Hydraulic actuation of the compressor slide valve capacity control system.

The oil-gas mixture discharged from the compressor is routed to the oil separator vessel (PK-361-V1 or PK-362-V1), where the lubricating oil is separated from the process gas and collected in the oil sump section.

The oil separator serves as the main oil reservoir for the package and incorporates multi-stage oil separation elements to minimize oil carryover to the downstream gas system. The separator is equipped with an electric oil heater used during shutdown periods to maintain oil temperature and reduce gas absorption into the lubricant.

Oil circulation is provided by two positive displacement oil pumps (1 operating + 1 standby), which draw oil from the separator and circulate it through the oil cooling and filtration system before returning it to the compressor.

Each lubrication system includes:

- Oil separator and oil reservoir (PK-361-V1 / PK-362-V1).
- Oil heater (X-361 / X-362).
- Dual oil pumps (PK-361-P1A/B or PK-362-P1A/B).
- Air-cooled oil cooler (PK-361-E1 or PK-362-E1).
- Duplex oil filters (1 operating + 1 standby).
- Oil pressure regulating / bypass valve (PDCV).

After leaving the oil pumps, the oil passes through the air-cooled oil cooler where excess heat is removed. The cooled oil is then routed through the duplex oil filters to ensure a clean oil supply to the compressor bearings, rotor injection ports and slide valve hydraulic system.

The lubrication system is continuously monitored by the control system through oil pressure, oil temperature and oil level measurements. Adequate oil differential pressure is required to permit compressor operation and to ensure proper lubrication and slide valve movement.

Loss of oil pressure, low oil level, excessive oil temperature or other critical lubrication faults generate compressor shutdown signals in accordance with the package protection philosophy.

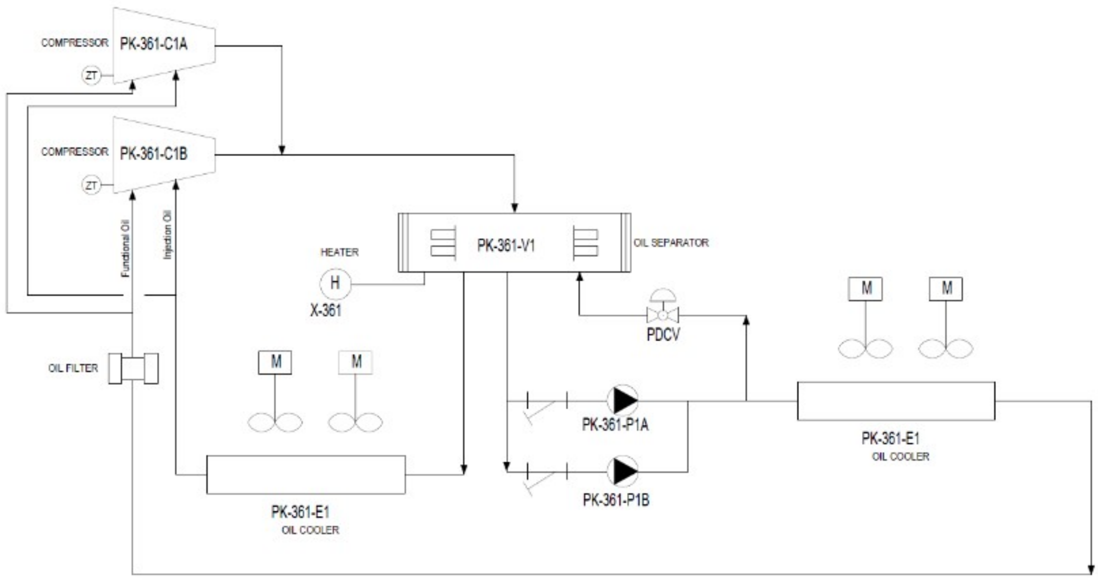


Figure 9.6: PK-361 lubrication oil system

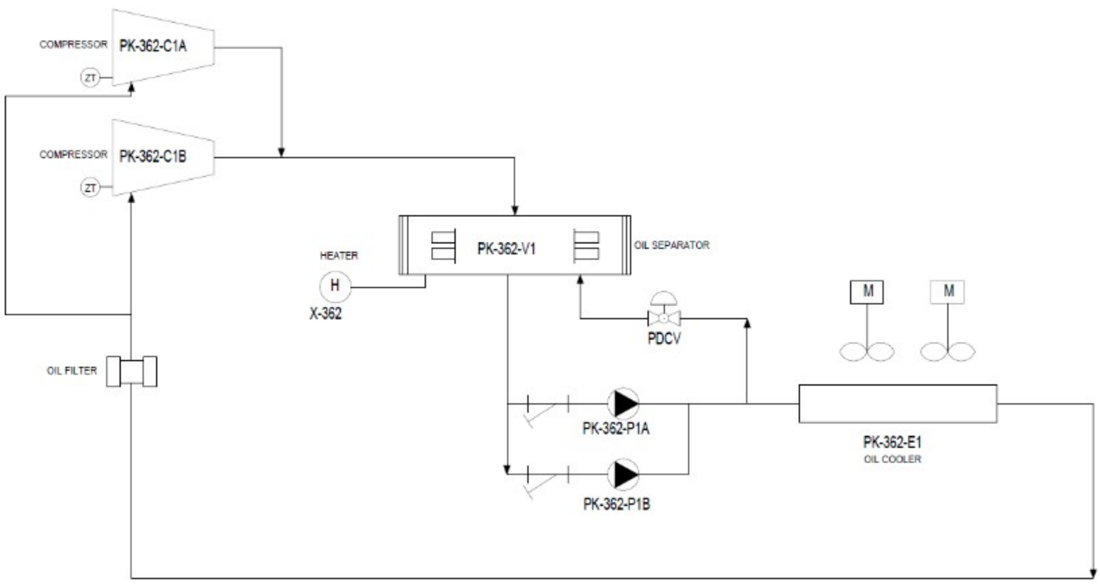


Figure 9.7: PK-362 lubrication oil system

9.7.6 Magnetic Coupling

The lube oil pump (P-341-1) is equipped with a magnetic coupling, which transmits torque from the motor to the pump without a direct mechanical shaft connection. This design eliminates the need for dynamic seals and ensures leak-free operation, which is particularly suitable for ammonia service.

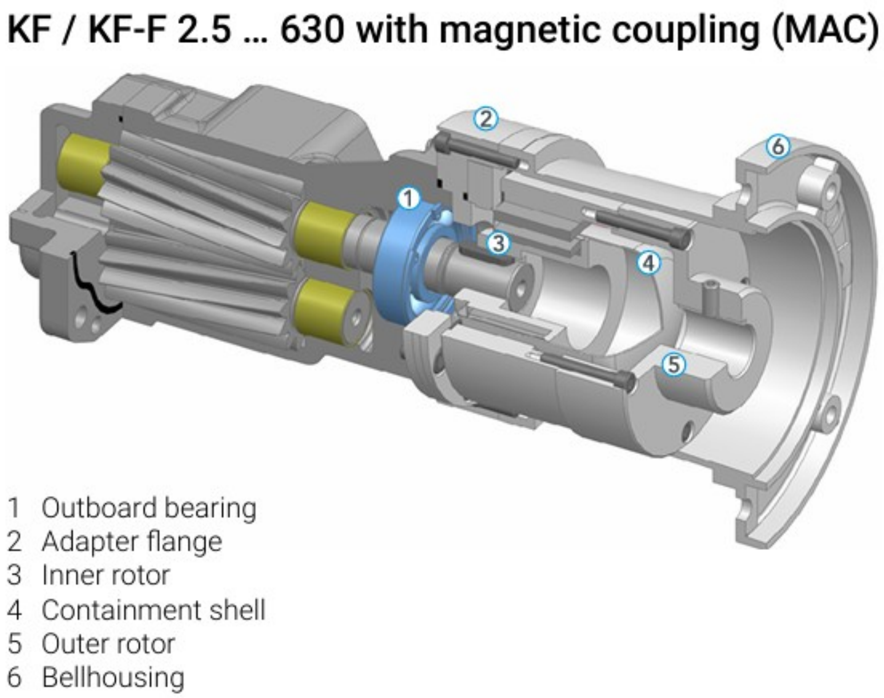


Figure 9.8: Oil pump magnetic coupling

The magnetic coupling consists of an outer rotor connected to the motor shaft and an inner rotor connected to the pump shaft. Torque is transmitted through magnetic forces across a containment shell, allowing complete separation between the driven and driving components.

This configuration provides several advantages, including improved reliability, reduced maintenance requirements, and elimination of potential leakage points. It is particularly suitable for hazardous fluids, as it minimizes the risk of external release.

Operationally, the magnetic coupling has a maximum torque transmission limit. In case of overload, magnetic decoupling may occur, resulting in the pump rotating without effective torque transmission. This condition may lead to loss of oil circulation and must be detected promptly. Dry running conditions shall be strictly avoided, as they can damage internal pump components.

9.7.7 Compressor Start-Up Permissives

Before initiating the automatic start-up sequence of any BOG compressor package (PK-361-C1A/B or PK-362-C1A/B), the following start-up permissives shall be confirmed:

- The selected compressor motor to be available in **AUTO MODE**.
- At least one lube oil pump to be available in **AUTO MODE**.
- All oil cooler fans to be available in **AUTO MODE**.
- The compressor capacity control / slide valve to be in **AUTO MODE.**
- Compressor suction and discharge isolation valves to be confirmed **open**.
- Lube oil temperature to be within the normal operating range.
- The compressor slide valve to be in the **unloaded position** before start.
- Associated capacity control solenoid valves to be in **AUTO MODE**.
- No active shutdowns, trips, or critical alarms present on the selected package.
- Motor restart inhibit / cool-down timer to be expired.

If any of the above permissives is not satisfied, the compressor shall remain unavailable for automatic start and the start-up sequence shall be inhibited.

These permissives ensure that the lubrication system, oil cooling system, compressor capacity control system, and process isolation valves are in their required operating condition before the compressor motor is energized.

9.7.8 Compressor Start-Up Sequence

After verification of all compressor start-up permissives, the selected compressor package (PK-361-C1A/B or PK-362-C1A/B) can be started automatically from the PCS.

The automatic start-up sequence establishes lubrication oil circulation, verifies oil differential pressure, starts the compressor in unloaded condition, stabilizes operation at minimum load, and then enables automatic capacity control.

Table 9.14 — Unit 360 – Compressor automatic start-up sequence

Step	Tag	SP		Action	Description
		Value	Unit		
1	HS-36x-ST	–	–	OC	Start automatic sequence from PCS after all start-up permissives are satisfied.
2	PK-36x-P1A/B	–	–	SC	Start auxiliary lube oil pump and establish oil circulation through the lubrication system.
3	PDI-36x	1.5	bar	SC	Verify lube oil differential pressure is greater than or equal to 1.5 bar.
4	ZT-36x	<5	%	SC	Drive compressor slide valve to fully unloaded position and confirm feedback.
5	MC-36x	–	–	SC	Start compressor motor and accelerate to nominal operating speed.
6	ZT-36x	15	%	SC	Move slide valve to minimum load position.
7	PK-36x	30	s	CK	Maintain compressor at minimum load for stabilization period.
8	PK-36x-E1	–	–	SC	Enable oil cooler fans as required to maintain lube oil temperature within normal operating range.
9	PIC/TIC-36x	–	–	SC	Enable automatic capacity control loops. Compressor capacity is automatically adjusted through the slide valve.
10	PK-36x	–	–	CK	Compressor reaches normal operating mode with lubrication, cooling and capacity control systems in service.

9.7.9 Compressor Shutdown Sequence

After confirmation of stable operating conditions and reduction of gas compression demand, the compressor package (PK-361-C1A/B or PK-362-C1A/B) can be stopped following the automatic shutdown sequence executed by the PCS.

The shutdown sequence unloads the compressor prior to motor stop in order to reduce mechanical load and starting torque requirements for the next start-up. Following motor shutdown, the lubrication system remains in operation for a defined post-lubrication period to ensure adequate bearing and rotor protection during coast-down.

Table 9.15 — Unit 360 – Compressor automatic shutdown sequence

Step	Tag	SP		Action	Description
		Value	Unit		
1	HS-36x-SP	–	–	OC	Start automatic shutdown sequence from PCS,

Step	Tag	SP		Action	Description
		Value	Unit		
2	ZT-36x1	<5	%	SC	System command SET compressor slide valve to fully unloaded position and confirm feedback.
3	MC-36x1A/B	–	–	SC	System command STOP compressor main motor and confirm de-energization.
4	PT-36x1	–	–	CK	Verify suction and discharge pressures stabilize following compressor stop.
5	PK-36x-P1A/B	–	–	CK	Lubrication oil pump remains in service to provide post-lubrication to bearings, rotors and seals.
6	–	30	s	CK	Maintain post-lubrication period after compressor motor stop.
7	PK-36x-P1A/B	–	–	SC	System command STOP lubrication oil pump after completion of post-lubrication timer.
8	PDT-36x1	< Min	–	CK	Confirm oil differential pressure decays normally after oil pump stop.
9	MC-36x1A/B	–	–	CK	Confirm compressor fully stopped and no start command active.
10	PK-36x	–	–	CK	Compressor package shutdown completed. All shutdown permissives are reset.

9.8 Troubleshooting

Table 9.16 — Unit 360 – PK-36x Troubleshooting

TROUBLE	POSSIBLE CAUSE	SOLUTION
Complete loss of power	Main power switch disconnected	Check all power wiring
	Main fuse failure	Replace the burned-out fuse
Partial loss of power	Auxiliary fuse failure	Replace the burned-out fuse
	Power disconnected to auxiliary equipment	Check wiring to auxiliary equipment
Oil pump will not run in HAND mode	Overload open	Reset overload
	Compressor is running	Oil pumps must be in AUTO while compressor is operating
	Main power off	Check starter disconnect and power supply

TROUBLE	POSSIBLE CAUSE	SOLUTION
	Panel control power off	Turn panel control power on
	Motor burned out	Inspect and repair or replace motor
	Pump rotation reversed	Check motor rotation and correct if required
Oil pump will not run in AUTO mode	Compressor start has not been initiated	Initiate compressor start sequence
	Pump not developing sufficient pressure	Adjust pressure differential control valve to 30–60 psid
	Oil too cold	Warm oil to approximately 100°F
	Pump rotation reversed	Check motor rotation and correct if required
Oil pump runs in AUTO mode but compressor will not start	Lube oil supply pressure too low	Adjust pressure differential control valve to 30–60 psid
	Compressor motor wired incorrectly	Check and correct motor wiring
	Slide valve not at less than 5% position	Confirm compressor is fully unloaded and hydraulic oil block valves are open
Oil pump runs in AUTO mode, compressor starts, but stops immediately	Motor starter or auxiliary contact wired incorrectly	Check and correct wiring; auxiliary contact must seal within 3 seconds
	Error in compressor start logic	Check and correct control logic
High or low lube oil temperature	High suction temperature	Check operating conditions and correct
	Thermal expansion valve not functioning properly	Check and correct valve operation
Low lube oil pressure	Oil pump not developing enough pressure	Adjust PDCV to 50–60 psid at regulator inlet; isolate PDCV and open bypass to verify pressure build-up
	Oil too cold	Run pump in HAND mode to warm oil to at least 90°F; close compressor suction valve while warming
	Oil pressure switch defective or incorrectly set	Check switch setting

TROUBLE	POSSIBLE CAUSE	SOLUTION
	Control panel contact not closing properly or loose wiring	Check and correct
	Lube oil filter clogged	Check and replace filter if necessary
	Low oil level in oil reservoir	Add oil to vessel
Bad oil quality	Solids in oil	Drain oil and refill with clean oil; flush system if necessary; analyze oil and eliminate source of contamination
High oil carryover	Coalescer element loose	Tighten and mount coalescer properly; verify differential pressure across coalescer is not less than 0.25 psid
High differential pressure across oil filter	Coalescer drain line choked	Check for oil accumulation in sight glass and open globe valve further to restore oil flow
	Oil filter plugged	Check and replace oil filter if necessary
Decrease in oil level in oil reservoir	Oil leakage from system	Check and correct leaks
	Coalescers not tight enough to seal	Tighten coalescers
	Oil return line from oil separator blocked	Adjust or clear block valve / return line
	Excess leakage from compressor seal	Check and correct
	Oil leaving system through a relief valve	Check all relief valve discharge piping for oil presence
Abnormal noise	Bearings damaged	Replace bearings
	Loose connections	Stop compressor and tighten connections
	Foreign material entered compressor	Stop compressor; check suction strainer and compressor internals
	Misalignment or damaged shaft coupling	Stop compressor; check alignment and coupling condition
	Rotors rubbing against casing inner surface	Stop and disassemble compressor; replace damaged parts and correct internal clearances

10 Unit 370 - Vaporizer Unit (E-371, E-372)

10.1 Operating Summary

CATEGORY	DESCRIPTION
Inputs	Heavy-hydrocarbon condensate intermittently separated in liquefaction cold boxes V-356A and V-356B.
	Nitrogen-rich overhead gas from NRU distillation column T-357.
	Boil-Off Gas (BOG) routed from UNIT 360.
	Regeneration / sweet gas routed from dehydration system (UNIT 310).
	Startup fuel gas supplied from raw gas vessel V-201 (startup conditions only).
	Hot oil heating medium supplied from UNIT 410 for all vaporization duties.
Outputs	Fully vaporized and thermally conditioned fuel gas routed to the plant fuel gas headers.
	High-pressure fuel gas delivered to critical consumers after final conditioning via electric heater X-373.
Key Operating Specs	Plate-type hot-oil vaporizers E-371 (hydrocarbon-rich streams) and E-372 (NRU overhead gas).
	Passive vaporization service with outlet temperature controlled exclusively via hot oil circuits (TV-3710 and TV-3720).
	Electric HP fuel gas heater X-373 providing final temperature conditioning downstream of Skid S-371.
	Independent high-high temperature protection on X-373 (TAHH-X373).
	Design prevents introduction of cold or two-phase fluids into fuel gas headers under all operating scenarios.
Major Equipment	E-371 NGL Vaporizer Heater (hot-oil heated, plate-type).
	E-372 NRU Vaporizer Heater (hot-oil heated, plate-type).
	Skid S-371 integrating vaporizers, instrumentation, and shutdown valves.
	X-373 Electric High-Pressure Fuel Gas Heater.
	Associated pressure control valves, shutdown valves, and mechanical relief devices.

10.2 General Arrangement and Risk Location

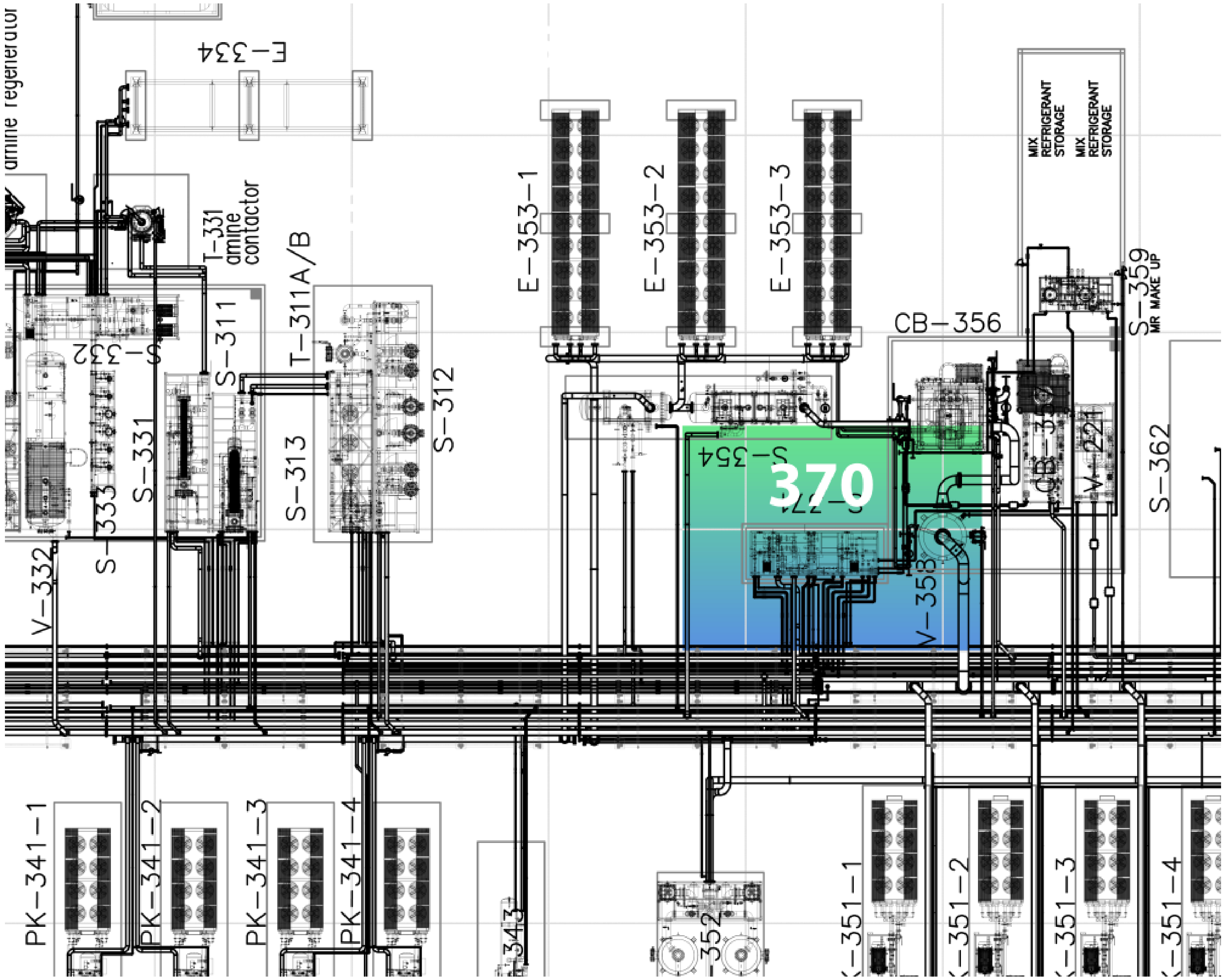


Figure 10.1: Unit 370 GA - Risk location

10.3 Relevant Documents

Table 10.2 — Unit 370 – Relevant Documents

ID	DOC NO	TITLE
1	PRO02-ING-FR02	PROCESS DESCRIPTION
2	PRO10-ING-PR03	PROCESS FLOW DIAGRAM (GAS LIQUEFACTION)
3	PRO13-ING-PR27	UNIT 370 VAPORIZER SKID S-371
4	PRO13-ING-PR28	HOT OIL UNIT (INTERFACE WITH E-371 / E-372)
5	SAF12-ING-FR01	CAUSE & EFFECT MATRIX
6	SAF05-ING-FR01	RELIEF AND BLOWDOWN PHILOSOPHY
7	SAF01-ING-FR01	SAFETY AND ISOLATION PHILOSOPHY

10.4 Process Description

Unit 370 provides thermal conditioning and vaporization of selected hydrocarbon-rich and inert gas streams prior to their introduction into the plant fuel gas network. The unit ensures that all contributing streams are delivered in a fully vaporized state and within acceptable temperature limits, preventing the ingress of cold or two-phase fluids into the fuel gas headers and downstream consumers.

The unit receives and conditions heavy-hydrocarbon condensate separated in the liquefaction cold boxes V-356A and V-356B, when present, regeneration or sweet gas routed from the dehydration system via dedicated tie-ins, boil-off gas routed from the BOG handling system in Unit 360, and nitrogen-rich overhead gas from the Nitrogen Rejection Unit. All heating duties within Unit 370 are provided by the hot oil system, Unit 410.

E-371 – NGL Vaporizer Heater

E-371 is a plate-type hot-oil heat exchanger installed on Skid S-371 and dedicated to the vaporization and heating of hydrocarbon-rich streams prior to their entry into the fuel gas network.

The exchanger receives heavy-hydrocarbon condensate from the liquefaction cold boxes via control valves LV-3561A and LV-3561B. Vaporization of this stream occurs only when condensate is formed and available, as continuous liquid service is not assumed under normal operating conditions.

E-371 also receives startup fuel gas supplied as raw gas from separator V-201. This line is intended exclusively for startup conditions; following plant startup, the associated manual isolation valves and control valve are closed. Additional hydrocarbon gas streams may be routed to the skid in accordance with the applicable P&IDs, including tie-ins from the dehydration and BOG handling systems.

Process flow rates to E-371 are controlled upstream by the respective source units. The exchanger operates with passive thermal duty, while outlet temperature is regulated by the hot oil circuit through temperature control valve TV-3710.

E-372 – NRU Vaporizer Heater

E-372 is a plate-type hot-oil heat exchanger installed on Skid S-371 and dedicated to heating the nitrogen-rich overhead gas from the Nitrogen Rejection Unit prior to its introduction into the fuel gas system.

In addition to the NRU overhead gas, E-372 receives auxiliary gas streams routed to the skid, including tie-ins from the dehydration system, as defined in the P&IDs. The heating medium for E-372 is hot oil supplied from Unit 410.

The primary function of E-372 is to prevent low-temperature gas from entering the fuel gas headers, thereby avoiding cold gas breakthrough and operational disturbances to fuel gas consumers. The outlet temperature is controlled via the hot oil circuit using control valve TV-3720. The conditioned gas is then routed to the fuel gas manifold(s) as indicated on the process diagrams.

HP Fuel Gas Heater – X-373

Downstream of Skid S-371, the high-pressure fuel gas network includes an electrically heated fuel gas heater, X-373, which provides final temperature conditioning of the HP fuel gas prior to delivery to critical consumers such as the power generation units.

The heater outlet temperature is controlled by temperature indicator controller TIC-3721, while high-high temperature protection is provided by TAHH-X373. This heater ensures compliance with minimum fuel gas temperature requirements under cold ambient conditions and during transient operating scenarios, providing additional operational robustness to the fuel gas system.

10.5 Technological Objects

10.5.1 Main Equipment List

Table 10.3 — Unit 370 – Main Equipment List

ID	TAG	EQUIPMENT
1	E-371	NGL vaporizer heater
2	E-372	NRU vaporizer heater
3	X-373	HP fuel gas heater

10.5.2 Electrical Loads

Table 10.4 — Unit 370 – Electrical Loads

ID	TAG	SERVICE	TYPE	VOLTAGE	POWER
1	X-373	HP FUEL GAS HEATER	DOL	690	3.00

10.5.3 Valves

Table 10.5 — Unit 370 – Valves

ID	TAG	SERVICE	SYSTEM	NOTES
1	LEV-3561A	V-356A LIQUID LEVEL CONTROL	ESD	-
2	LEV-3561B	V-356B LIQUID LEVEL CONTROL	ESD	-
3	LV-3561A	V-356A LIQUID LEVEL CONTROL	-	-
4	LV-3561B	V-356B LIQUID LEVEL CONTROL	-	-
5	PEV-3731	FUEL GAS PRESSURE CONTROL	ESD	-
6	PEV-3732	FUEL GAS PRESSURE CONTROL	ESD	-
7	PV-3731	FUEL GAS PRESSURE CONTROL	-	-
8	PV-3732	FUEL GAS PRESSURE CONTROL	-	-
9	TV-3710	FUEL GAS OUTLET FROM E-371 TEMPERATURE CONTROL	-	-
10	TV-3720	FUEL GAS OUTLET FROM E-372 TEMPERATURE CONTROL	-	-
11	SDEV-3701	E-371 INLET SHUT DOWN	ESD	-
12	SDEV-3702	E-372 INLET SHUT DOWN	ESD	-
13	SDV-3701	E-371 INLET SHUT DOWN	-	-
14	SDV-3702	E-372 INLET SHUT DOWN	-	-
15	PSV-3701A	E-371 INLET PRESSURE RELIEF	-	SET 10 barg
16	PSV-3701B	E-371 INLET PRESSURE RELIEF	-	SET 10 barg
17	PSV-4801A	HP FUEL GAS HEADER PRESSURE RELIEF	-	SET 10 barg
18	PSV-4801B	HP FUEL GAS HEADER PRESSURE RELIEF	-	SET 10 barg

10.5.4 Instrumentation

Table 10.6 — Unit 370 – Instrument List

ID	TAG	SERVICE	RANGE	OP	UNITS
1	LVY-3561A	V-356A LIQUID LEVEL CONTROL	-	-	-
2	LVY-3561B	V-356B LIQUID LEVEL CONTROL	-	-	-
3	PG-3701	FUEL GAS INLET TO E-371 PRESSURE	0..10	-	barg
4	PG-3702	FUEL GAS OUTLET FROM E-371 PRESSURE	0..10	-	barg
5	PG-3703	HOT OIL INLET TO E-371 PRESSURE	0..10	-	barg
6	PG-3704	HOT OIL RETURN FROM E-371 PRESSURE	0..10	-	barg
7	PG-3705	FUEL GAS INLET TO E-372 PRESSURE	0..10	-	barg
8	PG-3706	FUEL GAS OUTLET FROM E-372 PRESSURE	0..10	-	barg

ID	TAG	SERVICE	RANGE	OP	UNITS
9	PG-3707	HOT OIL INLET TO E-372 PRESSURE	0..10	-	barg
10	PG-3708	HOT OIL RETURN FROM E-372 PRESSURE	0..10	-	barg
11	PT-3602	FUEL GAS TO LP MANIFOLD PRESSURE	0..10	-	barg
12	PT-3702	FUEL GAS TO LP MANIFOLD PRESSURE	0..10	-	barg
13	PVY-3731	FUEL GAS PRESSURE CONTROL	-	-	-
14	PVY-3732	FUEL GAS PRESSURE CONTROL	-	-	-
15	TAHH-X-373	X-373 HEATER HIGH HIGH TEMPERATURE SWITCH	150	-	°C
16	TT-3701	FUEL GAS TO LP MANIFOLD TEMPERATURE	-50..120	-	°C
17	TT-3702	FUEL GAS OUTLET FROM E-372 TEMPERATURE	-50..120	-	°C
18	TT-3710	FUEL GAS OUTLET FROM E-371 TEMPERATURE	-50..200	-	°C
19	TT-3720	FUEL GAS OUTLET FROM E-372 TEMPERATURE	-50..200	-	°C
20	TT-3721	HP FUEL GAS TO POWER GENERATORS TEMPERATURE	-15..120	-	°C
21	TVY-3710	FUEL GAS OUTLET FROM E-371 TEMPERATURE CONTROL	-	-	-
22	TVY-3720	FUEL GAS OUTLET FROM E-372 TEMPERATURE CONTROL	-	-	-
23	ZSH-3701	SDV-3701 OPEN POSITION INDICATION	-	-	-
24	ZSH-3702	SDV-3702 OPEN POSITION INDICATION	-	-	-
25	ZSL-3701	SDV-3701 CLOSED POSITION INDICATION	-	-	-
26	ZSL-3702	SDV-3702 CLOSED POSITION INDICATION	-	-	-

10.5.5 Junction Boxes

Table 10.7 — Unit 370 – Junction Boxes

ID	TAG	SERVICE	CONNECTION	TYPE
1	P-PN-OS371.1	VAPORIZER	COPPER-COPPER	STANDARD

10.5.6 Controllers

Table 10.8 — Unit 370 – Controllers

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
LIC-3561A	LIQUID LEVEL IN NGL SEPARATOR V-356A	LV-3561A	0..100%	—
LIC-3561B	LIQUID LEVEL IN NGL SEPARATOR V-356B	LV-3561B	0..100%	—
PIC-3602	FUEL GAS MANIFOLD PRESSURE	PV-3731 / PV-3732	0..10 barg	—
TIC-3710	E-371 OUTLET TEMPERATURE	TV-3710	-50..200°C	—

LOOP	DESCRIPTION	ELEMENT	RANGE	SP
TIC-3720	E-372 OUTLET TEMPERATURE	TV-3720	-50..200°C	—
TIC-3721	X-373 HP FUEL GAS TEMPERATURE	X-373	-15..120°C	—

10.6 Operating Interface

10.6.1 PCS Operating Screens – Unit 370

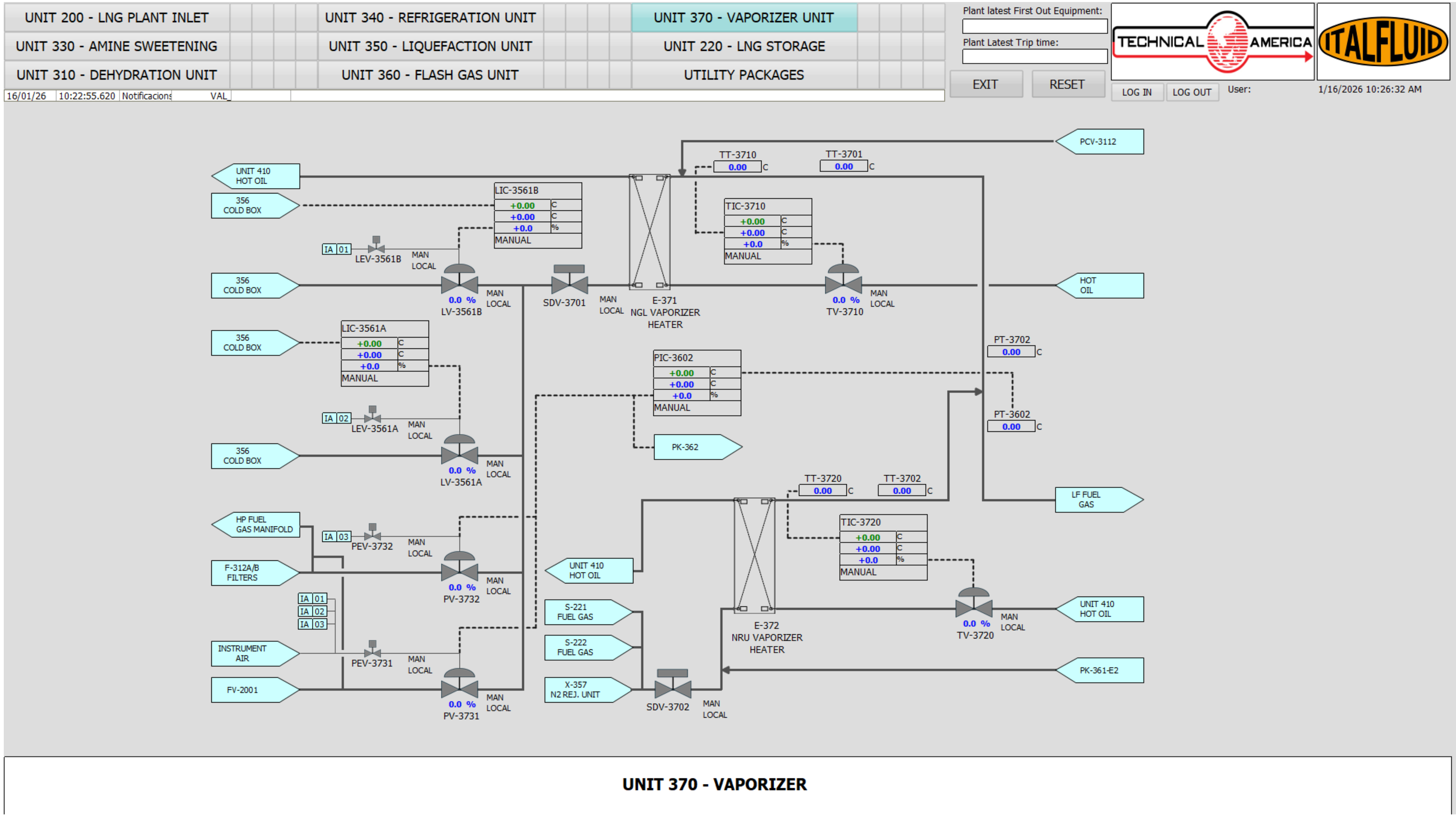


Figure 10.2: UNIT 360 - HMI Operating Screen

10.6.2 Critical Alarms and Trips

Table 10.9 — Unit 370 – Critical Alarms

ALARM	INSTRUMENT	TYPE	SP	DESCRIPTION
LALL-3561A	LT-3561A	LSD	—	LOW LOW LIQUID LEVEL ON V-356A
LALL-3561B	LT-3561B	LSD	—	LOW LOW LIQUID LEVEL ON V-356B
PAHH-3702	PT-3702	LSD	—	HIGH HIGH PRESSURE ON E-371
TAHH-X373	TAHH-X-373	LSD	—	HIGH HIGH TEMPERATURE ON X-373
TALL-3701	TT-3701	PSD	—	LOW LOW TEMPERATURE ON FUEL GAS
TALL-3702	TT-3702	PSD	—	LOW LOW TEMPERATURE ON FUEL GAS

10.6.3 Unit Cause and Effect Matrix

Table 10.10 — Unit 370 – Cause & Effect Matrix

ALARM	INSTRUMENT	LEV-3561A	LEV-3561B	X-373	SDV-3113	SDV-3701	PEV-3731	PEV-3732
LALL-3561A	LT-3561A	C						
LALL-3561B	LT-3561B		C					
PAHH-3702	PT-3702	C	C		T	T	T	T
TAHH-X373	TT-X373			T				

11 Plant Operating Procedures

11.1 Black Start

The Tendrara LNG plant is designed to operate as a self-generated power facility, where the availability, stability, and distribution of electrical power constitute a primary constraint during plant start-up. The power generation system comprises seven (7) gas engine generator sets, arranged into two distribution groups feeding the main plant busbars. In addition, the facility is equipped with two diesel generator sets dedicated to the Black Start procedure, providing the initial power source required to energize essential auxiliaries, close tie-breakers, and enable the sequential start-up of the gas engine generators in the absence of external power.

Depending on the selected busbar configuration (e.g., AB–CD, ABC–D, A–BCD), the electrical network topology changes; therefore, the start-up sequence of major consumers must be adapted accordingly. The selected configuration directly influences load distribution, generator dispatch, and the permissible sequence for starting high-demand equipment such as compressor packages and utility systems.

Compressor packages represent by far the largest electrical loads in the facility and therefore constitute the primary constraint during plant ramp-up. Several of these high-demand consumers are connected to specific bus sections; consequently, the feasibility of starting a given compression train is not only a process decision but fundamentally an electrical stability decision.

Although all compressor packages are started through variable frequency drives (VFDs), which reduce direct inrush currents, they still impose significant dynamic load ramps on the generation system. Therefore, plant ramp-up must be validated through start-up configuration matrices to ensure that adequate operating margins between total generation capacity and process demand are maintained at every step.

The start-up sequence is not fixed and depends on the combination of generator sets available (including units temporarily out of service for maintenance) together with the compressor trains selected for operation. In addition, operating philosophy requires a balanced distribution of running hours among generators and major rotating equipment; consequently, continuous rotation of units is operationally mandatory.

As a result, equipment availability and rotation requirements directly affect the electrical configuration and permissible load path. Operators must clearly understand that any change in generator status or compressor allocation modifies the applicable start-up matrix and sequence. The operating matrices presented in this chapter therefore provide a structured framework that must be adapted to each specific configuration to prevent generation instability, system collapse, or total plant blackout.

The ramp-up sequence illustrated in the operating matrix consists of twenty-three (23) progressive steps, beginning with the Black Start configuration and culminating in full plant load stabilization.

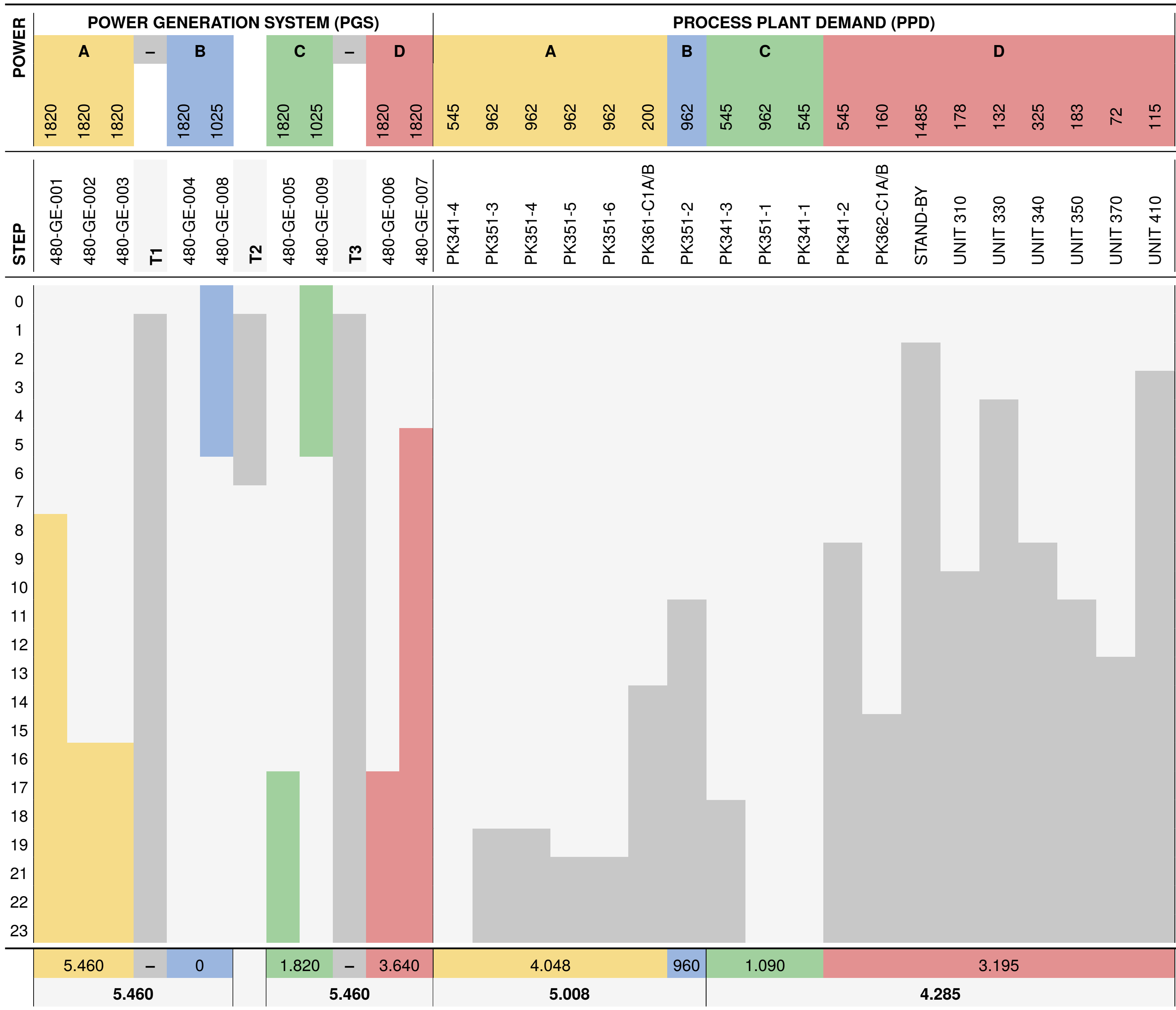
The sequence starts with the energization of the system using the two diesel generator sets (480-GE-008 and 480-GE-009), which are operated in parallel to establish a stable initial power source. Once the busbars are energized and the tie-breakers configured, essential auxiliary systems are progressively started, including air compression, hot oil circulation, dehydration, and gas sweetening units.

As process demand increases, gas engine generators are sequentially started and synchronized to the respective bus sections. Load transfer from diesel generators to gas generators is performed in a controlled manner to maintain adequate spinning reserve and voltage stability. Once sufficient gas generation capacity is available, the diesel generators are stopped and isolated from the system.

Throughout the sequence, tie-breaker positions are adjusted according to the selected busbar configuration (AB–CD), ensuring proper load distribution between sections and maintaining electrical stability. High-demand equipment, particularly compressor packages, is introduced progressively in coordination with available generation capacity.

The final steps of the sequence involve the staged start-up of the main compression trains and process units, cold box cool-down at controlled temperature gradients, activation of the nitrogen rejection system, and gradual increase of plant throughput.

At every step, the operating matrix verifies that total power generation (PGS) exceeds process plant demand (PPD) with adequate operating margin, thereby preventing overload conditions and ensuring a stable and secure ramp-up toward full plant capacity.



OPERATING POINTS			OPERATING PROCEDURE			
PGS [kW]	**PPD [kW]**	**OP [%]**	**STEP**	**UNIT**	**TAG**	**ACTION**
0	0	0	0	480	480-GE-008/009	START BOTH DIESEL GENERATORS ON SYNC
2.050	0	0	1	480	480-GE-008/009	TI/T2/T3 CLOSED - OPERATING IN PARALLEL.
2.050	855	20	2	500	PK-501A/B/C	START AIR COMPRESSOR PACKAGE AND STAND BY
2.050	970	20	3	410	B-411A/B	START HOT OIL PACKAGE
2.050	1.102	20	4	330	P-332A/B	START GAS SWEETENING UNIT
3.870	1.102	20	5	480	480-GE-005/006/007	START ONE OF THREE
1.820	1.102	20	6	480	480-GE-008/009	STOP BOTH DIESEL GENSETS
1.820	1.102	20	7	480		OPEN TIE-IN BREAKER T2
3.640	1.102	40	8	480	480-GE-001/002/003/004	START ONE OF FOUR
3.640	1.177	40	9	340	PK341-2/3/4	START ONE OF THREE
2.050	1.955	20	10	310	T-311A/B	START DEHYDRATION UNIT
3.640	2.987	40	11	350	PK351-2/3/4/5/6	START ONE OF FIVE
3.640	2.987	40	12	330	XC-3561B	COLD BOX COOL DOWN (3 °C/h)
3.640	3.059	40	13	370	X-357	START N2 REJECTION REBOILER
3.642	3.159	40	14	360	PK361-C1A/B	START ONE OF TWO
3.642	3.329	40	15	360	PK362-C1A/B	START ONE OF TWO
7.280	3.399	60	16	480	480-GE-002/003/004	START TWO OF THREE
10.920	3.399	80	17	480	480-GE-005/006	START TWO OF TWO
10.920	3.944	80	18	340	PK341-3/4	START ONE OF TWO
10.920	5.903	80	19	350	PK351-3/4/5/6	START TWO OF FOUR
10.920	7.967	100	21	350	PK351-1/5/6	START TWO OF THREE
10.920	8.512	100	22	340	PK341-4	START PK-341-4
10.920	9.293	100	23	200	FIC-2001	INCREASE PLANT CAPACITY
10.920	**9.293**	**100**				

Table 11.1 — Ramp-up operating configuration matrix – power generation and process plant operating points - BUSBAR configuration AB-CD

11.2 Normal Start-up Sequence

11.3 Normal Operation Tasks

11.4 Normal Shutdown